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Staple cartridge electrical contacts

Abstract

A staple cartridge assembly for use with a surgical stapler and surgical stapling systems are disclosed. The staple cartridge comprises a cartridge body comprising a proximal end. A plurality of staples. A plurality of electrical contacts positioned at the proximal end of the cartridge body and electrically coupleable to the electrical connector upon the installation of the replaceable staple cartridge assembly in the end effector along a distal-to-proximal installation motion and electrically decoupleable from the electrical connector upon removal of the replaceable staple cartridge assembly from the end effector along a proximal-to-distal removal motion.

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4500024	12/1984	DiGiovanni et al.	N/A	N/A
D278081	12/1984	Green	N/A	N/A
4503842	12/1984	Takayama	N/A	N/A
4505272	12/1984	Utyamyshev et al.	N/A	N/A
4505273	12/1984	Braun et al.	N/A	N/A
4505414	12/1984	Filipi	N/A	N/A
4506671	12/1984	Green	N/A	N/A
4512038	12/1984	Alexander et al.	N/A	N/A
4514477	12/1984	Kobayashi	N/A	N/A
4520817	12/1984	Green	N/A	N/A

4522327	12/1984	Korthoff et al.	N/A	N/A
4523707	12/1984	Blake, III et al.	N/A	N/A
4526174	12/1984	Froehlich	N/A	N/A
4527724	12/1984	Chow et al.	N/A	N/A
4530357	12/1984	Pawloski et al.	N/A	N/A
4530453	12/1984	Green	N/A	N/A
4531522	12/1984	Bedi et al.	N/A	N/A
4532927	12/1984	Miksza, Jr.	N/A	N/A
4540202	12/1984	Amphoux et al.	N/A	N/A
4548202	12/1984	Duncan	N/A	N/A
4556058	12/1984	Green	N/A	N/A
4560915	12/1984	Soultanian	N/A	N/A
4565109	12/1985	Tsay	N/A	N/A
4565189	12/1985	Mabuchi	N/A	N/A
4566620	12/1985	Green et al.	N/A	N/A
4569346	12/1985	Poirier	N/A	N/A
4569469	12/1985	Mongeon et al.	N/A	N/A
4571213	12/1985	Ishimoto	N/A	N/A
4573468	12/1985	Conta et al.	N/A	N/A
4573469	12/1985	Golden et al.	N/A	N/A
4573622	12/1985	Green et al.	N/A	N/A
4576165	12/1985	Green et al.	N/A	N/A
4576167	12/1985	Noiles	N/A	N/A
4580712	12/1985	Green	N/A	N/A
4585153	12/1985	Failla et al.	N/A	N/A
4586501	12/1985	Claracq	N/A	N/A
4586502	12/1985	Bedi et al.	N/A	N/A
4589416	12/1985	Green	N/A	N/A
4589582	12/1985	Bilotti	N/A	N/A
4589870	12/1985	Citrin et al.	N/A	N/A
4591085	12/1985	Di Giovanni	N/A	N/A
RE32214	12/1985	Schramm	N/A	N/A
4597753	12/1985	Turley	N/A	N/A
4600037	12/1985	Hatten	N/A	N/A
4604786	12/1985	Howie, Jr.	N/A	N/A
4605001	12/1985	Rothfuss et al.	N/A	N/A
4605004	12/1985	Di Giovanni et al.	N/A	N/A
4606343	12/1985	Conta et al.	N/A	N/A
4607636	12/1985	Kula et al.	N/A	N/A
4607638	12/1985	Crainich	N/A	N/A
4608980	12/1985	Aihara	N/A	N/A
4608981	12/1985	Rothfuss et al.	N/A	N/A
4610250	12/1985	Green	N/A	N/A
4610383	12/1985	Rothfuss et al.	N/A	N/A
4612933	12/1985	Brinkerhoff et al.	N/A	N/A
D286180	12/1985	Korthoff	N/A	N/A
D286442	12/1985	Korthoff et al.	N/A	N/A
4617893	12/1985	Donner et al.	N/A	N/A
4617914	12/1985	Ueda	N/A	N/A
4617935	12/1985	Cartmell et al.	N/A	N/A

4619262	12/1985	Taylor	N/A	N/A
4619391	12/1985	Sharkany et al.	N/A	N/A
4624401	12/1985	Gassner et al.	N/A	N/A
D287278	12/1985	Spreckelmeier	N/A	N/A
4628459	12/1985	Shinohara et al.	N/A	N/A
4628636	12/1985	Folger	N/A	N/A
4629107	12/1985	Fedotov et al.	N/A	N/A
4632290	12/1985	Green et al.	N/A	N/A
4633861	12/1986	Chow et al.	N/A	N/A
4633874	12/1986	Chow et al.	N/A	N/A
4634419	12/1986	Kreizman et al.	N/A	N/A
4635638	12/1986	Weintraub et al.	N/A	N/A
4641076	12/1986	Linden	N/A	N/A
4642618	12/1986	Johnson et al.	N/A	N/A
4642738	12/1986	Meller	N/A	N/A
4643173	12/1986	Bell et al.	N/A	N/A
4643731	12/1986	Eckenhoff	N/A	N/A
4646722	12/1986	Silverstein et al.	N/A	N/A
4646745	12/1986	Noiles	N/A	N/A
4651734	12/1986	Doss et al.	N/A	N/A
4652820	12/1986	Maresca	N/A	N/A
4654028	12/1986	Suma	N/A	N/A
4655222	12/1986	Florez et al.	N/A	N/A
4662555	12/1986	Thornton	N/A	N/A
4663874	12/1986	Sano et al.	N/A	N/A
4664305	12/1986	Blake, III et al.	N/A	N/A
4665916	12/1986	Green	N/A	N/A
4667674	12/1986	Korthoff et al.	N/A	N/A
4669647	12/1986	Storace	N/A	N/A
4671278	12/1986	Chin	N/A	N/A
4671280	12/1986	Dorband et al.	N/A	N/A
4671445	12/1986	Barker et al.	N/A	N/A
4672964	12/1986	Dee et al.	N/A	N/A
4675944	12/1986	Wells	N/A	N/A
4676245	12/1986	Fukuda	N/A	N/A
4679460	12/1986	Yoshigai	N/A	N/A
4679719	12/1986	Kramer	N/A	N/A
4684051	12/1986	Akopov et al.	N/A	N/A
4688555	12/1986	Wardle	N/A	N/A
4691703	12/1986	Auth et al.	N/A	N/A
4693248	12/1986	Failla	N/A	N/A
4698579	12/1986	Richter et al.	N/A	N/A
4700703	12/1986	Resnick et al.	N/A	N/A
4705038	12/1986	Sjostrom et al.	N/A	N/A
4708141	12/1986	Inoue et al.	N/A	N/A
4709120	12/1986	Pearson	N/A	N/A
4715520	12/1986	Roehr, Jr. et al.	N/A	N/A
4719917	12/1987	Barrows et al.	N/A	N/A
4721099	12/1987	Chikama	N/A	N/A
4722340	12/1987	Takayama et al.	N/A	N/A

4724840	12/1987	McVay et al.	N/A	N/A
4726247	12/1987	Hormann	N/A	N/A
4727308	12/1987	Huljak et al.	N/A	N/A
4728020	12/1987	Green et al.	N/A	N/A
4728876	12/1987	Mongeon et al.	N/A	N/A
4729260	12/1987	Dudden	N/A	N/A
4730726	12/1987	Holzwarth	N/A	N/A
4741336	12/1987	Failla et al.	N/A	N/A
4743214	12/1987	Tai-Cheng	N/A	N/A
4744363	12/1987	Hasson	N/A	N/A
4747820	12/1987	Hornlein et al.	N/A	N/A
4750902	12/1987	Wuchinich et al.	N/A	N/A
4752024	12/1987	Green et al.	N/A	N/A
4754909	12/1987	Barker et al.	N/A	N/A
4755070	12/1987	Cerutti	N/A	N/A
4761326	12/1987	Barnes et al.	N/A	N/A
4763669	12/1987	Jaeger	N/A	N/A
4767044	12/1987	Green	N/A	N/A
D297764	12/1987	Hunt et al.	N/A	N/A
4773420	12/1987	Green	N/A	N/A
4777780	12/1987	Holzwarth	N/A	N/A
4781186	12/1987	Simpson et al.	N/A	N/A
4784137	12/1987	Kulik et al.	N/A	N/A
4787387	12/1987	Burbank, III et al.	N/A	N/A
4788485	12/1987	Kawagishi et al.	N/A	N/A
D298967	12/1987	Hunt	N/A	N/A
4788978	12/1987	Strekopytov et al.	N/A	N/A
4790225	12/1987	Moody et al.	N/A	N/A
4790314	12/1987	Weaver	N/A	N/A
4805617	12/1988	Bedi et al.	N/A	N/A
4805823	12/1988	Rothfuss	N/A	N/A
4807628	12/1988	Peters et al.	N/A	N/A
4809695	12/1988	Gwathmey et al.	N/A	N/A
4815460	12/1988	Porat et al.	N/A	N/A
4817643	12/1988	Olson	N/A	N/A
4817847	12/1988	Redtenbacher et al.	N/A	N/A
4819495	12/1988	Hormann	N/A	N/A
4819853	12/1988	Green	N/A	N/A
4821939	12/1988	Green	N/A	N/A
4827552	12/1988	Bojar et al.	N/A	N/A
4827911	12/1988	Broadwin et al.	N/A	N/A
4828542	12/1988	Hermann	N/A	N/A
4828944	12/1988	Yabe et al.	N/A	N/A
4830855	12/1988	Stewart	N/A	N/A
4832158	12/1988	Farrar et al.	N/A	N/A
4833937	12/1988	Nagano	N/A	N/A
4834096	12/1988	Oh et al.	N/A	N/A
4834720	12/1988	Blinkhorn	N/A	N/A
4838859	12/1988	Strassmann	N/A	N/A
4844068	12/1988	Arata et al.	N/A	N/A

4848637	12/1988	Pruitt	N/A	N/A
4856078	12/1988	Konopka	N/A	N/A
4860644	12/1988	Kohl et al.	N/A	N/A
4862891	12/1988	Smith	N/A	N/A
4863423	12/1988	Wallace	N/A	N/A
4865030	12/1988	Polyak	N/A	N/A
4868530	12/1988	Ahs	N/A	N/A
4868958	12/1988	Suzuki et al.	N/A	N/A
4869414	12/1988	Green et al.	N/A	N/A
4869415	12/1988	Fox	N/A	N/A
4870966	12/1988	Dellon et al.	N/A	N/A
4873977	12/1988	Avant et al.	N/A	N/A
4875486	12/1988	Rapoport et al.	N/A	N/A
4880015	12/1988	Nierman	N/A	N/A
4890613	12/1989	Golden et al.	N/A	N/A
4892244	12/1989	Fox et al.	N/A	N/A
4893622	12/1989	Green et al.	N/A	N/A
4894051	12/1989	Shiber	N/A	N/A
4896584	12/1989	Stoll et al.	N/A	N/A
4896678	12/1989	Ogawa	N/A	N/A
4900303	12/1989	Lemelson	N/A	N/A
4903697	12/1989	Resnick et al.	N/A	N/A
4909789	12/1989	Taguchi et al.	N/A	N/A
4915100	12/1989	Green	N/A	N/A
4919039	12/1989	Nutter	N/A	N/A
4919679	12/1989	Averill et al.	N/A	N/A
4921479	12/1989	Grayzel	N/A	N/A
4925082	12/1989	Kim	N/A	N/A
4928699	12/1989	Sasai	N/A	N/A
4930503	12/1989	Pruitt	N/A	N/A
4930674	12/1989	Barak	N/A	N/A
4931047	12/1989	Broadwin et al.	N/A	N/A
4931737	12/1989	Hishiki	N/A	N/A
4932960	12/1989	Green et al.	N/A	N/A
4933800	12/1989	Yang	N/A	N/A
4933843	12/1989	Scheller et al.	N/A	N/A
D309350	12/1989	Sutherland et al.	N/A	N/A
4938408	12/1989	Bedi et al.	N/A	N/A
4941623	12/1989	Pruitt	N/A	N/A
4943182	12/1989	Hoblingre	N/A	N/A
4944443	12/1989	Oddsens et al.	N/A	N/A
4946067	12/1989	Kelsall	N/A	N/A
4948327	12/1989	Crupi, Jr.	N/A	N/A
4949707	12/1989	LeVahn et al.	N/A	N/A
4949927	12/1989	Madocks et al.	N/A	N/A
4950268	12/1989	Rink	N/A	N/A
4951860	12/1989	Peters et al.	N/A	N/A
4951861	12/1989	Schulze et al.	N/A	N/A
4954960	12/1989	Lo et al.	N/A	N/A
4955959	12/1989	Tompkins et al.	N/A	N/A

4957212	12/1989	Duck et al.	N/A	N/A
4962681	12/1989	Yang	N/A	N/A
4962877	12/1989	Hervas	N/A	N/A
4964559	12/1989	Deniega et al.	N/A	N/A
4964863	12/1989	Kanshin et al.	N/A	N/A
4965709	12/1989	Ngo	N/A	N/A
4970656	12/1989	Lo et al.	N/A	N/A
4973274	12/1989	Hirukawa	N/A	N/A
4973302	12/1989	Armour et al.	N/A	N/A
4976173	12/1989	Yang	N/A	N/A
4978049	12/1989	Green	N/A	N/A
4978333	12/1989	Broadwin et al.	N/A	N/A
4979952	12/1989	Kubota et al.	N/A	N/A
4984564	12/1990	Yuen	N/A	N/A
4986808	12/1990	Broadwin et al.	N/A	N/A
4987049	12/1990	Komamura et al.	N/A	N/A
4988334	12/1990	Hornlein et al.	N/A	N/A
4995877	12/1990	Ams et al.	N/A	N/A
4995959	12/1990	Metzner	N/A	N/A
4996975	12/1990	Nakamura	N/A	N/A
5001649	12/1990	Lo et al.	N/A	N/A
5002543	12/1990	Bradshaw et al.	N/A	N/A
5002553	12/1990	Shiber	N/A	N/A
5005754	12/1990	Van Overloop	N/A	N/A
5009222	12/1990	Her	N/A	N/A
5009661	12/1990	Michelson	N/A	N/A
5012411	12/1990	PolICASTRO et al.	N/A	N/A
D316875	12/1990	Momot et al.	N/A	N/A
5014898	12/1990	Heidrich	N/A	N/A
5014899	12/1990	Presty et al.	N/A	N/A
5015227	12/1990	Broadwin et al.	N/A	N/A
5018515	12/1990	Gilman	N/A	N/A
5018657	12/1990	Pedlick et al.	N/A	N/A
5019077	12/1990	De Bastiani et al.	N/A	N/A
5024652	12/1990	Dumenek et al.	N/A	N/A
5024671	12/1990	Tu et al.	N/A	N/A
5025559	12/1990	McCullough	N/A	N/A
5027834	12/1990	Pruitt	N/A	N/A
5030226	12/1990	Green et al.	N/A	N/A
5031814	12/1990	Tompkins et al.	N/A	N/A
5033552	12/1990	Hu	N/A	N/A
5035040	12/1990	Kerrigan et al.	N/A	N/A
5037018	12/1990	Matsuda et al.	N/A	N/A
5038109	12/1990	Goble et al.	N/A	N/A
5038247	12/1990	Kelley et al.	N/A	N/A
5040715	12/1990	Green et al.	N/A	N/A
5042707	12/1990	Taheri	N/A	N/A
5056953	12/1990	Marot et al.	N/A	N/A
5060658	12/1990	Dejter, Jr. et al.	N/A	N/A
5061269	12/1990	Muller	N/A	N/A

5062491	12/1990	Takeshima et al.	N/A	N/A
5062563	12/1990	Green et al.	N/A	N/A
5065929	12/1990	Schulze et al.	N/A	N/A
5071052	12/1990	Rodak et al.	N/A	N/A
5071430	12/1990	de Salis et al.	N/A	N/A
5074454	12/1990	Peters	N/A	N/A
5077506	12/1990	Krause	N/A	N/A
5079006	12/1991	Urquhart	N/A	N/A
5080556	12/1991	Carreno	N/A	N/A
5083695	12/1991	Foslien et al.	N/A	N/A
5084057	12/1991	Green et al.	N/A	N/A
5088979	12/1991	Filipi et al.	N/A	N/A
5088997	12/1991	Delahuerga et al.	N/A	N/A
5089606	12/1991	Cole et al.	N/A	N/A
5094247	12/1991	Hernandez et al.	N/A	N/A
5098004	12/1991	Kerrigan	N/A	N/A
5098360	12/1991	Hirota	N/A	N/A
5100042	12/1991	Gravener et al.	N/A	N/A
5100420	12/1991	Green et al.	N/A	N/A
5100422	12/1991	Berguer et al.	N/A	N/A
5104025	12/1991	Main et al.	N/A	N/A
5104397	12/1991	Vasconcelos et al.	N/A	N/A
5104400	12/1991	Berguer et al.	N/A	N/A
5106008	12/1991	Tompkins et al.	N/A	N/A
5108368	12/1991	Hammerslag et al.	N/A	N/A
5109722	12/1991	Hufnagle et al.	N/A	N/A
5111987	12/1991	Moeinzadeh et al.	N/A	N/A
5116349	12/1991	Aranyi	N/A	N/A
D327323	12/1991	Hunt	N/A	N/A
5119009	12/1991	McCaleb et al.	N/A	N/A
5122156	12/1991	Granger et al.	N/A	N/A
5124990	12/1991	Williamson	N/A	N/A
5129570	12/1991	Schulze et al.	N/A	N/A
5135483	12/1991	Wagner et al.	N/A	N/A
5137198	12/1991	Nobis et al.	N/A	N/A
5139513	12/1991	Segato	N/A	N/A
5141144	12/1991	Foslien et al.	N/A	N/A
5142932	12/1991	Moya et al.	N/A	N/A
5151102	12/1991	Kamiyama et al.	N/A	N/A
5155941	12/1991	Takahashi et al.	N/A	N/A
5156151	12/1991	Imran	N/A	N/A
5156315	12/1991	Green et al.	N/A	N/A
5156609	12/1991	Nakao et al.	N/A	N/A
5156614	12/1991	Green et al.	N/A	N/A
5158222	12/1991	Green et al.	N/A	N/A
5158567	12/1991	Green	N/A	N/A
D330699	12/1991	Gill	N/A	N/A
5163598	12/1991	Peters et al.	N/A	N/A
5163842	12/1991	Nonomura	N/A	N/A
5164652	12/1991	Johnson et al.	N/A	N/A

5168605	12/1991	Bartlett	N/A	N/A
5170925	12/1991	Madden et al.	N/A	N/A
5171247	12/1991	Hughett et al.	N/A	N/A
5171249	12/1991	Stefanchik et al.	N/A	N/A
5171253	12/1991	Klieman	N/A	N/A
5173053	12/1991	Swanson et al.	N/A	N/A
5173133	12/1991	Morin et al.	N/A	N/A
5176677	12/1992	Wuchinich	N/A	N/A
5176688	12/1992	Narayan et al.	N/A	N/A
5180375	12/1992	Feibus	N/A	N/A
5181514	12/1992	Solomon et al.	N/A	N/A
5187422	12/1992	Izenbaard et al.	N/A	N/A
5188102	12/1992	Idemoto et al.	N/A	N/A
5188111	12/1992	Yates et al.	N/A	N/A
5188126	12/1992	Fabian et al.	N/A	N/A
5190517	12/1992	Zieve et al.	N/A	N/A
5190544	12/1992	Chapman et al.	N/A	N/A
5190560	12/1992	Woods et al.	N/A	N/A
5190657	12/1992	Heagle et al.	N/A	N/A
5192288	12/1992	Thompson et al.	N/A	N/A
5193731	12/1992	Aranyi	N/A	N/A
5195505	12/1992	Josefsen	N/A	N/A
5195968	12/1992	Lundquist et al.	N/A	N/A
5197648	12/1992	Gingold	N/A	N/A
5197649	12/1992	Bessler et al.	N/A	N/A
5197966	12/1992	Sommerkamp	N/A	N/A
5197970	12/1992	Green et al.	N/A	N/A
5200280	12/1992	Karasa	N/A	N/A
5201750	12/1992	Hocherl et al.	N/A	N/A
5205459	12/1992	Brinkerhoff et al.	N/A	N/A
5207672	12/1992	Roth et al.	N/A	N/A
5207697	12/1992	Carusillo et al.	N/A	N/A
5209747	12/1992	Knoepfler	N/A	N/A
5209756	12/1992	Seedhom et al.	N/A	N/A
5211649	12/1992	Kohler et al.	N/A	N/A
5211655	12/1992	Hasson	N/A	N/A
5217457	12/1992	Delahuerga et al.	N/A	N/A
5217478	12/1992	Rexroth	N/A	N/A
5219111	12/1992	Bilotti et al.	N/A	N/A
5220269	12/1992	Chen et al.	N/A	N/A
5221036	12/1992	Takase	N/A	N/A
5221281	12/1992	Klicek	N/A	N/A
5222945	12/1992	Basnight	N/A	N/A
5222963	12/1992	Brinkerhoff et al.	N/A	N/A
5222975	12/1992	Crainich	N/A	N/A
5222976	12/1992	Yoon	N/A	N/A
5223675	12/1992	Taft	N/A	N/A
D337962	12/1992	Avitan	N/A	N/A
D338729	12/1992	Sprecklemeier et al.	N/A	N/A
5234447	12/1992	Kaster et al.	N/A	N/A

5236269	12/1992	Handy	N/A	N/A
5236424	12/1992	Imran	N/A	N/A
5236440	12/1992	Hlavacek	N/A	N/A
5236629	12/1992	Mahabadi et al.	N/A	N/A
5239981	12/1992	Anapliotis	N/A	N/A
5240163	12/1992	Stein et al.	N/A	N/A
5242456	12/1992	Nash et al.	N/A	N/A
5242457	12/1992	Akopov et al.	N/A	N/A
5244462	12/1992	Delahuerga et al.	N/A	N/A
5246156	12/1992	Rothfuss et al.	N/A	N/A
5246443	12/1992	Mai	N/A	N/A
5251801	12/1992	Ruckdeschel et al.	N/A	N/A
5253793	12/1992	Green et al.	N/A	N/A
5258007	12/1992	Spetzler et al.	N/A	N/A
5258008	12/1992	Wilk	N/A	N/A
5258009	12/1992	Conners	N/A	N/A
5258010	12/1992	Green et al.	N/A	N/A
5258012	12/1992	Luscombe et al.	N/A	N/A
5259366	12/1992	Reydel et al.	N/A	N/A
5259835	12/1992	Clark et al.	N/A	N/A
5260637	12/1992	Pizzi	N/A	N/A
5261135	12/1992	Mitchell	N/A	N/A
5261877	12/1992	Fine et al.	N/A	N/A
5261922	12/1992	Hood	N/A	N/A
5263629	12/1992	Trumbull et al.	N/A	N/A
5263937	12/1992	Shipp	N/A	N/A
5263973	12/1992	Cook	N/A	N/A
5264218	12/1992	Rogozinski	N/A	N/A
5268622	12/1992	Philipp	N/A	N/A
5269794	12/1992	Rexroth	N/A	N/A
5271543	12/1992	Grant et al.	N/A	N/A
5271544	12/1992	Fox et al.	N/A	N/A
RE34519	12/1993	Fox et al.	N/A	N/A
5275322	12/1993	Brinkerhoff et al.	N/A	N/A
5275323	12/1993	Schulze et al.	N/A	N/A
5275608	12/1993	Forman et al.	N/A	N/A
5279416	12/1993	Malec et al.	N/A	N/A
5281216	12/1993	Klicek	N/A	N/A
5281400	12/1993	Berry, Jr.	N/A	N/A
5282806	12/1993	Haber et al.	N/A	N/A
5282826	12/1993	Quadri	N/A	N/A
5282829	12/1993	Hermes	N/A	N/A
5284128	12/1993	Hart	N/A	N/A
5285381	12/1993	Iskarous et al.	N/A	N/A
5285945	12/1993	Brinkerhoff et al.	N/A	N/A
5286253	12/1993	Fucci	N/A	N/A
5289963	12/1993	McGarry et al.	N/A	N/A
5290271	12/1993	Jernberg	N/A	N/A
5290310	12/1993	Makower et al.	N/A	N/A
5291133	12/1993	Gokhale et al.	N/A	N/A

5292053	12/1993	Bilotti et al.	N/A	N/A
5293024	12/1993	Sugahara et al.	N/A	N/A
5297714	12/1993	Kramer	N/A	N/A
5300087	12/1993	Knoepfler	N/A	N/A
5302148	12/1993	Heinz	N/A	N/A
5303606	12/1993	Kokinda	N/A	N/A
5304204	12/1993	Bregen	N/A	N/A
D347474	12/1993	Olson	N/A	N/A
5307976	12/1993	Olson et al.	N/A	N/A
5308353	12/1993	Beurrier	N/A	N/A
5308358	12/1993	Bond et al.	N/A	N/A
5308576	12/1993	Green et al.	N/A	N/A
5309387	12/1993	Mori et al.	N/A	N/A
5309927	12/1993	Welch	N/A	N/A
5312023	12/1993	Green et al.	N/A	N/A
5312024	12/1993	Grant et al.	N/A	N/A
5312329	12/1993	Beaty et al.	N/A	N/A
5313935	12/1993	Kortenbach et al.	N/A	N/A
5313967	12/1993	Lieber et al.	N/A	N/A
5314424	12/1993	Nicholas	N/A	N/A
5314445	12/1993	Heidmueller et al.	N/A	N/A
5314466	12/1993	Stern et al.	N/A	N/A
5318221	12/1993	Green et al.	N/A	N/A
5318589	12/1993	Lichtman	N/A	N/A
5320627	12/1993	Sorensen et al.	N/A	N/A
D348930	12/1993	Olson	N/A	N/A
5326013	12/1993	Green et al.	N/A	N/A
5329923	12/1993	Lundquist	N/A	N/A
5330486	12/1993	Wilk	N/A	N/A
5330487	12/1993	Thornton et al.	N/A	N/A
5330502	12/1993	Hassler et al.	N/A	N/A
5331971	12/1993	Bales et al.	N/A	N/A
5332142	12/1993	Robinson et al.	N/A	N/A
5333422	12/1993	Warren et al.	N/A	N/A
5333772	12/1993	Rothfuss et al.	N/A	N/A
5333773	12/1993	Main et al.	N/A	N/A
5334183	12/1993	Wuchinich	N/A	N/A
5336130	12/1993	Ray	N/A	N/A
5336229	12/1993	Noda	N/A	N/A
5336232	12/1993	Green et al.	N/A	N/A
5338317	12/1993	Hasson et al.	N/A	N/A
5339799	12/1993	Kami et al.	N/A	N/A
5341724	12/1993	Vatel	N/A	N/A
5341807	12/1993	Nardella	N/A	N/A
5341810	12/1993	Dardel	N/A	N/A
5342380	12/1993	Hood	N/A	N/A
5342381	12/1993	Tidemand	N/A	N/A
5342385	12/1993	Norelli et al.	N/A	N/A
5342395	12/1993	Jarrett et al.	N/A	N/A
5342396	12/1993	Cook	N/A	N/A

5343382	12/1993	Hale et al.	N/A	N/A
5343391	12/1993	Mushabac	N/A	N/A
5344059	12/1993	Green et al.	N/A	N/A
5344060	12/1993	Gravener et al.	N/A	N/A
5344454	12/1993	Clarke et al.	N/A	N/A
5346504	12/1993	Ortiz et al.	N/A	N/A
5348259	12/1993	Blanco et al.	N/A	N/A
5350104	12/1993	Main et al.	N/A	N/A
5350355	12/1993	Sklar	N/A	N/A
5350388	12/1993	Epstein	N/A	N/A
5350391	12/1993	Iacovelli	N/A	N/A
5350400	12/1993	Esposito et al.	N/A	N/A
5352229	12/1993	Goble et al.	N/A	N/A
5352235	12/1993	Koros et al.	N/A	N/A
5352238	12/1993	Green et al.	N/A	N/A
5353798	12/1993	Sieben	N/A	N/A
5354215	12/1993	Viracola	N/A	N/A
5354250	12/1993	Christensen	N/A	N/A
5354303	12/1993	Spaeth et al.	N/A	N/A
5355897	12/1993	Pietrafitta et al.	N/A	N/A
5356006	12/1993	Alpern et al.	N/A	N/A
5356064	12/1993	Green et al.	N/A	N/A
5358506	12/1993	Green et al.	N/A	N/A
5358510	12/1993	Luscombe et al.	N/A	N/A
5359231	12/1993	Flowers et al.	N/A	N/A
D352780	12/1993	Glaeser et al.	N/A	N/A
5359993	12/1993	Slater et al.	N/A	N/A
5360305	12/1993	Kerrigan	N/A	N/A
5360428	12/1993	Hutchinson, Jr.	N/A	N/A
5361902	12/1993	Abidin et al.	N/A	N/A
5364001	12/1993	Bryan	N/A	N/A
5364002	12/1993	Green et al.	N/A	N/A
5364003	12/1993	Williamson, IV	N/A	N/A
5366133	12/1993	Geiste	N/A	N/A
5366134	12/1993	Green et al.	N/A	N/A
5366479	12/1993	McGarry et al.	N/A	N/A
5368015	12/1993	Wilk	N/A	N/A
5368592	12/1993	Stern et al.	N/A	N/A
5368599	12/1993	Hirsch et al.	N/A	N/A
5369565	12/1993	Chen et al.	N/A	N/A
5370645	12/1993	Klicek et al.	N/A	N/A
5372124	12/1993	Takayama et al.	N/A	N/A
5372596	12/1993	Klicek et al.	N/A	N/A
5372602	12/1993	Burke	N/A	N/A
5374277	12/1993	Hassler	N/A	N/A
5375588	12/1993	Yoon	N/A	N/A
5376095	12/1993	Ortiz	N/A	N/A
5379933	12/1994	Green et al.	N/A	N/A
5381649	12/1994	Webb	N/A	N/A
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5381943	12/1994	Allen et al.	N/A	N/A
5382247	12/1994	Cimino et al.	N/A	N/A
5383460	12/1994	Jang et al.	N/A	N/A
5383738	12/1994	Herbermann	N/A	N/A
5383874	12/1994	Jackson et al.	N/A	N/A
5383880	12/1994	Hooven	N/A	N/A
5383881	12/1994	Green et al.	N/A	N/A
5383882	12/1994	Buess et al.	N/A	N/A
5383888	12/1994	Zvenyatsky et al.	N/A	N/A
5383895	12/1994	Holmes et al.	N/A	N/A
5388568	12/1994	van der Heide	N/A	N/A
5388748	12/1994	Davignon et al.	N/A	N/A
5389072	12/1994	Imran	N/A	N/A
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5389102	12/1994	Green et al.	N/A	N/A
5389104	12/1994	Hahnen et al.	N/A	N/A
5391180	12/1994	Tovey et al.	N/A	N/A
5392979	12/1994	Green et al.	N/A	N/A
5395030	12/1994	Kuramoto et al.	N/A	N/A
5395033	12/1994	Byrne et al.	N/A	N/A
5395034	12/1994	Allen et al.	N/A	N/A
5395312	12/1994	Desai	N/A	N/A
5395384	12/1994	Duthoit et al.	N/A	N/A
5397046	12/1994	Savage et al.	N/A	N/A
5397324	12/1994	Carroll et al.	N/A	N/A
5400267	12/1994	Denen et al.	N/A	N/A
5403276	12/1994	Schechter et al.	N/A	N/A
5403312	12/1994	Yates et al.	N/A	N/A
5404106	12/1994	Matsuda	N/A	N/A
5404870	12/1994	Brinkerhoff et al.	N/A	N/A
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5405072	12/1994	Zlock et al.	N/A	N/A
5405073	12/1994	Porter	N/A	N/A
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5405360	12/1994	Tovey	N/A	N/A
5407293	12/1994	Crainich	N/A	N/A
5408409	12/1994	Glassman et al.	N/A	N/A
5409498	12/1994	Braddock et al.	N/A	N/A
5409703	12/1994	McAnalley et al.	N/A	N/A
D357981	12/1994	Green et al.	N/A	N/A
5411481	12/1994	Allen et al.	N/A	N/A
5411508	12/1994	Bessler et al.	N/A	N/A
5413107	12/1994	Oakley et al.	N/A	N/A
5413267	12/1994	Solyntjes et al.	N/A	N/A
5413268	12/1994	Green	227/176.1	A61B 17/07207
5413272	12/1994	Green et al.	N/A	N/A
5413573	12/1994	Koivukangas	N/A	N/A
5415334	12/1994	Williamson et al.	N/A	N/A
5415335	12/1994	Knodell, Jr.	N/A	N/A

5417203	12/1994	Tovey et al.	N/A	N/A
5417361	12/1994	Williamson, IV	N/A	N/A
5419766	12/1994	Chang et al.	N/A	N/A
5421829	12/1994	Olichney et al.	N/A	N/A
5422567	12/1994	Matsunaga	N/A	N/A
5423471	12/1994	Mastri et al.	N/A	N/A
5423809	12/1994	Klicek	N/A	N/A
5423835	12/1994	Green et al.	N/A	N/A
5425355	12/1994	Kulick	N/A	N/A
5425745	12/1994	Green et al.	N/A	N/A
5427298	12/1994	Tegtmeier	N/A	N/A
5431322	12/1994	Green	227/176.1	A61B 17/07207
5431323	12/1994	Smith et al.	N/A	N/A
5431645	12/1994	Smith et al.	N/A	N/A
5431654	12/1994	Nic	N/A	N/A
5431666	12/1994	Sauer et al.	N/A	N/A
5431668	12/1994	Burbank, III et al.	N/A	N/A
5433721	12/1994	Hooven et al.	N/A	N/A
5437681	12/1994	Meade et al.	N/A	N/A
5438302	12/1994	Goble	N/A	N/A
5438997	12/1994	Sieben et al.	N/A	N/A
5439155	12/1994	Viola	N/A	N/A
5439156	12/1994	Grant et al.	N/A	N/A
5439479	12/1994	Shichman et al.	N/A	N/A
5441191	12/1994	Linden	N/A	N/A
5441193	12/1994	Gravener	N/A	N/A
5441483	12/1994	Avitall	N/A	N/A
5441494	12/1994	Ortiz	N/A	N/A
5441499	12/1994	Fritzschi	N/A	N/A
5443197	12/1994	Malis et al.	N/A	N/A
5443198	12/1994	Viola et al.	N/A	N/A
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5444113	12/1994	Sinclair et al.	N/A	N/A
5445155	12/1994	Sieben	N/A	N/A
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5445604	12/1994	Lang	N/A	N/A
5445644	12/1994	Pietrafitta et al.	N/A	N/A
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5447417	12/1994	Kuhl et al.	N/A	N/A
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5449365	12/1994	Green et al.	N/A	N/A
5449370	12/1994	Vaitekunas	N/A	N/A
5452836	12/1994	Huitema et al.	N/A	N/A
5452837	12/1994	Williamson, IV et al.	N/A	N/A
5454378	12/1994	Palmer et al.	N/A	N/A
5454822	12/1994	Schob et al.	N/A	N/A
5454824	12/1994	Fontayne et al.	N/A	N/A

5454827	12/1994	Aust et al.	N/A	N/A
5456401	12/1994	Green et al.	N/A	N/A
5456917	12/1994	Wise et al.	N/A	N/A
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5458579	12/1994	Chodorow et al.	N/A	N/A
5462215	12/1994	Viola et al.	N/A	N/A
5464013	12/1994	Lemelson	N/A	N/A
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5464300	12/1994	Crainich	N/A	N/A
5465819	12/1994	Weilant et al.	N/A	N/A
5465894	12/1994	Clark et al.	N/A	N/A
5465895	12/1994	Knodel et al.	N/A	N/A
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5470006	12/1994	Rodak	N/A	N/A
5470007	12/1994	Plyley et al.	N/A	N/A
5470008	12/1994	Rodak	N/A	N/A
5470009	12/1994	Rodak	N/A	N/A
5470010	12/1994	Rothfuss et al.	N/A	N/A
5471129	12/1994	Mann	N/A	N/A
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5472442	12/1994	Klicek	N/A	N/A
5473204	12/1994	Temple	N/A	N/A
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5476206	12/1994	Green et al.	N/A	N/A
5476479	12/1994	Green et al.	N/A	N/A
5476481	12/1994	Schondorf	N/A	N/A
5478003	12/1994	Green et al.	N/A	N/A
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5480409	12/1995	Riza	N/A	N/A
5482197	12/1995	Green et al.	N/A	N/A
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5484095	12/1995	Green et al.	N/A	N/A
5484398	12/1995	Stoddard	N/A	N/A
5484451	12/1995	Akopov et al.	N/A	N/A
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5490819	12/1995	Nicholas et al.	N/A	N/A
5492671	12/1995	Krafft	N/A	N/A
5496312	12/1995	Klicek	N/A	N/A
5496317	12/1995	Goble et al.	N/A	N/A
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5507426	12/1995	Young et al.	N/A	N/A
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5509596	12/1995	Green et al.	N/A	N/A
5509916	12/1995	Taylor	N/A	N/A
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5511564	12/1995	Wilk	N/A	N/A
5514129	12/1995	Smith	N/A	N/A
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5518164	12/1995	Hooven	N/A	N/A
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5544802	12/1995	Crainich	N/A	N/A
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5551622	12/1995	Yoon	N/A	N/A
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5554148	12/1995	Aebischer et al.	N/A	N/A
5554169	12/1995	Green et al.	N/A	N/A
5556020	12/1995	Hou	N/A	N/A
5556416	12/1995	Clark et al.	N/A	N/A
5558533	12/1995	Hashizawa et al.	N/A	N/A
5558665	12/1995	Kieturakis	N/A	N/A
5558671	12/1995	Yates	N/A	N/A
5560530	12/1995	Bolanos et al.	N/A	N/A
5560532	12/1995	DeFonzo et al.	N/A	N/A
5561881	12/1995	Klinger et al.	N/A	N/A
5562239	12/1995	Boiarski et al.	N/A	N/A
5562241	12/1995	Knodel et al.	N/A	N/A
5562682	12/1995	Oberlin et al.	N/A	N/A
5562690	12/1995	Green et al.	N/A	N/A
5562694	12/1995	Sauer et al.	N/A	N/A
5562701	12/1995	Huitema et al.	N/A	N/A
5562702	12/1995	Huitema et al.	N/A	N/A
5563481	12/1995	Krause	N/A	N/A
5564615	12/1995	Bishop et al.	N/A	N/A
5569161	12/1995	Ebling et al.	N/A	N/A
5569270	12/1995	Weng	N/A	N/A
5569284	12/1995	Young et al.	N/A	N/A
5571090	12/1995	Sherts	N/A	N/A
5571100	12/1995	Goble et al.	N/A	N/A
5571116	12/1995	Bolanos et al.	N/A	N/A
5571285	12/1995	Chow et al.	N/A	N/A
5571488	12/1995	Beerstecher et al.	N/A	N/A
5573169	12/1995	Green et al.	N/A	N/A
5573543	12/1995	Akopov et al.	N/A	N/A
5574431	12/1995	McKeown et al.	N/A	N/A
5575054	12/1995	Klinzing et al.	N/A	N/A

5575789	12/1995	Bell et al.	N/A	N/A
5575799	12/1995	Bolanos et al.	N/A	N/A
5575803	12/1995	Cooper et al.	N/A	N/A
5575805	12/1995	Li	N/A	N/A
5577654	12/1995	Bishop	N/A	N/A
5578052	12/1995	Koros et al.	N/A	N/A
5579978	12/1995	Green et al.	N/A	N/A
5580067	12/1995	Hamblin et al.	N/A	N/A
5582611	12/1995	Tsuruta et al.	N/A	N/A
5582617	12/1995	Klieman et al.	N/A	N/A
5582907	12/1995	Pall	N/A	N/A
5583114	12/1995	Barrows et al.	N/A	N/A
5584425	12/1995	Savage et al.	N/A	N/A
5586711	12/1995	Plyley et al.	N/A	N/A
5588579	12/1995	Schnut et al.	N/A	N/A
5588580	12/1995	Paul et al.	N/A	N/A
5588581	12/1995	Conlon et al.	N/A	N/A
5591170	12/1996	Spievack et al.	N/A	N/A
5591187	12/1996	Dekel	N/A	N/A
5597107	12/1996	Knodel et al.	N/A	N/A
5599151	12/1996	Daum et al.	N/A	N/A
5599279	12/1996	Slotman et al.	N/A	N/A
5599344	12/1996	Paterson	N/A	N/A
5599350	12/1996	Schulze et al.	N/A	N/A
5599852	12/1996	Scopelianos et al.	N/A	N/A
5601224	12/1996	Bishop et al.	N/A	N/A
5601573	12/1996	Fogelberg et al.	N/A	N/A
5601604	12/1996	Vincent	N/A	N/A
5602449	12/1996	Krause et al.	N/A	N/A
5603443	12/1996	Clark et al.	N/A	N/A
5605272	12/1996	Witt et al.	N/A	N/A
5605273	12/1996	Hamblin et al.	N/A	N/A
D378500	12/1996	Nakai et al.	N/A	N/A
5607094	12/1996	Clark et al.	N/A	N/A
5607095	12/1996	Smith et al.	N/A	N/A
5607303	12/1996	Nakamura	N/A	N/A
5607433	12/1996	Polla et al.	N/A	N/A
5607436	12/1996	Pratt et al.	N/A	N/A
5607450	12/1996	Zvenyatsky et al.	N/A	N/A
5607474	12/1996	Athanasiou et al.	N/A	N/A
5609285	12/1996	Grant et al.	N/A	N/A
5609601	12/1996	Kolesa et al.	N/A	N/A
5611709	12/1996	McAnulty	N/A	N/A
5611813	12/1996	Lichtman	N/A	N/A
5613499	12/1996	Palmer et al.	N/A	N/A
5613937	12/1996	Garrison et al.	N/A	N/A
5613966	12/1996	Makower et al.	N/A	N/A
5614887	12/1996	Buchbinder	N/A	N/A
5615820	12/1996	Viola	N/A	N/A
5618294	12/1996	Aust et al.	N/A	N/A

5618303	12/1996	Marlow et al.	N/A	N/A
5618307	12/1996	Donlon et al.	N/A	N/A
5619992	12/1996	Guthrie et al.	N/A	N/A
5620289	12/1996	Curry	N/A	N/A
5620326	12/1996	Younker	N/A	N/A
5620415	12/1996	Lucey et al.	N/A	N/A
5620452	12/1996	Yoon	N/A	N/A
5624398	12/1996	Smith et al.	N/A	N/A
5624452	12/1996	Yates	N/A	N/A
5626587	12/1996	Bishop et al.	N/A	N/A
5626595	12/1996	Sklar et al.	N/A	N/A
5626979	12/1996	Mitsui et al.	N/A	N/A
5628446	12/1996	Geiste et al.	N/A	N/A
5628743	12/1996	Cimino	N/A	N/A
5628745	12/1996	Bek	N/A	N/A
5630539	12/1996	Plyley et al.	N/A	N/A
5630540	12/1996	Blewett	N/A	N/A
5630541	12/1996	Williamson, IV et al.	N/A	N/A
5630782	12/1996	Adair	N/A	N/A
5631973	12/1996	Green	N/A	N/A
5632432	12/1996	Schulze	227/176.1	A61B 17/07207
5632433	12/1996	Grant et al.	N/A	N/A
5633374	12/1996	Humphrey et al.	N/A	N/A
5634584	12/1996	Okorochoa et al.	N/A	N/A
5636779	12/1996	Palmer	N/A	N/A
5636780	12/1996	Green et al.	N/A	N/A
5637110	12/1996	Pennybacker et al.	N/A	N/A
5638582	12/1996	Klatt et al.	N/A	N/A
5639008	12/1996	Gallagher et al.	N/A	N/A
D381077	12/1996	Hunt	N/A	N/A
5643291	12/1996	Pier et al.	N/A	N/A
5643293	12/1996	Kogasaka et al.	N/A	N/A
5643294	12/1996	Tovey et al.	N/A	N/A
5643319	12/1996	Green et al.	N/A	N/A
5645209	12/1996	Green et al.	N/A	N/A
5647526	12/1996	Green et al.	N/A	N/A
5647869	12/1996	Goble et al.	N/A	N/A
5649937	12/1996	Bito et al.	N/A	N/A
5649956	12/1996	Jensen et al.	N/A	N/A
5651491	12/1996	Heaton et al.	N/A	N/A
5651762	12/1996	Bridges	N/A	N/A
5651821	12/1996	Uchida	N/A	N/A
5653373	12/1996	Green et al.	N/A	N/A
5653374	12/1996	Young et al.	N/A	N/A
5653677	12/1996	Okada et al.	N/A	N/A
5653721	12/1996	Knodel et al.	N/A	N/A
5653748	12/1996	Strecker	N/A	N/A
5655698	12/1996	Yoon	N/A	N/A
5656917	12/1996	Theobald	N/A	N/A

5657417	12/1996	Di Troia	N/A	N/A
5657429	12/1996	Wang et al.	N/A	N/A
5657921	12/1996	Young et al.	N/A	N/A
5658238	12/1996	Suzuki et al.	N/A	N/A
5658281	12/1996	Heard	N/A	N/A
5658298	12/1996	Vincent et al.	N/A	N/A
5658300	12/1996	Bito et al.	N/A	N/A
5658307	12/1996	Exconde	N/A	N/A
5662258	12/1996	Knodel et al.	N/A	N/A
5662260	12/1996	Yoon	N/A	N/A
5662662	12/1996	Bishop et al.	N/A	N/A
5662667	12/1996	Knodel	N/A	N/A
5664404	12/1996	Ivanov et al.	N/A	N/A
5665085	12/1996	Nardella	N/A	N/A
5667517	12/1996	Hooven	N/A	N/A
5667526	12/1996	Levin	N/A	N/A
5667527	12/1996	Cook	N/A	N/A
5667864	12/1996	Landoll	N/A	N/A
5669544	12/1996	Schulze et al.	N/A	N/A
5669904	12/1996	Platt, Jr. et al.	N/A	N/A
5669907	12/1996	Platt, Jr. et al.	N/A	N/A
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5681341	12/1996	Lunsford et al.	N/A	N/A
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5686090	12/1996	Schilder et al.	N/A	N/A
5688270	12/1996	Yates et al.	N/A	N/A
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5697542	12/1996	Knodel et al.	N/A	N/A

5697543	12/1996	Burdorff	N/A	N/A
5697909	12/1996	Eggers et al.	N/A	N/A
5697943	12/1996	Sauer et al.	N/A	N/A
5700265	12/1996	Romano	N/A	N/A
5700270	12/1996	Peyser et al.	N/A	N/A
5700276	12/1996	Benecke	N/A	N/A
5702387	12/1996	Arts et al.	N/A	N/A
5702408	12/1996	Wales et al.	N/A	N/A
5702409	12/1996	Rayburn et al.	N/A	N/A
5704087	12/1997	Strub	N/A	N/A
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5704792	12/1997	Sobhani	N/A	N/A
5706997	12/1997	Green et al.	N/A	N/A
5706998	12/1997	Plyley et al.	N/A	N/A
5707392	12/1997	Kortenbach	N/A	N/A
5709334	12/1997	Sorrentino et al.	N/A	N/A
5709335	12/1997	Heck	N/A	N/A
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5709706	12/1997	Kienzle et al.	N/A	N/A
5711472	12/1997	Bryan	N/A	N/A
5711960	12/1997	Shikinami	N/A	N/A
5712460	12/1997	Carr et al.	N/A	N/A
5713128	12/1997	Schrenk et al.	N/A	N/A
5713505	12/1997	Huitema	N/A	N/A
5713895	12/1997	Lontine et al.	N/A	N/A
5713896	12/1997	Nardella	N/A	N/A
5713920	12/1997	Bezwada et al.	N/A	N/A
5715604	12/1997	Lanzoni	N/A	N/A
5715836	12/1997	Kliegis et al.	N/A	N/A
5715987	12/1997	Kelley et al.	N/A	N/A
5715988	12/1997	Palmer	N/A	N/A
5716352	12/1997	Viola et al.	N/A	N/A
5716366	12/1997	Yates	N/A	N/A
5718359	12/1997	Palmer et al.	N/A	N/A
5718360	12/1997	Green et al.	N/A	N/A
5718548	12/1997	Cotellessa	N/A	N/A
5718714	12/1997	Livneh	N/A	N/A
5720744	12/1997	Eggleston et al.	N/A	N/A
D393067	12/1997	Geary et al.	N/A	N/A
5724025	12/1997	Tavori	N/A	N/A
5725536	12/1997	Oberlin et al.	N/A	N/A
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5732712	12/1997	Adair	N/A	N/A
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5732872	12/1997	Bolduc et al.	N/A	N/A

5733308	12/1997	Daugherty et al.	N/A	N/A
5735445	12/1997	Vidal et al.	N/A	N/A
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5736271	12/1997	Cisar et al.	N/A	N/A
5738474	12/1997	Blewett	N/A	N/A
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5743456	12/1997	Jones et al.	N/A	N/A
5746770	12/1997	Zeitels et al.	N/A	N/A
5747953	12/1997	Philipp	N/A	N/A
5749889	12/1997	Bacich et al.	N/A	N/A
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5752970	12/1997	Yoon	N/A	N/A
5752973	12/1997	Kieturakis	N/A	N/A
5755717	12/1997	Yates et al.	N/A	N/A
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D395645	12/1997	Cappa et al.	N/A	N/A
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5766188	12/1997	Igaki	N/A	N/A
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5769892	12/1997	Kingwell	N/A	N/A
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5772379	12/1997	Evensen	N/A	N/A
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5779132	12/1997	Knodel et al.	N/A	N/A
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5787897	12/1997	Kieturakis	N/A	N/A
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5796188	12/1997	Bays	N/A	N/A
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5797637	12/1997	Ervin	N/A	N/A
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5807393	12/1997	Williamson, IV et al.	N/A	N/A
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5810240	12/1997	Robertson	N/A	N/A
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5812188	12/1997	Adair	N/A	N/A
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5816471	12/1997	Plyley et al.	N/A	N/A
5817084	12/1997	Jensen	N/A	N/A
5817091	12/1997	Nardella et al.	N/A	N/A
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5827323	12/1997	Klieman et al.	N/A	N/A
5829662	12/1997	Allen et al.	N/A	N/A
5830598	12/1997	Patterson	N/A	N/A
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5833695	12/1997	Yoon	N/A	N/A
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5843021	12/1997	Edwards et al.	N/A	N/A
5843096	12/1997	Igaki et al.	N/A	N/A
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5849020	12/1997	Long et al.	N/A	N/A
5849023	12/1997	Mericle	N/A	N/A
5851179	12/1997	Ritson et al.	N/A	N/A
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5853366	12/1997	Dowlatshahi	N/A	N/A
5855311	12/1998	Hamblin et al.	N/A	N/A
5855583	12/1998	Wang et al.	N/A	N/A
5860581	12/1998	Robertson et al.	N/A	N/A
5860975	12/1998	Goble et al.	N/A	N/A
5865361	12/1998	Milliman et al.	N/A	N/A
5865638	12/1998	Trafton	N/A	N/A
5868361	12/1998	Rinderer	N/A	N/A
5868664	12/1998	Speier et al.	N/A	N/A
5868760	12/1998	McGuckin, Jr.	N/A	N/A
5868790	12/1998	Vincent et al.	N/A	N/A
5871135	12/1998	Williamson IV et al.	N/A	N/A
5873885	12/1998	Weidenbenner	N/A	N/A
5876401	12/1998	Schulze et al.	N/A	N/A
5878193	12/1998	Wang et al.	N/A	N/A
5878607	12/1998	Nunes et al.	N/A	N/A
5878937	12/1998	Green et al.	N/A	N/A
5878938	12/1998	Bittner et al.	N/A	N/A
5881777	12/1998	Bassi et al.	N/A	N/A
5881943	12/1998	Heck et al.	N/A	N/A
5891094	12/1998	Masterson et al.	N/A	N/A

5891160	12/1998	Williamson, IV et al.	N/A	N/A
5891558	12/1998	Bell et al.	N/A	N/A
5893506	12/1998	Powell	N/A	N/A
5893835	12/1998	Witt et al.	N/A	N/A
5893855	12/1998	Jacobs	N/A	N/A
5893863	12/1998	Yoon	N/A	N/A
5893878	12/1998	Pierce	N/A	N/A
5894979	12/1998	Powell	N/A	N/A
5897552	12/1998	Edwards et al.	N/A	N/A
5897562	12/1998	Bolanos et al.	N/A	N/A
5899824	12/1998	Kurtz et al.	N/A	N/A
5899914	12/1998	Zirps et al.	N/A	N/A
5901895	12/1998	Heaton et al.	N/A	N/A
5902312	12/1998	Frater et al.	N/A	N/A
5903117	12/1998	Gregory	N/A	N/A
5904647	12/1998	Ouchi	N/A	N/A
5904693	12/1998	Dicesare et al.	N/A	N/A
5904702	12/1998	Ek et al.	N/A	N/A
5906577	12/1998	Beane et al.	N/A	N/A
5906625	12/1998	Bito et al.	N/A	N/A
5907211	12/1998	Hall et al.	N/A	N/A
5907664	12/1998	Wang et al.	N/A	N/A
5908149	12/1998	Welch et al.	N/A	N/A
5908402	12/1998	Blythe	N/A	N/A
5908427	12/1998	McKean et al.	N/A	N/A
5909062	12/1998	Krietzman	N/A	N/A
5911353	12/1998	Bolanos et al.	N/A	N/A
5915616	12/1998	Viola et al.	N/A	N/A
5916225	12/1998	Kugel	N/A	N/A
5918791	12/1998	Sorrentino et al.	N/A	N/A
5919198	12/1998	Graves, Jr. et al.	N/A	N/A
5921956	12/1998	Grinberg et al.	N/A	N/A
5922001	12/1998	Yoon	N/A	N/A
5922003	12/1998	Anctil et al.	N/A	N/A
5924864	12/1998	Loge et al.	N/A	N/A
5928137	12/1998	Green	N/A	N/A
5928256	12/1998	Riza	N/A	N/A
5931847	12/1998	Bittner et al.	N/A	N/A
5931853	12/1998	McEwen et al.	N/A	N/A
5937951	12/1998	Izuchukwu et al.	N/A	N/A
5938667	12/1998	Peyser et al.	N/A	N/A
5941442	12/1998	Geiste et al.	N/A	N/A
5941890	12/1998	Voegele et al.	N/A	N/A
5944172	12/1998	Hannula	N/A	N/A
5944715	12/1998	Goble et al.	N/A	N/A
5946978	12/1998	Yamashita	N/A	N/A
5947984	12/1998	Whipple	N/A	N/A
5947996	12/1998	Logeman	N/A	N/A
5948030	12/1998	Miller et al.	N/A	N/A
5948429	12/1998	Bell et al.	N/A	N/A

5951301	12/1998	Younker	N/A	N/A
5951516	12/1998	Bunyan	N/A	N/A
5951552	12/1998	Long et al.	N/A	N/A
5951574	12/1998	Stefanchik et al.	N/A	N/A
5951575	12/1998	Bolduc et al.	N/A	N/A
5951581	12/1998	Saadat et al.	N/A	N/A
5954259	12/1998	Viola et al.	N/A	N/A
5957831	12/1998	Adair	N/A	N/A
5964394	12/1998	Robertson	N/A	N/A
5964774	12/1998	McKean et al.	N/A	N/A
5966126	12/1998	Szabo	N/A	N/A
5971916	12/1998	Koren	N/A	N/A
5973221	12/1998	Collyer et al.	N/A	N/A
D416089	12/1998	Barton et al.	N/A	N/A
5976122	12/1998	Madhani et al.	N/A	N/A
5977746	12/1998	Hershberger et al.	N/A	N/A
5980248	12/1998	Kusakabe et al.	N/A	N/A
5980569	12/1998	Scirica	N/A	N/A
5984949	12/1998	Levin	N/A	N/A
5988479	12/1998	Palmer	N/A	N/A
5990379	12/1998	Gregory	N/A	N/A
5993464	12/1998	Knodel	N/A	N/A
5993466	12/1998	Yoon	N/A	N/A
5997528	12/1998	Bisch et al.	N/A	N/A
5997552	12/1998	Person et al.	N/A	N/A
6001108	12/1998	Wang et al.	N/A	N/A
6003517	12/1998	Sheffield et al.	N/A	N/A
6004319	12/1998	Goble et al.	N/A	N/A
6004335	12/1998	Vaitekunas et al.	N/A	N/A
6007521	12/1998	Bidwell et al.	N/A	N/A
6010054	12/1999	Johnson et al.	N/A	N/A
6010513	12/1999	Tormala et al.	N/A	N/A
6010520	12/1999	Pattison	N/A	N/A
6012494	12/1999	Balazs	N/A	N/A
6013076	12/1999	Goble et al.	N/A	N/A
6013991	12/1999	Philipp	N/A	N/A
6015406	12/1999	Goble et al.	N/A	N/A
6015417	12/1999	Reynolds, Jr.	N/A	N/A
6017322	12/1999	Snoke et al.	N/A	N/A
6017354	12/1999	Culp et al.	N/A	N/A
6017356	12/1999	Frederick et al.	N/A	N/A
6018227	12/1999	Kumar et al.	N/A	N/A
6019745	12/1999	Gray	N/A	N/A
6019780	12/1999	Lombardo et al.	N/A	N/A
6022352	12/1999	Vandewalle	N/A	N/A
6023275	12/1999	Horvitz et al.	N/A	N/A
6023641	12/1999	Thompson	N/A	N/A
6024708	12/1999	Bales et al.	N/A	N/A
6024741	12/1999	Williamson, IV et al.	N/A	N/A
6024748	12/1999	Manzo et al.	N/A	N/A

6024750	12/1999	Mastri et al.	N/A	N/A
6024764	12/1999	Schroeppel	N/A	N/A
6027501	12/1999	Goble et al.	N/A	N/A
6030384	12/1999	Nezhat	N/A	N/A
6031148	12/1999	Hayes et al.	N/A	N/A
6032849	12/1999	Mastri et al.	N/A	N/A
6033105	12/1999	Barker et al.	N/A	N/A
6033378	12/1999	Lundquist et al.	N/A	N/A
6033399	12/1999	Gines	N/A	N/A
6033427	12/1999	Lee	N/A	N/A
6036641	12/1999	Taylor et al.	N/A	N/A
6036667	12/1999	Manna et al.	N/A	N/A
6037724	12/1999	Buss et al.	N/A	N/A
6037927	12/1999	Rosenberg	N/A	N/A
6039126	12/1999	Hsieh	N/A	N/A
6039733	12/1999	Buyse et al.	N/A	N/A
6039734	12/1999	Goble	N/A	N/A
6042601	12/1999	Smith	N/A	N/A
6042607	12/1999	Williamson, IV et al.	N/A	N/A
6043626	12/1999	Snyder et al.	N/A	N/A
D422545	12/1999	Palalau et al.	N/A	N/A
6045560	12/1999	McKean et al.	N/A	N/A
6047861	12/1999	Vidal et al.	N/A	N/A
6049145	12/1999	Austin et al.	N/A	N/A
6050172	12/1999	Corves et al.	N/A	N/A
6050472	12/1999	Shibata	N/A	N/A
6050989	12/1999	Fox et al.	N/A	N/A
6050990	12/1999	Tankovich et al.	N/A	N/A
6050996	12/1999	Schmaltz et al.	N/A	N/A
6053390	12/1999	Green et al.	N/A	N/A
6053899	12/1999	Slanda et al.	N/A	N/A
6053922	12/1999	Krause et al.	N/A	N/A
6054142	12/1999	Li et al.	N/A	N/A
6055062	12/1999	Dina et al.	N/A	N/A
RE36720	12/1999	Green et al.	N/A	N/A
6056735	12/1999	Okada et al.	N/A	N/A
6056746	12/1999	Goble et al.	N/A	N/A
6059806	12/1999	Hoegerle	N/A	N/A
6062360	12/1999	Shields	N/A	N/A
6063020	12/1999	Jones et al.	N/A	N/A
6063025	12/1999	Bridges et al.	N/A	N/A
6063050	12/1999	Manna et al.	N/A	N/A
6063095	12/1999	Wang et al.	N/A	N/A
6063097	12/1999	Oi et al.	N/A	N/A
6063098	12/1999	Houser et al.	N/A	N/A
6065679	12/1999	Levie et al.	N/A	N/A
6065919	12/1999	Peck	N/A	N/A
6066132	12/1999	Chen et al.	N/A	N/A
6066144	12/1999	Wolf et al.	N/A	N/A
6066151	12/1999	Miyawaki et al.	N/A	N/A

6068627	12/1999	Orszulak et al.	N/A	N/A
6071233	12/1999	Ishikawa et al.	N/A	N/A
6072299	12/1999	Kurle et al.	N/A	N/A
6074386	12/1999	Goble et al.	N/A	N/A
6074401	12/1999	Gardiner et al.	N/A	N/A
6075441	12/1999	Maloney	N/A	N/A
6077280	12/1999	Fossum	N/A	N/A
6077286	12/1999	Cuschieri et al.	N/A	N/A
6077290	12/1999	Marini	N/A	N/A
6079606	12/1999	Milliman et al.	N/A	N/A
6080181	12/1999	Jensen et al.	N/A	N/A
6082577	12/1999	Coates et al.	N/A	N/A
6083191	12/1999	Rose	N/A	N/A
6083223	12/1999	Baker	N/A	N/A
6083234	12/1999	Nicholas et al.	N/A	N/A
6083242	12/1999	Cook	N/A	N/A
6086544	12/1999	Hibner et al.	N/A	N/A
6086600	12/1999	Kortenbach	N/A	N/A
6090106	12/1999	Goble et al.	N/A	N/A
6090123	12/1999	Culp et al.	N/A	N/A
6093186	12/1999	Goble	N/A	N/A
6094021	12/1999	Noro et al.	N/A	N/A
D429252	12/1999	Haitani et al.	N/A	N/A
6099537	12/1999	Sugai et al.	N/A	N/A
6099551	12/1999	Gabbay	N/A	N/A
6102271	12/1999	Longo et al.	N/A	N/A
6102926	12/1999	Tartaglia et al.	N/A	N/A
6104162	12/1999	Sainsbury et al.	N/A	N/A
6104304	12/1999	Clark et al.	N/A	N/A
6106511	12/1999	Jensen	N/A	N/A
6109500	12/1999	Alli et al.	N/A	N/A
6110187	12/1999	Donlon	N/A	N/A
6113618	12/1999	Nic	N/A	N/A
6117148	12/1999	Ravo et al.	N/A	N/A
6117158	12/1999	Measamer et al.	N/A	N/A
6119913	12/1999	Adams et al.	N/A	N/A
6120433	12/1999	Mizuno et al.	N/A	N/A
6120462	12/1999	Hibner et al.	N/A	N/A
6123241	12/1999	Walter et al.	N/A	N/A
6123701	12/1999	Nezhat	N/A	N/A
H1904	12/1999	Yates et al.	N/A	N/A
RE36923	12/1999	Hiroi et al.	N/A	N/A
6126058	12/1999	Adams et al.	N/A	N/A
6126359	12/1999	Dittrich et al.	N/A	N/A
6126670	12/1999	Walker et al.	N/A	N/A
6131789	12/1999	Schulze et al.	N/A	N/A
6131790	12/1999	Piraka	N/A	N/A
6132368	12/1999	Cooper	N/A	N/A
6134962	12/1999	Sugitani	N/A	N/A
6139546	12/1999	Koenig et al.	N/A	N/A

6142149	12/1999	Steen	N/A	N/A
6142933	12/1999	Longo et al.	N/A	N/A
6147135	12/1999	Yuan et al.	N/A	N/A
6148979	12/1999	Roach et al.	N/A	N/A
6149660	12/1999	Laufer et al.	N/A	N/A
6151323	12/1999	O'Connell et al.	N/A	N/A
6152935	12/1999	Kammerer et al.	N/A	N/A
6155083	12/1999	Goeser et al.	N/A	N/A
6155473	12/1999	Tompkins et al.	N/A	N/A
6156056	12/1999	Kearns et al.	N/A	N/A
6157169	12/1999	Lee	N/A	N/A
6157303	12/1999	Bodie et al.	N/A	N/A
6159146	12/1999	El Gazayerli	N/A	N/A
6159200	12/1999	Verdura et al.	N/A	N/A
6159224	12/1999	Yoon	N/A	N/A
6162208	12/1999	Hipps	N/A	N/A
6162220	12/1999	Nezhat	N/A	N/A
6162537	12/1999	Martin et al.	N/A	N/A
6165175	12/1999	Wampler et al.	N/A	N/A
6165184	12/1999	Verdura et al.	N/A	N/A
6165188	12/1999	Saadat et al.	N/A	N/A
6167185	12/1999	Smiley et al.	N/A	N/A
6168605	12/2000	Measamer et al.	N/A	N/A
6171305	12/2000	Sherman	N/A	N/A
6171316	12/2000	Kovac et al.	N/A	N/A
6171330	12/2000	Benchetrit	N/A	N/A
6173074	12/2000	Russo	N/A	N/A
6174308	12/2000	Goble et al.	N/A	N/A
6174309	12/2000	Wrublewski et al.	N/A	N/A
6174318	12/2000	Bates et al.	N/A	N/A
6175290	12/2000	Forsythe et al.	N/A	N/A
6179195	12/2000	Adams et al.	N/A	N/A
6179776	12/2000	Adams et al.	N/A	N/A
6181105	12/2000	Cutolo et al.	N/A	N/A
6182673	12/2000	Kindermann et al.	N/A	N/A
6185356	12/2000	Parker et al.	N/A	N/A
6186142	12/2000	Schmidt et al.	N/A	N/A
6186957	12/2000	Milam	N/A	N/A
6187003	12/2000	Buyse et al.	N/A	N/A
6190386	12/2000	Rydell	N/A	N/A
6193129	12/2000	Bittner et al.	N/A	N/A
6197042	12/2000	Ginn et al.	N/A	N/A
6200311	12/2000	Danek et al.	N/A	N/A
6200330	12/2000	Benderev et al.	N/A	N/A
6202914	12/2000	Geiste et al.	N/A	N/A
6206894	12/2000	Thompson et al.	N/A	N/A
6206897	12/2000	Jamiolkowski et al.	N/A	N/A
6206903	12/2000	Ramans	N/A	N/A
6206904	12/2000	Ouchi	N/A	N/A
6209414	12/2000	Uneme	N/A	N/A

6210403	12/2000	Klicek	N/A	N/A
6211626	12/2000	Lys et al.	N/A	N/A
6213999	12/2000	Platt, Jr. et al.	N/A	N/A
6214028	12/2000	Yoon et al.	N/A	N/A
6220368	12/2000	Ark et al.	N/A	N/A
6221007	12/2000	Green	N/A	N/A
6221023	12/2000	Matsuba et al.	N/A	N/A
6223100	12/2000	Green	N/A	N/A
6223835	12/2000	Habedank et al.	N/A	N/A
6224617	12/2000	Saadat et al.	N/A	N/A
6228080	12/2000	Gines	N/A	N/A
6228081	12/2000	Goble	N/A	N/A
6228083	12/2000	Lands et al.	N/A	N/A
6228084	12/2000	Kirwan, Jr.	N/A	N/A
6228089	12/2000	Wahrburg	N/A	N/A
6228098	12/2000	Kayan et al.	N/A	N/A
6231565	12/2000	Tovey et al.	N/A	N/A
6234178	12/2000	Goble et al.	N/A	N/A
6235036	12/2000	Gardner et al.	N/A	N/A
6237604	12/2000	Burnside et al.	N/A	N/A
6238384	12/2000	Peer	N/A	N/A
6241139	12/2000	Milliman et al.	N/A	N/A
6241140	12/2000	Adams et al.	N/A	N/A
6241723	12/2000	Heim et al.	N/A	N/A
6245084	12/2000	Mark et al.	N/A	N/A
6248116	12/2000	Chevillon et al.	N/A	N/A
6248117	12/2000	Blatter	N/A	N/A
6249076	12/2000	Madden et al.	N/A	N/A
6249105	12/2000	Andrews et al.	N/A	N/A
6250532	12/2000	Green et al.	N/A	N/A
6251485	12/2000	Harris et al.	N/A	N/A
D445745	12/2000	Norman	N/A	N/A
6254534	12/2000	Butler et al.	N/A	N/A
6254619	12/2000	Garabet et al.	N/A	N/A
6254642	12/2000	Taylor	N/A	N/A
6258107	12/2000	Balazs et al.	N/A	N/A
6261246	12/2000	Pantages et al.	N/A	N/A
6261286	12/2000	Goble et al.	N/A	N/A
6261679	12/2000	Chen et al.	N/A	N/A
6264086	12/2000	McGuckin, Jr.	N/A	N/A
6264087	12/2000	Whitman	N/A	N/A
6264617	12/2000	Bales et al.	N/A	N/A
6269997	12/2000	Balazs et al.	N/A	N/A
6270508	12/2000	Klieman et al.	N/A	N/A
6270916	12/2000	Sink et al.	N/A	N/A
6273252	12/2000	Mitchell	N/A	N/A
6273876	12/2000	Klima et al.	N/A	N/A
6273897	12/2000	Dalessandro et al.	N/A	N/A
6277114	12/2000	Bullivant et al.	N/A	N/A
6280407	12/2000	Manna et al.	N/A	N/A

6283981	12/2000	Beaupre	N/A	N/A
6293927	12/2000	McGuckin, Jr.	N/A	N/A
6293942	12/2000	Goble et al.	N/A	N/A
6296607	12/2000	Milbocker	N/A	N/A
6296640	12/2000	Wampler et al.	N/A	N/A
6302311	12/2000	Adams et al.	N/A	N/A
6302743	12/2000	Chiu et al.	N/A	N/A
6305891	12/2000	Burlingame	N/A	N/A
6306134	12/2000	Goble et al.	N/A	N/A
6306149	12/2000	Meade	N/A	N/A
6306424	12/2000	Vyakarnam et al.	N/A	N/A
6309397	12/2000	Julian et al.	N/A	N/A
6309400	12/2000	Beaupre	N/A	N/A
6309403	12/2000	Minor et al.	N/A	N/A
6312435	12/2000	Wallace et al.	N/A	N/A
6315184	12/2000	Whitman	N/A	N/A
6317616	12/2000	Glossop	N/A	N/A
6319510	12/2000	Yates	N/A	N/A
6320123	12/2000	Reimers	N/A	N/A
6322494	12/2000	Bullivant et al.	N/A	N/A
6324339	12/2000	Hudson et al.	N/A	N/A
6325799	12/2000	Goble	N/A	N/A
6325805	12/2000	Ogilvie et al.	N/A	N/A
6325810	12/2000	Hamilton et al.	N/A	N/A
6328498	12/2000	Mersch	N/A	N/A
6330965	12/2000	Milliman et al.	N/A	N/A
6331181	12/2000	Tierney et al.	N/A	N/A
6331761	12/2000	Kumar et al.	N/A	N/A
6333029	12/2000	Vyakarnam et al.	N/A	N/A
6334860	12/2001	Dorn	N/A	N/A
6334861	12/2001	Chandler et al.	N/A	N/A
6336926	12/2001	Goble	N/A	N/A
6338737	12/2001	Toledano	N/A	N/A
6338738	12/2001	Bellotti et al.	N/A	N/A
6343731	12/2001	Adams et al.	N/A	N/A
6346077	12/2001	Taylor et al.	N/A	N/A
6348061	12/2001	Whitman	N/A	N/A
6349868	12/2001	Mattingly et al.	N/A	N/A
D454951	12/2001	Bon	N/A	N/A
6352503	12/2001	Matsui et al.	N/A	N/A
6352532	12/2001	Kramer et al.	N/A	N/A
6355699	12/2001	Vyakarnam et al.	N/A	N/A
6356072	12/2001	Chass	N/A	N/A
6358224	12/2001	Tims et al.	N/A	N/A
6358263	12/2001	Mark et al.	N/A	N/A
6358459	12/2001	Ziegler et al.	N/A	N/A
6361542	12/2001	Dimitriu et al.	N/A	N/A
6364828	12/2001	Yeung et al.	N/A	N/A
6364877	12/2001	Goble et al.	N/A	N/A
6364888	12/2001	Niemeyer et al.	N/A	N/A

6366441	12/2001	Ozawa et al.	N/A	N/A
6370981	12/2001	Watarai	N/A	N/A
6371114	12/2001	Schmidt et al.	N/A	N/A
6373152	12/2001	Wang et al.	N/A	N/A
6377011	12/2001	Ben-Ur	N/A	N/A
6383201	12/2001	Dong	N/A	N/A
6387092	12/2001	Burnside et al.	N/A	N/A
6387113	12/2001	Hawkins et al.	N/A	N/A
6387114	12/2001	Adams	N/A	N/A
6391038	12/2001	Vargas et al.	N/A	N/A
6392854	12/2001	O'Gorman	N/A	N/A
6394998	12/2001	Wallace et al.	N/A	N/A
6398779	12/2001	Buyse et al.	N/A	N/A
6398781	12/2001	Goble et al.	N/A	N/A
6398797	12/2001	Bombard et al.	N/A	N/A
6402766	12/2001	Bowman et al.	N/A	N/A
6402780	12/2001	Williamson, IV et al.	N/A	N/A
6406440	12/2001	Stefanchik	N/A	N/A
6406472	12/2001	Jensen	N/A	N/A
6409724	12/2001	Penny et al.	N/A	N/A
H2037	12/2001	Yates et al.	N/A	N/A
6412639	12/2001	Hickey	N/A	N/A
6413274	12/2001	Pedros	N/A	N/A
6415542	12/2001	Bates et al.	N/A	N/A
6416486	12/2001	Wampler	N/A	N/A
6416509	12/2001	Goble et al.	N/A	N/A
6419695	12/2001	Gabbay	N/A	N/A
6423079	12/2001	Blake, III	N/A	N/A
6424885	12/2001	Niemeyer et al.	N/A	N/A
RE37814	12/2001	Allgeyer	N/A	N/A
6428070	12/2001	Takanashi et al.	N/A	N/A
6428487	12/2001	Burdorff et al.	N/A	N/A
6429611	12/2001	Li	N/A	N/A
6430298	12/2001	Kettl et al.	N/A	N/A
6432065	12/2001	Burdorff et al.	N/A	N/A
6436097	12/2001	Nardella	N/A	N/A
6436107	12/2001	Wang et al.	N/A	N/A
6436110	12/2001	Bowman et al.	N/A	N/A
6436115	12/2001	Beaupre	N/A	N/A
6436122	12/2001	Frank et al.	N/A	N/A
6439439	12/2001	Rickard et al.	N/A	N/A
6439446	12/2001	Perry et al.	N/A	N/A
6440146	12/2001	Nicholas et al.	N/A	N/A
6441577	12/2001	Blumenkranz et al.	N/A	N/A
D462758	12/2001	Epstein et al.	N/A	N/A
6443973	12/2001	Whitman	N/A	N/A
6445530	12/2001	Baker	N/A	N/A
6447518	12/2001	Krause et al.	N/A	N/A
6447523	12/2001	Middleman et al.	N/A	N/A
6447799	12/2001	Ullman	N/A	N/A

6447864	12/2001	Johnson et al.	N/A	N/A
6450391	12/2001	Kayan et al.	N/A	N/A
6450989	12/2001	Dubrul et al.	N/A	N/A
6454656	12/2001	Brissette et al.	N/A	N/A
6454781	12/2001	Witt et al.	N/A	N/A
6457338	12/2001	Frenken	N/A	N/A
6457625	12/2001	Tormala et al.	N/A	N/A
6458077	12/2001	Boebel et al.	N/A	N/A
6458142	12/2001	Faller et al.	N/A	N/A
6458147	12/2001	Cruise et al.	N/A	N/A
6460627	12/2001	Below et al.	N/A	N/A
6463824	12/2001	Prell et al.	N/A	N/A
6468275	12/2001	Wampler et al.	N/A	N/A
6468286	12/2001	Mastri et al.	N/A	N/A
6471106	12/2001	Reining	N/A	N/A
6471659	12/2001	Eggers et al.	N/A	N/A
6478210	12/2001	Adams et al.	N/A	N/A
6482063	12/2001	Frigard	N/A	N/A
6482200	12/2001	Shippert	N/A	N/A
6482217	12/2001	Pintor et al.	N/A	N/A
6485490	12/2001	Wampler et al.	N/A	N/A
6485503	12/2001	Jacobs et al.	N/A	N/A
6485667	12/2001	Tan	N/A	N/A
6486286	12/2001	McGall et al.	N/A	N/A
6488196	12/2001	Fenton, Jr.	N/A	N/A
6488197	12/2001	Whitman	N/A	N/A
6488659	12/2001	Rosenman	N/A	N/A
6491201	12/2001	Whitman	N/A	N/A
6491690	12/2001	Goble et al.	N/A	N/A
6491701	12/2001	Tierney et al.	N/A	N/A
6491702	12/2001	Heilbrun et al.	N/A	N/A
6492785	12/2001	Kasten et al.	N/A	N/A
6494882	12/2001	Lebouitz et al.	N/A	N/A
6494885	12/2001	Dhindsa	N/A	N/A
6494888	12/2001	Laufer et al.	N/A	N/A
6494896	12/2001	D'Alessio et al.	N/A	N/A
6498480	12/2001	Manara	N/A	N/A
6500176	12/2001	Truckai et al.	N/A	N/A
6500189	12/2001	Lang et al.	N/A	N/A
6500194	12/2001	Benderev et al.	N/A	N/A
D468749	12/2002	Friedman	N/A	N/A
6503139	12/2002	Coral	N/A	N/A
6503257	12/2002	Grant et al.	N/A	N/A
6503259	12/2002	Huxel et al.	N/A	N/A
6505768	12/2002	Whitman	N/A	N/A
6506197	12/2002	Rollero et al.	N/A	N/A
6506399	12/2002	Donovan	N/A	N/A
6510854	12/2002	Goble	N/A	N/A
6511468	12/2002	Cragg et al.	N/A	N/A
6512360	12/2002	Goto et al.	N/A	N/A

6514252	12/2002	Nezhat et al.	N/A	N/A
6516073	12/2002	Schulz et al.	N/A	N/A
6517528	12/2002	Pantages et al.	N/A	N/A
6517535	12/2002	Edwards	N/A	N/A
6517565	12/2002	Whitman et al.	N/A	N/A
6517566	12/2002	Hovland et al.	N/A	N/A
6520971	12/2002	Perry et al.	N/A	N/A
6520972	12/2002	Peters	N/A	N/A
6522101	12/2002	Malackowski	N/A	N/A
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6525499	12/2002	Naganuma	N/A	N/A
D471206	12/2002	Buzzard et al.	N/A	N/A
6527782	12/2002	Hogg et al.	N/A	N/A
6527785	12/2002	Sancoff et al.	N/A	N/A
6530942	12/2002	Fogarty et al.	N/A	N/A
6532958	12/2002	Buan et al.	N/A	N/A
6533157	12/2002	Whitman	N/A	N/A
6533723	12/2002	Lockery et al.	N/A	N/A
6533784	12/2002	Truckai et al.	N/A	N/A
6535764	12/2002	Imran et al.	N/A	N/A
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D473239	12/2002	Cockerill	N/A	N/A
6539816	12/2002	Kogiso et al.	N/A	N/A
6540737	12/2002	Bacher et al.	N/A	N/A
6543456	12/2002	Freeman	N/A	N/A
6545384	12/2002	Pelrine et al.	N/A	N/A
6547786	12/2002	Goble	N/A	N/A
6550546	12/2002	Thurler et al.	N/A	N/A
6551333	12/2002	Kuhns et al.	N/A	N/A
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6555770	12/2002	Kawase	N/A	N/A
6558378	12/2002	Sherman et al.	N/A	N/A
6558379	12/2002	Batchelor et al.	N/A	N/A
6558429	12/2002	Taylor	N/A	N/A
6561187	12/2002	Schmidt et al.	N/A	N/A
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6566619	12/2002	Gillman et al.	N/A	N/A
6569085	12/2002	Kortenbach et al.	N/A	N/A
6569171	12/2002	DeGuillebon et al.	N/A	N/A
6569173	12/2002	Blatter et al.	N/A	N/A
6572629	12/2002	Kalloo et al.	N/A	N/A
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6578751	12/2002	Hartwick	N/A	N/A
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6587750	12/2002	Gerbi et al.	N/A	N/A
6588277	12/2002	Giordano et al.	N/A	N/A
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6588931	12/2002	Betzner et al.	N/A	N/A
6589118	12/2002	Soma et al.	N/A	N/A
6589164	12/2002	Flaherty	N/A	N/A
6592538	12/2002	Hotchkiss et al.	N/A	N/A
6592572	12/2002	Suzuta	N/A	N/A
6592597	12/2002	Grant et al.	N/A	N/A
6594552	12/2002	Nowlin et al.	N/A	N/A
6595914	12/2002	Kato	N/A	N/A
6596296	12/2002	Nelson et al.	N/A	N/A
6596304	12/2002	Bayon et al.	N/A	N/A
6596432	12/2002	Kawakami et al.	N/A	N/A
6599295	12/2002	Tornier et al.	N/A	N/A
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D478665	12/2002	Isaacs et al.	N/A	N/A
D478986	12/2002	Johnston et al.	N/A	N/A
6601749	12/2002	Sullivan et al.	N/A	N/A
6602252	12/2002	Mollenauer	606/49	A61B 18/085
6602262	12/2002	Griego et al.	N/A	N/A
6603050	12/2002	Heaton	N/A	N/A
6605078	12/2002	Adams	N/A	N/A
6605669	12/2002	Awokola et al.	N/A	N/A
6605911	12/2002	Klesing	N/A	N/A
6607475	12/2002	Doyle et al.	N/A	N/A
6611793	12/2002	Burnside et al.	N/A	N/A
6613069	12/2002	Boyd et al.	N/A	N/A
6616686	12/2002	Coleman et al.	N/A	N/A
6619529	12/2002	Green et al.	N/A	N/A
6620111	12/2002	Stephens et al.	N/A	N/A
6620161	12/2002	Schulze et al.	N/A	N/A
6620166	12/2002	Wenstrom, Jr. et al.	N/A	N/A
6623482	12/2002	Pendekanti et al.	N/A	N/A
6625517	12/2002	Bogdanov et al.	N/A	N/A
6626834	12/2002	Dunne et al.	N/A	N/A
6626901	12/2002	Treat et al.	N/A	N/A
6626938	12/2002	Butaric et al.	N/A	N/A
H2086	12/2002	Amsler	N/A	N/A
6629630	12/2002	Adams	N/A	N/A
6629974	12/2002	Penny et al.	N/A	N/A
6629988	12/2002	Weadock	N/A	N/A
6635838	12/2002	Kornelson	N/A	N/A
6636412	12/2002	Smith	N/A	N/A
6638108	12/2002	Tachi	N/A	N/A
6638285	12/2002	Gabbay	N/A	N/A
6638297	12/2002	Huitema	N/A	N/A
RE38335	12/2002	Aust et al.	N/A	N/A

6641528	12/2002	Torii	N/A	N/A
6644532	12/2002	Green et al.	N/A	N/A
6645201	12/2002	Utley et al.	N/A	N/A
6646307	12/2002	Yu et al.	N/A	N/A
6648816	12/2002	Irion et al.	N/A	N/A
6648901	12/2002	Fleischman et al.	N/A	N/A
6652595	12/2002	Nicolo	N/A	N/A
D484243	12/2002	Ryan et al.	N/A	N/A
D484595	12/2002	Ryan et al.	N/A	N/A
D484596	12/2002	Ryan et al.	N/A	N/A
6656177	12/2002	Truckai et al.	N/A	N/A
6656193	12/2002	Grant et al.	N/A	N/A
6659940	12/2002	Adler	N/A	N/A
6660008	12/2002	Foerster et al.	N/A	N/A
6663623	12/2002	Oyama et al.	N/A	N/A
6663641	12/2002	Kovac et al.	N/A	N/A
6666854	12/2002	Lange	N/A	N/A
6666860	12/2002	Takahashi	N/A	N/A
6666875	12/2002	Sakurai et al.	N/A	N/A
6667825	12/2002	Lu et al.	N/A	N/A
6669073	12/2002	Milliman et al.	N/A	N/A
6670806	12/2002	Wendt et al.	N/A	N/A
6671185	12/2002	Duval	N/A	N/A
D484977	12/2003	Ryan et al.	N/A	N/A
6676660	12/2003	Wampler et al.	N/A	N/A
6677687	12/2003	Ho et al.	N/A	N/A
6679269	12/2003	Swanson	N/A	N/A
6679410	12/2003	Wursch et al.	N/A	N/A
6681978	12/2003	Geiste et al.	N/A	N/A
6681979	12/2003	Whitman	N/A	N/A
6682527	12/2003	Strul	N/A	N/A
6682528	12/2003	Frazier et al.	N/A	N/A
6682544	12/2003	Mastri et al.	N/A	N/A
6685698	12/2003	Morley et al.	N/A	N/A
6685727	12/2003	Fisher et al.	N/A	N/A
6689153	12/2003	Skiba	N/A	N/A
6692507	12/2003	Pugsley et al.	N/A	N/A
6692692	12/2003	Stetzel	N/A	N/A
6695198	12/2003	Adams et al.	N/A	N/A
6695199	12/2003	Whitman	N/A	N/A
6695774	12/2003	Hale et al.	N/A	N/A
6695849	12/2003	Michelson	N/A	N/A
6696814	12/2003	Henderson et al.	N/A	N/A
6697048	12/2003	Rosenberg et al.	N/A	N/A
6698643	12/2003	Whitman	N/A	N/A
6699177	12/2003	Wang et al.	N/A	N/A
6699214	12/2003	Gellman	N/A	N/A
6699235	12/2003	Wallace et al.	N/A	N/A
6704210	12/2003	Myers	N/A	N/A
6705503	12/2003	Pedicini et al.	N/A	N/A

6709445	12/2003	Boebel et al.	N/A	N/A
6712773	12/2003	Viola	N/A	N/A
6716215	12/2003	David et al.	N/A	N/A
6716223	12/2003	Leopold et al.	N/A	N/A
6716232	12/2003	Vidal et al.	N/A	N/A
6716233	12/2003	Whitman	N/A	N/A
6720734	12/2003	Norris	N/A	N/A
6722550	12/2003	Ricordi et al.	N/A	N/A
6722552	12/2003	Fenton, Jr.	N/A	N/A
6723087	12/2003	O'Neill et al.	N/A	N/A
6723091	12/2003	Goble et al.	N/A	N/A
6723106	12/2003	Charles et al.	N/A	N/A
6723109	12/2003	Solingen	N/A	N/A
6726651	12/2003	Robinson et al.	N/A	N/A
6726697	12/2003	Nicholas et al.	N/A	N/A
6726705	12/2003	Peterson et al.	N/A	N/A
6726706	12/2003	Dominguez	N/A	N/A
6729119	12/2003	Schnipke et al.	N/A	N/A
6731976	12/2003	Penn et al.	N/A	N/A
6736810	12/2003	Hoey et al.	N/A	N/A
6736825	12/2003	Blatter et al.	N/A	N/A
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6743230	12/2003	Lutze et al.	N/A	N/A
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6747121	12/2003	Gogolewski	N/A	N/A
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6749560	12/2003	Konstorum et al.	N/A	N/A
6749600	12/2003	Levy	N/A	N/A
6752768	12/2003	Burdorff et al.	N/A	N/A
6752816	12/2003	Culp et al.	N/A	N/A
6754959	12/2003	Guiette, III et al.	N/A	N/A
6755195	12/2003	Lemke et al.	N/A	N/A
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6755825	12/2003	Shoenman et al.	N/A	N/A
6755843	12/2003	Chung et al.	N/A	N/A
6756705	12/2003	Pulford, Jr.	N/A	N/A
6758846	12/2003	Goble et al.	N/A	N/A
6761685	12/2003	Adams et al.	N/A	N/A
6762339	12/2003	Klun et al.	N/A	N/A
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6764445	12/2003	Ramans et al.	N/A	N/A
6766957	12/2003	Matsuura et al.	N/A	N/A
6767352	12/2003	Field et al.	N/A	N/A
6767356	12/2003	Kanner et al.	N/A	N/A
6769590	12/2003	Vresh et al.	N/A	N/A
6769594	12/2003	Orban, III	N/A	N/A
6770027	12/2003	Banik et al.	N/A	N/A
6770070	12/2003	Balbierz	N/A	N/A
6770072	12/2003	Truckai et al.	N/A	N/A

6770078	12/2003	Bonutti	N/A	N/A
6773409	12/2003	Truckai et al.	N/A	N/A
6773437	12/2003	Ogilvie et al.	N/A	N/A
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6778846	12/2003	Martinez et al.	N/A	N/A
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6780180	12/2003	Goble et al.	N/A	N/A
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6788018	12/2003	Blumenkranz	N/A	N/A
6790173	12/2003	Saadat et al.	N/A	N/A
6793652	12/2003	Whitman et al.	N/A	N/A
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6793663	12/2003	Kneifel et al.	N/A	N/A
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6799669	12/2003	Fukumura et al.	N/A	N/A
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6802822	12/2003	Dodge	N/A	N/A
6802843	12/2003	Truckai et al.	N/A	N/A
6802844	12/2003	Ferree	N/A	N/A
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6808525	12/2003	Latterell et al.	N/A	N/A
6810359	12/2003	Sakaguchi	N/A	N/A
6814154	12/2003	Chou	N/A	N/A
6814741	12/2003	Bowman et al.	N/A	N/A
6817508	12/2003	Racenet et al.	N/A	N/A
6817509	12/2003	Geiste et al.	N/A	N/A
6817974	12/2003	Cooper et al.	N/A	N/A
6818018	12/2003	Sawhney	N/A	N/A
6820791	12/2003	Adams	N/A	N/A
6821273	12/2003	Mollenauer	N/A	N/A
6821282	12/2003	Perry et al.	N/A	N/A
6821284	12/2003	Sturtz et al.	N/A	N/A
6827246	12/2003	Sullivan et al.	N/A	N/A
6827712	12/2003	Tovey et al.	N/A	N/A
6827725	12/2003	Batchelor et al.	N/A	N/A
6828902	12/2003	Casden	N/A	N/A
6830174	12/2003	Hillstead et al.	N/A	N/A
6831629	12/2003	Nishino et al.	N/A	N/A
6832998	12/2003	Goble	N/A	N/A
6834001	12/2003	Myono	N/A	N/A

6835173	12/2003	Couvillion, Jr.	N/A	N/A
6835199	12/2003	McGuckin, Jr. et al.	N/A	N/A
6835336	12/2003	Watt	N/A	N/A
6836611	12/2003	Popovic et al.	N/A	N/A
6837846	12/2004	Jaffe et al.	N/A	N/A
6837883	12/2004	Moll et al.	N/A	N/A
6838493	12/2004	Williams et al.	N/A	N/A
6840423	12/2004	Adams et al.	N/A	N/A
6840938	12/2004	Morley et al.	N/A	N/A
6841967	12/2004	Kim et al.	N/A	N/A
6843403	12/2004	Whitman	N/A	N/A
6843789	12/2004	Goble	N/A	N/A
6843793	12/2004	Brock et al.	N/A	N/A
6846307	12/2004	Whitman et al.	N/A	N/A
6846308	12/2004	Whitman et al.	N/A	N/A
6846309	12/2004	Whitman et al.	N/A	N/A
6847190	12/2004	Schaefer et al.	N/A	N/A
6849071	12/2004	Whitman et al.	N/A	N/A
6850817	12/2004	Green	N/A	N/A
6852122	12/2004	Rush	N/A	N/A
6852330	12/2004	Bowman et al.	N/A	N/A
6853879	12/2004	Sunaoshi	N/A	N/A
6858005	12/2004	Ohline et al.	N/A	N/A
6859882	12/2004	Fung	N/A	N/A
RE38708	12/2004	Bolanos et al.	N/A	N/A
D502994	12/2004	Blake, III	N/A	N/A
6860169	12/2004	Shinozaki	N/A	N/A
6861142	12/2004	Wilkie et al.	N/A	N/A
6861954	12/2004	Levin	N/A	N/A
6863668	12/2004	Gillespie et al.	N/A	N/A
6863694	12/2004	Boyce et al.	N/A	N/A
6863924	12/2004	Ranganathan et al.	N/A	N/A
6866178	12/2004	Adams et al.	N/A	N/A
6866668	12/2004	Giannetti et al.	N/A	N/A
6866671	12/2004	Tierney et al.	N/A	N/A
6867248	12/2004	Martin et al.	N/A	N/A
6869430	12/2004	Balbierz et al.	N/A	N/A
6869435	12/2004	Blake, III	N/A	N/A
6872214	12/2004	Sonnenschein et al.	N/A	N/A
6874669	12/2004	Adams et al.	N/A	N/A
6876850	12/2004	Maeshima et al.	N/A	N/A
6877647	12/2004	Green et al.	N/A	N/A
6878106	12/2004	Herrmann	N/A	N/A
6882127	12/2004	Konigbauer	N/A	N/A
6883199	12/2004	Lundell et al.	N/A	N/A
6884392	12/2004	Malkin et al.	N/A	N/A
6884428	12/2004	Binette et al.	N/A	N/A
6886730	12/2004	Fujisawa et al.	N/A	N/A
6887244	12/2004	Walker et al.	N/A	N/A
6887710	12/2004	Call et al.	N/A	N/A

6889116	12/2004	Jinno	N/A	N/A
6893435	12/2004	Goble	N/A	N/A
6894140	12/2004	Roby	N/A	N/A
6895176	12/2004	Archer et al.	N/A	N/A
6899538	12/2004	Matoba	N/A	N/A
6899593	12/2004	Moeller et al.	N/A	N/A
6899705	12/2004	Niemeyer	N/A	N/A
6899915	12/2004	Yelick et al.	N/A	N/A
6905057	12/2004	Swayze et al.	N/A	N/A
6905497	12/2004	Truckai et al.	N/A	N/A
6905498	12/2004	Hooven	N/A	N/A
6908472	12/2004	Wiener et al.	N/A	N/A
6911033	12/2004	de Guillebon et al.	N/A	N/A
6911916	12/2004	Wang et al.	N/A	N/A
6913579	12/2004	Truckai et al.	N/A	N/A
6913608	12/2004	Liddicoat et al.	N/A	N/A
6913613	12/2004	Schwarz et al.	N/A	N/A
6921397	12/2004	Corcoran et al.	N/A	N/A
6921412	12/2004	Black et al.	N/A	N/A
6923093	12/2004	Ullah	N/A	N/A
6923803	12/2004	Goble	N/A	N/A
6923819	12/2004	Meade et al.	N/A	N/A
6925849	12/2004	Jairam	N/A	N/A
6926716	12/2004	Baker et al.	N/A	N/A
6927315	12/2004	Heinecke et al.	N/A	N/A
6928902	12/2004	Eyssallenne	N/A	N/A
6929641	12/2004	Goble et al.	N/A	N/A
6929644	12/2004	Truckai et al.	N/A	N/A
6931830	12/2004	Liao	N/A	N/A
6932218	12/2004	Kosann et al.	N/A	N/A
6932810	12/2004	Ryan	N/A	N/A
6936042	12/2004	Wallace et al.	N/A	N/A
6936948	12/2004	Bell et al.	N/A	N/A
D509297	12/2004	Wells	N/A	N/A
D509589	12/2004	Wells	N/A	N/A
6938706	12/2004	Ng	N/A	N/A
6939358	12/2004	Palacios et al.	N/A	N/A
6942662	12/2004	Goble et al.	N/A	N/A
6942674	12/2004	Belef et al.	N/A	N/A
6945444	12/2004	Gresham et al.	N/A	N/A
6945981	12/2004	Donofrio et al.	N/A	N/A
6949196	12/2004	Schmitz et al.	N/A	N/A
6951562	12/2004	Zwirnmann	N/A	N/A
6953138	12/2004	Dworak et al.	N/A	N/A
6953139	12/2004	Milliman et al.	N/A	N/A
6953461	12/2004	McClurken et al.	N/A	N/A
6957758	12/2004	Aranyi	N/A	N/A
6958035	12/2004	Friedman et al.	N/A	N/A
6958070	12/2004	Witt et al.	N/A	N/A
D511525	12/2004	Hernandez et al.	N/A	N/A

6959851	12/2004	Heinrich	N/A	N/A
6959852	12/2004	Shelton, IV et al.	N/A	N/A
6960107	12/2004	Schaub et al.	N/A	N/A
6960163	12/2004	Ewers et al.	N/A	N/A
6960220	12/2004	Marino et al.	N/A	N/A
6962587	12/2004	Johnson et al.	N/A	N/A
6963792	12/2004	Green	N/A	N/A
6964363	12/2004	Wales et al.	N/A	N/A
6966907	12/2004	Goble	N/A	N/A
6966909	12/2004	Marshall et al.	N/A	N/A
6968908	12/2004	Tokunaga et al.	N/A	N/A
6969385	12/2004	Moreyra	N/A	N/A
6969395	12/2004	Eskuri	N/A	N/A
6971988	12/2004	Orban, III	N/A	N/A
6972199	12/2004	Lebouitz et al.	N/A	N/A
6974435	12/2004	Daw et al.	N/A	N/A
6974462	12/2004	Sater	N/A	N/A
6978921	12/2004	Shelton, IV et al.	N/A	N/A
6978922	12/2004	Bilotti et al.	N/A	N/A
6981628	12/2005	Wales	N/A	N/A
6981941	12/2005	Whitman et al.	N/A	N/A
6981978	12/2005	Gannoe	N/A	N/A
6984203	12/2005	Tartaglia et al.	N/A	N/A
6984231	12/2005	Goble et al.	N/A	N/A
6986451	12/2005	Mastri et al.	N/A	N/A
6988649	12/2005	Shelton, IV et al.	N/A	N/A
6988650	12/2005	Schwemberger et al.	N/A	N/A
6989034	12/2005	Hammer et al.	N/A	N/A
6990731	12/2005	Haytayan	N/A	N/A
6990796	12/2005	Schnipke et al.	N/A	N/A
6991146	12/2005	Sinisi et al.	N/A	N/A
6993200	12/2005	Tastl et al.	N/A	N/A
6993413	12/2005	Sunaoshi	N/A	N/A
6994708	12/2005	Manzo	N/A	N/A
6995729	12/2005	Govari et al.	N/A	N/A
6996433	12/2005	Burbank et al.	N/A	N/A
6997931	12/2005	Sauer et al.	N/A	N/A
6997935	12/2005	Anderson et al.	N/A	N/A
6998736	12/2005	Lee et al.	N/A	N/A
6998816	12/2005	Wieck et al.	N/A	N/A
6999821	12/2005	Jenney et al.	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7000819	12/2005	Swayze et al.	N/A	N/A
7000911	12/2005	McCormick et al.	N/A	N/A
7001380	12/2005	Goble	N/A	N/A
7001408	12/2005	Knodel et al.	N/A	N/A
7004174	12/2005	Eggers et al.	N/A	N/A
7005828	12/2005	Karikomi	N/A	N/A
7007176	12/2005	Goodfellow et al.	N/A	N/A
7008433	12/2005	Voellmicke et al.	N/A	N/A

7008435	12/2005	Cummins	N/A	N/A
7009039	12/2005	Yayon et al.	N/A	N/A
7011213	12/2005	Clark et al.	N/A	N/A
7011657	12/2005	Truckai et al.	N/A	N/A
7014640	12/2005	Kemppainen et al.	N/A	N/A
7018357	12/2005	Emmons	N/A	N/A
7018390	12/2005	Turovskiy et al.	N/A	N/A
7021399	12/2005	Driessen	N/A	N/A
7021669	12/2005	Lindermeir et al.	N/A	N/A
7022131	12/2005	Derowe et al.	N/A	N/A
7023159	12/2005	Gorti et al.	N/A	N/A
7025064	12/2005	Wang et al.	N/A	N/A
7025732	12/2005	Thompson et al.	N/A	N/A
7025743	12/2005	Mann et al.	N/A	N/A
7025774	12/2005	Freeman et al.	N/A	N/A
7025775	12/2005	Gadberry et al.	N/A	N/A
7028570	12/2005	Ohta et al.	N/A	N/A
7029435	12/2005	Nakao	N/A	N/A
7029439	12/2005	Roberts et al.	N/A	N/A
7030904	12/2005	Adair et al.	N/A	N/A
7032798	12/2005	Whitman et al.	N/A	N/A
7032799	12/2005	Viola et al.	N/A	N/A
7033356	12/2005	Latterell et al.	N/A	N/A
7033378	12/2005	Smith et al.	N/A	N/A
7035716	12/2005	Harris et al.	N/A	N/A
7035762	12/2005	Menard et al.	N/A	N/A
7036680	12/2005	Flannery	N/A	N/A
7037314	12/2005	Armstrong	N/A	N/A
7037344	12/2005	Kagan et al.	N/A	N/A
7038421	12/2005	Trifilo	N/A	N/A
7041088	12/2005	Nawrocki et al.	N/A	N/A
7041102	12/2005	Truckai et al.	N/A	N/A
7041868	12/2005	Greene et al.	N/A	N/A
7043852	12/2005	Hayashida et al.	N/A	N/A
7044350	12/2005	Kameyama et al.	N/A	N/A
7044352	12/2005	Shelton, IV et al.	N/A	N/A
7044353	12/2005	Mastri et al.	N/A	N/A
7046082	12/2005	Komiya et al.	N/A	N/A
7048165	12/2005	Haramiishi	N/A	N/A
7048687	12/2005	Reuss et al.	N/A	N/A
7048716	12/2005	Kucharczyk et al.	N/A	N/A
7048745	12/2005	Tierney et al.	N/A	N/A
7052454	12/2005	Taylor	N/A	N/A
7052494	12/2005	Goble et al.	N/A	N/A
7052499	12/2005	Steger et al.	N/A	N/A
7055730	12/2005	Ehrenfels et al.	N/A	N/A
7055731	12/2005	Shelton, IV et al.	N/A	N/A
7056123	12/2005	Gregorio et al.	N/A	N/A
7056284	12/2005	Martone et al.	N/A	N/A
7056330	12/2005	Gayton	N/A	N/A

7059331	12/2005	Adams et al.	N/A	N/A
7059508	12/2005	Shelton, IV et al.	N/A	N/A
7063671	12/2005	Couvillon, Jr.	N/A	N/A
7063712	12/2005	Vargas et al.	N/A	N/A
7064509	12/2005	Fu et al.	N/A	N/A
7066879	12/2005	Fowler et al.	N/A	N/A
7066944	12/2005	Laufer et al.	N/A	N/A
7067038	12/2005	Trokhan et al.	N/A	N/A
7070083	12/2005	Jankowski	N/A	N/A
7070559	12/2005	Adams et al.	N/A	N/A
7070597	12/2005	Truckai et al.	N/A	N/A
7071287	12/2005	Rhine et al.	N/A	N/A
7075412	12/2005	Reynolds et al.	N/A	N/A
7075770	12/2005	Smith	N/A	N/A
7077856	12/2005	Whitman	N/A	N/A
7080769	12/2005	Vresh et al.	N/A	N/A
7081114	12/2005	Rashidi	N/A	N/A
7081318	12/2005	Lee et al.	N/A	N/A
7083073	12/2005	Yoshie et al.	N/A	N/A
7083075	12/2005	Swayze et al.	N/A	N/A
7083571	12/2005	Wang et al.	N/A	N/A
7083615	12/2005	Peterson et al.	N/A	N/A
7083619	12/2005	Truckai et al.	N/A	N/A
7083620	12/2005	Jahns et al.	N/A	N/A
7083626	12/2005	Hart et al.	N/A	N/A
7086267	12/2005	Dworak et al.	N/A	N/A
7087049	12/2005	Nowlin et al.	N/A	N/A
7087054	12/2005	Truckai et al.	N/A	N/A
7087071	12/2005	Nicholas et al.	N/A	N/A
7090637	12/2005	Danitz et al.	N/A	N/A
7090673	12/2005	Dycus et al.	N/A	N/A
7090683	12/2005	Brock et al.	N/A	N/A
7090684	12/2005	McGuckin, Jr. et al.	N/A	N/A
7091191	12/2005	Laredo et al.	N/A	N/A
7091412	12/2005	Wang et al.	N/A	N/A
7093492	12/2005	Treiber et al.	N/A	N/A
7094202	12/2005	Nobis et al.	N/A	N/A
7094247	12/2005	Monassevitch et al.	N/A	N/A
7094916	12/2005	DeLuca et al.	N/A	N/A
7096972	12/2005	Orozco, Jr.	N/A	N/A
7097089	12/2005	Marczyk	N/A	N/A
7097644	12/2005	Long	N/A	N/A
7097650	12/2005	Weller et al.	N/A	N/A
7098794	12/2005	Lindsay et al.	N/A	N/A
7100949	12/2005	Williams et al.	N/A	N/A
7101187	12/2005	Deconinck et al.	N/A	N/A
7101363	12/2005	Nishizawa et al.	N/A	N/A
7101371	12/2005	Dycus et al.	N/A	N/A
7101394	12/2005	Hamm et al.	N/A	N/A
7104741	12/2005	Krohn	N/A	N/A

7108695	12/2005	Witt et al.	N/A	N/A
7108701	12/2005	Evens et al.	N/A	N/A
7108709	12/2005	Cummins	N/A	N/A
7111768	12/2005	Cummins et al.	N/A	N/A
7111769	12/2005	Wales et al.	N/A	N/A
7112201	12/2005	Truckai et al.	N/A	N/A
7112214	12/2005	Peterson et al.	N/A	N/A
RE39358	12/2005	Goble	N/A	N/A
D530339	12/2005	Hernandez et al.	N/A	N/A
7114642	12/2005	Whitman	N/A	N/A
7116100	12/2005	Mock et al.	N/A	N/A
7118020	12/2005	Lee et al.	N/A	N/A
7118528	12/2005	Piskun	N/A	N/A
7118563	12/2005	Weckwerth et al.	N/A	N/A
7118582	12/2005	Wang et al.	N/A	N/A
7119534	12/2005	Butzmann	N/A	N/A
7121446	12/2005	Arad et al.	N/A	N/A
7121773	12/2005	Mikiya et al.	N/A	N/A
7122028	12/2005	Looper et al.	N/A	N/A
7125403	12/2005	Julian et al.	N/A	N/A
7125409	12/2005	Truckai et al.	N/A	N/A
7126303	12/2005	Farritor et al.	N/A	N/A
7126879	12/2005	Snyder	N/A	N/A
7128253	12/2005	Mastri et al.	N/A	N/A
7128254	12/2005	Shelton, IV et al.	N/A	N/A
7128748	12/2005	Mooradian et al.	N/A	N/A
7131445	12/2005	Amoah	N/A	N/A
7133601	12/2005	Phillips et al.	N/A	N/A
7134364	12/2005	Kageler et al.	N/A	N/A
7134587	12/2005	Schwemberger et al.	N/A	N/A
7135027	12/2005	Delmotte	N/A	N/A
7137980	12/2005	Buyse et al.	N/A	N/A
7137981	12/2005	Long	N/A	N/A
7139016	12/2005	Squilla et al.	N/A	N/A
7140527	12/2005	Ehrenfels et al.	N/A	N/A
7140528	12/2005	Shelton, IV	N/A	N/A
7141055	12/2005	Abrams et al.	N/A	N/A
7143923	12/2005	Shelton, IV et al.	N/A	N/A
7143924	12/2005	Scirica et al.	N/A	N/A
7143925	12/2005	Shelton, IV et al.	N/A	N/A
7143926	12/2005	Shelton, IV et al.	N/A	N/A
7146191	12/2005	Kerner et al.	N/A	N/A
7147138	12/2005	Shelton, IV	N/A	N/A
7147139	12/2005	Schwemberger et al.	N/A	N/A
7147140	12/2005	Wukusick et al.	N/A	N/A
7147637	12/2005	Goble	N/A	N/A
7147648	12/2005	Lin	N/A	N/A
7147650	12/2005	Lee	N/A	N/A
7150748	12/2005	Ebbutt et al.	N/A	N/A
7153300	12/2005	Goble	N/A	N/A

7153314	12/2005	Laufer et al.	N/A	N/A
7155316	12/2005	Sutherland et al.	N/A	N/A
7156846	12/2006	Dycus et al.	N/A	N/A
7156863	12/2006	Sonnenschein et al.	N/A	N/A
7159750	12/2006	Racenet et al.	N/A	N/A
7160296	12/2006	Pearson et al.	N/A	N/A
7160299	12/2006	Baily	N/A	N/A
7160311	12/2006	Blatter et al.	N/A	N/A
7161036	12/2006	Oikawa et al.	N/A	N/A
7161580	12/2006	Bailey et al.	N/A	N/A
7162758	12/2006	Skinner	N/A	N/A
7163563	12/2006	Schwartz et al.	N/A	N/A
7166117	12/2006	Hellenkamp	N/A	N/A
7166133	12/2006	Evans et al.	N/A	N/A
7168604	12/2006	Milliman et al.	N/A	N/A
7169146	12/2006	Truckai et al.	N/A	N/A
7170910	12/2006	Chen et al.	N/A	N/A
7171279	12/2006	Buckingham et al.	N/A	N/A
7172104	12/2006	Scirica et al.	N/A	N/A
7172593	12/2006	Trieu et al.	N/A	N/A
7172615	12/2006	Morriss et al.	N/A	N/A
7174202	12/2006	Bladen et al.	N/A	N/A
7174636	12/2006	Lowe	N/A	N/A
7177533	12/2006	McFarlin et al.	N/A	N/A
7179223	12/2006	Motoki et al.	N/A	N/A
7179267	12/2006	Nolan et al.	N/A	N/A
7182239	12/2006	Myers	N/A	N/A
7182763	12/2006	Nardella	N/A	N/A
7183737	12/2006	Kitagawa	N/A	N/A
7187960	12/2006	Abreu	N/A	N/A
7188758	12/2006	Viola et al.	N/A	N/A
7189207	12/2006	Viola	N/A	N/A
7190147	12/2006	Gileff et al.	N/A	N/A
7193199	12/2006	Jang	N/A	N/A
7195627	12/2006	Amoah et al.	N/A	N/A
7196911	12/2006	Takano et al.	N/A	N/A
D541418	12/2006	Schechter et al.	N/A	N/A
7197965	12/2006	Anderson	N/A	N/A
7199537	12/2006	Okamura et al.	N/A	N/A
7199545	12/2006	Oleynikov et al.	N/A	N/A
7202576	12/2006	Dechene et al.	N/A	N/A
7202653	12/2006	Pai	N/A	N/A
7204404	12/2006	Nguyen et al.	N/A	N/A
7204835	12/2006	Latterell et al.	N/A	N/A
7205959	12/2006	Henriksson	N/A	N/A
7206626	12/2006	Quaid, III	N/A	N/A
7207233	12/2006	Wadge	N/A	N/A
7207471	12/2006	Heinrich et al.	N/A	N/A
7207472	12/2006	Wukusick et al.	N/A	N/A
7207556	12/2006	Saitoh et al.	N/A	N/A

7208005	12/2006	Frecker et al.	N/A	N/A
7210609	12/2006	Leiboff et al.	N/A	N/A
7211081	12/2006	Goble	N/A	N/A
7211084	12/2006	Goble et al.	N/A	N/A
7211092	12/2006	Hughett	N/A	N/A
7211979	12/2006	Khatib et al.	N/A	N/A
7213736	12/2006	Wales et al.	N/A	N/A
7214224	12/2006	Goble	N/A	N/A
7215517	12/2006	Takamatsu	N/A	N/A
7217285	12/2006	Vargas et al.	N/A	N/A
7220260	12/2006	Fleming et al.	N/A	N/A
7220272	12/2006	Weadock	N/A	N/A
7225959	12/2006	Patton et al.	N/A	N/A
7225963	12/2006	Scirica	N/A	N/A
7225964	12/2006	Mastri et al.	N/A	N/A
7226450	12/2006	Athanasiou et al.	N/A	N/A
7226467	12/2006	Lucatero et al.	N/A	N/A
7228505	12/2006	Shimazu et al.	N/A	N/A
7229408	12/2006	Douglas et al.	N/A	N/A
7234624	12/2006	Gresham et al.	N/A	N/A
7235072	12/2006	Sartor et al.	N/A	N/A
7235089	12/2006	McGuckin, Jr.	N/A	N/A
7235302	12/2006	Jing et al.	N/A	N/A
7237708	12/2006	Guy et al.	N/A	N/A
7238195	12/2006	Viola	N/A	N/A
7238901	12/2006	Kim et al.	N/A	N/A
7239657	12/2006	Gunnarsson	N/A	N/A
7241288	12/2006	Braun	N/A	N/A
7241289	12/2006	Braun	N/A	N/A
7246734	12/2006	Shelton, IV	N/A	N/A
7247161	12/2006	Johnston et al.	N/A	N/A
7249267	12/2006	Chapuis	N/A	N/A
7252641	12/2006	Thompson et al.	N/A	N/A
7252660	12/2006	Kunz	N/A	N/A
7255012	12/2006	Hedtke	N/A	N/A
7255696	12/2006	Goble et al.	N/A	N/A
7256695	12/2006	Hamel et al.	N/A	N/A
7258262	12/2006	Mastri et al.	N/A	N/A
7258546	12/2006	Beier et al.	N/A	N/A
7260431	12/2006	Libbus et al.	N/A	N/A
7265374	12/2006	Lee et al.	N/A	N/A
7267677	12/2006	Johnson et al.	N/A	N/A
7267679	12/2006	McGuckin, Jr. et al.	N/A	N/A
7272002	12/2006	Drapeau	N/A	N/A
7273483	12/2006	Wiener et al.	N/A	N/A
7273488	12/2006	Nakamura et al.	N/A	N/A
D552623	12/2006	Vong et al.	N/A	N/A
7275674	12/2006	Racenet et al.	N/A	N/A
7276044	12/2006	Ferry et al.	N/A	N/A
7276068	12/2006	Johnson et al.	N/A	N/A

7278562	12/2006	Mastri et al.	N/A	N/A
7278563	12/2006	Green	N/A	N/A
7278949	12/2006	Bader	N/A	N/A
7278994	12/2006	Goble	N/A	N/A
7282048	12/2006	Goble et al.	N/A	N/A
7283096	12/2006	Geisheimer et al.	N/A	N/A
7286850	12/2006	Frielink et al.	N/A	N/A
7287682	12/2006	Ezzat et al.	N/A	N/A
7289139	12/2006	Amling et al.	N/A	N/A
7293685	12/2006	Ehrenfels et al.	N/A	N/A
7295893	12/2006	Sunaoshi	N/A	N/A
7295907	12/2006	Lu et al.	N/A	N/A
7296722	12/2006	Ivanko	N/A	N/A
7296724	12/2006	Green et al.	N/A	N/A
7297149	12/2006	Vitali et al.	N/A	N/A
7300373	12/2006	Jinno et al.	N/A	N/A
7300431	12/2006	Dubrovsky	N/A	N/A
7300450	12/2006	Vleugels et al.	N/A	N/A
7303106	12/2006	Milliman et al.	N/A	N/A
7303107	12/2006	Milliman et al.	N/A	N/A
7303108	12/2006	Shelton, IV	N/A	N/A
7303502	12/2006	Thompson	N/A	N/A
7303556	12/2006	Metzger	N/A	N/A
7306597	12/2006	Manzo	N/A	N/A
7308998	12/2006	Mastri et al.	N/A	N/A
7311238	12/2006	Liu	N/A	N/A
7311709	12/2006	Truckai et al.	N/A	N/A
7313430	12/2006	Urquhart et al.	N/A	N/A
7314473	12/2007	Jinno et al.	N/A	N/A
7317955	12/2007	McGreevy	N/A	N/A
7320704	12/2007	Lashinski et al.	N/A	N/A
7322859	12/2007	Evans	N/A	N/A
7322975	12/2007	Goble et al.	N/A	N/A
7322994	12/2007	Nicholas et al.	N/A	N/A
7324572	12/2007	Chang	N/A	N/A
7326203	12/2007	Papineau et al.	N/A	N/A
7326213	12/2007	Benderev et al.	N/A	N/A
7328828	12/2007	Ortiz et al.	N/A	N/A
7328829	12/2007	Arad et al.	N/A	N/A
7330004	12/2007	DeJonge et al.	N/A	N/A
7331340	12/2007	Barney	N/A	N/A
7331343	12/2007	Schmidt et al.	N/A	N/A
7331403	12/2007	Berry et al.	N/A	N/A
7331406	12/2007	Wottreng, Jr. et al.	N/A	N/A
7331969	12/2007	Inganas et al.	N/A	N/A
7334717	12/2007	Rethy et al.	N/A	N/A
7334718	12/2007	McAlister et al.	N/A	N/A
7335199	12/2007	Goble et al.	N/A	N/A
7335401	12/2007	Finke et al.	N/A	N/A
7336045	12/2007	Clermonts	N/A	N/A

7336048	12/2007	Lohr	N/A	N/A
7336183	12/2007	Reddy et al.	N/A	N/A
7336184	12/2007	Smith et al.	N/A	N/A
7337774	12/2007	Webb	N/A	N/A
7338505	12/2007	Belson	N/A	N/A
7338513	12/2007	Lee et al.	N/A	N/A
7341554	12/2007	Sekine et al.	N/A	N/A
7341555	12/2007	Ootawara et al.	N/A	N/A
7341591	12/2007	Grinberg	N/A	N/A
7343920	12/2007	Toby et al.	N/A	N/A
7344532	12/2007	Goble et al.	N/A	N/A
7344533	12/2007	Pearson et al.	N/A	N/A
7346344	12/2007	Fontaine	N/A	N/A
7346406	12/2007	Brotto et al.	N/A	N/A
7348763	12/2007	Reinhart et al.	N/A	N/A
7348875	12/2007	Hughes et al.	N/A	N/A
RE40237	12/2007	Bilotti et al.	N/A	N/A
7351258	12/2007	Ricotta et al.	N/A	N/A
7354398	12/2007	Kanazawa	N/A	N/A
7354440	12/2007	Truckal et al.	N/A	N/A
7354447	12/2007	Shelton, IV et al.	N/A	N/A
7354502	12/2007	Polat et al.	N/A	N/A
7357287	12/2007	Shelton, IV et al.	N/A	N/A
7357806	12/2007	Rivera et al.	N/A	N/A
7361168	12/2007	Makower et al.	N/A	N/A
7361195	12/2007	Schwartz et al.	N/A	N/A
7362062	12/2007	Schneider et al.	N/A	N/A
7364060	12/2007	Milliman	N/A	N/A
7364061	12/2007	Swayze et al.	N/A	N/A
7367485	12/2007	Shelton, IV et al.	N/A	N/A
7367973	12/2007	Manzo et al.	N/A	N/A
7368124	12/2007	Chun et al.	N/A	N/A
7371210	12/2007	Brock et al.	N/A	N/A
7371403	12/2007	McCarthy et al.	N/A	N/A
7375493	12/2007	Calhoon et al.	N/A	N/A
7377918	12/2007	Amoah	N/A	N/A
7377928	12/2007	Zubik et al.	N/A	N/A
7378817	12/2007	Calhoon et al.	N/A	N/A
RE40388	12/2007	Gines	N/A	N/A
D570868	12/2007	Hosokawa et al.	N/A	N/A
7380695	12/2007	Doll et al.	N/A	N/A
7380696	12/2007	Shelton, IV et al.	N/A	N/A
7384403	12/2007	Sherman	N/A	N/A
7384417	12/2007	Cucin	N/A	N/A
7386365	12/2007	Nixon	N/A	N/A
7386730	12/2007	Uchikubo	N/A	N/A
7388217	12/2007	Buschbeck et al.	N/A	N/A
7388484	12/2007	Hsu	N/A	N/A
7391173	12/2007	Schena	N/A	N/A
7394190	12/2007	Huang	N/A	N/A

7396356	12/2007	Mollenauer	N/A	N/A
7397364	12/2007	Govari	N/A	N/A
7398707	12/2007	Morley et al.	N/A	N/A
7398907	12/2007	Racenet et al.	N/A	N/A
7398908	12/2007	Holsten et al.	N/A	N/A
7400107	12/2007	Schneider et al.	N/A	N/A
7400752	12/2007	Zacharias	N/A	N/A
7401000	12/2007	Nakamura	N/A	N/A
7401721	12/2007	Holsten et al.	N/A	N/A
7404449	12/2007	Birmingham et al.	N/A	N/A
7404508	12/2007	Smith et al.	N/A	N/A
7404509	12/2007	Ortiz et al.	N/A	N/A
7404822	12/2007	Viart et al.	N/A	N/A
D575793	12/2007	Ording	N/A	N/A
7407074	12/2007	Ortiz et al.	N/A	N/A
7407075	12/2007	Holsten et al.	N/A	N/A
7407076	12/2007	Racenet et al.	N/A	N/A
7407077	12/2007	Ortiz et al.	N/A	N/A
7407078	12/2007	Shelton, IV et al.	N/A	N/A
7408310	12/2007	Hong et al.	N/A	N/A
7410085	12/2007	Wolf et al.	N/A	N/A
7410086	12/2007	Ortiz et al.	N/A	N/A
7410483	12/2007	Danitz et al.	N/A	N/A
7413563	12/2007	Corcoran et al.	N/A	N/A
7416101	12/2007	Shelton, IV et al.	N/A	N/A
7418078	12/2007	Blanz et al.	N/A	N/A
RE40514	12/2007	Mastri et al.	N/A	N/A
7419080	12/2007	Smith et al.	N/A	N/A
7419081	12/2007	Ehrenfels et al.	N/A	N/A
7419321	12/2007	Tereschouk	N/A	N/A
7419495	12/2007	Menn et al.	N/A	N/A
7422136	12/2007	Marczyk	N/A	N/A
7422138	12/2007	Bilotti et al.	N/A	N/A
7422139	12/2007	Shelton, IV et al.	N/A	N/A
7422582	12/2007	Malackowski et al.	N/A	N/A
7424965	12/2007	Racenet et al.	N/A	N/A
7427607	12/2007	Suzuki	N/A	N/A
D578644	12/2007	Shumer et al.	N/A	N/A
7430772	12/2007	Van Es	N/A	N/A
7430849	12/2007	Coutts et al.	N/A	N/A
7431188	12/2007	Marczyk	N/A	N/A
7431189	12/2007	Shelton, IV et al.	N/A	N/A
7431230	12/2007	McPherson et al.	N/A	N/A
7431694	12/2007	Stefanchik et al.	N/A	N/A
7431730	12/2007	Viola	N/A	N/A
7434715	12/2007	Shelton, IV et al.	N/A	N/A
7434717	12/2007	Shelton, IV et al.	N/A	N/A
7435249	12/2007	Buyse et al.	N/A	N/A
7438209	12/2007	Hess et al.	N/A	N/A
7438718	12/2007	Milliman et al.	N/A	N/A

7439354	12/2007	Lenges et al.	N/A	N/A
7441684	12/2007	Shelton, IV et al.	N/A	N/A
7441685	12/2007	Boudreaux	N/A	N/A
7442201	12/2007	Pugsley et al.	N/A	N/A
7443547	12/2007	Moreno et al.	N/A	N/A
D580942	12/2007	Oshiro et al.	N/A	N/A
7446131	12/2007	Liu et al.	N/A	N/A
7448525	12/2007	Shelton, IV et al.	N/A	N/A
7450010	12/2007	Gravelle et al.	N/A	N/A
7450991	12/2007	Smith et al.	N/A	N/A
7451904	12/2007	Shelton, IV	N/A	N/A
7455208	12/2007	Wales et al.	N/A	N/A
7455676	12/2007	Holsten et al.	N/A	N/A
7455682	12/2007	Viola	N/A	N/A
7455687	12/2007	Saunders et al.	N/A	N/A
D582934	12/2007	Byeon	N/A	N/A
7461767	12/2007	Viola et al.	N/A	N/A
7462187	12/2007	Johnston et al.	N/A	N/A
7464845	12/2007	Chou	N/A	N/A
7464846	12/2007	Shelton, IV et al.	N/A	N/A
7464847	12/2007	Viola et al.	N/A	N/A
7464848	12/2007	Green et al.	N/A	N/A
7464849	12/2007	Shelton, IV et al.	N/A	N/A
7467740	12/2007	Shelton, IV et al.	N/A	N/A
7467849	12/2007	Silverbrook et al.	N/A	N/A
7472814	12/2008	Mastri et al.	N/A	N/A
7472815	12/2008	Shelton, IV et al.	N/A	N/A
7472816	12/2008	Holsten et al.	N/A	N/A
7473221	12/2008	Ewers et al.	N/A	N/A
7473253	12/2008	Dycus et al.	N/A	N/A
7473263	12/2008	Johnston et al.	N/A	N/A
7476237	12/2008	Taniguchi et al.	N/A	N/A
7479147	12/2008	Honeycutt et al.	N/A	N/A
7479608	12/2008	Smith	N/A	N/A
7481347	12/2008	Roy	N/A	N/A
7481348	12/2008	Marczyk	N/A	N/A
7481349	12/2008	Holsten et al.	N/A	N/A
7481824	12/2008	Boudreaux et al.	N/A	N/A
7485124	12/2008	Kuhns et al.	N/A	N/A
7485133	12/2008	Cannon et al.	N/A	N/A
7485142	12/2008	Milo	N/A	N/A
7487899	12/2008	Shelton, IV et al.	N/A	N/A
7489055	12/2008	Jeong et al.	N/A	N/A
7490749	12/2008	Schall et al.	N/A	N/A
7491232	12/2008	Bolduc et al.	N/A	N/A
7492261	12/2008	Cambre et al.	N/A	N/A
7494039	12/2008	Racenet et al.	N/A	N/A
7494460	12/2008	Haarstad et al.	N/A	N/A
7494499	12/2008	Nagase et al.	N/A	N/A
7494501	12/2008	Ahlberg et al.	N/A	N/A

7497137	12/2008	Tellenbach et al.	N/A	N/A
7500979	12/2008	Hueil et al.	N/A	N/A
7501198	12/2008	Barlev et al.	N/A	N/A
7503474	12/2008	Hillstead et al.	N/A	N/A
7506790	12/2008	Shelton, IV	N/A	N/A
7506791	12/2008	Omaits et al.	N/A	N/A
7507202	12/2008	Schoellhorn	N/A	N/A
7510107	12/2008	Timm et al.	N/A	N/A
7510534	12/2008	Burdorff et al.	N/A	N/A
7510566	12/2008	Jacobs et al.	N/A	N/A
7513407	12/2008	Chang	N/A	N/A
7513408	12/2008	Shelton, IV et al.	N/A	N/A
7517356	12/2008	Heinrich	N/A	N/A
7524320	12/2008	Tierney et al.	N/A	N/A
7527632	12/2008	Houghton et al.	N/A	N/A
7530984	12/2008	Sonnenschein et al.	N/A	N/A
7530985	12/2008	Takemoto et al.	N/A	N/A
7533790	12/2008	Knodel et al.	N/A	N/A
7533906	12/2008	Luetngen et al.	N/A	N/A
7534259	12/2008	Lashinski et al.	N/A	N/A
7540867	12/2008	Jinno et al.	N/A	N/A
7540872	12/2008	Schechter et al.	N/A	N/A
7542807	12/2008	Bertolero et al.	N/A	N/A
7543730	12/2008	Marczyk	N/A	N/A
7544197	12/2008	Kelsch et al.	N/A	N/A
7546939	12/2008	Adams et al.	N/A	N/A
7546940	12/2008	Milliman et al.	N/A	N/A
7547287	12/2008	Boecker et al.	N/A	N/A
7547312	12/2008	Bauman et al.	N/A	N/A
7549563	12/2008	Mather et al.	N/A	N/A
7549564	12/2008	Boudreaux	N/A	N/A
7549998	12/2008	Braun	N/A	N/A
7552854	12/2008	Wixey et al.	N/A	N/A
7553173	12/2008	Kowalick	N/A	N/A
7553275	12/2008	Padget et al.	N/A	N/A
7554343	12/2008	Bromfield	N/A	N/A
7556185	12/2008	Viola	N/A	N/A
7556186	12/2008	Milliman	N/A	N/A
7556647	12/2008	Drews et al.	N/A	N/A
7559449	12/2008	Viola	N/A	N/A
7559450	12/2008	Wales et al.	N/A	N/A
7559452	12/2008	Wales et al.	N/A	N/A
7559937	12/2008	de la Torre et al.	N/A	N/A
7561637	12/2008	Jonsson et al.	N/A	N/A
7562910	12/2008	Kertesz et al.	N/A	N/A
7563269	12/2008	Hashiguchi	N/A	N/A
7563862	12/2008	Sieg et al.	N/A	N/A
7565993	12/2008	Milliman et al.	N/A	N/A
7566300	12/2008	Devierre et al.	N/A	N/A
7567045	12/2008	Fristedt	N/A	N/A

7568603	12/2008	Shelton, IV et al.	N/A	N/A
7568604	12/2008	Ehrenfels et al.	N/A	N/A
7568619	12/2008	Todd et al.	N/A	N/A
7572285	12/2008	Frey et al.	N/A	N/A
7572298	12/2008	Roller et al.	N/A	N/A
7575144	12/2008	Ortiz et al.	N/A	N/A
7578825	12/2008	Huebner	N/A	N/A
D600703	12/2008	LaManna et al.	N/A	N/A
D600712	12/2008	LaManna et al.	N/A	N/A
7582086	12/2008	Privitera et al.	N/A	N/A
7583063	12/2008	Dooley	N/A	N/A
7584880	12/2008	Racenet et al.	N/A	N/A
7586289	12/2008	Andruk et al.	N/A	N/A
7588174	12/2008	Holsten et al.	N/A	N/A
7588175	12/2008	Timm et al.	N/A	N/A
7588176	12/2008	Timm et al.	N/A	N/A
7588177	12/2008	Racenet	N/A	N/A
7591783	12/2008	Boulais et al.	N/A	N/A
7591818	12/2008	Bertolero et al.	N/A	N/A
7593766	12/2008	Faber et al.	N/A	N/A
7595642	12/2008	Doyle	N/A	N/A
D601578	12/2008	Poulet et al.	N/A	N/A
7597229	12/2008	Boudreaux et al.	N/A	N/A
7597230	12/2008	Racenet et al.	N/A	N/A
7597693	12/2008	Garrison	N/A	N/A
7597699	12/2008	Rogers	N/A	N/A
7598972	12/2008	Tomita	N/A	N/A
7600663	12/2008	Green	N/A	N/A
7604118	12/2008	Iio et al.	N/A	N/A
7604150	12/2008	Boudreaux	N/A	N/A
7604151	12/2008	Hess et al.	N/A	N/A
7604668	12/2008	Farnsworth et al.	N/A	N/A
7605826	12/2008	Sauer	N/A	N/A
7607557	12/2008	Shelton, IV et al.	N/A	N/A
7608091	12/2008	Goldfarb et al.	N/A	N/A
D604325	12/2008	Ebeling et al.	N/A	N/A
7611038	12/2008	Racenet et al.	N/A	N/A
7611474	12/2008	Hibner et al.	N/A	N/A
7615003	12/2008	Stefanchik et al.	N/A	N/A
7615006	12/2008	Abe	N/A	N/A
7615067	12/2008	Lee et al.	N/A	N/A
7617961	12/2008	Viola	N/A	N/A
7618427	12/2008	Ortiz et al.	N/A	N/A
D605201	12/2008	Lorenz et al.	N/A	N/A
D606992	12/2008	Liu et al.	N/A	N/A
D607010	12/2008	Kocmick	N/A	N/A
7624902	12/2008	Marczyk et al.	N/A	N/A
7624903	12/2008	Green et al.	N/A	N/A
7625370	12/2008	Hart et al.	N/A	N/A
7625388	12/2008	Boukhny et al.	N/A	N/A

7625662	12/2008	Vaisnys et al.	N/A	N/A
7630841	12/2008	Comisky et al.	N/A	N/A
7631793	12/2008	Rethy et al.	N/A	N/A
7631794	12/2008	Rethy et al.	N/A	N/A
7635074	12/2008	Olson et al.	N/A	N/A
7635922	12/2008	Becker	N/A	N/A
7637409	12/2008	Marczyk	N/A	N/A
7637410	12/2008	Marczyk	N/A	N/A
7638958	12/2008	Philipp et al.	N/A	N/A
7641091	12/2009	Olson et al.	N/A	N/A
7641092	12/2009	Kruszynski et al.	N/A	N/A
7641093	12/2009	Doll et al.	N/A	N/A
7641095	12/2009	Viola	N/A	N/A
7641671	12/2009	Crainich	N/A	N/A
7644016	12/2009	Nycz et al.	N/A	N/A
7644484	12/2009	Vereschagin	N/A	N/A
7644783	12/2009	Roberts et al.	N/A	N/A
7644848	12/2009	Swayze et al.	N/A	N/A
7645230	12/2009	Mikkaichi et al.	N/A	N/A
7648055	12/2009	Marczyk	N/A	N/A
7648457	12/2009	Stefanchik et al.	N/A	N/A
7648519	12/2009	Lee et al.	N/A	N/A
7650185	12/2009	Maile et al.	N/A	N/A
7651017	12/2009	Ortiz et al.	N/A	N/A
7651498	12/2009	Shifrin et al.	N/A	N/A
7654431	12/2009	Hueil et al.	N/A	N/A
7655003	12/2009	Lorang et al.	N/A	N/A
7655004	12/2009	Long	N/A	N/A
7655288	12/2009	Bauman et al.	N/A	N/A
7655584	12/2009	Biran et al.	N/A	N/A
7656131	12/2009	Embrey et al.	N/A	N/A
7658311	12/2009	Boudreaux	N/A	N/A
7658312	12/2009	Vidal et al.	N/A	N/A
7658705	12/2009	Melvin et al.	N/A	N/A
7659219	12/2009	Biran et al.	N/A	N/A
7661448	12/2009	Kim et al.	N/A	N/A
7662161	12/2009	Briganti et al.	N/A	N/A
7665646	12/2009	Prommersberger	N/A	N/A
7665647	12/2009	Shelton, IV et al.	N/A	N/A
7666195	12/2009	Kelleher et al.	N/A	N/A
7669746	12/2009	Shelton, IV	N/A	N/A
7669747	12/2009	Weisenburgh, II et al.	N/A	N/A
7670334	12/2009	Hueil et al.	N/A	N/A
7670337	12/2009	Young	N/A	N/A
7673780	12/2009	Shelton, IV et al.	N/A	N/A
7673781	12/2009	Swayze et al.	N/A	N/A
7673782	12/2009	Hess et al.	N/A	N/A
7673783	12/2009	Morgan et al.	N/A	N/A
7674253	12/2009	Fisher et al.	N/A	N/A
7674255	12/2009	Braun	N/A	N/A

7674263	12/2009	Ryan	N/A	N/A
7674270	12/2009	Layer	N/A	N/A
7678121	12/2009	Knodel	N/A	N/A
7682307	12/2009	Danitz et al.	N/A	N/A
7682367	12/2009	Shah et al.	N/A	N/A
7682686	12/2009	Curro et al.	N/A	N/A
7686201	12/2009	Csiky	N/A	N/A
7686804	12/2009	Johnson et al.	N/A	N/A
7686826	12/2009	Lee et al.	N/A	N/A
7688028	12/2009	Phillips et al.	N/A	N/A
7690547	12/2009	Racenet et al.	N/A	N/A
7691098	12/2009	Wallace et al.	N/A	N/A
7691103	12/2009	Fernandez et al.	N/A	N/A
7691106	12/2009	Schenberger et al.	N/A	N/A
7694864	12/2009	Okada et al.	N/A	N/A
7694865	12/2009	Scirica	N/A	N/A
7695485	12/2009	Whitman et al.	N/A	N/A
7695493	12/2009	Saadat et al.	N/A	N/A
7699204	12/2009	Viola	N/A	N/A
7699835	12/2009	Lee et al.	N/A	N/A
7699844	12/2009	Utley et al.	N/A	N/A
7699846	12/2009	Ryan	N/A	N/A
7699856	12/2009	Van Wyk et al.	N/A	N/A
7699859	12/2009	Bombard et al.	N/A	N/A
7699860	12/2009	Huitema et al.	N/A	N/A
7699868	12/2009	Frank et al.	N/A	N/A
7703653	12/2009	Shah et al.	N/A	N/A
7705559	12/2009	Powell et al.	N/A	N/A
7706853	12/2009	Hacker et al.	N/A	N/A
7708180	12/2009	Murray et al.	N/A	N/A
7708181	12/2009	Cole et al.	N/A	N/A
7708182	12/2009	Viola	N/A	N/A
7708758	12/2009	Lee et al.	N/A	N/A
7708768	12/2009	Danek et al.	N/A	N/A
7709136	12/2009	Touhton et al.	N/A	N/A
7712182	12/2009	Zeiler et al.	N/A	N/A
7713190	12/2009	Brock et al.	N/A	N/A
7713542	12/2009	Xu et al.	N/A	N/A
7714239	12/2009	Smith	N/A	N/A
7714334	12/2009	Lin	N/A	N/A
7717312	12/2009	Beetel	N/A	N/A
7717313	12/2009	Criscuolo et al.	N/A	N/A
7717846	12/2009	Zirps et al.	N/A	N/A
7717873	12/2009	Swick	N/A	N/A
7717915	12/2009	Miyazawa	N/A	N/A
7717926	12/2009	Whitfield et al.	N/A	N/A
7718180	12/2009	Karp	N/A	N/A
7718556	12/2009	Matsuda et al.	N/A	N/A
7721930	12/2009	Mckenna et al.	N/A	N/A
7721931	12/2009	Shelton, IV et al.	N/A	N/A

7721932	12/2009	Cole et al.	N/A	N/A
7721933	12/2009	Ehrenfels et al.	N/A	N/A
7721934	12/2009	Shelton, IV et al.	N/A	N/A
7721936	12/2009	Shalton, IV et al.	N/A	N/A
7722527	12/2009	Bouchier et al.	N/A	N/A
7722607	12/2009	Dumbauld et al.	N/A	N/A
7722610	12/2009	Viola et al.	N/A	N/A
7725214	12/2009	Diolaiti	N/A	N/A
7726171	12/2009	Langlotz et al.	N/A	N/A
7726537	12/2009	Olson et al.	N/A	N/A
7726538	12/2009	Holsten et al.	N/A	N/A
7726539	12/2009	Holsten et al.	N/A	N/A
7727954	12/2009	McKay	N/A	N/A
7728553	12/2009	Carrier et al.	N/A	N/A
7729742	12/2009	Govari	N/A	N/A
7731072	12/2009	Timm et al.	N/A	N/A
7731073	12/2009	Wixey et al.	N/A	N/A
7731724	12/2009	Huitema et al.	N/A	N/A
7735703	12/2009	Morgan et al.	N/A	N/A
7735704	12/2009	Bilotti	N/A	N/A
7736254	12/2009	Schena	N/A	N/A
7736306	12/2009	Brustad et al.	N/A	N/A
7736356	12/2009	Cooper et al.	N/A	N/A
7736374	12/2009	Vaughan et al.	N/A	N/A
7738971	12/2009	Swayze et al.	N/A	N/A
7740159	12/2009	Shelton, IV et al.	N/A	N/A
7742036	12/2009	Grant et al.	N/A	N/A
7743960	12/2009	Whitman et al.	N/A	N/A
7744624	12/2009	Bettuchi	N/A	N/A
7744627	12/2009	Orban, III et al.	N/A	N/A
7744628	12/2009	Viola	N/A	N/A
7747146	12/2009	Milano et al.	N/A	N/A
7748587	12/2009	Haramiishi et al.	N/A	N/A
7748632	12/2009	Coleman et al.	N/A	N/A
7749204	12/2009	Dhanaraj et al.	N/A	N/A
7749240	12/2009	Takahashi et al.	N/A	N/A
7751870	12/2009	Whitman	N/A	N/A
7753245	12/2009	Boudreaux et al.	N/A	N/A
7753246	12/2009	Scirica	N/A	N/A
7753904	12/2009	Shelton, IV et al.	N/A	N/A
7757924	12/2009	Gerbi et al.	N/A	N/A
7758594	12/2009	Lamson et al.	N/A	N/A
7758612	12/2009	Shipp	N/A	N/A
7758613	12/2009	Whitman	N/A	N/A
7762462	12/2009	Gelbman	N/A	N/A
7762998	12/2009	Birk et al.	N/A	N/A
D622286	12/2009	Umezawa	N/A	N/A
7766207	12/2009	Mather et al.	N/A	N/A
7766209	12/2009	Baxter, III et al.	N/A	N/A
7766210	12/2009	Shelton, IV et al.	N/A	N/A

7766821	12/2009	Brunnen et al.	N/A	N/A
7766894	12/2009	Weitzner et al.	N/A	N/A
7770658	12/2009	Ito et al.	N/A	N/A
7770773	12/2009	Whitman et al.	N/A	N/A
7770774	12/2009	Mastri et al.	N/A	N/A
7770775	12/2009	Shelton, IV et al.	N/A	N/A
7770776	12/2009	Chen et al.	N/A	N/A
7771396	12/2009	Stefanchik et al.	N/A	N/A
7772720	12/2009	McGee et al.	N/A	N/A
7772725	12/2009	Siman-Tov	N/A	N/A
7775972	12/2009	Brock et al.	N/A	N/A
7776037	12/2009	Odom	N/A	N/A
7776060	12/2009	Mooradian et al.	N/A	N/A
7776065	12/2009	Griffiths et al.	N/A	N/A
7778004	12/2009	Nerheim et al.	N/A	N/A
7779614	12/2009	McGonagle et al.	N/A	N/A
7779737	12/2009	Newman, Jr. et al.	N/A	N/A
7780054	12/2009	Wales	N/A	N/A
7780055	12/2009	Scirica et al.	N/A	N/A
7780309	12/2009	McMillan et al.	N/A	N/A
7780651	12/2009	Madhani et al.	N/A	N/A
7780663	12/2009	Yates et al.	N/A	N/A
7780685	12/2009	Hunt et al.	N/A	N/A
7782382	12/2009	Fujimura	N/A	N/A
7784662	12/2009	Wales et al.	N/A	N/A
7784663	12/2009	Shelton, IV	N/A	N/A
7787256	12/2009	Chan et al.	N/A	N/A
7789283	12/2009	Shah	N/A	N/A
7789875	12/2009	Brock et al.	N/A	N/A
7789883	12/2009	Takashino et al.	N/A	N/A
7789889	12/2009	Zubik et al.	N/A	N/A
7793812	12/2009	Moore et al.	N/A	N/A
7794475	12/2009	Hess et al.	N/A	N/A
7798386	12/2009	Schall et al.	N/A	N/A
7799039	12/2009	Shelton, IV et al.	N/A	N/A
7799044	12/2009	Johnston et al.	N/A	N/A
7799965	12/2009	Patel et al.	N/A	N/A
7803151	12/2009	Whitman	N/A	N/A
7806871	12/2009	Li et al.	N/A	N/A
7806891	12/2009	Nowlin et al.	N/A	N/A
7810690	12/2009	Bilotti et al.	N/A	N/A
7810691	12/2009	Boyden et al.	N/A	N/A
7810692	12/2009	Hall et al.	N/A	N/A
7810693	12/2009	Broehl et al.	N/A	N/A
7811275	12/2009	Birk et al.	N/A	N/A
7814816	12/2009	Alberti et al.	N/A	N/A
7815092	12/2009	Whitman et al.	N/A	N/A
7815565	12/2009	Stefanchik et al.	N/A	N/A
7815662	12/2009	Spivey et al.	N/A	N/A
7819296	12/2009	Hueil et al.	N/A	N/A

7819297	12/2009	Doll et al.	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7819299	12/2009	Shelton, IV et al.	N/A	N/A
7819799	12/2009	Merril et al.	N/A	N/A
7819884	12/2009	Lee et al.	N/A	N/A
7819885	12/2009	Cooper	N/A	N/A
7819886	12/2009	Whitfield et al.	N/A	N/A
7819894	12/2009	Mitsuishi et al.	N/A	N/A
7823076	12/2009	Borovsky et al.	N/A	N/A
7823592	12/2009	Bettuchi et al.	N/A	N/A
7823760	12/2009	Zemlok et al.	N/A	N/A
7824401	12/2009	Manzo et al.	N/A	N/A
7824422	12/2009	Benchetrit	N/A	N/A
7824426	12/2009	Racenet et al.	N/A	N/A
7828189	12/2009	Holsten et al.	N/A	N/A
7828794	12/2009	Sartor	N/A	N/A
7828808	12/2009	Hinman et al.	N/A	N/A
7829416	12/2009	Kudou et al.	N/A	N/A
7831292	12/2009	Quaid et al.	N/A	N/A
7832408	12/2009	Shelton, IV et al.	N/A	N/A
7832611	12/2009	Boyden et al.	N/A	N/A
7832612	12/2009	Baxter, III et al.	N/A	N/A
7833234	12/2009	Bailly et al.	N/A	N/A
7835823	12/2009	Sillman et al.	N/A	N/A
7836400	12/2009	May et al.	N/A	N/A
7837079	12/2009	Holsten et al.	N/A	N/A
7837080	12/2009	Schwemberger	N/A	N/A
7837081	12/2009	Holsten et al.	N/A	N/A
7837425	12/2009	Saeki et al.	N/A	N/A
7837685	12/2009	Weinberg et al.	N/A	N/A
7837687	12/2009	Harp	N/A	N/A
7837694	12/2009	Tethrake et al.	N/A	N/A
7838789	12/2009	Stoffers et al.	N/A	N/A
7839109	12/2009	Carmen, Jr. et al.	N/A	N/A
7840253	12/2009	Tremblay et al.	N/A	N/A
7841503	12/2009	Sonnenschein et al.	N/A	N/A
7842025	12/2009	Coleman et al.	N/A	N/A
7842028	12/2009	Lee	N/A	N/A
7843158	12/2009	Prisco	N/A	N/A
7845533	12/2009	Marczyk et al.	N/A	N/A
7845534	12/2009	Viola et al.	N/A	N/A
7845535	12/2009	Scircia	N/A	N/A
7845536	12/2009	Viola et al.	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7845538	12/2009	Whitman	N/A	N/A
7845912	12/2009	Sung et al.	N/A	N/A
7846085	12/2009	Silverman et al.	N/A	N/A
7846149	12/2009	Jankowski	N/A	N/A
7846161	12/2009	Dumbauld et al.	N/A	N/A
7848066	12/2009	Yanagishima	N/A	N/A

7850623	12/2009	Griffin et al.	N/A	N/A
7850642	12/2009	Moll et al.	N/A	N/A
7850982	12/2009	Stopek et al.	N/A	N/A
7853813	12/2009	Lee	N/A	N/A
7854735	12/2009	Houser et al.	N/A	N/A
7854736	12/2009	Ryan	N/A	N/A
7857183	12/2009	Shelton, IV	N/A	N/A
7857184	12/2009	Viola	N/A	N/A
7857185	12/2009	Swayze et al.	N/A	N/A
7857186	12/2009	Baxter, III et al.	N/A	N/A
7857813	12/2009	Schmitz et al.	N/A	N/A
7861906	12/2010	Doll et al.	N/A	N/A
7862502	12/2010	Pool et al.	N/A	N/A
7862546	12/2010	Conlon et al.	N/A	N/A
7862579	12/2010	Ortiz et al.	N/A	N/A
7866525	12/2010	Scirica	N/A	N/A
7866527	12/2010	Hall et al.	N/A	N/A
7866528	12/2010	Olson et al.	N/A	N/A
7870989	12/2010	Viola et al.	N/A	N/A
7871418	12/2010	Thompson et al.	N/A	N/A
7871440	12/2010	Schwartz et al.	N/A	N/A
7875055	12/2010	Cichocki, Jr.	N/A	N/A
7877869	12/2010	Mehdizadeh et al.	N/A	N/A
7879063	12/2010	Khosravi	N/A	N/A
7879070	12/2010	Ortiz et al.	N/A	N/A
7879367	12/2010	Heublein et al.	N/A	N/A
7883461	12/2010	Albrecht et al.	N/A	N/A
7883465	12/2010	Donofrio et al.	N/A	N/A
7883540	12/2010	Niwa et al.	N/A	N/A
7886951	12/2010	Hessler	N/A	N/A
7886952	12/2010	Scirica et al.	N/A	N/A
7887530	12/2010	Zemlok et al.	N/A	N/A
7887535	12/2010	Lands et al.	N/A	N/A
7887536	12/2010	Johnson et al.	N/A	N/A
7887563	12/2010	Cummins	N/A	N/A
7887755	12/2010	Mingerink et al.	N/A	N/A
7891531	12/2010	Ward	N/A	N/A
7891532	12/2010	Mastri et al.	N/A	N/A
7892200	12/2010	Birk et al.	N/A	N/A
7892245	12/2010	Liddicoat et al.	N/A	N/A
7893586	12/2010	West et al.	N/A	N/A
7896214	12/2010	Farascioni	N/A	N/A
7896215	12/2010	Adams et al.	N/A	N/A
7896671	12/2010	Kim et al.	N/A	N/A
7896869	12/2010	DiSilvestro et al.	N/A	N/A
7896877	12/2010	Hall et al.	N/A	N/A
7896895	12/2010	Boudreaux et al.	N/A	N/A
7896897	12/2010	Gresham et al.	N/A	N/A
7896900	12/2010	Frank et al.	N/A	N/A
7898198	12/2010	Murphree	N/A	N/A

7900805	12/2010	Shelton, IV et al.	N/A	N/A
7900806	12/2010	Chen et al.	N/A	N/A
7901381	12/2010	Birk et al.	N/A	N/A
7905380	12/2010	Shelton, IV et al.	N/A	N/A
7905381	12/2010	Baxter, III	227/19	A61B 17/068
7905881	12/2010	Masuda et al.	N/A	N/A
7905889	12/2010	Catanese, III et al.	N/A	N/A
7905890	12/2010	Whitfield et al.	N/A	N/A
7905902	12/2010	Huitema et al.	N/A	N/A
7909039	12/2010	Hur	N/A	N/A
7909191	12/2010	Baker et al.	N/A	N/A
7909220	12/2010	Viola	N/A	N/A
7909221	12/2010	Viola et al.	N/A	N/A
7909224	12/2010	Prommersberger	N/A	N/A
7913891	12/2010	Doll et al.	N/A	N/A
7913893	12/2010	Mastri et al.	N/A	N/A
7914521	12/2010	Wang et al.	N/A	N/A
7914543	12/2010	Roth et al.	N/A	N/A
7914551	12/2010	Ortiz et al.	N/A	N/A
7918230	12/2010	Whitman et al.	N/A	N/A
7918376	12/2010	Knodel et al.	N/A	N/A
7918377	12/2010	Measamer et al.	N/A	N/A
7918845	12/2010	Saadat et al.	N/A	N/A
7918848	12/2010	Lau et al.	N/A	N/A
7918861	12/2010	Brock et al.	N/A	N/A
7918867	12/2010	Dana et al.	N/A	N/A
7922061	12/2010	Shelton, IV et al.	N/A	N/A
7922063	12/2010	Zemlok et al.	N/A	N/A
7922743	12/2010	Heinrich et al.	N/A	N/A
7923144	12/2010	Kohn et al.	N/A	N/A
7926691	12/2010	Viola et al.	N/A	N/A
7926692	12/2010	Racenet et al.	N/A	N/A
7927328	12/2010	Orszulak et al.	N/A	N/A
7928281	12/2010	Augustine	N/A	N/A
7930040	12/2010	Kelsch et al.	N/A	N/A
7930065	12/2010	Larkin et al.	N/A	N/A
7931660	12/2010	Aranyi et al.	N/A	N/A
7931695	12/2010	Ringeisen	N/A	N/A
7931877	12/2010	Steffens et al.	N/A	N/A
D638028	12/2010	Cook et al.	N/A	N/A
7934630	12/2010	Shelton, IV et al.	N/A	N/A
7934631	12/2010	Balbierz et al.	N/A	N/A
7934896	12/2010	Schnier	N/A	N/A
7935130	12/2010	Williams	N/A	N/A
7935773	12/2010	Hadba et al.	N/A	N/A
7936142	12/2010	Otsuka et al.	N/A	N/A
7938307	12/2010	Bettuchi	N/A	N/A
7939152	12/2010	Haskin et al.	N/A	N/A
7941865	12/2010	Seman, Jr. et al.	N/A	N/A

7942300	12/2010	Rethy et al.	N/A	N/A
7942303	12/2010	Shah	N/A	N/A
7942890	12/2010	D'Agostino et al.	N/A	N/A
7944175	12/2010	Mori et al.	N/A	N/A
7945792	12/2010	Cherpantier	N/A	N/A
7945798	12/2010	Carlson et al.	N/A	N/A
7946453	12/2010	Voegele et al.	N/A	N/A
7947011	12/2010	Birk et al.	N/A	N/A
7948381	12/2010	Lindsay et al.	N/A	N/A
7950560	12/2010	Zemlok et al.	N/A	N/A
7950561	12/2010	Aranyi	N/A	N/A
7950562	12/2010	Beardsley et al.	N/A	N/A
7951071	12/2010	Whitman et al.	N/A	N/A
7951166	12/2010	Orban, III et al.	N/A	N/A
7952464	12/2010	Nikitin et al.	N/A	N/A
7954682	12/2010	Giordano et al.	N/A	N/A
7954684	12/2010	Boudreaux	N/A	N/A
7954685	12/2010	Viola	N/A	N/A
7954686	12/2010	Baxter, III et al.	N/A	N/A
7954687	12/2010	Zemlok	227/176.1	A61B 17/064
7954688	12/2010	Argentine et al.	N/A	N/A
7955253	12/2010	Ewers et al.	N/A	N/A
7955257	12/2010	Frasier et al.	N/A	N/A
7955322	12/2010	Devengenzo et al.	N/A	N/A
7955327	12/2010	Sartor et al.	N/A	N/A
7955380	12/2010	Chu et al.	N/A	N/A
7959050	12/2010	Smith et al.	N/A	N/A
7959051	12/2010	Smith et al.	N/A	N/A
7959052	12/2010	Sonnenschein et al.	N/A	N/A
7963432	12/2010	Knodel et al.	N/A	N/A
7963433	12/2010	Whitman et al.	N/A	N/A
7963913	12/2010	Devengenzo et al.	N/A	N/A
7963963	12/2010	Francischelli et al.	N/A	N/A
7963964	12/2010	Santilli et al.	N/A	N/A
7964206	12/2010	Suokas et al.	N/A	N/A
7966236	12/2010	Noriega et al.	N/A	N/A
7966269	12/2010	Bauer et al.	N/A	N/A
7966799	12/2010	Morgan et al.	N/A	N/A
7967178	12/2010	Scirica et al.	N/A	N/A
7967179	12/2010	Olson et al.	N/A	N/A
7967180	12/2010	Scirica	N/A	N/A
7967181	12/2010	Viola et al.	N/A	N/A
7967791	12/2010	Franer et al.	N/A	N/A
7967839	12/2010	Flock et al.	N/A	N/A
7972298	12/2010	Wallace et al.	N/A	N/A
7972315	12/2010	Birk et al.	N/A	N/A
7976213	12/2010	Bertolotti et al.	N/A	N/A
7976508	12/2010	Hoag	N/A	N/A
7976563	12/2010	Summerer	N/A	N/A

7979137	12/2010	Tracey et al.	N/A	N/A
7980443	12/2010	Scheib et al.	N/A	N/A
7981025	12/2010	Pool et al.	N/A	N/A
7981102	12/2010	Patel et al.	N/A	N/A
7981132	12/2010	Dubrul et al.	N/A	N/A
7987405	12/2010	Turner et al.	N/A	N/A
7988015	12/2010	Mason, II et al.	N/A	N/A
7988026	12/2010	Knodel et al.	N/A	N/A
7988027	12/2010	Olson et al.	N/A	N/A
7988028	12/2010	Farascioni et al.	N/A	N/A
7988779	12/2010	Disalvo et al.	N/A	N/A
7992757	12/2010	Wheeler et al.	N/A	N/A
7993360	12/2010	Hacker et al.	N/A	N/A
7994670	12/2010	Ji	N/A	N/A
7997054	12/2010	Bertsch et al.	N/A	N/A
7997468	12/2010	Farascioni	N/A	N/A
7997469	12/2010	Olson et al.	N/A	N/A
8002696	12/2010	Suzuki	N/A	N/A
8002784	12/2010	Jinno et al.	N/A	N/A
8002785	12/2010	Weiss et al.	N/A	N/A
8002795	12/2010	Beetel	N/A	N/A
8006365	12/2010	Levin et al.	N/A	N/A
8006885	12/2010	Marczyk	N/A	N/A
8006889	12/2010	Adams et al.	N/A	N/A
8007370	12/2010	Hirsch et al.	N/A	N/A
8007465	12/2010	Birk et al.	N/A	N/A
8007479	12/2010	Birk et al.	N/A	N/A
8007511	12/2010	Brock et al.	N/A	N/A
8007513	12/2010	Nalagatla et al.	N/A	N/A
8008598	12/2010	Whitman et al.	N/A	N/A
8010180	12/2010	Quaid et al.	N/A	N/A
8011550	12/2010	Aranyi et al.	N/A	N/A
8011551	12/2010	Marczyk et al.	N/A	N/A
8011553	12/2010	Mastri et al.	N/A	N/A
8011555	12/2010	Tarinelli et al.	N/A	N/A
8012170	12/2010	Whitman et al.	N/A	N/A
8016176	12/2010	Kasvikis et al.	N/A	N/A
8016177	12/2010	Bettuchi et al.	N/A	N/A
8016178	12/2010	Olson et al.	N/A	N/A
8016849	12/2010	Wenchell	N/A	N/A
8016855	12/2010	Whitman et al.	N/A	N/A
8016858	12/2010	Whitman	N/A	N/A
8016881	12/2010	Furst	N/A	N/A
8020741	12/2010	Cole et al.	N/A	N/A
8020742	12/2010	Marczyk	N/A	N/A
8020743	12/2010	Shelton, IV	N/A	N/A
8021375	12/2010	Aldrich et al.	N/A	N/A
8025199	12/2010	Whitman et al.	N/A	N/A
8025896	12/2010	Malaviya et al.	N/A	N/A
8028835	12/2010	Yasuda et al.	N/A	N/A

8028882	12/2010	Viola	N/A	N/A
8028883	12/2010	Stopek	N/A	N/A
8028884	12/2010	Sniffin et al.	N/A	N/A
8028885	12/2010	Smith et al.	N/A	N/A
8029510	12/2010	Hoegerle	N/A	N/A
8031069	12/2010	Cohn et al.	N/A	N/A
8033438	12/2010	Scirica	N/A	N/A
8033439	12/2010	Racenet et al.	N/A	N/A
8033440	12/2010	Wenchell et al.	N/A	N/A
8033442	12/2010	Racenet et al.	N/A	N/A
8034077	12/2010	Smith et al.	N/A	N/A
8034337	12/2010	Simard	N/A	N/A
8034363	12/2010	Li et al.	N/A	N/A
8035487	12/2010	Malackowski	N/A	N/A
8037591	12/2010	Spivey et al.	N/A	N/A
8038044	12/2010	Viola	N/A	N/A
8038045	12/2010	Bettuchi et al.	N/A	N/A
8038046	12/2010	Smith et al.	N/A	N/A
8038686	12/2010	Huitema et al.	N/A	N/A
8043207	12/2010	Adams	N/A	N/A
8043328	12/2010	Hahnen et al.	N/A	N/A
8044536	12/2010	Nguyen et al.	N/A	N/A
8044604	12/2010	Hagino et al.	N/A	N/A
8047236	12/2010	Perry	N/A	N/A
8048503	12/2010	Farnsworth et al.	N/A	N/A
8052636	12/2010	Moll et al.	N/A	N/A
8052697	12/2010	Phillips	N/A	N/A
8056787	12/2010	Boudreaux et al.	N/A	N/A
8056788	12/2010	Mastri et al.	N/A	N/A
8056789	12/2010	White et al.	N/A	N/A
8057508	12/2010	Shelton, IV	N/A	N/A
8058771	12/2010	Giordano et al.	N/A	N/A
8060250	12/2010	Reiland et al.	N/A	N/A
8061014	12/2010	Smith et al.	N/A	N/A
8061576	12/2010	Cappola	N/A	N/A
8062236	12/2010	Soltz	N/A	N/A
8062306	12/2010	Nobis et al.	N/A	N/A
8062330	12/2010	Prommersberger et al.	N/A	N/A
8063619	12/2010	Zhu et al.	N/A	N/A
8066158	12/2010	Vogel et al.	N/A	N/A
8066166	12/2010	Demmy et al.	N/A	N/A
8066167	12/2010	Measamer et al.	N/A	N/A
8066168	12/2010	Vidal et al.	N/A	N/A
8066720	12/2010	Knodel et al.	N/A	N/A
D650074	12/2010	Hunt et al.	N/A	N/A
D650789	12/2010	Arnold	N/A	N/A
8070033	12/2010	Milliman et al.	N/A	N/A
8070034	12/2010	Knodel	N/A	N/A
8070035	12/2010	Holsten et al.	N/A	N/A

8070743	12/2010	Kagan et al.	N/A	N/A
8074858	12/2010	Marczyk	N/A	N/A
8074859	12/2010	Kostrzewski	N/A	N/A
8074861	12/2010	Ehrenfels et al.	N/A	N/A
8075476	12/2010	Vargas	N/A	N/A
8075571	12/2010	Vitali et al.	N/A	N/A
8079950	12/2010	Stern et al.	N/A	N/A
8079989	12/2010	Birk et al.	N/A	N/A
8080004	12/2010	Downey et al.	N/A	N/A
8083118	12/2010	Milliman et al.	N/A	N/A
8083119	12/2010	Prommersberger	N/A	N/A
8083120	12/2010	Shelton, IV et al.	N/A	N/A
8084001	12/2010	Burns et al.	N/A	N/A
8084969	12/2010	David et al.	N/A	N/A
8085013	12/2010	Wei et al.	N/A	N/A
D652048	12/2011	Joseph	N/A	N/A
8087562	12/2011	Manoux et al.	N/A	N/A
8087563	12/2011	Milliman et al.	N/A	N/A
8089509	12/2011	Chatenever et al.	N/A	N/A
8091753	12/2011	Viola	N/A	N/A
8091756	12/2011	Viola	N/A	N/A
8092443	12/2011	Bischoff	N/A	N/A
8092932	12/2011	Phillips et al.	N/A	N/A
8093572	12/2011	Kuduvalli	N/A	N/A
8096458	12/2011	Hessler	N/A	N/A
8096459	12/2011	Ortiz et al.	N/A	N/A
8097017	12/2011	Viola	N/A	N/A
8100310	12/2011	Zemlok	N/A	N/A
8100824	12/2011	Hegeman et al.	N/A	N/A
8100872	12/2011	Patel	N/A	N/A
8102138	12/2011	Sekine et al.	N/A	N/A
8102278	12/2011	Deck et al.	N/A	N/A
8105320	12/2011	Manzo	N/A	N/A
8105350	12/2011	Lee et al.	N/A	N/A
8107925	12/2011	Natsuno et al.	N/A	N/A
8108033	12/2011	Drew et al.	N/A	N/A
8108072	12/2011	Zhao et al.	N/A	N/A
8109426	12/2011	Milliman et al.	N/A	N/A
8110208	12/2011	Hen	N/A	N/A
8113405	12/2011	Milliman	N/A	N/A
8113407	12/2011	Holsten et al.	N/A	N/A
8113408	12/2011	Wenchell et al.	N/A	N/A
8113410	12/2011	Hall et al.	N/A	N/A
8114017	12/2011	Bacher	N/A	N/A
8114100	12/2011	Smith et al.	N/A	N/A
8114345	12/2011	Dlugos, Jr. et al.	N/A	N/A
8118206	12/2011	Zand et al.	N/A	N/A
8118207	12/2011	Racenet et al.	N/A	N/A
8120301	12/2011	Goldberg et al.	N/A	N/A
8122128	12/2011	Burke, II et al.	N/A	N/A

8123103	12/2011	Milliman	N/A	N/A
8123523	12/2011	Carron et al.	N/A	N/A
8123766	12/2011	Bauman et al.	N/A	N/A
8123767	12/2011	Bauman et al.	N/A	N/A
8125168	12/2011	Johnson et al.	N/A	N/A
8127975	12/2011	Olson et al.	N/A	N/A
8127976	12/2011	Scirica et al.	N/A	N/A
8128624	12/2011	Couture et al.	N/A	N/A
8128643	12/2011	Aranyi et al.	N/A	N/A
8128645	12/2011	Sonnenschein et al.	N/A	N/A
8128662	12/2011	Altarac et al.	N/A	N/A
8132703	12/2011	Milliman et al.	N/A	N/A
8132705	12/2011	Viola et al.	N/A	N/A
8132706	12/2011	Marczyk et al.	N/A	N/A
8133500	12/2011	Ringeisen et al.	N/A	N/A
8134306	12/2011	Drader et al.	N/A	N/A
8136711	12/2011	Beardsley et al.	N/A	N/A
8136712	12/2011	Zingman	N/A	N/A
8136713	12/2011	Hathaway et al.	N/A	N/A
8137339	12/2011	Jinno et al.	N/A	N/A
8140417	12/2011	Shibata	N/A	N/A
8141762	12/2011	Bedi et al.	N/A	N/A
8141763	12/2011	Milliman	N/A	N/A
8142200	12/2011	Crunkilton et al.	N/A	N/A
8142425	12/2011	Eggers	N/A	N/A
8142461	12/2011	Houser et al.	N/A	N/A
8142515	12/2011	Therin et al.	N/A	N/A
8143520	12/2011	Cutler	N/A	N/A
8146790	12/2011	Milliman	N/A	N/A
8147421	12/2011	Farquhar et al.	N/A	N/A
8147456	12/2011	Fisher et al.	N/A	N/A
8147485	12/2011	Wham et al.	N/A	N/A
8152041	12/2011	Kostrzewski	N/A	N/A
8152756	12/2011	Webster et al.	N/A	N/A
8154239	12/2011	Katsuki et al.	N/A	N/A
8157145	12/2011	Shelton, IV et al.	N/A	N/A
8157148	12/2011	Scirica	N/A	N/A
8157151	12/2011	Ingmanson et al.	N/A	N/A
8157152	12/2011	Holsten et al.	N/A	N/A
8157153	12/2011	Shelton, IV et al.	N/A	N/A
8157793	12/2011	Omori et al.	N/A	N/A
8157834	12/2011	Conlon	N/A	N/A
8161977	12/2011	Shelton, IV et al.	N/A	N/A
8162138	12/2011	Bettenhausen et al.	N/A	N/A
8162197	12/2011	Mastri et al.	N/A	N/A
8162668	12/2011	Toly	N/A	N/A
8162933	12/2011	Francischelli et al.	N/A	N/A
8162965	12/2011	Reschke et al.	N/A	N/A
8167185	12/2011	Shelton, IV et al.	N/A	N/A
8167622	12/2011	Zhou	N/A	N/A

8167895	12/2011	D'Agostino et al.	N/A	N/A
8167898	12/2011	Schaller et al.	N/A	N/A
8170241	12/2011	Roe et al.	N/A	N/A
8172004	12/2011	Ho	N/A	N/A
8172120	12/2011	Boyden et al.	N/A	N/A
8172122	12/2011	Kasvikis et al.	N/A	N/A
8172124	12/2011	Shelton, IV et al.	N/A	N/A
8177776	12/2011	Humayun et al.	N/A	N/A
8177797	12/2011	Shimoji et al.	N/A	N/A
8179705	12/2011	Chapuis	N/A	N/A
8180458	12/2011	Kane et al.	N/A	N/A
8181839	12/2011	Beetel	N/A	N/A
8181840	12/2011	Milliman	N/A	N/A
8182422	12/2011	Bayer et al.	N/A	N/A
8182444	12/2011	Uber, III et al.	N/A	N/A
8183807	12/2011	Tsai et al.	N/A	N/A
8186555	12/2011	Shelton, IV et al.	N/A	N/A
8186556	12/2011	Viola	N/A	N/A
8186558	12/2011	Sapienza	N/A	N/A
8186560	12/2011	Hess et al.	N/A	N/A
8190238	12/2011	Moll et al.	N/A	N/A
D661314	12/2011	Marchetti	N/A	N/A
D661315	12/2011	Marchetti et al.	N/A	N/A
8191752	12/2011	Scirica	N/A	N/A
8192350	12/2011	Ortiz et al.	N/A	N/A
8192460	12/2011	Orban, III et al.	N/A	N/A
8192651	12/2011	Young et al.	N/A	N/A
8193129	12/2011	Tagawa et al.	N/A	N/A
8196795	12/2011	Moore et al.	N/A	N/A
8196796	12/2011	Shelton, IV et al.	N/A	N/A
8197501	12/2011	Shadeck et al.	N/A	N/A
8197502	12/2011	Smith et al.	N/A	N/A
8197837	12/2011	Jamiolkowski et al.	N/A	N/A
8201720	12/2011	Hessler	N/A	N/A
8201721	12/2011	Zemlok et al.	N/A	N/A
8202549	12/2011	Stucky et al.	N/A	N/A
8205779	12/2011	Ma et al.	N/A	N/A
8205780	12/2011	Sorrentino et al.	N/A	N/A
8205781	12/2011	Baxter, III et al.	N/A	N/A
8207863	12/2011	Neubauer et al.	N/A	N/A
8210411	12/2011	Yates et al.	N/A	N/A
8210413	12/2011	Whitman et al.	N/A	N/A
8210414	12/2011	Bettuchi et al.	N/A	N/A
8210415	12/2011	Ward	N/A	N/A
8210416	12/2011	Milliman et al.	N/A	N/A
8210721	12/2011	Chen et al.	N/A	N/A
8211125	12/2011	Spivey	N/A	N/A
8214019	12/2011	Govari et al.	N/A	N/A
8215531	12/2011	Shelton, IV et al.	N/A	N/A
8215532	12/2011	Marczyk	N/A	N/A

8215533	12/2011	Viola et al.	N/A	N/A
8220468	12/2011	Cooper et al.	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8220690	12/2011	Hess et al.	N/A	N/A
8221402	12/2011	Francischelli et al.	N/A	N/A
8221424	12/2011	Cha	N/A	N/A
8221433	12/2011	Lozier et al.	N/A	N/A
8225799	12/2011	Bettuchi	N/A	N/A
8225979	12/2011	Farascioni et al.	N/A	N/A
8226553	12/2011	Shelton, IV et al.	N/A	N/A
8226635	12/2011	Petrie et al.	N/A	N/A
8226675	12/2011	Houser et al.	N/A	N/A
8226715	12/2011	Hwang et al.	N/A	N/A
8227946	12/2011	Kim	N/A	N/A
8228020	12/2011	Shin et al.	N/A	N/A
8228048	12/2011	Spencer	N/A	N/A
8229549	12/2011	Whitman et al.	N/A	N/A
8230235	12/2011	Goodman et al.	N/A	N/A
8231040	12/2011	Zemlok et al.	N/A	N/A
8231042	12/2011	Hessler et al.	N/A	N/A
8231043	12/2011	Tarinelli et al.	N/A	N/A
8235272	12/2011	Nicholas et al.	N/A	N/A
8235274	12/2011	Cappola	N/A	N/A
8236010	12/2011	Ortiz et al.	N/A	N/A
8236011	12/2011	Harris et al.	N/A	N/A
8236020	12/2011	Smith et al.	N/A	N/A
8237388	12/2011	Jinno et al.	N/A	N/A
8240536	12/2011	Marczyk	N/A	N/A
8240537	12/2011	Marczyk	N/A	N/A
8241271	12/2011	Millman et al.	N/A	N/A
8241284	12/2011	Dycus et al.	N/A	N/A
8241308	12/2011	Kortenbach et al.	N/A	N/A
8241322	12/2011	Whitman et al.	N/A	N/A
8245594	12/2011	Rogers et al.	N/A	N/A
8245898	12/2011	Smith et al.	N/A	N/A
8245899	12/2011	Swensgard et al.	N/A	N/A
8245900	12/2011	Scirica	N/A	N/A
8245901	12/2011	Stopek	N/A	N/A
8246608	12/2011	Omori et al.	N/A	N/A
8246637	12/2011	Viola et al.	N/A	N/A
8252009	12/2011	Weller et al.	N/A	N/A
D667450	12/2011	Eby et al.	N/A	N/A
8256654	12/2011	Bettuchi et al.	N/A	N/A
8256655	12/2011	Sniffin et al.	N/A	N/A
8256656	12/2011	Milliman et al.	N/A	N/A
8257251	12/2011	Shelton, IV et al.	N/A	N/A
8257356	12/2011	Bleich et al.	N/A	N/A
8257386	12/2011	Lee et al.	N/A	N/A
8257391	12/2011	Orban, III et al.	N/A	N/A
8257634	12/2011	Scirica	N/A	N/A

8258745	12/2011	Smith et al.	N/A	N/A
8261958	12/2011	Knodel	N/A	N/A
8262560	12/2011	Whitman	N/A	N/A
8262655	12/2011	Ghabrial et al.	N/A	N/A
8266232	12/2011	Piper et al.	N/A	N/A
8267300	12/2011	Boudreaux	N/A	N/A
8267849	12/2011	Wazer et al.	N/A	N/A
8267924	12/2011	Zemlok et al.	N/A	N/A
8267946	12/2011	Whitfield et al.	N/A	N/A
8267951	12/2011	Whayne et al.	N/A	N/A
8268344	12/2011	Ma et al.	N/A	N/A
8269121	12/2011	Smith	N/A	N/A
8272553	12/2011	Mastri et al.	N/A	N/A
8272554	12/2011	Whitman et al.	N/A	N/A
8272918	12/2011	Lam	N/A	N/A
8273404	12/2011	Dave et al.	N/A	N/A
8276594	12/2011	Shah	N/A	N/A
8276801	12/2011	Zemlok et al.	N/A	N/A
8276802	12/2011	Kostrzewski	N/A	N/A
8277473	12/2011	Sunaoshi et al.	N/A	N/A
8281446	12/2011	Moskovich	N/A	N/A
8281973	12/2011	Wenchell et al.	N/A	N/A
8281974	12/2011	Hessler et al.	N/A	N/A
8282654	12/2011	Ferrari et al.	N/A	N/A
8285367	12/2011	Hyde et al.	N/A	N/A
8286723	12/2011	Puzio et al.	N/A	N/A
8286845	12/2011	Perry et al.	N/A	N/A
8286846	12/2011	Smith et al.	N/A	N/A
8286847	12/2011	Taylor	N/A	N/A
8287487	12/2011	Estes	N/A	N/A
8287522	12/2011	Moses et al.	N/A	N/A
8287561	12/2011	Nunez et al.	N/A	N/A
8288984	12/2011	Yang	N/A	N/A
8289403	12/2011	Dobashi et al.	N/A	N/A
8290883	12/2011	Takeuchi et al.	N/A	N/A
8292147	12/2011	Viola	N/A	N/A
8292148	12/2011	Viola	N/A	N/A
8292150	12/2011	Bryant	N/A	N/A
8292151	12/2011	Viola	N/A	N/A
8292152	12/2011	Milliman et al.	N/A	N/A
8292155	12/2011	Shelton, IV et al.	N/A	N/A
8292157	12/2011	Smith et al.	N/A	N/A
8292158	12/2011	Sapienza	N/A	N/A
8292801	12/2011	Dejima et al.	N/A	N/A
8292888	12/2011	Whitman	N/A	N/A
8292906	12/2011	Taylor et al.	N/A	N/A
8294399	12/2011	Suzuki et al.	N/A	N/A
8298161	12/2011	Vargas	N/A	N/A
8298189	12/2011	Fisher et al.	N/A	N/A
8298233	12/2011	Mueller	N/A	N/A

8298677	12/2011	Wiesner et al.	N/A	N/A
8302323	12/2011	Fortier et al.	N/A	N/A
8303621	12/2011	Miyamoto et al.	N/A	N/A
8308040	12/2011	Huang et al.	N/A	N/A
8308041	12/2011	Kostrzewski	N/A	N/A
8308042	12/2011	Aranyi	N/A	N/A
8308043	12/2011	Bindra et al.	N/A	N/A
8308046	12/2011	Prommersberger	N/A	N/A
8308659	12/2011	Scheibe et al.	N/A	N/A
8308725	12/2011	Bell et al.	N/A	N/A
8310188	12/2011	Nakai	N/A	N/A
8313496	12/2011	Sauer et al.	N/A	N/A
8313499	12/2011	Magnusson et al.	N/A	N/A
8313509	12/2011	Kostrzewski	N/A	N/A
8317070	12/2011	Hueil et al.	N/A	N/A
8317071	12/2011	Knodel	N/A	N/A
8317074	12/2011	Ortiz et al.	N/A	N/A
8317437	12/2011	Merkley et al.	N/A	N/A
8317744	12/2011	Kirschenman	N/A	N/A
8317790	12/2011	Bell et al.	N/A	N/A
8319002	12/2011	Daniels et al.	N/A	N/A
D672784	12/2011	Clanton et al.	N/A	N/A
8322455	12/2011	Shelton, IV et al.	N/A	N/A
8322589	12/2011	Boudreaux	N/A	N/A
8322590	12/2011	Patel et al.	N/A	N/A
8322901	12/2011	Michelotti	N/A	N/A
8323271	12/2011	Humayun et al.	N/A	N/A
8323789	12/2011	Rozhin et al.	N/A	N/A
8324585	12/2011	McBroom et al.	N/A	N/A
8327514	12/2011	Kim	N/A	N/A
8328061	12/2011	Kasvikis	N/A	N/A
8328062	12/2011	Viola	N/A	N/A
8328063	12/2011	Milliman et al.	N/A	N/A
8328064	12/2011	Racenet et al.	N/A	N/A
8328065	12/2011	Shah	N/A	N/A
8328802	12/2011	Deville et al.	N/A	N/A
8328823	12/2011	Aranyi et al.	N/A	N/A
8333313	12/2011	Boudreaux et al.	N/A	N/A
8333691	12/2011	Schaaf	N/A	N/A
8333764	12/2011	Francischelli et al.	N/A	N/A
8333779	12/2011	Smith et al.	N/A	N/A
8334468	12/2011	Palmer et al.	N/A	N/A
8336753	12/2011	Olson et al.	N/A	N/A
8336754	12/2011	Cappola et al.	N/A	N/A
8342377	12/2012	Milliman et al.	N/A	N/A
8342378	12/2012	Marczyk et al.	N/A	N/A
8342379	12/2012	Whitman et al.	N/A	N/A
8342380	12/2012	Viola	N/A	N/A
8343150	12/2012	Artale	N/A	N/A
8347978	12/2012	Forster et al.	N/A	N/A

8348118	12/2012	Segura	N/A	N/A
8348123	12/2012	Scirica et al.	N/A	N/A
8348124	12/2012	Scirica	N/A	N/A
8348125	12/2012	Viola et al.	N/A	N/A
8348126	12/2012	Olson et al.	N/A	N/A
8348127	12/2012	Marczyk	N/A	N/A
8348129	12/2012	Bedi	227/19	A61B 17/0686
8348130	12/2012	Shah et al.	N/A	N/A
8348131	12/2012	Omaits et al.	N/A	N/A
8348837	12/2012	Wenchell	N/A	N/A
8348948	12/2012	Bahney	N/A	N/A
8348959	12/2012	Wolford et al.	N/A	N/A
8348972	12/2012	Soltz et al.	N/A	N/A
8349987	12/2012	Kapiamba et al.	N/A	N/A
8352004	12/2012	Mannheimer et al.	N/A	N/A
8353437	12/2012	Boudreaux	N/A	N/A
8353438	12/2012	Baxter, III et al.	N/A	N/A
8353439	12/2012	Baxter, III et al.	N/A	N/A
8356740	12/2012	Knodel	N/A	N/A
8357144	12/2012	Whitman et al.	N/A	N/A
8357158	12/2012	McKenna et al.	N/A	N/A
8357161	12/2012	Mueller	N/A	N/A
8359174	12/2012	Nakashima et al.	N/A	N/A
8360296	12/2012	Zingman	N/A	N/A
8360297	12/2012	Shelton, IV et al.	N/A	N/A
8360298	12/2012	Farascioni et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8361501	12/2012	DiTizio et al.	N/A	N/A
D676866	12/2012	Chaudhri	N/A	N/A
8365972	12/2012	Aranyi et al.	N/A	N/A
8365973	12/2012	White et al.	N/A	N/A
8365975	12/2012	Manoux et al.	N/A	N/A
8365976	12/2012	Hess et al.	N/A	N/A
8366559	12/2012	Papenfuss et al.	N/A	N/A
8366719	12/2012	Markey et al.	N/A	N/A
8366787	12/2012	Brown et al.	N/A	N/A
8368327	12/2012	Benning et al.	N/A	N/A
8369056	12/2012	Senriuchi et al.	N/A	N/A
8371393	12/2012	Higuchi et al.	N/A	N/A
8371491	12/2012	Huitema et al.	N/A	N/A
8371492	12/2012	Aranyi et al.	N/A	N/A
8371493	12/2012	Aranyi et al.	N/A	N/A
8371494	12/2012	Racenet et al.	N/A	N/A
8372094	12/2012	Bettuchi et al.	N/A	N/A
8374723	12/2012	Zhao et al.	N/A	N/A
8376865	12/2012	Forster et al.	N/A	N/A
8377029	12/2012	Nagao et al.	N/A	N/A
8377044	12/2012	Coe et al.	N/A	N/A
8377059	12/2012	Deville et al.	N/A	N/A

8381828	12/2012	Whitman et al.	N/A	N/A
8381834	12/2012	Barhitte et al.	N/A	N/A
8382773	12/2012	Whitfield et al.	N/A	N/A
8382790	12/2012	Uenohara et al.	N/A	N/A
D677273	12/2012	Randall et al.	N/A	N/A
8387848	12/2012	Johnson et al.	N/A	N/A
8388633	12/2012	Rousseau et al.	N/A	N/A
8389588	12/2012	Ringeisen et al.	N/A	N/A
8393513	12/2012	Jankowski	N/A	N/A
8393514	12/2012	Shelton, IV et al.	N/A	N/A
8393516	12/2012	Kostrzewski	N/A	N/A
8397832	12/2012	Blickle et al.	N/A	N/A
8397971	12/2012	Yates et al.	N/A	N/A
8397972	12/2012	Kostrzewski	N/A	N/A
8397973	12/2012	Hausen	N/A	N/A
8398633	12/2012	Mueller	N/A	N/A
8398669	12/2012	Kim	N/A	N/A
8398673	12/2012	Hinchliffe et al.	N/A	N/A
8398674	12/2012	Prestel	N/A	N/A
8400108	12/2012	Powell et al.	N/A	N/A
8400851	12/2012	Byun	N/A	N/A
8403138	12/2012	Weisshaupt et al.	N/A	N/A
8403195	12/2012	Beardsley et al.	N/A	N/A
8403196	12/2012	Beardsley et al.	N/A	N/A
8403198	12/2012	Sorrentino et al.	N/A	N/A
8403832	12/2012	Cunningham et al.	N/A	N/A
8403926	12/2012	Nobis et al.	N/A	N/A
8403945	12/2012	Whitfield et al.	N/A	N/A
8403946	12/2012	Whitfield et al.	N/A	N/A
8403950	12/2012	Palmer et al.	N/A	N/A
D679726	12/2012	Kobayashi	N/A	N/A
D680646	12/2012	Hunt et al.	N/A	N/A
8408439	12/2012	Huang et al.	N/A	N/A
8408442	12/2012	Racenet et al.	N/A	N/A
8409079	12/2012	Okamoto et al.	N/A	N/A
8409174	12/2012	Omori	N/A	N/A
8409175	12/2012	Lee et al.	N/A	N/A
8409211	12/2012	Baroud	N/A	N/A
8409222	12/2012	Whitfield et al.	N/A	N/A
8409223	12/2012	Sorrentino et al.	N/A	N/A
8409234	12/2012	Stahler et al.	N/A	N/A
8411500	12/2012	Gapihan et al.	N/A	N/A
8413661	12/2012	Rousseau et al.	N/A	N/A
8413870	12/2012	Pastorelli et al.	N/A	N/A
8413871	12/2012	Racenet et al.	N/A	N/A
8413872	12/2012	Patel	N/A	N/A
8414469	12/2012	Diolaiti	N/A	N/A
8414577	12/2012	Boudreaux et al.	N/A	N/A
8414598	12/2012	Brock et al.	N/A	N/A
8418073	12/2012	Mohr et al.	N/A	N/A

8418906	12/2012	Farascioni et al.	N/A	N/A
8418907	12/2012	Johnson et al.	N/A	N/A
8418908	12/2012	Beardsley	N/A	N/A
8418909	12/2012	Kostrzewski	N/A	N/A
8419635	12/2012	Shelton, IV et al.	N/A	N/A
8419717	12/2012	Diolaiti et al.	N/A	N/A
8419747	12/2012	Hinman et al.	N/A	N/A
8419754	12/2012	Laby et al.	N/A	N/A
8419755	12/2012	Deem et al.	N/A	N/A
8423182	12/2012	Robinson et al.	N/A	N/A
8424737	12/2012	Scirica	N/A	N/A
8424739	12/2012	Racenet et al.	N/A	N/A
8424740	12/2012	Shelton, IV et al.	N/A	N/A
8424741	12/2012	McGuckin, Jr. et al.	N/A	N/A
8424742	12/2012	Bettuchi	N/A	N/A
8425600	12/2012	Maxwell	N/A	N/A
8427430	12/2012	Lee et al.	N/A	N/A
8430292	12/2012	Patel et al.	N/A	N/A
8430892	12/2012	Bindra et al.	N/A	N/A
8430898	12/2012	Wiener et al.	N/A	N/A
D681674	12/2012	Koehn et al.	N/A	N/A
8435257	12/2012	Smith et al.	N/A	N/A
8439246	12/2012	Knodel	N/A	N/A
8439830	12/2012	McKinley et al.	N/A	N/A
8444036	12/2012	Shelton, IV	N/A	N/A
8444037	12/2012	Nicholas et al.	N/A	N/A
8444549	12/2012	Viola et al.	N/A	N/A
8449536	12/2012	Selig	N/A	N/A
8449560	12/2012	Roth et al.	N/A	N/A
8453904	12/2012	Eskaros et al.	N/A	N/A
8453906	12/2012	Huang et al.	N/A	N/A
8453907	12/2012	Laurent et al.	N/A	N/A
8453908	12/2012	Bedi et al.	N/A	N/A
8453912	12/2012	Mastri et al.	N/A	N/A
8453914	12/2012	Laurent et al.	N/A	N/A
8454495	12/2012	Kawano et al.	N/A	N/A
8454551	12/2012	Allen et al.	N/A	N/A
8454628	12/2012	Smith et al.	N/A	N/A
8454640	12/2012	Johnston et al.	N/A	N/A
8457757	12/2012	Cauler et al.	N/A	N/A
8459520	12/2012	Giordano et al.	N/A	N/A
8459521	12/2012	Zemlok et al.	N/A	N/A
8459524	12/2012	Pribanic et al.	N/A	N/A
8459525	12/2012	Yates et al.	N/A	N/A
8464922	12/2012	Marczyk	N/A	N/A
8464923	12/2012	Shelton, IV	N/A	N/A
8464924	12/2012	Gresham et al.	N/A	N/A
8464925	12/2012	Hull et al.	N/A	N/A
8465475	12/2012	Isbell, Jr.	N/A	N/A
8465502	12/2012	Zergiebel	N/A	N/A

8465515	12/2012	Drew et al.	N/A	N/A
8469254	12/2012	Czernik et al.	N/A	N/A
8469946	12/2012	Sugita	N/A	N/A
8469973	12/2012	Meade et al.	N/A	N/A
8470355	12/2012	Skalla et al.	N/A	N/A
D686240	12/2012	Lin	N/A	N/A
D686244	12/2012	Moriya et al.	N/A	N/A
8474677	12/2012	Woodard, Jr. et al.	N/A	N/A
8475453	12/2012	Marczyk et al.	N/A	N/A
8475454	12/2012	Alshemari	N/A	N/A
8475474	12/2012	Bombard et al.	N/A	N/A
8479968	12/2012	Hodgkinson et al.	N/A	N/A
8479969	12/2012	Shelton, IV	N/A	N/A
8480703	12/2012	Nicholas et al.	N/A	N/A
8483509	12/2012	Matsuzaka	N/A	N/A
8485412	12/2012	Shelton, IV et al.	N/A	N/A
8485413	12/2012	Scheib et al.	N/A	N/A
8485970	12/2012	Widenhouse et al.	N/A	N/A
8486047	12/2012	Stopek	N/A	N/A
8487199	12/2012	Palmer et al.	N/A	N/A
8487487	12/2012	Dietz et al.	N/A	N/A
8490851	12/2012	Blier et al.	N/A	N/A
8490852	12/2012	Viola	N/A	N/A
8490853	12/2012	Criscuolo et al.	N/A	N/A
8491581	12/2012	Deville et al.	N/A	N/A
8491603	12/2012	Yeung et al.	N/A	N/A
8491624	12/2012	Kerr et al.	N/A	N/A
8496153	12/2012	Demmy et al.	N/A	N/A
8496154	12/2012	Marczyk et al.	N/A	N/A
8496156	12/2012	Sniffin et al.	N/A	N/A
8496683	12/2012	Prommersberger et al.	N/A	N/A
8498691	12/2012	Moll et al.	N/A	N/A
8499673	12/2012	Keller	N/A	N/A
8499966	12/2012	Palmer et al.	N/A	N/A
8499992	12/2012	Whitman et al.	N/A	N/A
8499993	12/2012	Shelton, IV et al.	N/A	N/A
8499994	12/2012	D'Arcangelo	N/A	N/A
8500721	12/2012	Jinno	N/A	N/A
8500762	12/2012	Sholev et al.	N/A	N/A
8502091	12/2012	Palmer et al.	N/A	N/A
8505799	12/2012	Viola et al.	N/A	N/A
8505801	12/2012	Ehrenfels et al.	N/A	N/A
8505802	12/2012	Viola et al.	N/A	N/A
8506555	12/2012	Ruiz Morales	N/A	N/A
8506557	12/2012	Zemlok et al.	N/A	N/A
8506580	12/2012	Zergiebel et al.	N/A	N/A
8506581	12/2012	Wingardner, III et al.	N/A	N/A
8511308	12/2012	Hecox et al.	N/A	N/A
8512359	12/2012	Whitman et al.	N/A	N/A

8512402	12/2012	Marczyk et al.	N/A	N/A
8517239	12/2012	Scheib et al.	N/A	N/A
8517241	12/2012	Nicholas et al.	N/A	N/A
8517243	12/2012	Giordano et al.	N/A	N/A
8517244	12/2012	Shelton, IV et al.	N/A	N/A
8517938	12/2012	Eisenhardt et al.	N/A	N/A
8518024	12/2012	Williams et al.	N/A	N/A
8521273	12/2012	Kliman	N/A	N/A
8523042	12/2012	Masiakos et al.	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8523787	12/2012	Ludwin et al.	N/A	N/A
8523881	12/2012	Cabiri et al.	N/A	N/A
8523882	12/2012	Huitema et al.	N/A	N/A
8523900	12/2012	Jinno et al.	N/A	N/A
8529588	12/2012	Ahlberg et al.	N/A	N/A
8529599	12/2012	Holsten	N/A	N/A
8529600	12/2012	Woodard, Jr. et al.	N/A	N/A
8529819	12/2012	Ostapoff et al.	N/A	N/A
8531153	12/2012	Baarman et al.	N/A	N/A
8532747	12/2012	Nock et al.	N/A	N/A
8534527	12/2012	Brendel et al.	N/A	N/A
8534528	12/2012	Shelton, IV	N/A	N/A
8535304	12/2012	Sklar et al.	N/A	N/A
8535340	12/2012	Allen	N/A	N/A
8539866	12/2012	Nayak et al.	N/A	N/A
8540128	12/2012	Shelton, IV et al.	N/A	N/A
8540129	12/2012	Baxter, III et al.	N/A	N/A
8540130	12/2012	Moore et al.	N/A	N/A
8540131	12/2012	Swayze	N/A	N/A
8540133	12/2012	Bedi et al.	N/A	N/A
8540646	12/2012	Mendez-Coll	N/A	N/A
8540733	12/2012	Whitman et al.	N/A	N/A
8540735	12/2012	Mitelberg et al.	N/A	N/A
D690614	12/2012	Mascadri et al.	N/A	N/A
8550984	12/2012	Takemoto	N/A	N/A
8551076	12/2012	Duval et al.	N/A	N/A
8555660	12/2012	Takenaka et al.	N/A	N/A
8556151	12/2012	Viola	N/A	N/A
8556918	12/2012	Bauman et al.	N/A	N/A
8556935	12/2012	Knodel et al.	N/A	N/A
8560147	12/2012	Taylor et al.	N/A	N/A
8561617	12/2012	Lindh et al.	N/A	N/A
8561870	12/2012	Baxter, III et al.	N/A	N/A
8561871	12/2012	Rajappa et al.	N/A	N/A
8561873	12/2012	Ingmanson et al.	N/A	N/A
8562592	12/2012	Conlon et al.	N/A	N/A
8562598	12/2012	Falkenstein et al.	N/A	N/A
8567656	12/2012	Shelton, IV et al.	N/A	N/A
8568416	12/2012	Schmitz et al.	N/A	N/A
8568425	12/2012	Ross et al.	N/A	N/A

D692916	12/2012	Granchi et al.	N/A	N/A
8573459	12/2012	Smith et al.	N/A	N/A
8573461	12/2012	Shelton, IV et al.	N/A	N/A
8573462	12/2012	Smith et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8574199	12/2012	von Bulow et al.	N/A	N/A
8574263	12/2012	Mueller	N/A	N/A
8575880	12/2012	Grantz	N/A	N/A
8575895	12/2012	Garrastacho et al.	N/A	N/A
8579176	12/2012	Smith et al.	N/A	N/A
8579178	12/2012	Holsten et al.	N/A	N/A
8579897	12/2012	Vakharia et al.	N/A	N/A
8579937	12/2012	Gresham	N/A	N/A
8584919	12/2012	Hueil et al.	N/A	N/A
8584920	12/2012	Hodgkinson	N/A	N/A
8584921	12/2012	Scirica	N/A	N/A
8585583	12/2012	Sakaguchi et al.	N/A	N/A
8585598	12/2012	Razzaque et al.	N/A	N/A
8585721	12/2012	Kirsch	N/A	N/A
8590760	12/2012	Cummins et al.	N/A	N/A
8590762	12/2012	Hess et al.	N/A	N/A
8590764	12/2012	Hartwick et al.	N/A	N/A
8591400	12/2012	Sugiyama	N/A	N/A
D695310	12/2012	Jang et al.	N/A	N/A
8596515	12/2012	Okoniewski	N/A	N/A
8597745	12/2012	Farnsworth et al.	N/A	N/A
8599450	12/2012	Kubo et al.	N/A	N/A
8602125	12/2012	King	N/A	N/A
8602287	12/2012	Yates et al.	N/A	N/A
8602288	12/2012	Shelton, IV et al.	N/A	N/A
8603077	12/2012	Cooper et al.	N/A	N/A
8603089	12/2012	Viola	N/A	N/A
8603110	12/2012	Maruyama et al.	N/A	N/A
8603135	12/2012	Mueller	N/A	N/A
8608043	12/2012	Scirica	N/A	N/A
8608044	12/2012	Hueil et al.	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A
8608046	12/2012	Laurent et al.	N/A	N/A
8608745	12/2012	Guzman et al.	N/A	N/A
8613383	12/2012	Beckman et al.	N/A	N/A
8613384	12/2012	Pastorelli et al.	N/A	N/A
8616427	12/2012	Viola	N/A	N/A
8616431	12/2012	Timm et al.	N/A	N/A
8617155	12/2012	Johnson et al.	N/A	N/A
8620473	12/2012	Diolaiti et al.	N/A	N/A
8622274	12/2013	Yates et al.	N/A	N/A
8622275	12/2013	Baxter, III et al.	N/A	N/A
8627993	12/2013	Smith et al.	N/A	N/A
8627994	12/2013	Zemlok et al.	N/A	N/A
8627995	12/2013	Smith et al.	N/A	N/A

8628467	12/2013	Whitman et al.	N/A	N/A
8628518	12/2013	Blumenkranz et al.	N/A	N/A
8628544	12/2013	Farascioni	N/A	N/A
8628545	12/2013	Cabrera et al.	N/A	N/A
8631987	12/2013	Shelton, IV et al.	N/A	N/A
8631992	12/2013	Hausen et al.	N/A	N/A
8631993	12/2013	Kostrzewski	N/A	N/A
8632462	12/2013	Yoo et al.	N/A	N/A
8632525	12/2013	Kerr et al.	N/A	N/A
8632535	12/2013	Shelton, IV et al.	N/A	N/A
8632539	12/2013	Twomey et al.	N/A	N/A
8632563	12/2013	Nagase et al.	N/A	N/A
8636187	12/2013	Hueil et al.	N/A	N/A
8636190	12/2013	Zemlok et al.	N/A	N/A
8636191	12/2013	Meagher	N/A	N/A
8636193	12/2013	Whitman et al.	N/A	N/A
8636736	12/2013	Yates et al.	N/A	N/A
8636766	12/2013	Milliman et al.	N/A	N/A
8639936	12/2013	Hu et al.	N/A	N/A
8640788	12/2013	Dachs, II et al.	N/A	N/A
8646674	12/2013	Schulte et al.	N/A	N/A
8647258	12/2013	Aranyi et al.	N/A	N/A
8652120	12/2013	Giordano et al.	N/A	N/A
8652151	12/2013	Lehman et al.	N/A	N/A
8652155	12/2013	Houser et al.	N/A	N/A
8656929	12/2013	Miller et al.	N/A	N/A
8657174	12/2013	Yates et al.	N/A	N/A
8657175	12/2013	Sonnenschein et al.	N/A	N/A
8657176	12/2013	Shelton, IV et al.	N/A	N/A
8657177	12/2013	Scirica et al.	N/A	N/A
8657178	12/2013	Hueil et al.	N/A	N/A
8657482	12/2013	Malackowski et al.	N/A	N/A
8657808	12/2013	McPherson et al.	N/A	N/A
8657814	12/2013	Werneth et al.	N/A	N/A
8657821	12/2013	Palermo	N/A	N/A
D701238	12/2013	Lai et al.	N/A	N/A
8662370	12/2013	Takei	N/A	N/A
8663106	12/2013	Stivoric et al.	N/A	N/A
8663192	12/2013	Hester et al.	N/A	N/A
8663245	12/2013	Francischelli et al.	N/A	N/A
8663262	12/2013	Smith et al.	N/A	N/A
8663270	12/2013	Donnigan et al.	N/A	N/A
8664792	12/2013	Rebsdorf	N/A	N/A
8668129	12/2013	Olson	N/A	N/A
8668130	12/2013	Hess et al.	N/A	N/A
8672206	12/2013	Aranyi et al.	N/A	N/A
8672207	12/2013	Shelton, IV et al.	N/A	N/A
8672208	12/2013	Hess et al.	N/A	N/A
8672209	12/2013	Crainich	N/A	N/A
8672922	12/2013	Loh et al.	N/A	N/A

8672935	12/2013	Okada et al.	N/A	N/A
8672951	12/2013	Smith et al.	N/A	N/A
8673210	12/2013	Deshays	N/A	N/A
8675820	12/2013	Baic et al.	N/A	N/A
8678263	12/2013	Viola	N/A	N/A
8678994	12/2013	Sonnenschein et al.	N/A	N/A
8679093	12/2013	Farra	N/A	N/A
8679098	12/2013	Hart	N/A	N/A
8679114	12/2013	Chapman et al.	N/A	N/A
8679137	12/2013	Bauman et al.	N/A	N/A
8679154	12/2013	Smith et al.	N/A	N/A
8679156	12/2013	Smith et al.	N/A	N/A
8679454	12/2013	Guire et al.	N/A	N/A
8684248	12/2013	Milliman	N/A	N/A
8684249	12/2013	Racenet et al.	N/A	N/A
8684250	12/2013	Bettuchi et al.	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8684962	12/2013	Kirschenman et al.	N/A	N/A
8685004	12/2013	Zemlock et al.	N/A	N/A
8685020	12/2013	Weizman et al.	N/A	N/A
8690893	12/2013	Deitch et al.	N/A	N/A
8695866	12/2013	Leimbach et al.	N/A	N/A
8696665	12/2013	Hunt et al.	N/A	N/A
8701958	12/2013	Shelton, IV et al.	N/A	N/A
8701959	12/2013	Shah	N/A	N/A
8706316	12/2013	Hoevenaar	N/A	N/A
8708210	12/2013	Zemlok et al.	N/A	N/A
8708211	12/2013	Zemlok et al.	N/A	N/A
8708212	12/2013	Williams	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
8709012	12/2013	Muller	N/A	N/A
8712549	12/2013	Zdeblick et al.	N/A	N/A
8714352	12/2013	Farascioni et al.	N/A	N/A
8714429	12/2013	Demmy	N/A	N/A
8714430	12/2013	Natarajan et al.	N/A	N/A
8715256	12/2013	Greener	N/A	N/A
8715302	12/2013	Ibrahim et al.	N/A	N/A
8720766	12/2013	Hess et al.	N/A	N/A
8721630	12/2013	Ortiz et al.	N/A	N/A
8721666	12/2013	Schroeder et al.	N/A	N/A
8727197	12/2013	Hess et al.	N/A	N/A
8727199	12/2013	Wenchell	N/A	N/A
8727200	12/2013	Roy	N/A	N/A
8727961	12/2013	Ziv	N/A	N/A
8728099	12/2013	Cohn et al.	N/A	N/A
8728119	12/2013	Cummins	N/A	N/A
8733470	12/2013	Matthias et al.	N/A	N/A
8733611	12/2013	Milliman	N/A	N/A
8733612	12/2013	Ma	N/A	N/A
8733613	12/2013	Huitema et al.	N/A	N/A

8733614	12/2013	Ross et al.	N/A	N/A
8734336	12/2013	Bonadio et al.	N/A	N/A
8734359	12/2013	Ibanez et al.	N/A	N/A
8734478	12/2013	Widenhouse et al.	N/A	N/A
8734831	12/2013	Kim et al.	N/A	N/A
8739033	12/2013	Rosenberg	N/A	N/A
8739417	12/2013	Tokunaga et al.	N/A	N/A
8740034	12/2013	Morgan et al.	N/A	N/A
8740037	12/2013	Shelton, IV et al.	N/A	N/A
8740038	12/2013	Shelton, IV et al.	N/A	N/A
8740987	12/2013	Geremakis et al.	N/A	N/A
8746529	12/2013	Shelton, IV et al.	N/A	N/A
8746530	12/2013	Giordano et al.	N/A	N/A
8746533	12/2013	Whitman et al.	N/A	N/A
8746535	12/2013	Shelton, IV et al.	N/A	N/A
8747238	12/2013	Shelton, IV et al.	N/A	N/A
8747441	12/2013	Konieczynski et al.	N/A	N/A
8752264	12/2013	Ackley et al.	N/A	N/A
8752699	12/2013	Morgan et al.	N/A	N/A
8752747	12/2013	Shelton, IV et al.	N/A	N/A
8752748	12/2013	Whitman et al.	N/A	N/A
8752749	12/2013	Moore et al.	N/A	N/A
8753664	12/2013	Dao et al.	N/A	N/A
8757287	12/2013	Mak et al.	N/A	N/A
8757465	12/2013	Woodard, Jr. et al.	N/A	N/A
8758235	12/2013	Jaworek	N/A	N/A
8758366	12/2013	McLean et al.	N/A	N/A
8758391	12/2013	Swayze et al.	N/A	N/A
8758438	12/2013	Boyce et al.	N/A	N/A
8763875	12/2013	Morgan et al.	N/A	N/A
8763876	12/2013	Kostrzewski	N/A	N/A
8763877	12/2013	Schall et al.	N/A	N/A
8763879	12/2013	Shelton, IV et al.	N/A	N/A
8764732	12/2013	Hartwell	N/A	N/A
8765942	12/2013	Feraud et al.	N/A	N/A
8770458	12/2013	Scirica	N/A	N/A
8770459	12/2013	Racenet et al.	N/A	N/A
8770460	12/2013	Belzer	N/A	N/A
8771169	12/2013	Whitman et al.	N/A	N/A
8771260	12/2013	Conlon et al.	N/A	N/A
8777004	12/2013	Shelton, IV et al.	N/A	N/A
8777082	12/2013	Scirica	N/A	N/A
8777083	12/2013	Racenet et al.	N/A	N/A
8777898	12/2013	Suon et al.	N/A	N/A
8783541	12/2013	Shelton, IV et al.	N/A	N/A
8783542	12/2013	Riestenberg et al.	N/A	N/A
8783543	12/2013	Shelton, IV et al.	N/A	N/A
8784304	12/2013	Mikkaichi et al.	N/A	N/A
8784404	12/2013	Doyle et al.	N/A	N/A
8784415	12/2013	Malackowski et al.	N/A	N/A

8789737	12/2013	Hodgkinson et al.	N/A	N/A
8789739	12/2013	Swensgard	N/A	N/A
8789740	12/2013	Baxter, III et al.	N/A	N/A
8789741	12/2013	Baxter, III et al.	N/A	N/A
8790658	12/2013	Cigarini et al.	N/A	N/A
8790684	12/2013	Dave et al.	N/A	N/A
D711905	12/2013	Morrison et al.	N/A	N/A
8794098	12/2013	Long	N/A	N/A
8794496	12/2013	Scirica	N/A	N/A
8794497	12/2013	Zingman	N/A	N/A
8795159	12/2013	Moriyama	N/A	N/A
8795276	12/2013	Dietz et al.	N/A	N/A
8795308	12/2013	Valin	N/A	N/A
8795324	12/2013	Kawai et al.	N/A	N/A
8796995	12/2013	Cunanan et al.	N/A	N/A
8800681	12/2013	Rousson et al.	N/A	N/A
8800837	12/2013	Zemlok	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8800839	12/2013	Beetel	N/A	N/A
8800840	12/2013	Jankowski	N/A	N/A
8800841	12/2013	Ellerhorst et al.	N/A	N/A
8801710	12/2013	Ullrich et al.	N/A	N/A
8801734	12/2013	Shelton, IV et al.	N/A	N/A
8801735	12/2013	Shelton, IV et al.	N/A	N/A
8801752	12/2013	Fortier et al.	N/A	N/A
8801801	12/2013	Datta et al.	N/A	N/A
8806973	12/2013	Ross et al.	N/A	N/A
8807414	12/2013	Ross et al.	N/A	N/A
8808161	12/2013	Gregg et al.	N/A	N/A
8808164	12/2013	Hoffman et al.	N/A	N/A
8808274	12/2013	Hartwell	N/A	N/A
8808294	12/2013	Fox et al.	N/A	N/A
8808308	12/2013	Boukhny et al.	N/A	N/A
8808311	12/2013	Heinrich et al.	N/A	N/A
8808325	12/2013	Hess et al.	N/A	N/A
8810197	12/2013	Juergens	N/A	N/A
8811017	12/2013	Fujii et al.	N/A	N/A
8813866	12/2013	Suzuki	N/A	N/A
8814024	12/2013	Woodard, Jr. et al.	N/A	N/A
8814025	12/2013	Miller et al.	N/A	N/A
8814836	12/2013	Ignon et al.	N/A	N/A
8815594	12/2013	Harris et al.	N/A	N/A
8818523	12/2013	Olson et al.	N/A	N/A
8820603	12/2013	Shelton, IV et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8820606	12/2013	Hodgkinson	N/A	N/A
8820607	12/2013	Marczyk	N/A	N/A
8820608	12/2013	Miyamoto	N/A	N/A
8821514	12/2013	Aranyi	N/A	N/A
8822934	12/2013	Sayeh et al.	N/A	N/A

8825164	12/2013	Tweden et al.	N/A	N/A
8827133	12/2013	Shelton, IV et al.	N/A	N/A
8827134	12/2013	Viola et al.	N/A	N/A
8827903	12/2013	Shelton, IV et al.	N/A	N/A
8828046	12/2013	Stefanchik et al.	N/A	N/A
8831779	12/2013	Ortmaier et al.	N/A	N/A
8833219	12/2013	Pierce	N/A	N/A
8833630	12/2013	Milliman	N/A	N/A
8833632	12/2013	Swensgard	N/A	N/A
8834353	12/2013	Dejima et al.	N/A	N/A
8834465	12/2013	Ramstein et al.	N/A	N/A
8834498	12/2013	Byrum et al.	N/A	N/A
8834518	12/2013	Faller et al.	N/A	N/A
8840003	12/2013	Morgan et al.	N/A	N/A
8840004	12/2013	Holsten et al.	N/A	N/A
8840603	12/2013	Shelton, IV et al.	N/A	N/A
8840609	12/2013	Stuebe	N/A	N/A
8840876	12/2013	Eemeta et al.	N/A	N/A
8844789	12/2013	Shelton, IV et al.	N/A	N/A
8844790	12/2013	Demmy et al.	N/A	N/A
8845622	12/2013	Paik et al.	N/A	N/A
8851215	12/2013	Goto	N/A	N/A
8851354	12/2013	Swensgard et al.	N/A	N/A
8851355	12/2013	Aranyi et al.	N/A	N/A
8852174	12/2013	Burbank	N/A	N/A
8852185	12/2013	Twomey	N/A	N/A
8852199	12/2013	Deslauriers et al.	N/A	N/A
8852218	12/2013	Hughett, Sr. et al.	N/A	N/A
8852473	12/2013	Tan	N/A	N/A
8855822	12/2013	Bartol et al.	N/A	N/A
8857692	12/2013	Shima et al.	N/A	N/A
8857693	12/2013	Schuckmann et al.	N/A	N/A
8857694	12/2013	Shelton, IV et al.	N/A	N/A
8858538	12/2013	Belson et al.	N/A	N/A
8858547	12/2013	Brogna	N/A	N/A
8858571	12/2013	Shelton, IV et al.	N/A	N/A
8858590	12/2013	Shelton, IV et al.	N/A	N/A
8864007	12/2013	Widenhouse et al.	N/A	N/A
8864009	12/2013	Shelton, IV et al.	N/A	N/A
8864010	12/2013	Williams	N/A	N/A
8864750	12/2013	Ross et al.	N/A	N/A
8869912	12/2013	Roßkamp et al.	N/A	N/A
8869913	12/2013	Matthias et al.	N/A	N/A
8870049	12/2013	Amid et al.	N/A	N/A
8870050	12/2013	Hodgkinson	N/A	N/A
8870867	12/2013	Walberg et al.	N/A	N/A
8870912	12/2013	Brisson et al.	N/A	N/A
8871829	12/2013	Gerold et al.	N/A	N/A
D716820	12/2013	Wood	N/A	N/A
8875971	12/2013	Hall et al.	N/A	N/A

8875972	12/2013	Weisenburgh, II et al.	N/A	N/A
8876698	12/2013	Sakamoto et al.	N/A	N/A
8876857	12/2013	Burbank	N/A	N/A
8876858	12/2013	Braun	N/A	N/A
8882660	12/2013	Phee et al.	N/A	N/A
8882792	12/2013	Dietz et al.	N/A	N/A
8884560	12/2013	Ito	N/A	N/A
8887979	12/2013	Mastri et al.	N/A	N/A
8888688	12/2013	Julian et al.	N/A	N/A
8888695	12/2013	Piskun et al.	N/A	N/A
8888792	12/2013	Harris et al.	N/A	N/A
8888809	12/2013	Davison et al.	N/A	N/A
8893946	12/2013	Boudreaux et al.	N/A	N/A
8893949	12/2013	Shelton, IV et al.	N/A	N/A
8894647	12/2013	Beardsley et al.	N/A	N/A
8894654	12/2013	Anderson	N/A	N/A
8899460	12/2013	Wojcicki	N/A	N/A
8899461	12/2013	Farascioni	N/A	N/A
8899462	12/2013	Kostrzewski et al.	N/A	N/A
8899463	12/2013	Schall et al.	N/A	N/A
8899464	12/2013	Hueil et al.	N/A	N/A
8899465	12/2013	Shelton, IV et al.	N/A	N/A
8899466	12/2013	Baxter, III et al.	N/A	N/A
8900267	12/2013	Woolfson et al.	N/A	N/A
8905287	12/2013	Racenet et al.	N/A	N/A
8905977	12/2013	Shelton et al.	N/A	N/A
8910846	12/2013	Viola	N/A	N/A
8910847	12/2013	Nalagatla et al.	N/A	N/A
8911426	12/2013	Coppeta et al.	N/A	N/A
8911448	12/2013	Stein	N/A	N/A
8911460	12/2013	Neurohr et al.	N/A	N/A
8911471	12/2013	Spivey et al.	N/A	N/A
8912746	12/2013	Reid et al.	N/A	N/A
8915842	12/2013	Weisenburgh, II et al.	N/A	N/A
8920368	12/2013	Sandhu et al.	N/A	N/A
8920433	12/2013	Barrier et al.	N/A	N/A
8920435	12/2013	Smith et al.	N/A	N/A
8920438	12/2013	Aranyi et al.	N/A	N/A
8920443	12/2013	Hiles et al.	N/A	N/A
8920444	12/2013	Hiles et al.	N/A	N/A
8922163	12/2013	Macdonald	N/A	N/A
8925782	12/2014	Shelton, IV	N/A	N/A
8925783	12/2014	Zemlok et al.	N/A	N/A
8925788	12/2014	Hess et al.	N/A	N/A
8926506	12/2014	Widenhouse et al.	N/A	N/A
8926598	12/2014	Mollere et al.	N/A	N/A
8931576	12/2014	Iwata	N/A	N/A
8931679	12/2014	Kostrzewski	N/A	N/A
8931680	12/2014	Milliman	N/A	N/A
8931682	12/2014	Timm et al.	N/A	N/A

8931692	12/2014	Sancak	N/A	N/A
8936614	12/2014	Allen, IV	N/A	N/A
8937408	12/2014	Ganem et al.	N/A	N/A
8939343	12/2014	Milliman et al.	N/A	N/A
8939344	12/2014	Olson et al.	N/A	N/A
8939898	12/2014	Omoto	N/A	N/A
8944069	12/2014	Miller et al.	N/A	N/A
8945095	12/2014	Blumenkranz et al.	N/A	N/A
8945098	12/2014	Seibold et al.	N/A	N/A
8945163	12/2014	Voegele et al.	N/A	N/A
8955732	12/2014	Zemlok et al.	N/A	N/A
8956342	12/2014	Russo et al.	N/A	N/A
8956390	12/2014	Shah et al.	N/A	N/A
8958860	12/2014	Banerjee et al.	N/A	N/A
8960519	12/2014	Whitman et al.	N/A	N/A
8960520	12/2014	McCuen	N/A	N/A
8960521	12/2014	Kostrzewski	N/A	N/A
8961191	12/2014	Hanshew	N/A	N/A
8961504	12/2014	Hoarau et al.	N/A	N/A
8961542	12/2014	Whitfield et al.	N/A	N/A
8963714	12/2014	Medhal et al.	N/A	N/A
D725674	12/2014	Jung et al.	N/A	N/A
8967443	12/2014	McCuen	N/A	N/A
8967444	12/2014	Beetel	N/A	N/A
8967446	12/2014	Beardsley et al.	N/A	N/A
8967448	12/2014	Carter et al.	N/A	N/A
8968276	12/2014	Zemlok et al.	N/A	N/A
8968308	12/2014	Horner et al.	N/A	N/A
8968312	12/2014	Marczyk et al.	N/A	N/A
8968337	12/2014	Whitfield et al.	N/A	N/A
8968340	12/2014	Chowaniec et al.	N/A	N/A
8968355	12/2014	Malkowski et al.	N/A	N/A
8968358	12/2014	Reschke	N/A	N/A
8970507	12/2014	Holbein et al.	N/A	N/A
8973803	12/2014	Hall et al.	N/A	N/A
8973804	12/2014	Hess et al.	N/A	N/A
8973805	12/2014	Scirica et al.	N/A	N/A
8974440	12/2014	Farritor et al.	N/A	N/A
8974542	12/2014	Fujimoto et al.	N/A	N/A
8974932	12/2014	McGahan et al.	N/A	N/A
8978954	12/2014	Shelton, IV et al.	N/A	N/A
8978955	12/2014	Aronhalt et al.	N/A	N/A
8978956	12/2014	Schall et al.	N/A	N/A
8979843	12/2014	Timm et al.	N/A	N/A
8979890	12/2014	Boudreaux	N/A	N/A
8982195	12/2014	Claus et al.	N/A	N/A
8984711	12/2014	Ota et al.	N/A	N/A
8985240	12/2014	Winnard	N/A	N/A
8985429	12/2014	Balek et al.	N/A	N/A
8986302	12/2014	Aldridge et al.	N/A	N/A

8989903	12/2014	Weir et al.	N/A	N/A
8991676	12/2014	Hess et al.	N/A	N/A
8991677	12/2014	Moore et al.	N/A	N/A
8991678	12/2014	Wellman et al.	N/A	N/A
8992042	12/2014	Eichenholz	N/A	N/A
8992422	12/2014	Spivey et al.	N/A	N/A
8992565	12/2014	Brisson et al.	N/A	N/A
8996165	12/2014	Wang et al.	N/A	N/A
8998058	12/2014	Moore et al.	N/A	N/A
8998059	12/2014	Smith et al.	N/A	N/A
8998060	12/2014	Bruewer et al.	N/A	N/A
8998061	12/2014	Williams et al.	N/A	N/A
8998939	12/2014	Price et al.	N/A	N/A
9000720	12/2014	Stulen et al.	N/A	N/A
9002518	12/2014	Manzo et al.	N/A	N/A
9004339	12/2014	Park	N/A	N/A
9004799	12/2014	Tibbits	N/A	N/A
9005230	12/2014	Yates et al.	N/A	N/A
9005238	12/2014	DeSantis et al.	N/A	N/A
9005243	12/2014	Stopek et al.	N/A	N/A
9010606	12/2014	Aranyi et al.	N/A	N/A
9010608	12/2014	Casasanta, Jr. et al.	N/A	N/A
9010611	12/2014	Ross et al.	N/A	N/A
9011437	12/2014	Woodruff et al.	N/A	N/A
9011439	12/2014	Shalaby et al.	N/A	N/A
9011471	12/2014	Timm et al.	N/A	N/A
9014856	12/2014	Manzo et al.	N/A	N/A
9016539	12/2014	Kostrzewski et al.	N/A	N/A
9016540	12/2014	Whitman et al.	N/A	N/A
9016541	12/2014	Viola et al.	N/A	N/A
9016542	12/2014	Shelton, IV et al.	N/A	N/A
9016545	12/2014	Aranyi et al.	N/A	N/A
9017331	12/2014	Fox	N/A	N/A
9017355	12/2014	Smith et al.	N/A	N/A
9017369	12/2014	Renger et al.	N/A	N/A
9017371	12/2014	Whitman et al.	N/A	N/A
9017849	12/2014	Stulen et al.	N/A	N/A
9017851	12/2014	Felder et al.	N/A	N/A
D729274	12/2014	Clement et al.	N/A	N/A
D730393	12/2014	Bray et al.	N/A	N/A
9021684	12/2014	Lenker et al.	N/A	N/A
9023014	12/2014	Chowaniec et al.	N/A	N/A
9023069	12/2014	Kasvikis et al.	N/A	N/A
9023071	12/2014	Miller et al.	N/A	N/A
9026347	12/2014	Gadh et al.	N/A	N/A
9027817	12/2014	Milliman et al.	N/A	N/A
9028468	12/2014	Scarfogliero et al.	N/A	N/A
9028494	12/2014	Shelton, IV et al.	N/A	N/A
9028495	12/2014	Mueller et al.	N/A	N/A
9028510	12/2014	Miyamoto et al.	N/A	N/A

9028511	12/2014	Weller et al.	N/A	N/A
9028519	12/2014	Yates et al.	N/A	N/A
9028529	12/2014	Fox et al.	N/A	N/A
9030166	12/2014	Kano	N/A	N/A
9030169	12/2014	Christensen et al.	N/A	N/A
9033203	12/2014	Woodard, Jr. et al.	N/A	N/A
9033204	12/2014	Shelton, IV et al.	N/A	N/A
9034505	12/2014	Detry et al.	N/A	N/A
9038881	12/2014	Schaller et al.	N/A	N/A
9039690	12/2014	Kersten et al.	N/A	N/A
9039694	12/2014	Ross et al.	N/A	N/A
9039720	12/2014	Madan	N/A	N/A
9039736	12/2014	Scirica et al.	N/A	N/A
9040062	12/2014	Maeda et al.	N/A	N/A
9043027	12/2014	Durant et al.	N/A	N/A
9044227	12/2014	Shelton, IV et al.	N/A	N/A
9044228	12/2014	Woodard, Jr. et al.	N/A	N/A
9044229	12/2014	Scheib et al.	N/A	N/A
9044230	12/2014	Morgan et al.	N/A	N/A
9044238	12/2014	Orszulak	N/A	N/A
9044241	12/2014	Barner et al.	N/A	N/A
9044261	12/2014	Houser	N/A	N/A
9044281	12/2014	Pool et al.	N/A	N/A
9050083	12/2014	Yates et al.	N/A	N/A
9050084	12/2014	Schmid et al.	N/A	N/A
9050089	12/2014	Orszulak	N/A	N/A
9050100	12/2014	Yates et al.	N/A	N/A
9050120	12/2014	Swarup et al.	N/A	N/A
9050123	12/2014	Krause et al.	N/A	N/A
9050176	12/2014	Datta et al.	N/A	N/A
9050192	12/2014	Mansmann	N/A	N/A
9055941	12/2014	Schmid et al.	N/A	N/A
9055942	12/2014	Balbierz et al.	N/A	N/A
9055943	12/2014	Zemlok et al.	N/A	N/A
9055944	12/2014	Hodgkinson et al.	N/A	N/A
9055961	12/2014	Manzo et al.	N/A	N/A
9060770	12/2014	Shelton, IV et al.	N/A	N/A
9060776	12/2014	Yates et al.	N/A	N/A
9060794	12/2014	Kang et al.	N/A	N/A
9060894	12/2014	Wubbeling	N/A	N/A
9061392	12/2014	Forgues et al.	N/A	N/A
9070068	12/2014	Coveley et al.	N/A	N/A
D733727	12/2014	Cojuangco et al.	N/A	N/A
9072515	12/2014	Hall et al.	N/A	N/A
9072523	12/2014	Houser et al.	N/A	N/A
9072535	12/2014	Shelton, IV et al.	N/A	N/A
9072536	12/2014	Shelton, IV et al.	N/A	N/A
9078653	12/2014	Leimbach et al.	N/A	N/A
9078654	12/2014	Whitman et al.	N/A	N/A
9084586	12/2014	Hafner et al.	N/A	N/A

9084601	12/2014	Moore et al.	N/A	N/A
9084602	12/2014	Gleiman	N/A	N/A
9086875	12/2014	Harrat et al.	N/A	N/A
9089326	12/2014	Krumanaker et al.	N/A	N/A
9089330	12/2014	Widenhouse et al.	N/A	N/A
9089338	12/2014	Smith et al.	N/A	N/A
9089352	12/2014	Jeong	N/A	N/A
9089360	12/2014	Messerly et al.	N/A	N/A
9091588	12/2014	Lefler	N/A	N/A
D736792	12/2014	Brinda et al.	N/A	N/A
9095339	12/2014	Moore et al.	N/A	N/A
9095346	12/2014	Houser et al.	N/A	N/A
9095362	12/2014	Dachs, II et al.	N/A	N/A
9095367	12/2014	Olson et al.	N/A	N/A
9095642	12/2014	Harder et al.	N/A	N/A
9096033	12/2014	Holop et al.	N/A	N/A
9098153	12/2014	Shen et al.	N/A	N/A
9099863	12/2014	Smith et al.	N/A	N/A
9099877	12/2014	Banos et al.	N/A	N/A
9099922	12/2014	Toosky et al.	N/A	N/A
9101358	12/2014	Kerr et al.	N/A	N/A
9101359	12/2014	Smith et al.	N/A	N/A
9101385	12/2014	Shelton, IV et al.	N/A	N/A
9101475	12/2014	Wei et al.	N/A	N/A
9101621	12/2014	Zeldis	N/A	N/A
9107663	12/2014	Swensgard	N/A	N/A
9107667	12/2014	Hodgkinson	N/A	N/A
9107690	12/2014	Bales, Jr. et al.	N/A	N/A
9110587	12/2014	Kim et al.	N/A	N/A
9113862	12/2014	Morgan et al.	N/A	N/A
9113864	12/2014	Morgan et al.	N/A	N/A
9113865	12/2014	Shelton, IV et al.	N/A	N/A
9113866	12/2014	Felder et al.	N/A	N/A
9113868	12/2014	Felder et al.	N/A	N/A
9113873	12/2014	Marczyk et al.	N/A	N/A
9113874	12/2014	Shelton, IV et al.	N/A	N/A
9113875	12/2014	Viola et al.	N/A	N/A
9113876	12/2014	Zemlok et al.	N/A	N/A
9113877	12/2014	Whitman et al.	N/A	N/A
9113879	12/2014	Felder et al.	N/A	N/A
9113880	12/2014	Zemlok et al.	N/A	N/A
9113881	12/2014	Scirica	N/A	N/A
9113883	12/2014	Aronhalt et al.	N/A	N/A
9113884	12/2014	Shelton, IV et al.	N/A	N/A
9113887	12/2014	Behnke, II et al.	N/A	N/A
9119615	12/2014	Felder et al.	N/A	N/A
9119657	12/2014	Shelton, IV et al.	N/A	N/A
9119898	12/2014	Bayon et al.	N/A	N/A
9119957	12/2014	Gantz et al.	N/A	N/A
9123286	12/2014	Park	N/A	N/A

9124097	12/2014	Cruz	N/A	N/A
9125651	12/2014	Mandakolathur Vasudevan et al.	N/A	N/A
9125654	12/2014	Aronhalt et al.	N/A	N/A
9125662	12/2014	Shelton, IV	N/A	N/A
9126317	12/2014	Lawton et al.	N/A	N/A
9131835	12/2014	Widenhouse et al.	N/A	N/A
9131940	12/2014	Huitema et al.	N/A	N/A
9131950	12/2014	Matthew	N/A	N/A
9131957	12/2014	Skarbnik et al.	N/A	N/A
9138225	12/2014	Huang et al.	N/A	N/A
9138226	12/2014	Racenet et al.	N/A	N/A
9144455	12/2014	Kennedy et al.	N/A	N/A
D740414	12/2014	Katsura	N/A	N/A
D741882	12/2014	Shmilov et al.	N/A	N/A
9149274	12/2014	Spivey et al.	N/A	N/A
9149324	12/2014	Huang et al.	N/A	N/A
9149325	12/2014	Worrell et al.	N/A	N/A
9153994	12/2014	Wood et al.	N/A	N/A
9154189	12/2014	Von Novak et al.	N/A	N/A
9161753	12/2014	Prior	N/A	N/A
9161769	12/2014	Stoddard et al.	N/A	N/A
9161803	12/2014	Yates et al.	N/A	N/A
9161807	12/2014	Garrison	N/A	N/A
9161855	12/2014	Rousseau et al.	N/A	N/A
9164271	12/2014	Ebata et al.	N/A	N/A
9167960	12/2014	Yamaguchi et al.	N/A	N/A
9168038	12/2014	Shelton, IV et al.	N/A	N/A
9168039	12/2014	Knodel	N/A	N/A
9168042	12/2014	Milliman	N/A	N/A
9168054	12/2014	Turner et al.	N/A	N/A
9168144	12/2014	Rivin et al.	N/A	N/A
9171244	12/2014	Endou et al.	N/A	N/A
9179832	12/2014	Diolaiti	N/A	N/A
9179911	12/2014	Morgan et al.	N/A	N/A
9179912	12/2014	Yates et al.	N/A	N/A
9180223	12/2014	Yu et al.	N/A	N/A
9182244	12/2014	Luke et al.	N/A	N/A
9186046	12/2014	Ramamurthy et al.	N/A	N/A
9186137	12/2014	Farascioni et al.	N/A	N/A
9186140	12/2014	Hiles et al.	N/A	N/A
9186142	12/2014	Fanelli et al.	N/A	N/A
9186143	12/2014	Timm et al.	N/A	N/A
9186148	12/2014	Felder et al.	N/A	N/A
9186221	12/2014	Burbank	N/A	N/A
9192376	12/2014	Almodovar	N/A	N/A
9192380	12/2014	(Tarinelli) Racenet et al.	N/A	N/A
9192384	12/2014	Bettuchi	N/A	N/A
9192430	12/2014	Rachlin et al.	N/A	N/A

9192434	12/2014	Twomey et al.	N/A	N/A
9193045	12/2014	Saur et al.	N/A	N/A
9197079	12/2014	Yip et al.	N/A	N/A
D744528	12/2014	Agrawal	N/A	N/A
D746459	12/2014	Kaercher et al.	N/A	N/A
9198642	12/2014	Storz	N/A	N/A
9198644	12/2014	Balek et al.	N/A	N/A
9198661	12/2014	Swensgard	N/A	N/A
9198662	12/2014	Barton et al.	N/A	N/A
9198683	12/2014	Friedman et al.	N/A	N/A
9204830	12/2014	Zand et al.	N/A	N/A
9204877	12/2014	Whitman et al.	N/A	N/A
9204878	12/2014	Hall et al.	N/A	N/A
9204879	12/2014	Shelton, IV	N/A	N/A
9204880	12/2014	Baxter, III et al.	N/A	N/A
9204881	12/2014	Penna	N/A	N/A
9204923	12/2014	Manzo et al.	N/A	N/A
9204924	12/2014	Marczyk et al.	N/A	N/A
9211120	12/2014	Scheib et al.	N/A	N/A
9211121	12/2014	Hall et al.	N/A	N/A
9211122	12/2014	Hagerty et al.	N/A	N/A
9216013	12/2014	Scirica et al.	N/A	N/A
9216019	12/2014	Schmid et al.	N/A	N/A
9216020	12/2014	Zhang et al.	N/A	N/A
9216030	12/2014	Fan et al.	N/A	N/A
9216062	12/2014	Duque et al.	N/A	N/A
9220500	12/2014	Swayze et al.	N/A	N/A
9220501	12/2014	Baxter, III et al.	N/A	N/A
9220502	12/2014	Zemlok et al.	N/A	N/A
9220504	12/2014	Viola et al.	N/A	N/A
9220508	12/2014	Dannaher	N/A	N/A
9220559	12/2014	Worrell et al.	N/A	N/A
9220570	12/2014	Kim et al.	N/A	N/A
D746854	12/2015	Shardlow et al.	N/A	N/A
9226686	12/2015	Blair	N/A	N/A
9226750	12/2015	Weir et al.	N/A	N/A
9226751	12/2015	Shelton, IV et al.	N/A	N/A
9226754	12/2015	D'Agostino et al.	N/A	N/A
9226760	12/2015	Shelton, IV	N/A	N/A
9226761	12/2015	Burbank	N/A	N/A
9226767	12/2015	Stulen et al.	N/A	N/A
9226799	12/2015	Lightcap et al.	N/A	N/A
9232941	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9232945	12/2015	Zingman	N/A	N/A
9232979	12/2015	Parihar et al.	N/A	N/A
9233610	12/2015	Kim et al.	N/A	N/A
9237891	12/2015	Shelton, IV	N/A	N/A
9237892	12/2015	Hodgkinson	N/A	N/A
9237895	12/2015	McCarthy et al.	N/A	N/A

9237900	12/2015	Boudreaux et al.	N/A	N/A
9237921	12/2015	Messerly et al.	N/A	N/A
9239064	12/2015	Helbig et al.	N/A	N/A
9240740	12/2015	Zeng et al.	N/A	N/A
9241711	12/2015	Ivanko	N/A	N/A
9241712	12/2015	Zemlok et al.	N/A	N/A
9241714	12/2015	Timm et al.	N/A	N/A
9241716	12/2015	Whitman	N/A	N/A
9241731	12/2015	Boudreaux et al.	N/A	N/A
9241758	12/2015	Franer et al.	N/A	N/A
9244524	12/2015	Inoue et al.	N/A	N/A
D748668	12/2015	Kim et al.	N/A	N/A
D749128	12/2015	Perez et al.	N/A	N/A
D749623	12/2015	Gray et al.	N/A	N/A
D750122	12/2015	Shardlow et al.	N/A	N/A
D750129	12/2015	Kwon	N/A	N/A
9254131	12/2015	Soltz et al.	N/A	N/A
9254170	12/2015	Parihar et al.	N/A	N/A
9259265	12/2015	Harris et al.	N/A	N/A
9259268	12/2015	Behnke, II et al.	N/A	N/A
9259274	12/2015	Prisco	N/A	N/A
9259275	12/2015	Burbank	N/A	N/A
9261172	12/2015	Solomon et al.	N/A	N/A
9265500	12/2015	Sorrentino et al.	N/A	N/A
9265510	12/2015	Dietzel et al.	N/A	N/A
9265516	12/2015	Casey et al.	N/A	N/A
9265585	12/2015	Wingardner et al.	N/A	N/A
9271718	12/2015	Milad et al.	N/A	N/A
9271727	12/2015	McGuckin, Jr. et al.	N/A	N/A
9271753	12/2015	Butler et al.	N/A	N/A
9271799	12/2015	Shelton, IV et al.	N/A	N/A
9272406	12/2015	Aronhalt et al.	N/A	N/A
9274095	12/2015	Humayun et al.	N/A	N/A
9277919	12/2015	Timmer et al.	N/A	N/A
9277922	12/2015	Carter et al.	N/A	N/A
9277969	12/2015	Brannan et al.	N/A	N/A
9282962	12/2015	Schmid et al.	N/A	N/A
9282963	12/2015	Bryant	N/A	N/A
9282966	12/2015	Shelton, IV et al.	N/A	N/A
9282974	12/2015	Shelton, IV	N/A	N/A
9283028	12/2015	Johnson	N/A	N/A
9283045	12/2015	Rhee et al.	N/A	N/A
9283054	12/2015	Morgan et al.	N/A	N/A
9283334	12/2015	Mantell et al.	N/A	N/A
9289206	12/2015	Hess et al.	N/A	N/A
9289207	12/2015	Shelton, IV	N/A	N/A
9289210	12/2015	Baxter, III et al.	N/A	N/A
9289211	12/2015	Williams et al.	N/A	N/A
9289212	12/2015	Shelton, IV et al.	N/A	N/A
9289225	12/2015	Shelton, IV et al.	N/A	N/A

9289256	12/2015	Shelton, IV et al.	N/A	N/A
9293757	12/2015	Toussaint et al.	N/A	N/A
9295464	12/2015	Shelton, IV et al.	N/A	N/A
9295465	12/2015	Farascioni	N/A	N/A
9295466	12/2015	Hodgkinson et al.	N/A	N/A
9295467	12/2015	Scirica	N/A	N/A
9295468	12/2015	Heinrich et al.	N/A	N/A
9295514	12/2015	Shelton, IV et al.	N/A	N/A
9295522	12/2015	Kostrzewski	N/A	N/A
9295565	12/2015	McLean	N/A	N/A
9295784	12/2015	Eggert et al.	N/A	N/A
D753167	12/2015	Yu et al.	N/A	N/A
9301691	12/2015	Hufnagel et al.	N/A	N/A
9301752	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9301753	12/2015	Aldridge et al.	N/A	N/A
9301755	12/2015	Shelton, IV et al.	N/A	N/A
9301759	12/2015	Spivey et al.	N/A	N/A
9301811	12/2015	Goldberg et al.	N/A	N/A
9307965	12/2015	Ming et al.	N/A	N/A
9307986	12/2015	Hall et al.	N/A	N/A
9307987	12/2015	Swensgard et al.	N/A	N/A
9307988	12/2015	Shelton, IV	N/A	N/A
9307989	12/2015	Shelton, IV et al.	N/A	N/A
9307994	12/2015	Gresham et al.	N/A	N/A
9308009	12/2015	Madan et al.	N/A	N/A
9308011	12/2015	Chao et al.	N/A	N/A
9308646	12/2015	Lim et al.	N/A	N/A
9313915	12/2015	Niu et al.	N/A	N/A
9314246	12/2015	Shelton, IV et al.	N/A	N/A
9314247	12/2015	Shelton, IV et al.	N/A	N/A
9314261	12/2015	Bales, Jr. et al.	N/A	N/A
9314291	12/2015	Schall et al.	N/A	N/A
9314339	12/2015	Mansmann	N/A	N/A
9314908	12/2015	Tanimoto et al.	N/A	N/A
9320518	12/2015	Henderson et al.	N/A	N/A
9320520	12/2015	Shelton, IV et al.	N/A	N/A
9320521	12/2015	Shelton, IV et al.	N/A	N/A
9320523	12/2015	Shelton, IV et al.	N/A	N/A
9325516	12/2015	Pera et al.	N/A	N/A
D755196	12/2015	Meyers et al.	N/A	N/A
D756373	12/2015	Raskin et al.	N/A	N/A
D756377	12/2015	Connolly et al.	N/A	N/A
D757028	12/2015	Goldenberg et al.	N/A	N/A
9326767	12/2015	Koch, Jr. et al.	N/A	N/A
9326768	12/2015	Shelton, IV	N/A	N/A
9326769	12/2015	Shelton, IV et al.	N/A	N/A
9326770	12/2015	Shelton, IV et al.	N/A	N/A
9326771	12/2015	Baxter, III et al.	N/A	N/A
9326788	12/2015	Batross et al.	N/A	N/A

9326812	12/2015	Waller et al.	N/A	N/A
9326824	12/2015	Inoue et al.	N/A	N/A
9327061	12/2015	Govil et al.	N/A	N/A
9331721	12/2015	Martinez Nuevo et al.	N/A	N/A
9332890	12/2015	Ozawa	N/A	N/A
9332974	12/2015	Henderson et al.	N/A	N/A
9332984	12/2015	Weaner et al.	N/A	N/A
9332987	12/2015	Leimbach et al.	N/A	N/A
9333040	12/2015	Shellenberger et al.	N/A	N/A
9333082	12/2015	Wei et al.	N/A	N/A
9337668	12/2015	Yip	N/A	N/A
9339226	12/2015	Van Der Walt et al.	N/A	N/A
9339342	12/2015	Prisco et al.	N/A	N/A
9345477	12/2015	Anim et al.	N/A	N/A
9345479	12/2015	(Tarinelli) Racenet et al.	N/A	N/A
9345480	12/2015	Hessler et al.	N/A	N/A
9345481	12/2015	Hall et al.	N/A	N/A
9345503	12/2015	Ishida et al.	N/A	N/A
9351726	12/2015	Leimbach et al.	N/A	N/A
9351727	12/2015	Leimbach et al.	N/A	N/A
9351728	12/2015	Sniffin et al.	N/A	N/A
9351730	12/2015	Schmid et al.	N/A	N/A
9351731	12/2015	Carter et al.	N/A	N/A
9351732	12/2015	Hodgkinson	N/A	N/A
9352071	12/2015	Landgrebe et al.	N/A	N/A
D758433	12/2015	Lee et al.	N/A	N/A
D759063	12/2015	Chen	N/A	N/A
9358003	12/2015	Hall et al.	N/A	N/A
9358004	12/2015	Sniffin et al.	N/A	N/A
9358005	12/2015	Shelton, IV et al.	N/A	N/A
9358015	12/2015	Sorrentino et al.	N/A	N/A
9358031	12/2015	Manzo	N/A	N/A
9358065	12/2015	Ladtkow et al.	N/A	N/A
9364217	12/2015	Kostrzewski et al.	N/A	N/A
9364219	12/2015	Olson et al.	N/A	N/A
9364220	12/2015	Williams	N/A	N/A
9364223	12/2015	Scirica	N/A	N/A
9364226	12/2015	Zemlok et al.	N/A	N/A
9364228	12/2015	Straehnz et al.	N/A	N/A
9364229	12/2015	D'Agostino et al.	N/A	N/A
9364230	12/2015	Shelton, IV et al.	N/A	N/A
9364231	12/2015	Wenchell	N/A	N/A
9364233	12/2015	Alexander, III et al.	N/A	N/A
9364279	12/2015	Houser et al.	N/A	N/A
9368991	12/2015	Qahouq	N/A	N/A
9370341	12/2015	Ceniccola et al.	N/A	N/A
9370358	12/2015	Shelton, IV et al.	N/A	N/A
9370361	12/2015	Viola et al.	N/A	N/A

9370362	12/2015	Petty et al.	N/A	N/A
9370364	12/2015	Smith et al.	N/A	N/A
9370400	12/2015	Parihar	N/A	N/A
9375206	12/2015	Vidal et al.	N/A	N/A
9375218	12/2015	Wheeler et al.	N/A	N/A
9375230	12/2015	Ross et al.	N/A	N/A
9375232	12/2015	Hunt et al.	N/A	N/A
9375255	12/2015	Houser et al.	N/A	N/A
D761309	12/2015	Lee et al.	N/A	N/A
9381058	12/2015	Houser et al.	N/A	N/A
9383881	12/2015	Day et al.	N/A	N/A
9385640	12/2015	Sun et al.	N/A	N/A
9386983	12/2015	Swensgard et al.	N/A	N/A
9386984	12/2015	Aronhalt et al.	N/A	N/A
9386985	12/2015	Koch, Jr. et al.	N/A	N/A
9386988	12/2015	Baxter, III et al.	N/A	N/A
9387003	12/2015	Kaercher et al.	N/A	N/A
9392885	12/2015	Vogler et al.	N/A	N/A
9393015	12/2015	Laurent et al.	N/A	N/A
9393017	12/2015	Flanagan et al.	N/A	N/A
9393018	12/2015	Wang et al.	N/A	N/A
9393354	12/2015	Freedman et al.	N/A	N/A
9396369	12/2015	Whitehurst et al.	N/A	N/A
9396669	12/2015	Karkanias et al.	N/A	N/A
9398905	12/2015	Martin	N/A	N/A
9398911	12/2015	Auld	N/A	N/A
D763277	12/2015	Ahmed et al.	N/A	N/A
D764498	12/2015	Capela et al.	N/A	N/A
9402604	12/2015	Williams et al.	N/A	N/A
9402625	12/2015	Coleman et al.	N/A	N/A
9402626	12/2015	Ortiz et al.	N/A	N/A
9402627	12/2015	Stevenson et al.	N/A	N/A
9402629	12/2015	Ehrenfels et al.	N/A	N/A
9402679	12/2015	Ginnebaugh et al.	N/A	N/A
9402682	12/2015	Worrell et al.	N/A	N/A
9402688	12/2015	Min et al.	N/A	N/A
9408604	12/2015	Shelton, IV et al.	N/A	N/A
9408605	12/2015	Knodel et al.	N/A	N/A
9408606	12/2015	Shelton, IV	N/A	N/A
9408622	12/2015	Stulen et al.	N/A	N/A
9411370	12/2015	Benni et al.	N/A	N/A
9413128	12/2015	Tien et al.	N/A	N/A
9414838	12/2015	Shelton, IV et al.	N/A	N/A
9414849	12/2015	Nagashimada	N/A	N/A
9414880	12/2015	Monson et al.	N/A	N/A
9420967	12/2015	Zand et al.	N/A	N/A
9421003	12/2015	Williams et al.	N/A	N/A
9421014	12/2015	Ingmanson et al.	N/A	N/A
9421030	12/2015	Cole et al.	N/A	N/A
9421060	12/2015	Monson et al.	N/A	N/A

9421062	12/2015	Houser et al.	N/A	N/A
9421682	12/2015	McClaskey et al.	N/A	N/A
9427223	12/2015	Park et al.	N/A	N/A
9427231	12/2015	Racenet et al.	N/A	N/A
9429204	12/2015	Stefan et al.	N/A	N/A
D767624	12/2015	Lee et al.	N/A	N/A
9433411	12/2015	Racenet et al.	N/A	N/A
9433414	12/2015	Chen et al.	N/A	N/A
9433418	12/2015	Whitman et al.	N/A	N/A
9433419	12/2015	Gonzalez et al.	N/A	N/A
9433420	12/2015	Hodgkinson	N/A	N/A
9439649	12/2015	Shelton, IV et al.	N/A	N/A
9439650	12/2015	McGuckin, Jr. et al.	N/A	N/A
9439651	12/2015	Smith et al.	N/A	N/A
9439668	12/2015	Timm et al.	N/A	N/A
9445808	12/2015	Woodard, Jr. et al.	N/A	N/A
9445813	12/2015	Shelton, IV et al.	N/A	N/A
9445816	12/2015	Swayze et al.	N/A	N/A
9445817	12/2015	Bettuchi	N/A	N/A
9446226	12/2015	Zilberman	N/A	N/A
9451938	12/2015	Overes et al.	N/A	N/A
9451958	12/2015	Shelton, IV et al.	N/A	N/A
9452020	12/2015	Griffiths et al.	N/A	N/A
D768152	12/2015	Gutierrez et al.	N/A	N/A
D768156	12/2015	Frincke	N/A	N/A
D768167	12/2015	Jones et al.	N/A	N/A
D769315	12/2015	Scotti	N/A	N/A
D769930	12/2015	Agrawal	N/A	N/A
9461340	12/2015	Li et al.	N/A	N/A
9463012	12/2015	Bonutti et al.	N/A	N/A
9463040	12/2015	Jeong et al.	N/A	N/A
9463260	12/2015	Stopek	N/A	N/A
9468438	12/2015	Baber et al.	N/A	N/A
9468447	12/2015	Aman et al.	N/A	N/A
9470297	12/2015	Aranyi et al.	N/A	N/A
9471969	12/2015	Zeng et al.	N/A	N/A
9474506	12/2015	Magnin et al.	N/A	N/A
9474513	12/2015	Ishida et al.	N/A	N/A
9474523	12/2015	Meade et al.	N/A	N/A
9474528	12/2015	Marczyk	N/A	N/A
9474540	12/2015	Stokes et al.	N/A	N/A
9475180	12/2015	Eshleman et al.	N/A	N/A
9477649	12/2015	Davidson et al.	N/A	N/A
D770476	12/2015	Jitkoff et al.	N/A	N/A
D770515	12/2015	Cho et al.	N/A	N/A
D771116	12/2015	Dellinger et al.	N/A	N/A
D772905	12/2015	Ingenlath	N/A	N/A
9480476	12/2015	Aldridge et al.	N/A	N/A
9480492	12/2015	Aranyi et al.	N/A	N/A
9483095	12/2015	Tran et al.	N/A	N/A

9486186	12/2015	Fiebig et al.	N/A	N/A
9486213	12/2015	Altman et al.	N/A	N/A
9486214	12/2015	Shelton, IV	N/A	N/A
9486215	12/2015	Olson et al.	N/A	N/A
9486302	12/2015	Boey et al.	N/A	N/A
9488197	12/2015	Wi	N/A	N/A
9492146	12/2015	Kostrzewski et al.	N/A	N/A
9492167	12/2015	Shelton, IV et al.	N/A	N/A
9492170	12/2015	Bear et al.	N/A	N/A
9492172	12/2015	Weisshaupt et al.	N/A	N/A
9492189	12/2015	Williams et al.	N/A	N/A
9492192	12/2015	To et al.	N/A	N/A
9492237	12/2015	Kang et al.	N/A	N/A
9498213	12/2015	Marczyk et al.	N/A	N/A
9498219	12/2015	Moore et al.	N/A	N/A
9498231	12/2015	Haider et al.	N/A	N/A
9504455	12/2015	Whitman et al.	N/A	N/A
9504483	12/2015	Houser et al.	N/A	N/A
9504520	12/2015	Worrell et al.	N/A	N/A
9504521	12/2015	Deutmeyer et al.	N/A	N/A
9504528	12/2015	Ivinson et al.	N/A	N/A
9507399	12/2015	Chien	N/A	N/A
D774547	12/2015	Capela et al.	N/A	N/A
D775336	12/2015	Shelton, IV et al.	N/A	N/A
9510827	12/2015	Kostrzewski	N/A	N/A
9510828	12/2015	Yates et al.	N/A	N/A
9510830	12/2015	Shelton, IV et al.	N/A	N/A
9510846	12/2015	Sholev et al.	N/A	N/A
9510895	12/2015	Houser et al.	N/A	N/A
9510925	12/2015	Hotter et al.	N/A	N/A
9515366	12/2015	Herbsommer et al.	N/A	N/A
9517063	12/2015	Swayze et al.	N/A	N/A
9517065	12/2015	Simms et al.	N/A	N/A
9517068	12/2015	Shelton, IV et al.	N/A	N/A
9517326	12/2015	Hinman et al.	N/A	N/A
9521996	12/2015	Armstrong	N/A	N/A
9522003	12/2015	Weir et al.	N/A	N/A
9522005	12/2015	Williams et al.	N/A	N/A
9522014	12/2015	Nishizawa et al.	N/A	N/A
9522029	12/2015	Yates et al.	N/A	N/A
9526481	12/2015	Storz et al.	N/A	N/A
9526499	12/2015	Kostrzewski et al.	N/A	N/A
9526563	12/2015	Twomey	N/A	N/A
9526564	12/2015	Rusin	N/A	N/A
9526921	12/2015	Kimball et al.	N/A	N/A
D776683	12/2016	Gobinski et al.	N/A	N/A
D777773	12/2016	Shi	N/A	N/A
9532783	12/2016	Swayze et al.	N/A	N/A
9539060	12/2016	Lightcap et al.	N/A	N/A
9539726	12/2016	Simaan et al.	N/A	N/A

9545253	12/2016	Worrell et al.	N/A	N/A
9545258	12/2016	Smith et al.	N/A	N/A
9549732	12/2016	Yates et al.	N/A	N/A
9549733	12/2016	Knodel	N/A	N/A
9549735	12/2016	Shelton, IV et al.	N/A	N/A
9549750	12/2016	Shelton, IV et al.	N/A	N/A
9554794	12/2016	Baber et al.	N/A	N/A
9554796	12/2016	Kostrzewski	N/A	N/A
9554803	12/2016	Smith et al.	N/A	N/A
9554812	12/2016	Inkpen et al.	N/A	N/A
9554854	12/2016	Yates et al.	N/A	N/A
9559624	12/2016	Philipp	N/A	N/A
9561013	12/2016	Tsuchiya	N/A	N/A
9561029	12/2016	Scheib et al.	N/A	N/A
9561030	12/2016	Zhang et al.	N/A	N/A
9561031	12/2016	Heinrich et al.	N/A	N/A
9561032	12/2016	Shelton, IV et al.	N/A	N/A
9561038	12/2016	Shelton, IV et al.	N/A	N/A
9561045	12/2016	Hinman et al.	N/A	N/A
9561072	12/2016	Ko	N/A	N/A
9561082	12/2016	Yen et al.	N/A	N/A
9566061	12/2016	Aronhalt et al.	N/A	N/A
9566062	12/2016	Boudreaux	N/A	N/A
9566064	12/2016	Williams et al.	N/A	N/A
9566065	12/2016	Knodel	N/A	N/A
9566067	12/2016	Milliman et al.	N/A	N/A
9572552	12/2016	Bodor et al.	N/A	N/A
9572574	12/2016	Shelton, IV et al.	N/A	N/A
9572576	12/2016	Hodgkinson et al.	N/A	N/A
9572577	12/2016	Lloyd et al.	N/A	N/A
9572592	12/2016	Price et al.	N/A	N/A
9574644	12/2016	Parihar	N/A	N/A
9579039	12/2016	Jansen et al.	N/A	N/A
9579088	12/2016	Farritor et al.	N/A	N/A
9579143	12/2016	Ullrich et al.	N/A	N/A
9579158	12/2016	Brianza et al.	N/A	N/A
D780803	12/2016	Gill et al.	N/A	N/A
D781879	12/2016	Butcher et al.	N/A	N/A
D782530	12/2016	Paek et al.	N/A	N/A
9585550	12/2016	Abel et al.	N/A	N/A
9585657	12/2016	Shelton, IV et al.	N/A	N/A
9585658	12/2016	Shelton, IV	N/A	N/A
9585659	12/2016	Viola et al.	N/A	N/A
9585660	12/2016	Laurent et al.	N/A	N/A
9585662	12/2016	Shelton, IV et al.	N/A	N/A
9585663	12/2016	Shelton, IV et al.	N/A	N/A
9585672	12/2016	Bastia	N/A	N/A
9590433	12/2016	Li	N/A	N/A
9592050	12/2016	Schmid et al.	N/A	N/A
9592052	12/2016	Shelton, IV	N/A	N/A

9592053	12/2016	Shelton, IV et al.	N/A	N/A
9592054	12/2016	Schmid et al.	N/A	N/A
9597073	12/2016	Sorrentino et al.	N/A	N/A
9597075	12/2016	Shelton, IV et al.	N/A	N/A
9597078	12/2016	Scirica et al.	N/A	N/A
9597080	12/2016	Milliman et al.	N/A	N/A
9597104	12/2016	Nicholas et al.	N/A	N/A
9597143	12/2016	Madan et al.	N/A	N/A
9603595	12/2016	Shelton, IV et al.	N/A	N/A
9603598	12/2016	Shelton, IV et al.	N/A	N/A
9603599	12/2016	Miller et al.	N/A	N/A
9603991	12/2016	Shelton, IV et al.	N/A	N/A
D783658	12/2016	Hurst et al.	N/A	N/A
9610068	12/2016	Kappel et al.	N/A	N/A
9610079	12/2016	Kamei et al.	N/A	N/A
9610080	12/2016	Whitfield et al.	N/A	N/A
9610412	12/2016	Zemlok et al.	N/A	N/A
9614258	12/2016	Takahashi et al.	N/A	N/A
9615826	12/2016	Shelton, IV et al.	N/A	N/A
9622745	12/2016	Ingmanson et al.	N/A	N/A
9622746	12/2016	Simms et al.	N/A	N/A
9629623	12/2016	Lytle, IV et al.	N/A	N/A
9629626	12/2016	Soltz et al.	N/A	N/A
9629627	12/2016	Kostrzewski et al.	N/A	N/A
9629628	12/2016	Aranyi	N/A	N/A
9629629	12/2016	Leimbach et al.	N/A	N/A
9629631	12/2016	Nicholas et al.	N/A	N/A
9629632	12/2016	Linder et al.	N/A	N/A
9629652	12/2016	Mumaw et al.	N/A	N/A
9629814	12/2016	Widenhouse et al.	N/A	N/A
D785794	12/2016	Magno, Jr.	N/A	N/A
D786280	12/2016	Ma	N/A	N/A
D786896	12/2016	Kim et al.	N/A	N/A
D787547	12/2016	Basargin et al.	N/A	N/A
D788123	12/2016	Shan et al.	N/A	N/A
D788140	12/2016	Hemsley et al.	N/A	N/A
9636091	12/2016	Beardsley et al.	N/A	N/A
9636111	12/2016	Wenchell	N/A	N/A
9636112	12/2016	Penna et al.	N/A	N/A
9636113	12/2016	Wenchell	N/A	N/A
9636850	12/2016	Stopek et al.	N/A	N/A
9641122	12/2016	Romanowich et al.	N/A	N/A
9642620	12/2016	Baxter, III et al.	N/A	N/A
9642642	12/2016	Lim	N/A	N/A
9649096	12/2016	Sholev	N/A	N/A
9649110	12/2016	Parihar et al.	N/A	N/A
9649111	12/2016	Shelton, IV et al.	N/A	N/A
9649190	12/2016	Mathies	N/A	N/A
9651032	12/2016	Weaver et al.	N/A	N/A
9655613	12/2016	Schaller	N/A	N/A

9655614	12/2016	Swensgard et al.	N/A	N/A
9655615	12/2016	Knodel et al.	N/A	N/A
9655616	12/2016	Aranyi	N/A	N/A
9655624	12/2016	Shelton, IV et al.	N/A	N/A
9661991	12/2016	Glossop	N/A	N/A
9662108	12/2016	Williams	N/A	N/A
9662110	12/2016	Huang et al.	N/A	N/A
9662111	12/2016	Holsten et al.	N/A	N/A
9662116	12/2016	Smith et al.	N/A	N/A
9662130	12/2016	Bartels et al.	N/A	N/A
9662131	12/2016	Omori et al.	N/A	N/A
D788792	12/2016	Alessandri et al.	N/A	N/A
D789384	12/2016	Lin et al.	N/A	N/A
D790570	12/2016	Butcher et al.	N/A	N/A
9668728	12/2016	Williams et al.	N/A	N/A
9668729	12/2016	Williams et al.	N/A	N/A
9668732	12/2016	Patel et al.	N/A	N/A
9668733	12/2016	Williams	N/A	N/A
9668734	12/2016	Kostrzewski et al.	N/A	N/A
9668735	12/2016	Beetel	N/A	N/A
9675344	12/2016	Combrowski et al.	N/A	N/A
9675348	12/2016	Smith et al.	N/A	N/A
9675351	12/2016	Hodgkinson et al.	N/A	N/A
9675354	12/2016	Weir et al.	N/A	N/A
9675355	12/2016	Shelton, IV et al.	N/A	N/A
9675368	12/2016	Guo et al.	N/A	N/A
9675372	12/2016	Laurent et al.	N/A	N/A
9675375	12/2016	Houser et al.	N/A	N/A
9675405	12/2016	Trees et al.	N/A	N/A
9675819	12/2016	Dunbar et al.	N/A	N/A
9681870	12/2016	Baxter, III et al.	N/A	N/A
9681873	12/2016	Smith et al.	N/A	N/A
9681884	12/2016	Clem et al.	N/A	N/A
9687230	12/2016	Leimbach et al.	N/A	N/A
9687231	12/2016	Baxter, III et al.	N/A	N/A
9687232	12/2016	Shelton, IV et al.	N/A	N/A
9687233	12/2016	Fernandez et al.	N/A	N/A
9687236	12/2016	Leimbach et al.	N/A	N/A
9687237	12/2016	Schmid et al.	N/A	N/A
9687253	12/2016	Detry et al.	N/A	N/A
9689466	12/2016	Kanai et al.	N/A	N/A
9690362	12/2016	Leimbach et al.	N/A	N/A
9693772	12/2016	Ingmanson et al.	N/A	N/A
9693774	12/2016	Gettinger et al.	N/A	N/A
9693775	12/2016	Agarwal et al.	N/A	N/A
9693777	12/2016	Schellin et al.	N/A	N/A
9700309	12/2016	Jaworek et al.	N/A	N/A
9700310	12/2016	Morgan et al.	N/A	N/A
9700312	12/2016	Kostrzewski et al.	N/A	N/A
9700314	12/2016	Marczyk	N/A	N/A

9700315	12/2016	Chen et al.	N/A	N/A
9700317	12/2016	Aronhalt et al.	N/A	N/A
9700318	12/2016	Scirica et al.	N/A	N/A
9700319	12/2016	Motooka et al.	N/A	N/A
9700320	12/2016	Dinardo et al.	N/A	N/A
9700321	12/2016	Shelton, IV et al.	N/A	N/A
9700334	12/2016	Hinman et al.	N/A	N/A
9700381	12/2016	Amat Girbau	N/A	N/A
9702823	12/2016	Maher et al.	N/A	N/A
9706674	12/2016	Collins et al.	N/A	N/A
9706981	12/2016	Nicholas et al.	N/A	N/A
9706991	12/2016	Hess et al.	N/A	N/A
9706993	12/2016	Hessler et al.	N/A	N/A
9707003	12/2016	Hoell, Jr. et al.	N/A	N/A
9707005	12/2016	Strobl et al.	N/A	N/A
9707026	12/2016	Malackowski et al.	N/A	N/A
9707033	12/2016	Parihar et al.	N/A	N/A
9707043	12/2016	Bozung	N/A	N/A
9707684	12/2016	Ruiz Morales et al.	N/A	N/A
9713466	12/2016	Kostrzewski	N/A	N/A
9713468	12/2016	Harris et al.	N/A	N/A
9713470	12/2016	Scirica et al.	N/A	N/A
9713474	12/2016	Lorenz	N/A	N/A
D795919	12/2016	Bischoff et al.	N/A	N/A
9717497	12/2016	Zerkle et al.	N/A	N/A
9717498	12/2016	Aranyi et al.	N/A	N/A
9718190	12/2016	Larkin et al.	N/A	N/A
9722236	12/2016	Sathrum	N/A	N/A
9724091	12/2016	Shelton, IV et al.	N/A	N/A
9724092	12/2016	Baxter, III et al.	N/A	N/A
9724094	12/2016	Baber et al.	N/A	N/A
9724095	12/2016	Gupta et al.	N/A	N/A
9724096	12/2016	Thompson et al.	N/A	N/A
9724098	12/2016	Baxter, III et al.	N/A	N/A
9724118	12/2016	Schulte et al.	N/A	N/A
9724163	12/2016	Orban	N/A	N/A
9730692	12/2016	Shelton, IV et al.	N/A	N/A
9730695	12/2016	Leimbach et al.	N/A	N/A
9730697	12/2016	Morgan et al.	N/A	N/A
9730717	12/2016	Katsuki et al.	N/A	N/A
9730757	12/2016	Brudniok	N/A	N/A
9731410	12/2016	Hirabayashi et al.	N/A	N/A
9733663	12/2016	Leimbach et al.	N/A	N/A
9737297	12/2016	Racenet et al.	N/A	N/A
9737298	12/2016	Isbell, Jr.	N/A	N/A
9737299	12/2016	Yan	N/A	N/A
9737301	12/2016	Baber et al.	N/A	N/A
9737302	12/2016	Shelton, IV et al.	N/A	N/A
9737303	12/2016	Shelton, IV et al.	N/A	N/A
9737323	12/2016	Thapliyal et al.	N/A	N/A

9737365	12/2016	Hegeman et al.	N/A	N/A
9743927	12/2016	Whitman	N/A	N/A
9743928	12/2016	Shelton, IV et al.	N/A	N/A
9743929	12/2016	Leimbach et al.	N/A	N/A
D798319	12/2016	Bergstrand et al.	N/A	N/A
9750498	12/2016	Timm et al.	N/A	N/A
9750499	12/2016	Leimbach et al.	N/A	N/A
9750501	12/2016	Shelton, IV et al.	N/A	N/A
9750502	12/2016	Scirica et al.	N/A	N/A
9750503	12/2016	Milliman	N/A	N/A
9750639	12/2016	Barnes et al.	N/A	N/A
9751176	12/2016	McRoberts et al.	N/A	N/A
9757123	12/2016	Giordano et al.	N/A	N/A
9757124	12/2016	Schellin et al.	N/A	N/A
9757126	12/2016	Cappola	N/A	N/A
9757128	12/2016	Baber et al.	N/A	N/A
9757129	12/2016	Williams	N/A	N/A
9757130	12/2016	Shelton, IV	N/A	N/A
9763662	12/2016	Shelton, IV et al.	N/A	N/A
9763668	12/2016	Whitfield et al.	N/A	N/A
9770245	12/2016	Swayze et al.	N/A	N/A
9770274	12/2016	Pool et al.	N/A	N/A
D798886	12/2016	Prophete et al.	N/A	N/A
D800742	12/2016	Rhodes	N/A	N/A
D800744	12/2016	Jitkoff et al.	N/A	N/A
D800766	12/2016	Park et al.	N/A	N/A
D800904	12/2016	Leimbach et al.	N/A	N/A
9775608	12/2016	Aronhalt et al.	N/A	N/A
9775609	12/2016	Shelton, IV et al.	N/A	N/A
9775610	12/2016	Nicholas et al.	N/A	N/A
9775611	12/2016	Kostrzewski	N/A	N/A
9775613	12/2016	Shelton, IV et al.	N/A	N/A
9775614	12/2016	Shelton, IV et al.	N/A	N/A
9775618	12/2016	Bettuchi et al.	N/A	N/A
9775635	12/2016	Takei	N/A	N/A
9775678	12/2016	Lohmeier	N/A	N/A
9782169	12/2016	Kimsey et al.	N/A	N/A
9782170	12/2016	Zemlok et al.	N/A	N/A
9782180	12/2016	Smith et al.	N/A	N/A
9782187	12/2016	Zergiebel et al.	N/A	N/A
9782193	12/2016	Thistle	N/A	N/A
9782214	12/2016	Houser et al.	N/A	N/A
9788834	12/2016	Schmid et al.	N/A	N/A
9788835	12/2016	Morgan et al.	N/A	N/A
9788836	12/2016	Overmyer et al.	N/A	N/A
9788847	12/2016	Jinno	N/A	N/A
9788851	12/2016	Dannaher et al.	N/A	N/A
9788902	12/2016	Inoue et al.	N/A	N/A
9795379	12/2016	Leimbach et al.	N/A	N/A
9795380	12/2016	Shelton, IV et al.	N/A	N/A

9795381	12/2016	Shelton, IV	N/A	N/A
9795382	12/2016	Shelton, IV	N/A	N/A
9795383	12/2016	Aldridge et al.	N/A	N/A
9795384	12/2016	Weaner et al.	N/A	N/A
9797486	12/2016	Zergiebel et al.	N/A	N/A
9801626	12/2016	Parihar et al.	N/A	N/A
9801627	12/2016	Harris et al.	N/A	N/A
9801628	12/2016	Harris et al.	N/A	N/A
9801634	12/2016	Shelton, IV et al.	N/A	N/A
9801679	12/2016	Trees et al.	N/A	N/A
9802033	12/2016	Hibner et al.	N/A	N/A
9804618	12/2016	Leimbach et al.	N/A	N/A
D803234	12/2016	Day et al.	N/A	N/A
D803235	12/2016	Markson et al.	N/A	N/A
D803850	12/2016	Chang et al.	N/A	N/A
9808244	12/2016	Leimbach et al.	N/A	N/A
9808246	12/2016	Shelton, IV	N/A	A61B 17/07207
9808247	12/2016	Shelton, IV et al.	N/A	N/A
9808248	12/2016	Hoffman	N/A	N/A
9808249	12/2016	Shelton, IV	N/A	N/A
9814460	12/2016	Kimsey et al.	N/A	N/A
9814462	12/2016	Woodard, Jr. et al.	N/A	N/A
9814463	12/2016	Williams et al.	N/A	N/A
9814530	12/2016	Weir et al.	N/A	N/A
9814561	12/2016	Forsell	N/A	N/A
9815118	12/2016	Schmitt et al.	N/A	N/A
9820445	12/2016	Simpson et al.	N/A	N/A
9820737	12/2016	Beardsley et al.	N/A	N/A
9820738	12/2016	Lytle, IV et al.	N/A	N/A
9820741	12/2016	Kostrzewski	N/A	N/A
9820768	12/2016	Gee et al.	N/A	N/A
9825455	12/2016	Sandhu et al.	N/A	N/A
9826976	12/2016	Parihar et al.	N/A	N/A
9826977	12/2016	Leimbach et al.	N/A	N/A
9826978	12/2016	Shelton, IV et al.	N/A	N/A
9829698	12/2016	Haraguchi et al.	N/A	N/A
D806108	12/2016	Day	N/A	N/A
9833235	12/2016	Penna et al.	N/A	N/A
9833236	12/2016	Shelton, IV et al.	N/A	N/A
9833238	12/2016	Baxter, III et al.	N/A	N/A
9833239	12/2016	Yates et al.	N/A	N/A
9833241	12/2016	Huitema et al.	N/A	N/A
9833242	12/2016	Baxter, III et al.	N/A	N/A
9839420	12/2016	Shelton, IV et al.	N/A	N/A
9839421	12/2016	Zerkle et al.	N/A	N/A
9839422	12/2016	Schellin et al.	N/A	N/A
9839423	12/2016	Vendely et al.	N/A	N/A
9839427	12/2016	Swayze et al.	N/A	N/A
9839428	12/2016	Baxter, III et al.	N/A	N/A

9839429	12/2016	Weisenburgh, II et al.	N/A	N/A
9839480	12/2016	Pribanic et al.	N/A	N/A
9839481	12/2016	Blumenkranz et al.	N/A	N/A
9844313	12/2016	DiCarlo et al.	N/A	N/A
9844368	12/2016	Boudreaux et al.	N/A	N/A
9844369	12/2016	Huitema et al.	N/A	N/A
9844372	12/2016	Shelton, IV et al.	N/A	N/A
9844373	12/2016	Swayze et al.	N/A	N/A
9844374	12/2016	Lytle, IV et al.	N/A	N/A
9844375	12/2016	Overmyer et al.	N/A	N/A
9844376	12/2016	Baxter, III et al.	N/A	N/A
9844379	12/2016	Shelton, IV et al.	N/A	N/A
9848871	12/2016	Harris et al.	N/A	N/A
9848873	12/2016	Shelton, IV	N/A	N/A
9848875	12/2016	Aronhalt et al.	N/A	N/A
9848877	12/2016	Shelton, IV et al.	N/A	N/A
9850499	12/2016	Baylink et al.	N/A	N/A
9850994	12/2016	Schena	N/A	N/A
D808989	12/2017	Ayvazian et al.	N/A	N/A
9855039	12/2017	Racenet et al.	N/A	N/A
9855040	12/2017	Kostrzewski	N/A	N/A
9855662	12/2017	Ruiz Morales et al.	N/A	N/A
9861261	12/2017	Shahinian	N/A	N/A
9861359	12/2017	Shelton, IV et al.	N/A	N/A
9861361	12/2017	Aronhalt et al.	N/A	N/A
9861362	12/2017	Whitman et al.	N/A	N/A
9861366	12/2017	Aranyi	N/A	N/A
9861382	12/2017	Smith et al.	N/A	N/A
9861446	12/2017	Lang	N/A	N/A
9867612	12/2017	Parihar et al.	N/A	N/A
9867613	12/2017	Marczyk et al.	N/A	N/A
9867615	12/2017	Fanelli et al.	N/A	N/A
9867617	12/2017	Ma	N/A	N/A
9867618	12/2017	Hall et al.	N/A	N/A
9867620	12/2017	Fischvogt et al.	N/A	N/A
9868198	12/2017	Nicholas et al.	N/A	N/A
9872682	12/2017	Hess et al.	N/A	N/A
9872683	12/2017	Hopkins et al.	N/A	N/A
9872684	12/2017	Hall et al.	N/A	N/A
9872722	12/2017	Lech	N/A	N/A
9877721	12/2017	Schellin et al.	N/A	N/A
9877722	12/2017	Schellin et al.	N/A	N/A
9877723	12/2017	Hall et al.	N/A	N/A
9877776	12/2017	Boudreaux	N/A	N/A
D810099	12/2017	Riedel	N/A	N/A
9883843	12/2017	Garlow	N/A	N/A
9883860	12/2017	Leimbach et al.	N/A	N/A
9883861	12/2017	Shelton, IV et al.	N/A	N/A
9884456	12/2017	Schellin et al.	N/A	N/A
9888914	12/2017	Martin et al.	N/A	N/A

9888919	12/2017	Leimbach et al.	N/A	N/A
9888921	12/2017	Williams et al.	N/A	N/A
9888924	12/2017	Ebersole et al.	N/A	N/A
9889230	12/2017	Bennett et al.	N/A	N/A
9895147	12/2017	Shelton, IV	N/A	N/A
9895148	12/2017	Shelton, IV et al.	N/A	N/A
9895813	12/2017	Blumenkranz et al.	N/A	N/A
9901339	12/2017	Farascioni	N/A	N/A
9901341	12/2017	Kostrzewski	N/A	N/A
9901342	12/2017	Shelton, IV et al.	N/A	N/A
9901344	12/2017	Moore et al.	N/A	N/A
9901345	12/2017	Moore et al.	N/A	N/A
9901346	12/2017	Moore et al.	N/A	N/A
9901358	12/2017	Faller et al.	N/A	N/A
9901406	12/2017	State et al.	N/A	N/A
9901412	12/2017	Lathrop et al.	N/A	N/A
D813899	12/2017	Erant et al.	N/A	N/A
9907456	12/2017	Miyoshi	N/A	N/A
9907552	12/2017	Measamer et al.	N/A	N/A
9907553	12/2017	Cole et al.	N/A	N/A
9907600	12/2017	Stulen et al.	N/A	N/A
9907620	12/2017	Shelton, IV et al.	N/A	N/A
9913641	12/2017	Takemoto et al.	N/A	N/A
9913642	12/2017	Leimbach et al.	N/A	N/A
9913644	12/2017	McCuen	N/A	N/A
9913646	12/2017	Shelton, IV	N/A	N/A
9913647	12/2017	Weisenburgh, II et al.	N/A	N/A
9913648	12/2017	Shelton, IV et al.	N/A	N/A
9913694	12/2017	Brisson	N/A	N/A
9913733	12/2017	Piron et al.	N/A	N/A
9918704	12/2017	Shelton, IV et al.	N/A	N/A
9918714	12/2017	Gibbons, Jr.	N/A	N/A
9918715	12/2017	Menn	N/A	N/A
9918716	12/2017	Baxter, III et al.	N/A	N/A
9918717	12/2017	Czernik	N/A	N/A
9918730	12/2017	Trees et al.	N/A	N/A
9924941	12/2017	Burbank	N/A	N/A
9924942	12/2017	Swayze et al.	N/A	N/A
9924943	12/2017	Mohan Pinjala et al.	N/A	N/A
9924944	12/2017	Shelton, IV et al.	N/A	N/A
9924945	12/2017	Zheng et al.	N/A	N/A
9924946	12/2017	Vendely et al.	N/A	N/A
9924947	12/2017	Shelton, IV et al.	N/A	N/A
9924961	12/2017	Shelton, IV et al.	N/A	N/A
9931106	12/2017	Au et al.	N/A	N/A
9931116	12/2017	Racenet et al.	N/A	N/A
9931117	12/2017	Hathaway et al.	N/A	N/A
9931118	12/2017	Shelton, IV et al.	N/A	N/A
9931120	12/2017	Chen et al.	N/A	N/A
9936949	12/2017	Measamer et al.	N/A	N/A

9936950	12/2017	Shelton, IV et al.	N/A	N/A
9936951	12/2017	Hufnagel et al.	N/A	N/A
9936952	12/2017	Demmy	N/A	N/A
9936954	12/2017	Shelton, IV et al.	N/A	N/A
9937626	12/2017	Rockrohr	N/A	N/A
9943309	12/2017	Shelton, IV et al.	N/A	N/A
9943310	12/2017	Harris et al.	N/A	N/A
9943312	12/2017	Posada et al.	N/A	N/A
9949754	12/2017	Newhauser et al.	N/A	N/A
9953193	12/2017	Butler et al.	N/A	N/A
D819072	12/2017	Clediere	N/A	N/A
9955954	12/2017	Destoumieux et al.	N/A	N/A
9955965	12/2017	Chen et al.	N/A	N/A
9955966	12/2017	Zergiebel	N/A	N/A
9956677	12/2017	Baskar et al.	N/A	N/A
9962129	12/2017	Jerebko et al.	N/A	N/A
9962157	12/2017	Sapre	N/A	N/A
9962158	12/2017	Hall et al.	N/A	N/A
9962159	12/2017	Heinrich et al.	N/A	N/A
9962161	12/2017	Scheib et al.	N/A	N/A
9968354	12/2017	Shelton, IV et al.	N/A	N/A
9968355	12/2017	Shelton, IV et al.	N/A	N/A
9968356	12/2017	Shelton, IV et al.	N/A	N/A
9968397	12/2017	Taylor et al.	N/A	N/A
9974529	12/2017	Shelton, IV et al.	N/A	N/A
9974538	12/2017	Baxter, III et al.	N/A	N/A
9974539	12/2017	Yates et al.	N/A	N/A
9974541	12/2017	Calderoni	N/A	N/A
9974542	12/2017	Hodgkinson	N/A	N/A
9980713	12/2017	Aronhalt et al.	N/A	N/A
9980724	12/2017	Farascioni et al.	N/A	N/A
9980729	12/2017	Moore et al.	N/A	N/A
9980740	12/2017	Krause et al.	N/A	N/A
9980769	12/2017	Trees et al.	N/A	N/A
D819680	12/2017	Nguyen	N/A	N/A
D819682	12/2017	Howard et al.	N/A	N/A
D819684	12/2017	Dart	N/A	N/A
D820307	12/2017	Jian et al.	N/A	N/A
D820867	12/2017	Dickens et al.	N/A	N/A
9987000	12/2017	Shelton, IV et al.	N/A	N/A
9987003	12/2017	Timm et al.	N/A	N/A
9987006	12/2017	Morgan et al.	N/A	N/A
9987008	12/2017	Scirica et al.	N/A	N/A
9987095	12/2017	Chowaniec et al.	N/A	N/A
9987097	12/2017	van der Weide et al.	N/A	N/A
9987099	12/2017	Chen et al.	N/A	N/A
9993248	12/2017	Shelton, IV et al.	N/A	N/A
9993258	12/2017	Shelton, IV et al.	N/A	N/A
9993284	12/2017	Boudreaux	N/A	N/A
9999408	12/2017	Boudreaux et al.	N/A	N/A

9999423	12/2017	Schuckmann et al.	N/A	N/A
9999426	12/2017	Moore et al.	N/A	N/A
9999431	12/2017	Shelton, IV et al.	N/A	N/A
9999472	12/2017	Weir et al.	N/A	N/A
10004497	12/2017	Overmyer et al.	N/A	N/A
10004498	12/2017	Morgan et al.	N/A	N/A
10004500	12/2017	Shelton, IV et al.	N/A	N/A
10004501	12/2017	Shelton, IV et al.	N/A	N/A
10004505	12/2017	Moore et al.	N/A	N/A
10004506	12/2017	Shelton, IV et al.	N/A	N/A
10004552	12/2017	Kleyman et al.	N/A	N/A
D822206	12/2017	Shelton, IV et al.	N/A	N/A
D823858	12/2017	Li et al.	N/A	N/A
10010322	12/2017	Shelton, IV et al.	N/A	N/A
10010324	12/2017	Huitema et al.	N/A	N/A
10010395	12/2017	Puckett et al.	N/A	N/A
10013049	12/2017	Leimbach et al.	N/A	N/A
10016199	12/2017	Baber et al.	N/A	N/A
10016656	12/2017	Devor et al.	N/A	N/A
10022120	12/2017	Martin et al.	N/A	N/A
10022123	12/2017	Williams et al.	N/A	N/A
10022125	12/2017	(Prommersberger) Stopek et al.	N/A	N/A
10024407	12/2017	Aranyi et al.	N/A	N/A
10028742	12/2017	Shelton, IV et al.	N/A	N/A
10028743	12/2017	Shelton, IV et al.	N/A	N/A
10028744	12/2017	Shelton, IV et al.	N/A	N/A
10028761	12/2017	Leimbach et al.	N/A	N/A
10029108	12/2017	Powers et al.	N/A	N/A
10029125	12/2017	Shapiro et al.	N/A	N/A
10034344	12/2017	Yoshida	N/A	N/A
10034668	12/2017	Ebner	N/A	N/A
D826405	12/2017	Shelton, IV et al.	N/A	N/A
10039440	12/2017	Fenech et al.	N/A	N/A
10039529	12/2017	Kerr et al.	N/A	N/A
10039532	12/2017	Srinivas et al.	N/A	N/A
10039545	12/2017	Sadowski et al.	N/A	N/A
10041822	12/2017	Zemlok	N/A	N/A
10045769	12/2017	Aronhalt et al.	N/A	N/A
10045776	12/2017	Shelton, IV et al.	N/A	N/A
10045778	12/2017	Yates et al.	N/A	N/A
10045779	12/2017	Savage et al.	N/A	N/A
10045781	12/2017	Cropper et al.	N/A	N/A
10045782	12/2017	Murthy Aravalli	N/A	N/A
10045869	12/2017	Forsell	N/A	N/A
10046904	12/2017	Evans et al.	N/A	N/A
10052044	12/2017	Shelton, IV et al.	N/A	N/A
10052099	12/2017	Morgan et al.	N/A	N/A
10052100	12/2017	Morgan et al.	N/A	N/A
10052102	12/2017	Baxter, III et al.	N/A	N/A

10052104	12/2017	Shelton, IV et al.	N/A	N/A
10052164	12/2017	Overmyer	N/A	N/A
10058317	12/2017	Fan et al.	N/A	N/A
10058327	12/2017	Weisenburgh, II et al.	N/A	N/A
10058373	12/2017	Takashino et al.	N/A	N/A
10058395	12/2017	Devengenzo et al.	N/A	N/A
10058963	12/2017	Shelton, IV et al.	N/A	N/A
10064620	12/2017	Gettinger et al.	N/A	N/A
10064621	12/2017	Kerr et al.	N/A	N/A
10064622	12/2017	Murthy Aravalli	N/A	N/A
10064624	12/2017	Shelton, IV et al.	N/A	N/A
10064639	12/2017	Ishida et al.	N/A	N/A
10064642	12/2017	Marczyk et al.	N/A	N/A
10064649	12/2017	Golebieski et al.	N/A	N/A
10064688	12/2017	Shelton, IV et al.	N/A	N/A
10070861	12/2017	Spivey et al.	N/A	N/A
10070863	12/2017	Swayze et al.	N/A	N/A
10071452	12/2017	Shelton, IV et al.	N/A	N/A
10076325	12/2017	Huang et al.	N/A	N/A
10076326	12/2017	Yates et al.	N/A	N/A
10076340	12/2017	Belagali et al.	N/A	N/A
10080552	12/2017	Nicholas et al.	N/A	N/A
D830550	12/2017	Miller et al.	N/A	N/A
D831209	12/2017	Huitema et al.	N/A	N/A
D831676	12/2017	Park et al.	N/A	N/A
D832301	12/2017	Smith	N/A	N/A
10085624	12/2017	Isoda et al.	N/A	N/A
10085643	12/2017	Bandic et al.	N/A	N/A
10085728	12/2017	Jogasaki et al.	N/A	N/A
10085746	12/2017	Fischvogt	N/A	N/A
10085748	12/2017	Morgan et al.	N/A	N/A
10085749	12/2017	Cappola et al.	N/A	N/A
10085750	12/2017	Zergiebel et al.	N/A	N/A
10085751	12/2017	Overmyer et al.	N/A	N/A
10085754	12/2017	Sniffin et al.	N/A	N/A
10085806	12/2017	Hagn et al.	N/A	N/A
10092290	12/2017	Yigit et al.	N/A	N/A
10092292	12/2017	Boudreaux et al.	N/A	N/A
10098635	12/2017	Burbank	N/A	N/A
10098636	12/2017	Shelton, IV et al.	N/A	N/A
10098640	12/2017	Bertolero et al.	N/A	N/A
10098642	12/2017	Baxter, III et al.	N/A	N/A
10099303	12/2017	Yoshida et al.	N/A	N/A
10101861	12/2017	Kiyoto	N/A	N/A
10105126	12/2017	Sauer	N/A	N/A
10105128	12/2017	Cooper et al.	N/A	N/A
10105136	12/2017	Yates et al.	N/A	N/A
10105139	12/2017	Yates et al.	N/A	N/A
10105140	12/2017	Malinouskas et al.	N/A	N/A
10105142	12/2017	Baxter, III et al.	N/A	N/A

10105149	12/2017	Haider et al.	N/A	N/A
10106932	12/2017	Anderson et al.	N/A	N/A
10111657	12/2017	McCuen	N/A	N/A
10111658	12/2017	Chowaniec et al.	N/A	N/A
10111660	12/2017	Hemmann	N/A	N/A
10111665	12/2017	Aranyi et al.	N/A	N/A
10111679	12/2017	Baber et al.	N/A	N/A
10111698	12/2017	Scheib et al.	N/A	N/A
10111702	12/2017	Kostrzewski	N/A	N/A
D833608	12/2017	Miller et al.	N/A	N/A
10117649	12/2017	Baxter, III et al.	N/A	N/A
10117650	12/2017	Nicholas et al.	N/A	N/A
10117652	12/2017	Schmid et al.	N/A	N/A
10117653	12/2017	Leimbach et al.	N/A	N/A
10117654	12/2017	Ingmanson et al.	N/A	N/A
10123798	12/2017	Baxter, III et al.	N/A	N/A
10123845	12/2017	Yeung	N/A	N/A
10124493	12/2017	Rothfuss et al.	N/A	N/A
10130352	12/2017	Widenhouse et al.	N/A	N/A
10130359	12/2017	Hess et al.	N/A	N/A
10130360	12/2017	Olson et al.	N/A	N/A
10130361	12/2017	Yates et al.	N/A	N/A
10130363	12/2017	Huitema et al.	N/A	N/A
10130366	12/2017	Shelton, IV et al.	N/A	N/A
10130367	12/2017	Cappola et al.	N/A	N/A
10130382	12/2017	Gladstone	N/A	N/A
10130738	12/2017	Shelton, IV et al.	N/A	N/A
10130830	12/2017	Miret Carceller et al.	N/A	N/A
10133248	12/2017	Fitzsimmons et al.	N/A	N/A
10135242	12/2017	Baber et al.	N/A	N/A
10136879	12/2017	Ross et al.	N/A	N/A
10136887	12/2017	Shelton, IV et al.	N/A	N/A
10136889	12/2017	Shelton, IV et al.	N/A	N/A
10136890	12/2017	Shelton, IV et al.	N/A	N/A
10136891	12/2017	Shelton, IV et al.	N/A	N/A
10136949	12/2017	Felder et al.	N/A	N/A
D835659	12/2017	Anzures et al.	N/A	N/A
D836124	12/2017	Fan	N/A	N/A
10143474	12/2017	Bucciaglia et al.	N/A	N/A
10146423	12/2017	Reed et al.	N/A	N/A
10149679	12/2017	Shelton, IV et al.	N/A	N/A
10149680	12/2017	Parihar et al.	N/A	N/A
10149682	12/2017	Shelton, IV et al.	N/A	N/A
10149683	12/2017	Smith et al.	N/A	N/A
10149712	12/2017	Manwaring et al.	N/A	N/A
10152789	12/2017	Carnes et al.	N/A	N/A
10154841	12/2017	Weaner et al.	N/A	N/A
10159481	12/2017	Whitman et al.	N/A	N/A
10159482	12/2017	Swayze et al.	N/A	N/A
10159483	12/2017	Beckman et al.	N/A	N/A

10159506	12/2017	Boudreaux et al.	N/A	N/A
10161816	12/2017	Jackson et al.	N/A	N/A
10163065	12/2017	Koski et al.	N/A	N/A
10163589	12/2017	Zergiebel et al.	N/A	N/A
10164466	12/2017	Calderoni	N/A	N/A
D837244	12/2018	Kuo et al.	N/A	N/A
D837245	12/2018	Kuo et al.	N/A	N/A
10166023	12/2018	Vendely et al.	N/A	N/A
10166025	12/2018	Leimbach et al.	N/A	N/A
10166026	12/2018	Shelton, IV et al.	N/A	N/A
10172611	12/2018	Shelton, IV et al.	N/A	N/A
10172615	12/2018	Marczyk et al.	N/A	N/A
10172616	12/2018	Murray et al.	N/A	N/A
10172617	12/2018	Shelton, IV et al.	N/A	N/A
10172618	12/2018	Shelton, IV et al.	N/A	N/A
10172619	12/2018	Harris et al.	N/A	N/A
10172620	12/2018	Harris et al.	N/A	N/A
10172636	12/2018	Stulen et al.	N/A	N/A
10172669	12/2018	Felder et al.	N/A	N/A
10175127	12/2018	Collins et al.	N/A	N/A
10178992	12/2018	Wise et al.	N/A	N/A
10180463	12/2018	Beckman et al.	N/A	N/A
10182813	12/2018	Leimbach et al.	N/A	N/A
10182815	12/2018	Williams et al.	N/A	N/A
10182816	12/2018	Shelton, IV et al.	N/A	N/A
10182818	12/2018	Hensel et al.	N/A	N/A
10182819	12/2018	Shelton, IV	N/A	N/A
10182868	12/2018	Meier et al.	N/A	N/A
10188385	12/2018	Kerr et al.	N/A	N/A
10188389	12/2018	Vendely et al.	N/A	N/A
10188393	12/2018	Smith et al.	N/A	N/A
10188394	12/2018	Shelton, IV et al.	N/A	N/A
10190888	12/2018	Hryb et al.	N/A	N/A
D839900	12/2018	Gan	N/A	N/A
D841667	12/2018	Coren	N/A	N/A
10194801	12/2018	Elhawary et al.	N/A	N/A
10194904	12/2018	Viola et al.	N/A	N/A
10194907	12/2018	Marczyk et al.	N/A	N/A
10194908	12/2018	Duque et al.	N/A	N/A
10194910	12/2018	Shelton, IV et al.	N/A	N/A
10194911	12/2018	Miller et al.	N/A	N/A
10194912	12/2018	Scheib et al.	N/A	N/A
10194913	12/2018	Nalagatla et al.	N/A	N/A
10194976	12/2018	Boudreaux	N/A	N/A
10194992	12/2018	Robinson	N/A	N/A
10201348	12/2018	Scheib et al.	N/A	N/A
10201349	12/2018	Leimbach et al.	N/A	N/A
10201363	12/2018	Shelton, IV	N/A	N/A
10201364	12/2018	Leimbach et al.	N/A	N/A
10201365	12/2018	Boudreaux et al.	N/A	N/A

10201381	12/2018	Zergiebel et al.	N/A	N/A
10206605	12/2018	Shelton, IV et al.	N/A	N/A
10206676	12/2018	Shelton, IV	N/A	N/A
10206677	12/2018	Harris et al.	N/A	N/A
10206678	12/2018	Shelton, IV et al.	N/A	N/A
10206748	12/2018	Burbank	N/A	N/A
10210244	12/2018	Branavan et al.	N/A	N/A
10211586	12/2018	Adams et al.	N/A	N/A
10213198	12/2018	Aronhalt et al.	N/A	N/A
10213201	12/2018	Shelton, IV et al.	N/A	N/A
10213202	12/2018	Flanagan et al.	N/A	N/A
10213203	12/2018	Swayze et al.	N/A	N/A
10213204	12/2018	Aranyi et al.	N/A	N/A
10213262	12/2018	Shelton, IV et al.	N/A	N/A
D842328	12/2018	Jian et al.	N/A	N/A
10219811	12/2018	Haider et al.	N/A	N/A
10219832	12/2018	Bagwell et al.	N/A	N/A
10220522	12/2018	Rockrohr	N/A	N/A
10226239	12/2018	Nicholas et al.	N/A	N/A
10226249	12/2018	Jaworek et al.	N/A	N/A
10226250	12/2018	Beckman et al.	N/A	N/A
10226251	12/2018	Scheib et al.	N/A	N/A
10226274	12/2018	Worrell et al.	N/A	N/A
10231634	12/2018	Zand et al.	N/A	N/A
10231653	12/2018	Bohm et al.	N/A	N/A
10231734	12/2018	Thompson et al.	N/A	N/A
10231794	12/2018	Shelton, IV et al.	N/A	N/A
10238385	12/2018	Yates et al.	N/A	N/A
10238386	12/2018	Overmyer et al.	N/A	N/A
10238387	12/2018	Yates et al.	N/A	N/A
10238389	12/2018	Yates et al.	N/A	N/A
10238390	12/2018	Harris et al.	N/A	N/A
10238391	12/2018	Leimbach et al.	N/A	N/A
D844666	12/2018	Espeleta et al.	N/A	N/A
D844667	12/2018	Espeleta et al.	N/A	N/A
D845342	12/2018	Espeleta et al.	N/A	N/A
D847199	12/2018	Whitmore	N/A	N/A
10244991	12/2018	Shademan et al.	N/A	N/A
10245027	12/2018	Shelton, IV et al.	N/A	N/A
10245028	12/2018	Shelton, IV et al.	N/A	N/A
10245029	12/2018	Hunter et al.	N/A	N/A
10245030	12/2018	Hunter et al.	N/A	N/A
10245032	12/2018	Shelton, IV	N/A	N/A
10245033	12/2018	Overmyer et al.	N/A	N/A
10245034	12/2018	Shelton, IV et al.	N/A	N/A
10245035	12/2018	Swayze et al.	N/A	N/A
10245038	12/2018	Hopkins et al.	N/A	N/A
10245058	12/2018	Omori et al.	N/A	N/A
10251645	12/2018	Kostrzewski	N/A	N/A
10251648	12/2018	Harris et al.	N/A	N/A

10251649	12/2018	Schellin et al.	N/A	N/A
10251725	12/2018	Valentine et al.	N/A	N/A
10258322	12/2018	Fanton et al.	N/A	N/A
10258330	12/2018	Shelton, IV et al.	N/A	N/A
10258331	12/2018	Shelton, IV et al.	N/A	N/A
10258332	12/2018	Schmid et al.	N/A	N/A
10258333	12/2018	Shelton, IV et al.	N/A	N/A
10258336	12/2018	Baxter, III et al.	N/A	N/A
10258363	12/2018	Worrell et al.	N/A	N/A
10258418	12/2018	Shelton, IV et al.	N/A	N/A
10264797	12/2018	Zhang et al.	N/A	N/A
10265065	12/2018	Shelton, IV et al.	N/A	N/A
10265067	12/2018	Yates et al.	N/A	N/A
10265068	12/2018	Harris et al.	N/A	N/A
10265072	12/2018	Shelton, IV et al.	N/A	N/A
10265073	12/2018	Scheib et al.	N/A	N/A
10265074	12/2018	Shelton, IV et al.	N/A	N/A
10265090	12/2018	Ingmanson et al.	N/A	N/A
10271840	12/2018	Sapre	N/A	N/A
10271844	12/2018	Valentine et al.	N/A	N/A
10271845	12/2018	Shelton, IV	N/A	N/A
10271846	12/2018	Shelton, IV et al.	N/A	N/A
10271847	12/2018	Racenet et al.	N/A	N/A
10271849	12/2018	Vendely et al.	N/A	N/A
10271851	12/2018	Shelton, IV et al.	N/A	N/A
D847989	12/2018	Shelton, IV et al.	N/A	N/A
D848473	12/2018	Zhu et al.	N/A	N/A
D849046	12/2018	Kuo et al.	N/A	N/A
10278696	12/2018	Gurumurthy et al.	N/A	N/A
10278697	12/2018	Shelton, IV et al.	N/A	N/A
10278702	12/2018	Shelton, IV et al.	N/A	N/A
10278703	12/2018	Nativ et al.	N/A	N/A
10278707	12/2018	Thompson et al.	N/A	N/A
10278722	12/2018	Shelton, IV et al.	N/A	N/A
10278780	12/2018	Shelton, IV	N/A	N/A
10285694	12/2018	Viola et al.	N/A	N/A
10285695	12/2018	Jaworek et al.	N/A	N/A
10285699	12/2018	Vendely et al.	N/A	N/A
10285700	12/2018	Scheib	N/A	N/A
10285705	12/2018	Shelton, IV et al.	N/A	N/A
10285724	12/2018	Faller et al.	N/A	N/A
10285750	12/2018	Coulson et al.	N/A	N/A
10292701	12/2018	Scheib et al.	N/A	N/A
10292704	12/2018	Harris et al.	N/A	N/A
10292707	12/2018	Shelton, IV et al.	N/A	N/A
10293100	12/2018	Shelton, IV et al.	N/A	N/A
10293553	12/2018	Racenet et al.	N/A	N/A
10299787	12/2018	Shelton, IV	N/A	N/A
10299788	12/2018	Heinrich et al.	N/A	N/A
10299789	12/2018	Marczyk et al.	N/A	N/A

10299790	12/2018	Beardsley	N/A	N/A
10299792	12/2018	Huitema et al.	N/A	N/A
10299817	12/2018	Shelton, IV et al.	N/A	N/A
10299818	12/2018	Riva	N/A	N/A
10299878	12/2018	Shelton, IV et al.	N/A	N/A
10303851	12/2018	Nguyen et al.	N/A	N/A
D850617	12/2018	Shelton, IV et al.	N/A	N/A
D851676	12/2018	Foss et al.	N/A	N/A
D851762	12/2018	Shelton, IV et al.	N/A	N/A
10307159	12/2018	Harris et al.	N/A	N/A
10307160	12/2018	Vendely et al.	N/A	N/A
10307161	12/2018	Jankowski	N/A	N/A
10307163	12/2018	Moore et al.	N/A	N/A
10307170	12/2018	Parfett et al.	N/A	N/A
10307202	12/2018	Smith et al.	N/A	N/A
10314559	12/2018	Razzaque et al.	N/A	N/A
10314577	12/2018	Laurent et al.	N/A	N/A
10314578	12/2018	Leimbach et al.	N/A	N/A
10314579	12/2018	Chowaniec et al.	N/A	N/A
10314580	12/2018	Scheib et al.	N/A	N/A
10314582	12/2018	Shelton, IV et al.	N/A	N/A
10314584	12/2018	Scirica et al.	N/A	N/A
10314587	12/2018	Harris et al.	N/A	N/A
10314588	12/2018	Turner et al.	N/A	N/A
10314589	12/2018	Shelton, IV et al.	N/A	N/A
10314590	12/2018	Shelton, IV et al.	N/A	N/A
10315566	12/2018	Choi et al.	N/A	N/A
10321907	12/2018	Shelton, IV et al.	N/A	N/A
10321909	12/2018	Shelton, IV et al.	N/A	N/A
10321927	12/2018	Hinman	N/A	N/A
10327743	12/2018	St. Goar et al.	N/A	N/A
10327764	12/2018	Harris et al.	N/A	N/A
10327765	12/2018	Timm et al.	N/A	N/A
10327767	12/2018	Shelton, IV et al.	N/A	N/A
10327769	12/2018	Overmyer et al.	N/A	N/A
10327776	12/2018	Harris et al.	N/A	N/A
10327777	12/2018	Harris et al.	N/A	N/A
D854032	12/2018	Jones et al.	N/A	N/A
D854151	12/2018	Shelton, IV et al.	N/A	N/A
10335144	12/2018	Shelton, IV et al.	N/A	N/A
10335145	12/2018	Harris et al.	N/A	N/A
10335147	12/2018	Rector et al.	N/A	N/A
10335148	12/2018	Shelton, IV et al.	N/A	N/A
10335149	12/2018	Baxter, III et al.	N/A	N/A
10335150	12/2018	Shelton, IV	N/A	N/A
10335151	12/2018	Shelton, IV et al.	N/A	N/A
10337148	12/2018	Rouse et al.	N/A	N/A
10342533	12/2018	Shelton, IV et al.	N/A	N/A
10342535	12/2018	Scheib et al.	N/A	N/A
10342541	12/2018	Shelton, IV et al.	N/A	N/A

10342543	12/2018	Shelton, IV et al.	N/A	N/A
10342623	12/2018	Huelman et al.	N/A	N/A
10349937	12/2018	Williams	N/A	N/A
10349939	12/2018	Shelton, IV et al.	N/A	N/A
10349941	12/2018	Marczyk et al.	N/A	N/A
10349963	12/2018	Fiksen et al.	N/A	N/A
10350016	12/2018	Burbank et al.	N/A	N/A
10357246	12/2018	Shelton, IV et al.	N/A	N/A
10357247	12/2018	Shelton, IV et al.	N/A	N/A
10357248	12/2018	Dallessandro et al.	N/A	N/A
10357252	12/2018	Harris et al.	N/A	N/A
10363031	12/2018	Alexander, III et al.	N/A	N/A
10363033	12/2018	Timm et al.	N/A	N/A
10363036	12/2018	Yates et al.	N/A	N/A
10363037	12/2018	Aronhalt et al.	N/A	N/A
D855634	12/2018	Kim	N/A	N/A
D856359	12/2018	Huang et al.	N/A	N/A
10368838	12/2018	Williams et al.	N/A	N/A
10368861	12/2018	Baxter, III et al.	N/A	N/A
10368863	12/2018	Timm et al.	N/A	N/A
10368864	12/2018	Harris et al.	N/A	N/A
10368865	12/2018	Harris et al.	N/A	N/A
10368866	12/2018	Wang et al.	N/A	N/A
10368867	12/2018	Harris et al.	N/A	N/A
10368892	12/2018	Stulen et al.	N/A	N/A
10374544	12/2018	Yokoyama et al.	N/A	N/A
10376263	12/2018	Morgan et al.	N/A	N/A
10383626	12/2018	Soltz	N/A	N/A
10383628	12/2018	Kang et al.	N/A	N/A
10383629	12/2018	Ross et al.	N/A	N/A
10383630	12/2018	Shelton, IV et al.	N/A	N/A
10383631	12/2018	Collings et al.	N/A	N/A
10383633	12/2018	Shelton, IV et al.	N/A	N/A
10383634	12/2018	Shelton, IV et al.	N/A	N/A
10390823	12/2018	Shelton, IV et al.	N/A	N/A
10390825	12/2018	Shelton, IV et al.	N/A	N/A
10390828	12/2018	Vendely et al.	N/A	N/A
10390829	12/2018	Eckert et al.	N/A	N/A
10390830	12/2018	Schulz	N/A	N/A
10390841	12/2018	Shelton, IV et al.	N/A	N/A
10390897	12/2018	Kostrzewski	N/A	N/A
D859466	12/2018	Okada et al.	N/A	N/A
D860219	12/2018	Rasmussen et al.	N/A	N/A
D861035	12/2018	Park et al.	N/A	N/A
10398433	12/2018	Boudreaux et al.	N/A	N/A
10398434	12/2018	Shelton, IV et al.	N/A	N/A
10398436	12/2018	Shelton, IV et al.	N/A	N/A
10398460	12/2018	Overmyer	N/A	N/A
10404136	12/2018	Oktavec et al.	N/A	N/A
10405854	12/2018	Schmid et al.	N/A	N/A

10405857	12/2018	Shelton, IV et al.	N/A	N/A
10405859	12/2018	Harris et al.	N/A	N/A
10405863	12/2018	Wise et al.	N/A	N/A
10405914	12/2018	Manwaring et al.	N/A	N/A
10405932	12/2018	Overmyer	N/A	N/A
10405937	12/2018	Black et al.	N/A	N/A
10413155	12/2018	Inoue	N/A	N/A
10413291	12/2018	Worthington et al.	N/A	N/A
10413293	12/2018	Shelton, IV et al.	N/A	N/A
10413294	12/2018	Shelton, IV et al.	N/A	N/A
10413297	12/2018	Harris et al.	N/A	N/A
10413370	12/2018	Yates et al.	N/A	N/A
10413373	12/2018	Yates et al.	N/A	N/A
10420548	12/2018	Whitman et al.	N/A	N/A
10420549	12/2018	Yates et al.	N/A	N/A
10420550	12/2018	Shelton, IV	N/A	N/A
10420551	12/2018	Calderoni	N/A	N/A
10420552	12/2018	Shelton, IV et al.	N/A	N/A
10420553	12/2018	Shelton, IV et al.	N/A	N/A
10420554	12/2018	Collings et al.	N/A	N/A
10420555	12/2018	Shelton, IV et al.	N/A	N/A
10420558	12/2018	Nalagatla et al.	N/A	N/A
10420559	12/2018	Marczyk et al.	N/A	N/A
10420560	12/2018	Shelton, IV et al.	N/A	N/A
10420561	12/2018	Shelton, IV et al.	N/A	N/A
10420577	12/2018	Chowaniec et al.	N/A	N/A
D861707	12/2018	Yang	N/A	N/A
D862518	12/2018	Niven et al.	N/A	N/A
D863343	12/2018	Mazlish et al.	N/A	N/A
D864388	12/2018	Barber	N/A	N/A
D865174	12/2018	Auld et al.	N/A	N/A
D865175	12/2018	Widenhouse et al.	N/A	N/A
10426463	12/2018	Shelton, IV et al.	N/A	N/A
10426466	12/2018	Contini et al.	N/A	N/A
10426467	12/2018	Miller	N/A	A61B 5/0053
10426468	12/2018	Contini et al.	N/A	N/A
10426469	12/2018	Shelton, IV et al.	N/A	N/A
10426471	12/2018	Shelton, IV et al.	N/A	N/A
10426476	12/2018	Harris et al.	N/A	N/A
10426477	12/2018	Harris et al.	N/A	N/A
10426478	12/2018	Shelton, IV et al.	N/A	N/A
10426481	12/2018	Aronhalt et al.	N/A	N/A
10426555	12/2018	Crowley et al.	N/A	N/A
10433837	12/2018	Worthington et al.	N/A	N/A
10433839	12/2018	Scheib et al.	N/A	N/A
10433840	12/2018	Shelton, IV et al.	N/A	N/A
10433842	12/2018	Amariglio et al.	N/A	N/A
10433844	12/2018	Shelton, IV et al.	N/A	N/A
10433845	12/2018	Baxter, III et al.	N/A	N/A

10433846	12/2018	Vendely et al.	N/A	N/A
10433849	12/2018	Shelton, IV et al.	N/A	N/A
10433918	12/2018	Shelton, IV et al.	N/A	N/A
10441279	12/2018	Shelton, IV et al.	N/A	N/A
10441280	12/2018	Timm et al.	N/A	N/A
10441281	12/2018	Shelton, IV et al.	N/A	N/A
10441285	12/2018	Shelton, IV et al.	N/A	N/A
10441286	12/2018	Shelton, IV et al.	N/A	N/A
10441345	12/2018	Aldridge et al.	N/A	N/A
10441369	12/2018	Shelton, IV et al.	N/A	N/A
10448948	12/2018	Shelton, IV et al.	N/A	N/A
10448950	12/2018	Shelton, IV et al.	N/A	N/A
10448952	12/2018	Shelton, IV et al.	N/A	N/A
10456122	12/2018	Koltz et al.	N/A	N/A
10456132	12/2018	Gettinger et al.	N/A	N/A
10456133	12/2018	Yates et al.	N/A	N/A
10456137	12/2018	Vendely et al.	N/A	N/A
10456140	12/2018	Shelton, IV et al.	N/A	N/A
D865796	12/2018	Xu et al.	N/A	N/A
10463367	12/2018	Kostrzewski et al.	N/A	N/A
10463369	12/2018	Shelton, IV et al.	N/A	N/A
10463370	12/2018	Yates et al.	N/A	N/A
10463371	12/2018	Kostrzewski	N/A	N/A
10463372	12/2018	Shelton, IV et al.	N/A	N/A
10463373	12/2018	Mozdzierz et al.	N/A	N/A
10463382	12/2018	Ingmanson et al.	N/A	N/A
10463383	12/2018	Shelton, IV et al.	N/A	N/A
10463384	12/2018	Shelton, IV et al.	N/A	N/A
10470762	12/2018	Leimbach et al.	N/A	N/A
10470763	12/2018	Yates et al.	N/A	N/A
10470764	12/2018	Baxter, III et al.	N/A	N/A
10470767	12/2018	Gleiman et al.	N/A	N/A
10470768	12/2018	Harris et al.	N/A	N/A
10470769	12/2018	Shelton, IV et al.	N/A	N/A
10471282	12/2018	Kirk et al.	N/A	N/A
10471576	12/2018	Totsu	N/A	N/A
10471607	12/2018	Butt et al.	N/A	N/A
10478181	12/2018	Shelton, IV et al.	N/A	N/A
10478182	12/2018	Taylor	N/A	N/A
10478185	12/2018	Nicholas	N/A	N/A
10478187	12/2018	Shelton, IV et al.	N/A	N/A
10478188	12/2018	Harris et al.	N/A	N/A
10478189	12/2018	Bear et al.	N/A	N/A
10478190	12/2018	Miller et al.	N/A	N/A
10478207	12/2018	Lathrop	N/A	N/A
10482292	12/2018	Clouser et al.	N/A	N/A
10485536	12/2018	Ming et al.	N/A	N/A
10485537	12/2018	Yates et al.	N/A	N/A
10485539	12/2018	Shelton, IV et al.	N/A	N/A
10485541	12/2018	Shelton, IV et al.	N/A	N/A

10485542	12/2018	Shelton, IV et al.	N/A	N/A
10485543	12/2018	Shelton, IV et al.	N/A	N/A
10485546	12/2018	Shelton, IV et al.	N/A	N/A
10485547	12/2018	Shelton, IV et al.	N/A	N/A
D869655	12/2018	Shelton, IV et al.	N/A	N/A
D870742	12/2018	Cornell	N/A	N/A
10492783	12/2018	Shelton, IV et al.	N/A	N/A
10492785	12/2018	Overmyer et al.	N/A	N/A
10492787	12/2018	Smith et al.	N/A	N/A
10492814	12/2018	Snow et al.	N/A	N/A
10492847	12/2018	Godara et al.	N/A	N/A
10492851	12/2018	Hughett, Sr. et al.	N/A	N/A
10498269	12/2018	Zemlok et al.	N/A	N/A
10499890	12/2018	Shelton, IV et al.	N/A	N/A
10499914	12/2018	Huang et al.	N/A	N/A
10499917	12/2018	Scheib et al.	N/A	N/A
10499918	12/2018	Schellin et al.	N/A	N/A
10500000	12/2018	Swayze et al.	N/A	N/A
10500004	12/2018	Hanuschik et al.	N/A	N/A
10500309	12/2018	Shah et al.	N/A	N/A
10507034	12/2018	Timm	N/A	N/A
10508720	12/2018	Nicholas	N/A	N/A
10512461	12/2018	Gupta et al.	N/A	N/A
10512462	12/2018	Felder et al.	N/A	N/A
10512464	12/2018	Park et al.	N/A	N/A
10517590	12/2018	Giordano et al.	N/A	N/A
10517592	12/2018	Shelton, IV et al.	N/A	N/A
10517594	12/2018	Shelton, IV et al.	N/A	N/A
10517595	12/2018	Hunter et al.	N/A	N/A
10517596	12/2018	Hunter et al.	N/A	N/A
10517599	12/2018	Baxter, III et al.	N/A	N/A
10517682	12/2018	Giordano et al.	N/A	N/A
10524784	12/2019	Kostrzewski	N/A	N/A
10524787	12/2019	Shelton, IV et al.	N/A	N/A
10524788	12/2019	Vendely et al.	N/A	N/A
10524789	12/2019	Swayze et al.	N/A	N/A
10524790	12/2019	Shelton, IV et al.	N/A	N/A
10524795	12/2019	Nalagatla et al.	N/A	N/A
10524870	12/2019	Saraliev et al.	N/A	N/A
10531874	12/2019	Morgan et al.	N/A	N/A
10531887	12/2019	Shelton, IV et al.	N/A	N/A
10537324	12/2019	Shelton, IV et al.	N/A	N/A
10537325	12/2019	Bakos et al.	N/A	N/A
10537351	12/2019	Shelton, IV et al.	N/A	N/A
10542908	12/2019	Mei et al.	N/A	N/A
10542974	12/2019	Yates et al.	N/A	N/A
10542976	12/2019	Calderoni et al.	N/A	N/A
10542978	12/2019	Chowaniec et al.	N/A	N/A
10542979	12/2019	Shelton, IV et al.	N/A	N/A
10542982	12/2019	Beckman et al.	N/A	N/A

10542985	12/2019	Zhan et al.	N/A	N/A
10542988	12/2019	Schellin et al.	N/A	N/A
10542991	12/2019	Shelton, IV et al.	N/A	N/A
D876478	12/2019	Sakata et al.	N/A	N/A
10548504	12/2019	Shelton, IV et al.	N/A	N/A
10548593	12/2019	Shelton, IV et al.	N/A	N/A
10548600	12/2019	Shelton, IV et al.	N/A	N/A
10548673	12/2019	Harris et al.	N/A	N/A
10561412	12/2019	Bookbinder et al.	N/A	N/A
10561418	12/2019	Richard et al.	N/A	N/A
10561419	12/2019	Beardsley	N/A	N/A
10561420	12/2019	Harris et al.	N/A	N/A
10561422	12/2019	Schellin et al.	N/A	N/A
10561432	12/2019	Estrella et al.	N/A	N/A
10561474	12/2019	Adams et al.	N/A	N/A
10562160	12/2019	Iwata et al.	N/A	N/A
10568493	12/2019	Blase et al.	N/A	N/A
10568621	12/2019	Shelton, IV et al.	N/A	N/A
10568624	12/2019	Shelton, IV et al.	N/A	N/A
10568625	12/2019	Harris et al.	N/A	N/A
10568626	12/2019	Shelton, IV et al.	N/A	N/A
10568629	12/2019	Shelton, IV et al.	N/A	N/A
10568632	12/2019	Miller et al.	N/A	N/A
10568652	12/2019	Hess et al.	N/A	N/A
10569071	12/2019	Harris et al.	N/A	N/A
D879808	12/2019	Harris et al.	N/A	N/A
D879809	12/2019	Harris et al.	N/A	N/A
10575868	12/2019	Hall et al.	N/A	N/A
10580320	12/2019	Kamiguchi et al.	N/A	N/A
10582928	12/2019	Hunter et al.	N/A	N/A
10588231	12/2019	Sgroi, Jr. et al.	N/A	N/A
10588623	12/2019	Schmid et al.	N/A	N/A
10588625	12/2019	Weaner et al.	N/A	N/A
10588626	12/2019	Overmyer et al.	N/A	N/A
10588629	12/2019	Malinouskas et al.	N/A	N/A
10588630	12/2019	Shelton, IV et al.	N/A	N/A
10588631	12/2019	Shelton, IV et al.	N/A	N/A
10588632	12/2019	Shelton, IV et al.	N/A	N/A
10588633	12/2019	Shelton, IV et al.	N/A	N/A
10589410	12/2019	Aho	N/A	N/A
10595835	12/2019	Kerr et al.	N/A	N/A
10595862	12/2019	Shelton, IV et al.	N/A	N/A
10595882	12/2019	Parfett et al.	N/A	N/A
10595887	12/2019	Shelton, IV et al.	N/A	N/A
10595929	12/2019	Boudreaux et al.	N/A	N/A
10603036	12/2019	Hunter et al.	N/A	N/A
10603039	12/2019	Vendely et al.	N/A	N/A
10603041	12/2019	Miller et al.	N/A	N/A
10603117	12/2019	Schings et al.	N/A	N/A
10603128	12/2019	Zergiebel et al.	N/A	N/A

D882783	12/2019	Shelton, IV et al.	N/A	N/A
10610224	12/2019	Shelton, IV et al.	N/A	N/A
10610225	12/2019	Reed et al.	N/A	N/A
10610236	12/2019	Baril	N/A	N/A
10610313	12/2019	Bailey et al.	N/A	N/A
10610346	12/2019	Schwartz	N/A	N/A
10614184	12/2019	Solki	N/A	N/A
10617411	12/2019	Williams	N/A	N/A
10617412	12/2019	Shelton, IV et al.	N/A	N/A
10617413	12/2019	Shelton, IV et al.	N/A	N/A
10617414	12/2019	Shelton, IV et al.	N/A	N/A
10617416	12/2019	Leimbach et al.	N/A	N/A
10617417	12/2019	Baxter, III et al.	N/A	N/A
10617418	12/2019	Barton et al.	N/A	N/A
10617420	12/2019	Shelton, IV et al.	N/A	N/A
10617438	12/2019	O'Keefe et al.	N/A	N/A
10624616	12/2019	Mukherjee et al.	N/A	N/A
10624630	12/2019	Deville et al.	N/A	N/A
10624633	12/2019	Shelton, IV et al.	N/A	N/A
10624634	12/2019	Shelton, IV et al.	N/A	N/A
10624635	12/2019	Harris et al.	N/A	N/A
10624709	12/2019	Remm	N/A	N/A
10624861	12/2019	Widenhouse et al.	N/A	N/A
10625062	12/2019	Matlock et al.	N/A	N/A
10631857	12/2019	Kostrzewski	N/A	N/A
10631858	12/2019	Burbank	N/A	N/A
10631859	12/2019	Shelton, IV et al.	N/A	N/A
10631860	12/2019	Bakos et al.	N/A	N/A
10636104	12/2019	Mazar et al.	N/A	N/A
10639018	12/2019	Shelton, IV et al.	N/A	N/A
10639034	12/2019	Harris et al.	N/A	N/A
10639035	12/2019	Shelton, IV et al.	N/A	N/A
10639036	12/2019	Yates et al.	N/A	N/A
10639037	12/2019	Shelton, IV et al.	N/A	N/A
10639089	12/2019	Manwaring et al.	N/A	N/A
10639115	12/2019	Shelton, IV et al.	N/A	N/A
10642633	12/2019	Chopra et al.	N/A	N/A
10645905	12/2019	Gandola et al.	N/A	N/A
10646220	12/2019	Shelton, IV et al.	N/A	N/A
10646292	12/2019	Solomon et al.	N/A	N/A
10653413	12/2019	Worthington et al.	N/A	N/A
10653417	12/2019	Shelton, IV et al.	N/A	N/A
10653435	12/2019	Shelton, IV et al.	N/A	N/A
10660640	12/2019	Yates et al.	N/A	N/A
10667408	12/2019	Sgroi, Jr. et al.	N/A	N/A
D888953	12/2019	Baxter, III et al.	N/A	N/A
10667808	12/2019	Baxter, III et al.	N/A	N/A
10667809	12/2019	Bakos et al.	N/A	N/A
10667810	12/2019	Shelton, IV et al.	N/A	N/A
10667811	12/2019	Harris et al.	N/A	N/A

10667818	12/2019	McLain et al.	N/A	N/A
10674895	12/2019	Yeung et al.	N/A	N/A
10675021	12/2019	Harris et al.	N/A	N/A
10675024	12/2019	Shelton, IV et al.	N/A	N/A
10675025	12/2019	Swayze et al.	N/A	N/A
10675026	12/2019	Harris et al.	N/A	N/A
10675028	12/2019	Shelton, IV et al.	N/A	N/A
10675035	12/2019	Zingman	N/A	N/A
10675080	12/2019	Woloszko et al.	N/A	N/A
10675102	12/2019	Forgione et al.	N/A	N/A
10677035	12/2019	Balan et al.	N/A	N/A
10682134	12/2019	Shelton, IV et al.	N/A	N/A
10682136	12/2019	Harris et al.	N/A	N/A
10682137	12/2019	Stokes et al.	N/A	N/A
10682138	12/2019	Shelton, IV et al.	N/A	N/A
10682141	12/2019	Moore et al.	N/A	N/A
10682142	12/2019	Shelton, IV et al.	N/A	N/A
10687806	12/2019	Shelton, IV et al.	N/A	N/A
10687809	12/2019	Shelton, IV et al.	N/A	N/A
10687810	12/2019	Shelton, IV et al.	N/A	N/A
10687812	12/2019	Shelton, IV et al.	N/A	N/A
10687813	12/2019	Shelton, IV et al.	N/A	N/A
10687817	12/2019	Shelton, IV et al.	N/A	N/A
10687819	12/2019	Stokes et al.	N/A	N/A
10687904	12/2019	Harris et al.	N/A	N/A
10695053	12/2019	Hess et al.	N/A	N/A
10695055	12/2019	Shelton, IV et al.	N/A	N/A
10695057	12/2019	Shelton, IV et al.	N/A	N/A
10695058	12/2019	Lytle, IV et al.	N/A	N/A
10695062	12/2019	Leimbach et al.	N/A	N/A
10695063	12/2019	Morgan et al.	N/A	N/A
10695074	12/2019	Carusillo	N/A	N/A
10695081	12/2019	Shelton, IV et al.	N/A	N/A
10695119	12/2019	Smith	N/A	N/A
10695123	12/2019	Allen, IV	N/A	N/A
10695187	12/2019	Moskowitz et al.	N/A	N/A
D890784	12/2019	Shelton, IV et al.	N/A	N/A
D890805	12/2019	Echeverri et al.	N/A	N/A
10702266	12/2019	Parihar et al.	N/A	N/A
10702267	12/2019	Hess et al.	N/A	N/A
10702270	12/2019	Shelton, IV et al.	N/A	N/A
10702271	12/2019	Aranyi et al.	N/A	N/A
10705660	12/2019	Xiao	N/A	N/A
10709446	12/2019	Harris et al.	N/A	N/A
10709468	12/2019	Shelton, IV et al.	N/A	N/A
10709469	12/2019	Shelton, IV et al.	N/A	N/A
10709495	12/2019	Broderick et al.	N/A	N/A
10709496	12/2019	Moua et al.	N/A	N/A
10716563	12/2019	Shelton, IV et al.	N/A	N/A
10716565	12/2019	Shelton, IV et al.	N/A	N/A

10716568	12/2019	Hall et al.	N/A	N/A
10716614	12/2019	Yates et al.	N/A	N/A
10717179	12/2019	Koenig et al.	N/A	N/A
10722232	12/2019	Yates et al.	N/A	N/A
10722233	12/2019	Wellman	N/A	N/A
10722292	12/2019	Arya et al.	N/A	N/A
10722293	12/2019	Arya et al.	N/A	N/A
10722317	12/2019	Ward et al.	N/A	N/A
D893717	12/2019	Messerly et al.	N/A	N/A
10729432	12/2019	Shelton, IV et al.	N/A	N/A
10729434	12/2019	Harris et al.	N/A	N/A
10729435	12/2019	Richard	N/A	N/A
10729436	12/2019	Shelton, IV et al.	N/A	N/A
10729443	12/2019	Cabrera et al.	N/A	N/A
10729458	12/2019	Stoddard et al.	N/A	N/A
10729501	12/2019	Leimbach et al.	N/A	N/A
10729509	12/2019	Shelton, IV et al.	N/A	N/A
10736616	12/2019	Scheib et al.	N/A	N/A
10736628	12/2019	Yates et al.	N/A	N/A
10736629	12/2019	Shelton, IV et al.	N/A	N/A
10736630	12/2019	Huang et al.	N/A	N/A
10736633	12/2019	Vendely et al.	N/A	N/A
10736634	12/2019	Shelton, IV et al.	N/A	N/A
10736636	12/2019	Baxter, III et al.	N/A	N/A
10736644	12/2019	Windolf et al.	N/A	N/A
10736702	12/2019	Harris et al.	N/A	N/A
10737398	12/2019	Remirez et al.	N/A	N/A
10743849	12/2019	Shelton, IV et al.	N/A	N/A
10743850	12/2019	Hibner et al.	N/A	N/A
10743851	12/2019	Swayze et al.	N/A	N/A
10743868	12/2019	Shelton, IV et al.	N/A	N/A
10743870	12/2019	Hall et al.	N/A	N/A
10743872	12/2019	Leimbach et al.	N/A	N/A
10743873	12/2019	Overmyer et al.	N/A	N/A
10743874	12/2019	Shelton, IV et al.	N/A	N/A
10743875	12/2019	Shelton, IV et al.	N/A	N/A
10743877	12/2019	Shelton, IV et al.	N/A	N/A
10743930	12/2019	Nagtegaal	N/A	N/A
10751048	12/2019	Whitman et al.	N/A	N/A
10751051	12/2019	Weir et al.	N/A	N/A
10751053	12/2019	Harris et al.	N/A	N/A
10751076	12/2019	Laurent et al.	N/A	N/A
10751138	12/2019	Giordano et al.	N/A	N/A
10758226	12/2019	Weir et al.	N/A	N/A
10758229	12/2019	Shelton, IV et al.	N/A	N/A
10758230	12/2019	Shelton, IV et al.	N/A	N/A
10758232	12/2019	Shelton, IV et al.	N/A	N/A
10758233	12/2019	Scheib et al.	N/A	N/A
10758259	12/2019	Demmy et al.	N/A	N/A
10765425	12/2019	Yates et al.	N/A	N/A

10765427	12/2019	Shelton, IV et al.	N/A	N/A
10765429	12/2019	Leimbach et al.	N/A	N/A
10765430	12/2019	Wixey	N/A	N/A
10765432	12/2019	Moore et al.	N/A	N/A
10765442	12/2019	Strobl	N/A	N/A
10772625	12/2019	Shelton, IV et al.	N/A	N/A
10772628	12/2019	Chen et al.	N/A	N/A
10772629	12/2019	Shelton, IV et al.	N/A	N/A
10772630	12/2019	Wixey	N/A	N/A
10772631	12/2019	Zergiebel et al.	N/A	N/A
10772632	12/2019	Kostrzewski	N/A	N/A
10772651	12/2019	Shelton, IV et al.	N/A	N/A
10779818	12/2019	Zemlok et al.	N/A	N/A
10779820	12/2019	Harris et al.	N/A	N/A
10779821	12/2019	Harris et al.	N/A	N/A
10779822	12/2019	Yates et al.	N/A	N/A
10779823	12/2019	Shelton, IV et al.	N/A	N/A
10779824	12/2019	Shelton, IV et al.	N/A	N/A
10779825	12/2019	Shelton, IV et al.	N/A	N/A
10779826	12/2019	Shelton, IV et al.	N/A	N/A
10779903	12/2019	Wise et al.	N/A	N/A
10780539	12/2019	Shelton, IV et al.	N/A	N/A
10786248	12/2019	Rousseau et al.	N/A	N/A
10786253	12/2019	Shelton, IV et al.	N/A	N/A
10786255	12/2019	Hodgkinson et al.	N/A	N/A
D898767	12/2019	Shah et al.	N/A	N/A
D899455	12/2019	Rondoni et al.	N/A	N/A
10792038	12/2019	Becerra et al.	N/A	N/A
10796471	12/2019	Leimbach et al.	N/A	N/A
10799240	12/2019	Shelton, IV et al.	N/A	N/A
10799306	12/2019	Robinson et al.	N/A	N/A
10806448	12/2019	Shelton, IV et al.	N/A	N/A
10806449	12/2019	Shelton, IV et al.	N/A	N/A
10806450	12/2019	Yates et al.	N/A	N/A
10806451	12/2019	Harris et al.	N/A	N/A
10806453	12/2019	Chen et al.	N/A	N/A
10806479	12/2019	Shelton, IV et al.	N/A	N/A
10813638	12/2019	Shelton, IV et al.	N/A	N/A
10813639	12/2019	Shelton, IV et al.	N/A	N/A
10813640	12/2019	Adams et al.	N/A	N/A
10813641	12/2019	Setser et al.	N/A	N/A
10813683	12/2019	Baxter, III et al.	N/A	N/A
10813705	12/2019	Hares et al.	N/A	N/A
10813710	12/2019	Grubbs	N/A	N/A
10820939	12/2019	Sartor	N/A	N/A
10828028	12/2019	Harris et al.	N/A	N/A
10828030	12/2019	Weir et al.	N/A	N/A
10828032	12/2019	Leimbach et al.	N/A	N/A
10828033	12/2019	Shelton, IV et al.	N/A	N/A
10828089	12/2019	Clark et al.	N/A	N/A

10835245	12/2019	Swayze et al.	N/A	N/A
10835246	12/2019	Shelton, IV et al.	N/A	N/A
10835247	12/2019	Shelton, IV et al.	N/A	N/A
10835249	12/2019	Schellin et al.	N/A	N/A
10835251	12/2019	Shelton, IV et al.	N/A	N/A
10835330	12/2019	Shelton, IV et al.	N/A	N/A
10842357	12/2019	Moskowitz et al.	N/A	N/A
10842473	12/2019	Scheib et al.	N/A	N/A
10842488	12/2019	Swayze et al.	N/A	N/A
10842489	12/2019	Shelton, IV	N/A	N/A
10842490	12/2019	DiNardo et al.	N/A	N/A
10842491	12/2019	Shelton, IV et al.	N/A	N/A
10842492	12/2019	Shelton, IV et al.	N/A	N/A
10842523	12/2019	Shelton, IV et al.	N/A	N/A
D904612	12/2019	Wynn et al.	N/A	N/A
D904613	12/2019	Wynn et al.	N/A	N/A
D906355	12/2019	Messerly et al.	N/A	N/A
10849621	12/2019	Whitfield et al.	N/A	N/A
10849623	12/2019	Dunki-Jacobs et al.	N/A	N/A
10849697	12/2019	Yates et al.	N/A	N/A
10856866	12/2019	Shelton, IV et al.	N/A	N/A
10856867	12/2019	Shelton, IV et al.	N/A	N/A
10856868	12/2019	Shelton, IV et al.	N/A	N/A
10856869	12/2019	Shelton, IV et al.	N/A	N/A
10856870	12/2019	Harris et al.	N/A	N/A
10863981	12/2019	Overmyer et al.	N/A	N/A
10863984	12/2019	Shelton, IV et al.	N/A	N/A
10863986	12/2019	Yates et al.	N/A	N/A
10869663	12/2019	Shelton, IV et al.	N/A	N/A
10869664	12/2019	Shelton, IV	N/A	N/A
10869665	12/2019	Shelton, IV et al.	N/A	N/A
10869666	12/2019	Shelton, IV et al.	N/A	N/A
10869669	12/2019	Shelton, IV et al.	N/A	N/A
10874290	12/2019	Walen et al.	N/A	N/A
10874391	12/2019	Shelton, IV et al.	N/A	N/A
10874392	12/2019	Scirica et al.	N/A	N/A
10874393	12/2019	Satti, III et al.	N/A	N/A
10874396	12/2019	Moore et al.	N/A	N/A
10874399	12/2019	Zhang	N/A	N/A
10874474	12/2019	Wu et al.	N/A	N/A
10879275	12/2019	Li et al.	N/A	N/A
D907647	12/2020	Siebel et al.	N/A	N/A
D907648	12/2020	Siebel et al.	N/A	N/A
D908216	12/2020	Messerly et al.	N/A	N/A
10881339	12/2020	Peyser et al.	N/A	N/A
10881395	12/2020	Merchant et al.	N/A	N/A
10881396	12/2020	Shelton, IV et al.	N/A	N/A
10881398	12/2020	Whitman et al.	N/A	N/A
10881399	12/2020	Shelton, IV et al.	N/A	N/A
10881401	12/2020	Baber et al.	N/A	N/A

10881446	12/2020	Strobl	N/A	N/A
10888318	12/2020	Parihar et al.	N/A	N/A
10888321	12/2020	Shelton, IV et al.	N/A	N/A
10888322	12/2020	Morgan et al.	N/A	N/A
10888323	12/2020	Chen et al.	N/A	N/A
10888325	12/2020	Harris et al.	N/A	N/A
10888328	12/2020	Shelton, IV et al.	N/A	N/A
10888329	12/2020	Moore et al.	N/A	N/A
10888330	12/2020	Moore et al.	N/A	N/A
10888369	12/2020	Messerly et al.	N/A	N/A
10892899	12/2020	Shelton, IV et al.	N/A	N/A
10893853	12/2020	Shelton, IV et al.	N/A	N/A
10893863	12/2020	Shelton, IV et al.	N/A	N/A
10893864	12/2020	Harris et al.	N/A	N/A
10893867	12/2020	Leimbach et al.	N/A	N/A
10898183	12/2020	Shelton, IV et al.	N/A	N/A
10898184	12/2020	Yates et al.	N/A	N/A
10898185	12/2020	Overmyer et al.	N/A	N/A
10898186	12/2020	Bakos et al.	N/A	N/A
10898190	12/2020	Yates et al.	N/A	N/A
10898193	12/2020	Shelton, IV et al.	N/A	N/A
10898194	12/2020	Moore et al.	N/A	N/A
10898195	12/2020	Moore et al.	N/A	N/A
10903685	12/2020	Yates et al.	N/A	N/A
D910847	12/2020	Shelton, IV et al.	N/A	N/A
10905415	12/2020	DiNardo et al.	N/A	N/A
10905418	12/2020	Shelton, IV et al.	N/A	N/A
10905420	12/2020	Jasemian et al.	N/A	N/A
10905422	12/2020	Bakos et al.	N/A	N/A
10905423	12/2020	Baber et al.	N/A	N/A
10905426	12/2020	Moore et al.	N/A	N/A
10905427	12/2020	Moore et al.	N/A	N/A
10911515	12/2020	Biasi et al.	N/A	N/A
10912559	12/2020	Harris et al.	N/A	N/A
10912562	12/2020	Dunki-Jacobs et al.	N/A	N/A
10912575	12/2020	Shelton, IV et al.	N/A	N/A
10918364	12/2020	Applegate et al.	N/A	N/A
10918380	12/2020	Morgan et al.	N/A	N/A
10918385	12/2020	Overmyer et al.	N/A	N/A
10918386	12/2020	Shelton, IV et al.	N/A	N/A
10919156	12/2020	Roberts et al.	N/A	N/A
10925600	12/2020	McCuen	N/A	N/A
10925605	12/2020	Moore et al.	N/A	N/A
D914878	12/2020	Shelton, IV et al.	N/A	N/A
10932772	12/2020	Shelton, IV et al.	N/A	N/A
10932774	12/2020	Shelton, IV	N/A	N/A
10932775	12/2020	Shelton, IV et al.	N/A	N/A
10932778	12/2020	Smith et al.	N/A	N/A
10932779	12/2020	Vendely et al.	N/A	N/A

10932781	12/2020	Jones	N/A	A61B 17/2841
10932784	12/2020	Mozdzierz et al.	N/A	N/A
10932804	12/2020	Scheib et al.	N/A	N/A
10932806	12/2020	Shelton, IV et al.	N/A	N/A
10932872	12/2020	Shelton, IV et al.	N/A	N/A
10944728	12/2020	Wiener et al.	N/A	N/A
10945727	12/2020	Shelton, IV et al.	N/A	N/A
10945728	12/2020	Morgan et al.	N/A	N/A
10945729	12/2020	Shelton, IV et al.	N/A	N/A
10945731	12/2020	Baxter, III et al.	N/A	N/A
10952708	12/2020	Scheib et al.	N/A	N/A
10952726	12/2020	Chowaniec	N/A	N/A
10952727	12/2020	Giordano et al.	N/A	N/A
10952728	12/2020	Shelton, IV et al.	N/A	N/A
10952759	12/2020	Messerly et al.	N/A	N/A
10952767	12/2020	Kostrzewski et al.	N/A	N/A
10959722	12/2020	Morgan et al.	N/A	N/A
10959725	12/2020	Kerr et al.	N/A	N/A
10959726	12/2020	Williams et al.	N/A	N/A
10959727	12/2020	Hunter et al.	N/A	N/A
10959731	12/2020	Casasanta, Jr. et al.	N/A	N/A
10959744	12/2020	Shelton, IV et al.	N/A	N/A
10959797	12/2020	Licht et al.	N/A	N/A
D917500	12/2020	Siebel et al.	N/A	N/A
10966627	12/2020	Shelton, IV et al.	N/A	N/A
10966717	12/2020	Shah et al.	N/A	N/A
10966718	12/2020	Shelton, IV et al.	N/A	N/A
10966791	12/2020	Harris et al.	N/A	N/A
10973515	12/2020	Harris et al.	N/A	N/A
10973516	12/2020	Shelton, IV et al.	N/A	N/A
10973517	12/2020	Wixey	N/A	N/A
10973519	12/2020	Weir et al.	N/A	N/A
10973520	12/2020	Shelton, IV et al.	N/A	N/A
10980534	12/2020	Yates et al.	N/A	N/A
10980535	12/2020	Yates et al.	N/A	N/A
10980536	12/2020	Weaner et al.	N/A	N/A
10980537	12/2020	Shelton, IV et al.	N/A	N/A
10980538	12/2020	Nalagatla et al.	N/A	N/A
10980539	12/2020	Harris et al.	N/A	N/A
10980560	12/2020	Shelton, IV et al.	N/A	N/A
10983646	12/2020	Yoon et al.	N/A	N/A
10987102	12/2020	Gonzalez et al.	N/A	N/A
10987178	12/2020	Shelton, IV et al.	N/A	N/A
10993713	12/2020	Shelton, IV et al.	N/A	N/A
10993715	12/2020	Shelton, IV et al.	N/A	N/A
10993716	12/2020	Shelton, IV et al.	N/A	N/A
10993717	12/2020	Shelton, IV et al.	N/A	N/A
11000274	12/2020	Shelton, IV et al.	N/A	N/A
11000275	12/2020	Shelton, IV et al.	N/A	N/A

11000277	12/2020	Giordano et al.	N/A	N/A
11000278	12/2020	Shelton, IV et al.	N/A	N/A
11000279	12/2020	Shelton, IV et al.	N/A	N/A
11005291	12/2020	Calderoni	N/A	N/A
11006951	12/2020	Giordano et al.	N/A	N/A
11006955	12/2020	Shelton, IV et al.	N/A	N/A
11007004	12/2020	Shelton, IV et al.	N/A	N/A
11007022	12/2020	Shelton, IV et al.	N/A	N/A
11013511	12/2020	Huang et al.	N/A	N/A
11013552	12/2020	Widenhouse et al.	N/A	N/A
11013563	12/2020	Shelton, IV et al.	N/A	N/A
11020016	12/2020	Wallace et al.	N/A	N/A
11020112	12/2020	Shelton, IV et al.	N/A	N/A
11020113	12/2020	Shelton, IV et al.	N/A	N/A
11020114	12/2020	Shelton, IV et al.	N/A	N/A
11020115	12/2020	Scheib et al.	N/A	N/A
11020172	12/2020	Garrison	N/A	N/A
11026678	12/2020	Overmyer et al.	N/A	N/A
11026680	12/2020	Shelton, IV et al.	N/A	N/A
11026684	12/2020	Shelton, IV et al.	N/A	N/A
11026687	12/2020	Shelton, IV et al.	N/A	N/A
11026712	12/2020	Shelton, IV et al.	N/A	N/A
11026713	12/2020	Stokes et al.	N/A	N/A
11026751	12/2020	Shelton, IV et al.	N/A	N/A
11033267	12/2020	Shelton, IV et al.	N/A	N/A
11039834	12/2020	Harris et al.	N/A	N/A
11039836	12/2020	Shelton, IV et al.	N/A	N/A
11039837	12/2020	Shelton, IV et al.	N/A	N/A
11039849	12/2020	Bucciaglia et al.	N/A	N/A
11045189	12/2020	Yates et al.	N/A	N/A
11045191	12/2020	Shelton, IV et al.	N/A	N/A
11045192	12/2020	Harris et al.	N/A	N/A
11045196	12/2020	Olson et al.	N/A	N/A
11045197	12/2020	Shelton, IV et al.	N/A	N/A
11045199	12/2020	Mozdzierz et al.	N/A	N/A
11045270	12/2020	Shelton, IV et al.	N/A	N/A
D925563	12/2020	Melvin et al.	N/A	N/A
11051807	12/2020	Shelton, IV et al.	N/A	N/A
11051810	12/2020	Harris et al.	N/A	N/A
11051811	12/2020	Shelton, IV et al.	N/A	N/A
11051813	12/2020	Shelton, IV et al.	N/A	N/A
11051836	12/2020	Shelton, IV et al.	N/A	N/A
11051840	12/2020	Shelton, IV et al.	N/A	N/A
11051873	12/2020	Wiener et al.	N/A	N/A
11058418	12/2020	Shelton, IV et al.	N/A	N/A
11058420	12/2020	Shelton, IV et al.	N/A	N/A
11058422	12/2020	Harris et al.	N/A	N/A
11058423	12/2020	Shelton, IV et al.	N/A	N/A
11058424	12/2020	Shelton, IV et al.	N/A	N/A
11058425	12/2020	Widenhouse et al.	N/A	N/A

11058426	12/2020	Nalagatla et al.	N/A	N/A
11058498	12/2020	Shelton, IV et al.	N/A	N/A
11064997	12/2020	Shelton, IV et al.	N/A	N/A
11064998	12/2020	Shelton, IV	N/A	N/A
11065000	12/2020	Shankarsetty et al.	N/A	N/A
11065048	12/2020	Messerly et al.	N/A	N/A
11069012	12/2020	Shelton, IV et al.	N/A	N/A
11071542	12/2020	Chen et al.	N/A	N/A
11071543	12/2020	Shelton, IV et al.	N/A	N/A
11071545	12/2020	Baber et al.	N/A	N/A
11071554	12/2020	Parfett et al.	N/A	N/A
11071560	12/2020	Deck et al.	N/A	N/A
11076853	12/2020	Parfett et al.	N/A	N/A
11076854	12/2020	Baber et al.	N/A	N/A
11076921	12/2020	Shelton, IV et al.	N/A	N/A
11076929	12/2020	Shelton, IV et al.	N/A	N/A
11083452	12/2020	Schmid et al.	N/A	N/A
11083453	12/2020	Shelton, IV et al.	N/A	N/A
11083454	12/2020	Harris et al.	N/A	N/A
11083455	12/2020	Shelton, IV et al.	N/A	N/A
11083456	12/2020	Shelton, IV et al.	N/A	N/A
11083457	12/2020	Shelton, IV et al.	N/A	N/A
11083458	12/2020	Harris et al.	N/A	N/A
11090045	12/2020	Shelton, IV	N/A	N/A
11090046	12/2020	Shelton, IV et al.	N/A	N/A
11090047	12/2020	Shelton, IV et al.	N/A	N/A
11090048	12/2020	Fanelli et al.	N/A	N/A
11090049	12/2020	Bakos et al.	N/A	N/A
11090075	12/2020	Hunter et al.	N/A	N/A
11096687	12/2020	Flanagan et al.	N/A	N/A
11096688	12/2020	Shelton, IV et al.	N/A	N/A
11096689	12/2020	Overmyer et al.	N/A	N/A
11100631	12/2020	Yates et al.	N/A	N/A
11103241	12/2020	Yates et al.	N/A	N/A
11103248	12/2020	Shelton, IV et al.	N/A	N/A
11103268	12/2020	Shelton, IV et al.	N/A	N/A
11103269	12/2020	Shelton, IV et al.	N/A	N/A
11103301	12/2020	Messerly et al.	N/A	N/A
11109858	12/2020	Shelton, IV et al.	N/A	N/A
11109859	12/2020	Overmyer et al.	N/A	N/A
11109860	12/2020	Shelton, IV et al.	N/A	N/A
11109866	12/2020	Shelton, IV et al.	N/A	N/A
11109878	12/2020	Shelton, IV et al.	N/A	N/A
11109925	12/2020	Cooper et al.	N/A	N/A
11116485	12/2020	Scheib et al.	N/A	N/A
11116502	12/2020	Shelton, IV et al.	N/A	N/A
11116594	12/2020	Beardsley	N/A	N/A
11123069	12/2020	Baxter, III et al.	N/A	N/A
11123070	12/2020	Shelton, IV et al.	N/A	N/A
11129611	12/2020	Shelton, IV et al.	N/A	N/A

11129613	12/2020	Harris et al.	N/A	N/A
11129615	12/2020	Scheib et al.	N/A	N/A
11129616	12/2020	Shelton, IV et al.	N/A	N/A
11129634	12/2020	Scheib et al.	N/A	N/A
11129636	12/2020	Shelton, IV et al.	N/A	N/A
11129666	12/2020	Messerly et al.	N/A	N/A
11129680	12/2020	Shelton, IV et al.	N/A	N/A
11132462	12/2020	Shelton, IV et al.	N/A	N/A
11133106	12/2020	Shelton, IV et al.	N/A	N/A
11134938	12/2020	Timm et al.	N/A	N/A
11134940	12/2020	Shelton, IV et al.	N/A	N/A
11134942	12/2020	Harris et al.	N/A	N/A
11134943	12/2020	Giordano et al.	N/A	N/A
11134944	12/2020	Wise et al.	N/A	N/A
11134947	12/2020	Shelton, IV et al.	N/A	N/A
11135352	12/2020	Shelton, IV et al.	N/A	N/A
11141153	12/2020	Shelton, IV et al.	N/A	N/A
11141154	12/2020	Shelton, IV et al.	N/A	N/A
11141155	12/2020	Shelton, IV	N/A	N/A
11141156	12/2020	Shelton, IV	N/A	N/A
11141159	12/2020	Scheib et al.	N/A	N/A
11141160	12/2020	Shelton, IV et al.	N/A	N/A
11147547	12/2020	Shelton, IV et al.	N/A	N/A
11147549	12/2020	Timm et al.	N/A	N/A
11147551	12/2020	Shelton, IV	N/A	N/A
11147553	12/2020	Shelton, IV	N/A	N/A
11147554	12/2020	Aronhalt et al.	N/A	N/A
11154296	12/2020	Aronhalt et al.	N/A	N/A
11154297	12/2020	Swayze et al.	N/A	N/A
11154298	12/2020	Timm et al.	N/A	N/A
11154299	12/2020	Shelton, IV et al.	N/A	N/A
11154300	12/2020	Nalagatla et al.	N/A	N/A
11154301	12/2020	Beckman et al.	N/A	N/A
D936684	12/2020	Luo et al.	N/A	N/A
11160551	12/2020	Shelton, IV et al.	N/A	N/A
11160553	12/2020	Simms et al.	N/A	N/A
11160601	12/2020	Worrell et al.	N/A	N/A
11166716	12/2020	Shelton, IV et al.	N/A	N/A
11166717	12/2020	Shelton, IV et al.	N/A	N/A
11166720	12/2020	Giordano et al.	N/A	N/A
11166772	12/2020	Shelton, IV et al.	N/A	N/A
11166773	12/2020	Ragosta et al.	N/A	N/A
11172580	12/2020	Gaertner, II	N/A	N/A
11172927	12/2020	Shelton, IV	N/A	N/A
11172929	12/2020	Shelton, IV	N/A	N/A
11179150	12/2020	Yates et al.	N/A	N/A
11179151	12/2020	Shelton, IV et al.	N/A	N/A
11179152	12/2020	Morgan et al.	N/A	N/A
11179153	12/2020	Shelton, IV	N/A	N/A
11179155	12/2020	Shelton, IV et al.	N/A	N/A

11179208	12/2020	Yates et al.	N/A	N/A
11185325	12/2020	Shelton, IV et al.	N/A	N/A
11185330	12/2020	Huitema et al.	N/A	N/A
11191539	12/2020	Overmyer et al.	N/A	N/A
11191540	12/2020	Aronhalt et al.	N/A	N/A
11191543	12/2020	Overmyer et al.	N/A	N/A
11191545	12/2020	Vendely et al.	N/A	N/A
11197668	12/2020	Shelton, IV et al.	N/A	N/A
11197670	12/2020	Shelton, IV et al.	N/A	N/A
11197671	12/2020	Shelton, IV et al.	N/A	N/A
11197672	12/2020	Dunki-Jacobs et al.	N/A	N/A
11202570	12/2020	Shelton, IV et al.	N/A	N/A
11202631	12/2020	Shelton, IV et al.	N/A	N/A
11202633	12/2020	Harris et al.	N/A	N/A
11207064	12/2020	Shelton, IV et al.	N/A	N/A
11207065	12/2020	Harris et al.	N/A	N/A
11207067	12/2020	Shelton, IV et al.	N/A	N/A
11207089	12/2020	Kostrzewski et al.	N/A	N/A
11207090	12/2020	Shelton, IV et al.	N/A	N/A
11207146	12/2020	Shelton, IV et al.	N/A	N/A
11213293	12/2021	Worthington et al.	N/A	N/A
11213294	12/2021	Shelton, IV et al.	N/A	N/A
11213302	12/2021	Parfett et al.	N/A	N/A
11213359	12/2021	Shelton, IV et al.	N/A	N/A
11219453	12/2021	Shelton, IV et al.	N/A	N/A
11219455	12/2021	Shelton, IV et al.	N/A	N/A
11224423	12/2021	Shelton, IV et al.	N/A	N/A
11224426	12/2021	Shelton, IV et al.	N/A	N/A
11224427	12/2021	Shelton, IV et al.	N/A	N/A
11224428	12/2021	Scott et al.	N/A	N/A
11224454	12/2021	Shelton, IV et al.	N/A	N/A
11224497	12/2021	Shelton, IV et al.	N/A	N/A
11229436	12/2021	Shelton, IV et al.	N/A	N/A
11229437	12/2021	Shelton, IV et al.	N/A	N/A
11234698	12/2021	Shelton, IV et al.	N/A	N/A
11234700	12/2021	Ragosta et al.	N/A	N/A
11241229	12/2021	Shelton, IV et al.	N/A	N/A
11241230	12/2021	Shelton, IV et al.	N/A	N/A
11241235	12/2021	Shelton, IV et al.	N/A	N/A
11246590	12/2021	Swayze et al.	N/A	N/A
11246592	12/2021	Shelton, IV et al.	N/A	N/A
11246616	12/2021	Shelton, IV et al.	N/A	N/A
11246618	12/2021	Hall et al.	N/A	N/A
11246678	12/2021	Shelton, IV et al.	N/A	N/A
11253254	12/2021	Kimball et al.	N/A	N/A
11253256	12/2021	Harris et al.	N/A	N/A
D946025	12/2021	Vogler-Ivashchanka et al.	N/A	N/A
D946617	12/2021	Ahmed	N/A	N/A
11259799	12/2021	Overmyer et al.	N/A	N/A

11259803	12/2021	Shelton, IV et al.	N/A	N/A
11259805	12/2021	Shelton, IV et al.	N/A	N/A
11259806	12/2021	Shelton, IV et al.	N/A	N/A
11259807	12/2021	Shelton, IV et al.	N/A	N/A
11266405	12/2021	Shelton, IV et al.	N/A	N/A
11266406	12/2021	Leimbach et al.	N/A	N/A
11266409	12/2021	Huitema et al.	N/A	N/A
11266410	12/2021	Shelton, IV et al.	N/A	N/A
11266468	12/2021	Shelton, IV et al.	N/A	N/A
11272927	12/2021	Swayze et al.	N/A	N/A
11272928	12/2021	Shelton, IV	N/A	N/A
11272931	12/2021	Boudreaux et al.	N/A	N/A
11272938	12/2021	Shelton, IV et al.	N/A	N/A
11278279	12/2021	Morgan et al.	N/A	N/A
11278280	12/2021	Shelton, IV et al.	N/A	N/A
11278284	12/2021	Shelton, IV et al.	N/A	N/A
11278288	12/2021	Rector et al.	N/A	N/A
11284890	12/2021	Nalagatla et al.	N/A	N/A
11284891	12/2021	Shelton, IV et al.	N/A	N/A
11284898	12/2021	Baxter, III et al.	N/A	N/A
11284953	12/2021	Shelton, IV et al.	N/A	N/A
11291440	12/2021	Harris et al.	N/A	N/A
11291441	12/2021	Giordano et al.	N/A	N/A
11291442	12/2021	Wixey et al.	N/A	N/A
11291443	12/2021	Viola et al.	N/A	N/A
11291444	12/2021	Boudreaux et al.	N/A	N/A
11291445	12/2021	Shelton, IV et al.	N/A	N/A
11291447	12/2021	Shelton, IV et al.	N/A	N/A
11291449	12/2021	Swensgard et al.	N/A	N/A
11291451	12/2021	Shelton, IV	N/A	N/A
11291465	12/2021	Parihar et al.	N/A	N/A
11291510	12/2021	Shelton, IV et al.	N/A	N/A
11298125	12/2021	Ming et al.	N/A	N/A
11298127	12/2021	Shelton, IV	N/A	N/A
11298128	12/2021	Messerly et al.	N/A	N/A
11298129	12/2021	Bakos et al.	N/A	N/A
11298130	12/2021	Bakos et al.	N/A	N/A
11298132	12/2021	Shelton, IV et al.	N/A	N/A
11298134	12/2021	Huitema et al.	N/A	N/A
11304695	12/2021	Shelton, IV et al.	N/A	N/A
11304696	12/2021	Shelton, IV et al.	N/A	N/A
11304697	12/2021	Fanelli et al.	N/A	N/A
11304699	12/2021	Shelton, IV et al.	N/A	N/A
11304704	12/2021	Thomas et al.	N/A	N/A
11311290	12/2021	Shelton, IV et al.	N/A	N/A
11311292	12/2021	Shelton, IV et al.	N/A	N/A
11311294	12/2021	Swayze	N/A	H02H 7/20
11311295	12/2021	Wingardner et al.	N/A	N/A
11311342	12/2021	Parihar et al.	N/A	N/A
D950728	12/2021	Bakos et al.	N/A	N/A

D952144	12/2021	Boudreaux	N/A	N/A
11317910	12/2021	Miller et al.	N/A	N/A
11317912	12/2021	Jenkins et al.	N/A	N/A
11317913	12/2021	Shelton, IV et al.	N/A	N/A
11317915	12/2021	Boudreaux et al.	N/A	N/A
11317917	12/2021	Shelton, IV et al.	N/A	N/A
11317919	12/2021	Shelton, IV et al.	N/A	N/A
11317978	12/2021	Cameron et al.	N/A	N/A
11324501	12/2021	Shelton, IV et al.	N/A	N/A
11324503	12/2021	Shelton, IV et al.	N/A	N/A
11324506	12/2021	Beckman et al.	N/A	N/A
11324557	12/2021	Shelton, IV et al.	N/A	N/A
11331100	12/2021	Boudreaux et al.	N/A	N/A
11331101	12/2021	Harris et al.	N/A	N/A
11337691	12/2021	Widenhouse et al.	N/A	N/A
11337693	12/2021	Hess et al.	N/A	N/A
11337698	12/2021	Baxter, III et al.	N/A	N/A
11344299	12/2021	Yates et al.	N/A	N/A
11344303	12/2021	Shelton, IV et al.	N/A	N/A
D954736	12/2021	Teague et al.	N/A	N/A
11350843	12/2021	Shelton, IV et al.	N/A	N/A
11350916	12/2021	Shelton, IV et al.	N/A	N/A
11350928	12/2021	Shelton, IV et al.	N/A	N/A
11350929	12/2021	Giordano et al.	N/A	N/A
11350932	12/2021	Shelton, IV et al.	N/A	N/A
11350934	12/2021	Bakos et al.	N/A	N/A
11350935	12/2021	Shelton, IV et al.	N/A	N/A
11350938	12/2021	Shelton, IV et al.	N/A	N/A
11357503	12/2021	Bakos et al.	N/A	N/A
11361176	12/2021	Shelton, IV et al.	N/A	N/A
11364027	12/2021	Harris et al.	N/A	N/A
11364046	12/2021	Shelton, IV et al.	N/A	N/A
11369366	12/2021	Scheib et al.	N/A	N/A
11369368	12/2021	Shelton, IV et al.	N/A	N/A
11369376	12/2021	Simms et al.	N/A	N/A
11369377	12/2021	Boudreaux et al.	N/A	N/A
11373755	12/2021	Shelton, IV et al.	N/A	N/A
11376001	12/2021	Shelton, IV et al.	N/A	N/A
11376002	12/2021	Shelton, IV et al.	N/A	N/A
11376082	12/2021	Shelton, IV et al.	N/A	N/A
11376098	12/2021	Shelton, IV et al.	N/A	N/A
11382625	12/2021	Huitema et al.	N/A	N/A
11382626	12/2021	Shelton, IV et al.	N/A	N/A
11382627	12/2021	Huitema et al.	N/A	N/A
11382628	12/2021	Baxter, III et al.	N/A	N/A
11382638	12/2021	Harris et al.	N/A	N/A
11382697	12/2021	Shelton, IV et al.	N/A	N/A
11382704	12/2021	Overmyer et al.	N/A	N/A
11389160	12/2021	Shelton, IV et al.	N/A	N/A
11389161	12/2021	Shelton, IV et al.	N/A	N/A

11389162	12/2021	Baber et al.	N/A	N/A
11389164	12/2021	Yates et al.	N/A	N/A
11395651	12/2021	Shelton, IV et al.	N/A	N/A
11395652	12/2021	Parihar et al.	N/A	N/A
11399828	12/2021	Swayze et al.	N/A	N/A
11399829	12/2021	Leimbach et al.	N/A	N/A
11399831	12/2021	Overmyer et al.	N/A	N/A
11399837	12/2021	Shelton, IV et al.	N/A	N/A
11406377	12/2021	Schmid et al.	N/A	N/A
11406378	12/2021	Baxter, III et al.	N/A	N/A
11406380	12/2021	Yates et al.	N/A	N/A
11406381	12/2021	Parihar et al.	N/A	N/A
11406382	12/2021	Shelton, IV et al.	N/A	N/A
11406386	12/2021	Baber et al.	N/A	N/A
11406390	12/2021	Shelton, IV et al.	N/A	N/A
11406442	12/2021	Davison et al.	N/A	N/A
11410259	12/2021	Harris et al.	N/A	N/A
11413041	12/2021	Viola et al.	N/A	N/A
11413042	12/2021	Shelton, IV et al.	N/A	N/A
11413102	12/2021	Shelton, IV et al.	N/A	N/A
11419606	12/2021	Overmyer et al.	N/A	N/A
11419630	12/2021	Yates et al.	N/A	N/A
11424027	12/2021	Shelton, IV	N/A	N/A
11426160	12/2021	Shelton, IV et al.	N/A	N/A
11426167	12/2021	Shelton, IV et al.	N/A	N/A
11426251	12/2021	Kimball et al.	N/A	N/A
D962980	12/2021	Frenkler et al.	N/A	N/A
D964564	12/2021	Boudreaux	N/A	N/A
11432816	12/2021	Leimbach et al.	N/A	N/A
11432819	12/2021	Rector et al.	N/A	N/A
11432885	12/2021	Shelton, IV et al.	N/A	N/A
11439391	12/2021	Bruns et al.	N/A	N/A
11439470	12/2021	Spivey et al.	N/A	N/A
11446029	12/2021	Shelton, IV et al.	N/A	N/A
11446034	12/2021	Shelton, IV et al.	N/A	N/A
11452526	12/2021	Ross et al.	N/A	N/A
11452528	12/2021	Leimbach et al.	N/A	N/A
D966512	12/2021	Shelton, IV et al.	N/A	N/A
D967421	12/2021	Shelton, IV et al.	N/A	N/A
11457918	12/2021	Shelton, IV et al.	N/A	N/A
11464511	12/2021	Timm et al.	N/A	N/A
11464512	12/2021	Shelton, IV et al.	N/A	N/A
11464513	12/2021	Shelton, IV et al.	N/A	N/A
11464514	12/2021	Yates et al.	N/A	N/A
11464601	12/2021	Shelton, IV et al.	N/A	N/A
11471155	12/2021	Shelton, IV et al.	N/A	N/A
11471156	12/2021	Shelton, IV et al.	N/A	N/A
11471157	12/2021	Baxter, III et al.	N/A	N/A
11478241	12/2021	Shelton, IV et al.	N/A	N/A
11478242	12/2021	Shelton, IV et al.	N/A	N/A

11478244	12/2021	DiNardo et al.	N/A	N/A
D969849	12/2021	Stipech et al.	N/A	N/A
D971232	12/2021	Siebel et al.	N/A	N/A
11484307	12/2021	Hall et al.	N/A	N/A
11484309	12/2021	Harris et al.	N/A	N/A
11484310	12/2021	Shelton, IV et al.	N/A	N/A
11484311	12/2021	Shelton, IV et al.	N/A	N/A
11484312	12/2021	Shelton, IV et al.	N/A	N/A
11490889	12/2021	Overmyer et al.	N/A	N/A
11497488	12/2021	Leimbach et al.	N/A	N/A
11497489	12/2021	Baxter, III et al.	N/A	N/A
11497492	12/2021	Shelton, IV	N/A	N/A
11497499	12/2021	Shelton, IV et al.	N/A	N/A
11504116	12/2021	Schmid et al.	N/A	N/A
11504119	12/2021	Shelton, IV et al.	N/A	N/A
11504122	12/2021	Shelton, IV et al.	N/A	N/A
11504192	12/2021	Shelton, IV et al.	N/A	N/A
11510671	12/2021	Shelton, IV et al.	N/A	N/A
11510673	12/2021	Chen et al.	N/A	N/A
11510741	12/2021	Shelton, IV et al.	N/A	N/A
11517304	12/2021	Yates et al.	N/A	N/A
11517306	12/2021	Miller et al.	N/A	N/A
11517309	12/2021	Bakos et al.	N/A	N/A
11517311	12/2021	Lytle, IV et al.	N/A	N/A
11517315	12/2021	Huitema et al.	N/A	N/A
11517325	12/2021	Shelton, IV et al.	N/A	N/A
11517390	12/2021	Baxter, III	N/A	N/A
11523821	12/2021	Harris et al.	N/A	N/A
11523822	12/2021	Shelton, IV et al.	N/A	N/A
11523823	12/2021	Hunter et al.	N/A	N/A
11523824	12/2021	Williams	N/A	N/A
11523859	12/2021	Shelton, IV et al.	N/A	N/A
11529137	12/2021	Shelton, IV et al.	N/A	N/A
11529138	12/2021	Jaworek et al.	N/A	N/A
11529139	12/2021	Shelton, IV et al.	N/A	N/A
11529140	12/2021	Shelton, IV et al.	N/A	N/A
11529142	12/2021	Leimbach et al.	N/A	N/A
11534162	12/2021	Shelton, IV	N/A	N/A
11534259	12/2021	Leimbach et al.	N/A	N/A
D974560	12/2022	Shelton, IV et al.	N/A	N/A
D975278	12/2022	Shelton, IV et al.	N/A	N/A
D975850	12/2022	Shelton, IV et al.	N/A	N/A
D975851	12/2022	Shelton, IV et al.	N/A	N/A
D976401	12/2022	Shelton, IV et al.	N/A	N/A
11540824	12/2022	Shelton, IV et al.	N/A	N/A
11540829	12/2022	Shelton, IV et al.	N/A	N/A
11547403	12/2022	Shelton, IV et al.	N/A	N/A
11547404	12/2022	Shelton, IV et al.	N/A	N/A
11553911	12/2022	Shelton, IV et al.	N/A	N/A
11553916	12/2022	Vendely et al.	N/A	N/A

11553919	12/2022	Shelton, IV et al.	N/A	N/A
11553971	12/2022	Shelton, IV et al.	N/A	N/A
11559302	12/2022	Timm et al.	N/A	N/A
11559303	12/2022	Shelton, IV et al.	N/A	N/A
11559304	12/2022	Boudreaux et al.	N/A	N/A
11559307	12/2022	Shelton, IV et al.	N/A	N/A
11559308	12/2022	Yates et al.	N/A	N/A
11559496	12/2022	Widenhouse et al.	N/A	N/A
11564679	12/2022	Parihar et al.	N/A	N/A
11564682	12/2022	Timm et al.	N/A	N/A
11564686	12/2022	Yates et al.	N/A	N/A
11564688	12/2022	Swayze et al.	N/A	N/A
11564703	12/2022	Shelton, IV et al.	N/A	N/A
11564756	12/2022	Shelton, IV et al.	N/A	N/A
11571207	12/2022	Shelton, IV et al.	N/A	N/A
11571210	12/2022	Shelton, IV et al.	N/A	N/A
11571212	12/2022	Yates et al.	N/A	N/A
11571215	12/2022	Shelton, IV et al.	N/A	N/A
11571231	12/2022	Hess et al.	N/A	N/A
11576668	12/2022	Shelton, IV et al.	N/A	N/A
11576672	12/2022	Shelton, IV et al.	N/A	N/A
11576673	12/2022	Shelton, IV	N/A	N/A
11576677	12/2022	Shelton, IV et al.	N/A	N/A
11583274	12/2022	Widenhouse et al.	N/A	N/A
11583277	12/2022	Shelton, IV et al.	N/A	N/A
11583278	12/2022	Shelton, IV et al.	N/A	N/A
11583279	12/2022	Smith et al.	N/A	N/A
11589863	12/2022	Weir et al.	N/A	N/A
11589865	12/2022	Shelton, IV et al.	N/A	N/A
11589888	12/2022	Shelton, IV et al.	N/A	N/A
D980425	12/2022	Baxter, III	N/A	N/A
11596406	12/2022	Huitema et al.	N/A	N/A
11602340	12/2022	Schmid et al.	N/A	N/A
11602346	12/2022	Shelton, IV	N/A	N/A
11602366	12/2022	Shelton, IV et al.	N/A	N/A
11607219	12/2022	Shelton, IV et al.	N/A	N/A
11607239	12/2022	Swensgard et al.	N/A	N/A
11607278	12/2022	Shelton, IV et al.	N/A	N/A
11612393	12/2022	Morgan et al.	N/A	N/A
11612394	12/2022	Morgan et al.	N/A	N/A
11612395	12/2022	Yates	227/175.1	A61B 17/07207
11617575	12/2022	Yates et al.	N/A	N/A
11617576	12/2022	Yates et al.	N/A	N/A
11617577	12/2022	Huang	N/A	N/A
11622763	12/2022	Parihar et al.	N/A	N/A
11622766	12/2022	Shelton, IV	N/A	N/A
11622785	12/2022	Hess et al.	N/A	N/A
11627959	12/2022	Shelton, IV et al.	N/A	N/A
11627960	12/2022	Shelton, IV et al.	N/A	N/A

11628006	12/2022	Henderson et al.	N/A	N/A
11633183	12/2022	Parihar et al.	N/A	N/A
11633185	12/2022	Wilson et al.	N/A	N/A
D985009	12/2022	Barrett et al.	N/A	N/A
D985617	12/2022	Bahatyrevich et al.	N/A	N/A
11638581	12/2022	Parihar et al.	N/A	N/A
11638582	12/2022	Bakos et al.	N/A	N/A
11638583	12/2022	Yates et al.	N/A	N/A
11638587	12/2022	Shelton, IV et al.	N/A	N/A
11642125	12/2022	Harris et al.	N/A	N/A
11642128	12/2022	Shelton, IV et al.	N/A	N/A
11648005	12/2022	Yates et al.	N/A	N/A
11648006	12/2022	Timm et al.	N/A	N/A
11648008	12/2022	Shelton, IV et al.	N/A	N/A
11648009	12/2022	Jenkins	N/A	N/A
11648022	12/2022	Shelton, IV	N/A	N/A
11648024	12/2022	Shelton, IV et al.	N/A	N/A
11653914	12/2022	Shelton, IV et al.	N/A	N/A
11653915	12/2022	Shelton, IV et al.	N/A	N/A
11653917	12/2022	Scott et al.	N/A	N/A
11653918	12/2022	Swayze et al.	N/A	N/A
11653920	12/2022	Shelton, IV et al.	N/A	N/A
11659023	12/2022	Shelton, IV et al.	N/A	N/A
11660090	12/2022	Bakos et al.	N/A	N/A
11660110	12/2022	Shelton, IV et al.	N/A	N/A
11660163	12/2022	Shelton, IV et al.	N/A	N/A
11666327	12/2022	Whitman et al.	N/A	N/A
11666332	12/2022	Giordano et al.	N/A	N/A
11672531	12/2022	Timm et al.	N/A	N/A
11672532	12/2022	Shelton, IV et al.	N/A	N/A
11672536	12/2022	Shelton, IV et al.	N/A	N/A
11672605	12/2022	Messerly et al.	N/A	N/A
11678877	12/2022	Shelton, IV et al.	N/A	N/A
11678880	12/2022	Shelton, IV et al.	N/A	N/A
11678881	12/2022	Yates et al.	N/A	N/A
11678882	12/2022	Shelton, IV et al.	N/A	N/A
11684360	12/2022	Shelton, IV et al.	N/A	N/A
11684361	12/2022	Yates et al.	N/A	N/A
11684365	12/2022	Shelton, IV et al.	N/A	N/A
11684369	12/2022	Shelton, IV et al.	N/A	N/A
11684434	12/2022	Shelton, IV	N/A	N/A
11690615	12/2022	Parihar et al.	N/A	N/A
11690623	12/2022	Shelton, IV et al.	N/A	N/A
11696757	12/2022	Shelton, IV et al.	N/A	N/A
11696759	12/2022	Shelton, IV et al.	N/A	N/A
11696761	12/2022	Shelton, IV	N/A	N/A
11696778	12/2022	Shelton, IV et al.	N/A	N/A
11701110	12/2022	Yates et al.	N/A	N/A
11701111	12/2022	Shelton, IV et al.	N/A	N/A
11701113	12/2022	Shelton, IV et al.	N/A	N/A

11701114	12/2022	Shelton, IV et al.	N/A	N/A
11701115	12/2022	Harris et al.	N/A	N/A
11707273	12/2022	Kerr et al.	N/A	N/A
11712244	12/2022	Vendely et al.	N/A	N/A
11712303	12/2022	Shelton, IV et al.	N/A	N/A
11717285	12/2022	Yates et al.	N/A	N/A
11717289	12/2022	Leimbach	N/A	N/A
11717291	12/2022	Morgan et al.	N/A	N/A
11717294	12/2022	Huitema et al.	N/A	N/A
11717297	12/2022	Baber et al.	N/A	N/A
11723657	12/2022	Shelton, IV et al.	N/A	N/A
11723658	12/2022	Bakos et al.	N/A	N/A
11723662	12/2022	Leimbach et al.	N/A	N/A
11730474	12/2022	Shelton, IV	N/A	N/A
11730477	12/2022	Smith et al.	N/A	N/A
12193670	12/2024	Shelton, IV	N/A	A61B 17/07207
2001/0000531	12/2000	Casscells et al.	N/A	N/A
2001/0025183	12/2000	Shahidi	N/A	N/A
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2009/0001121	12/2008	Hess et al.	N/A	N/A
2009/0001130	12/2008	Hess et al.	N/A	N/A
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2009/0007014	12/2008	Coomer et al.	N/A	N/A
2009/0012534	12/2008	Madhani et al.	N/A	N/A
2009/0015195	12/2008	Loth-Krausser	N/A	N/A
2009/0020958	12/2008	Soul	N/A	N/A
2009/0030437	12/2008	Houser et al.	N/A	N/A
2009/0043253	12/2008	Podaima	N/A	N/A
2009/0048583	12/2008	Williams et al.	N/A	N/A
2009/0048589	12/2008	Takashino et al.	N/A	N/A
2009/0053288	12/2008	Eskridge, Jr. et al.	N/A	N/A
2009/0057369	12/2008	Smith et al.	N/A	N/A
2009/0069806	12/2008	De La Mora Levy et al.	N/A	N/A
2009/0076506	12/2008	Baker	N/A	N/A
2009/0076510	12/2008	Bell et al.	N/A	N/A
2009/0078736	12/2008	Van Lue	N/A	N/A
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2009/0088659	12/2008	Graham et al.	N/A	N/A
2009/0090763	12/2008	Zemlok et al.	N/A	N/A
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2009/0099876	12/2008	Whitman	N/A	N/A
2009/0110533	12/2008	Jinno	N/A	N/A
2009/0112234	12/2008	Crainich et al.	N/A	N/A
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2009/0167548	12/2008	Sugahara	N/A	N/A
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2009/0177226	12/2008	Reinprecht et al.	N/A	N/A
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2009/0242610	12/2008	Shelton, IV et al.	N/A	N/A
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2009/0247368	12/2008	Chiang	N/A	N/A
2009/0247901	12/2008	Zimmer	N/A	N/A
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2010/0147922	12/2009	Olson	N/A	N/A

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2011/0056717	12/2010	Herisse	N/A	N/A
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2011/0095067	12/2010	Ohdaira	N/A	N/A
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2011/0127945	12/2010	Yoneda	N/A	N/A
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2011/0218550	12/2010	Ma	N/A	N/A
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2011/0251606	12/2010	Kerr	N/A	N/A
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2011/0271186	12/2010	Owens	N/A	N/A
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2013/0075447	12/2012	Weisenburgh, II et al.	N/A	N/A
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2013/0090534	12/2012	Burns et al.	N/A	N/A
2013/0096568	12/2012	Justis	N/A	N/A
2013/0098968	12/2012	Aranyi et al.	N/A	N/A
2013/0098970	12/2012	Racenet et al.	N/A	N/A
2013/0106352	12/2012	Nagamine	N/A	N/A
2013/0112729	12/2012	Beardsley et al.	N/A	N/A
2013/0116669	12/2012	Shelton, IV et al.	N/A	N/A
2013/0123816	12/2012	Hodgkinson et al.	N/A	N/A
2013/0126202	12/2012	Oomori et al.	N/A	N/A
2013/0131476	12/2012	Siu et al.	N/A	N/A
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2013/0136969	12/2012	Yasui et al.	N/A	N/A
2013/0153639	12/2012	Hodgkinson et al.	N/A	N/A
2013/0153641	12/2012	Shelton, IV et al.	N/A	N/A
2013/0158390	12/2012	Tan et al.	N/A	N/A
2013/0162198	12/2012	Yokota et al.	N/A	N/A
2013/0165908	12/2012	Purdy et al.	N/A	N/A
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2013/0172713	12/2012	Kirschenman	N/A	N/A
2013/0172878	12/2012	Smith	N/A	N/A
2013/0175315	12/2012	Milliman	N/A	N/A
2013/0175317	12/2012	Yates et al.	N/A	N/A
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2013/0186936	12/2012	Shelton, IV	N/A	N/A
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2013/0215449	12/2012	Yamasaki	N/A	N/A
2013/0231681	12/2012	Robinson et al.	N/A	N/A
2013/0233906	12/2012	Hess et al.	N/A	N/A
2013/0238021	12/2012	Gross et al.	N/A	N/A
2013/0248578	12/2012	Arteaga Gonzalez	N/A	N/A
2013/0253480	12/2012	Kimball et al.	N/A	N/A
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2013/0277410	12/2012	Fernandez et al.	N/A	N/A
2013/0284792	12/2012	Ma	N/A	N/A
2013/0289565	12/2012	Hassler, Jr.	N/A	N/A
2013/0293353	12/2012	McPherson et al.	N/A	N/A
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2013/0310849	12/2012	Malkowski	N/A	N/A
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2015/0173789	12/2014	Baxter, III et al.	N/A	N/A
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2015/0201918	12/2014	Kumar et al.	N/A	N/A
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2017/0245880	12/2016	Honda et al.	N/A	N/A
2017/0245949	12/2016	Randle	N/A	N/A
2017/0249431	12/2016	Shelton, IV et al.	N/A	N/A
2017/0252060	12/2016	Ellingson et al.	N/A	N/A
2017/0255799	12/2016	Zhao et al.	N/A	N/A
2017/0258471	12/2016	DiNardo et al.	N/A	N/A
2017/0262110	12/2016	Polishchuk et al.	N/A	N/A
2017/0265774	12/2016	Johnson et al.	N/A	N/A
2017/0281186	12/2016	Shelton, IV et al.	N/A	N/A
2017/0296173	12/2016	Shelton, IV et al.	N/A	N/A
2017/0296185	12/2016	Swensgard et al.	N/A	N/A
2017/0303984	12/2016	Malackowski	N/A	N/A
2017/0308665	12/2016	Heck et al.	N/A	N/A
2017/0312042	12/2016	Giordano et al.	N/A	N/A
2017/0319047	12/2016	Poulsen et al.	N/A	N/A
2017/0319201	12/2016	Morgan et al.	N/A	N/A
2017/0333034	12/2016	Morgan et al.	N/A	N/A
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2017/0348010	12/2016	Chiang	N/A	N/A
2017/0348042	12/2016	Drochner et al.	N/A	N/A
2017/0348043	12/2016	Wang et al.	N/A	N/A
2017/0354413	12/2016	Chen et al.	N/A	N/A

2017/0358052	12/2016	Yuan	N/A	N/A
2017/0360441	12/2016	Sgroi	N/A	N/A
2018/0008265	12/2017	Hatanaka et al.	N/A	N/A
2018/0036024	12/2017	Allen, IV	N/A	N/A
2018/0036025	12/2017	Drochner et al.	N/A	N/A
2018/0042610	12/2017	Sgroi, Jr.	N/A	N/A
2018/0042689	12/2017	Mozdzierz et al.	N/A	N/A
2018/0049738	12/2017	Meloul et al.	N/A	N/A
2018/0049794	12/2017	Swayze et al.	N/A	N/A
2018/0051780	12/2017	Shelton, IV et al.	N/A	N/A
2018/0055501	12/2017	Zemlok et al.	N/A	N/A
2018/0067004	12/2017	Sgroi, Jr.	N/A	N/A
2018/0085117	12/2017	Shelton, IV et al.	N/A	N/A
2018/0085120	12/2017	Viola	N/A	N/A
2018/0092710	12/2017	Bosisio et al.	N/A	N/A
2018/0114591	12/2017	Pribanic et al.	N/A	N/A
2018/0116658	12/2017	Aronhalt, IV et al.	N/A	N/A
2018/0125481	12/2017	Yates et al.	N/A	N/A
2018/0125487	12/2017	Beardsley	N/A	N/A
2018/0125488	12/2017	Morgan et al.	N/A	N/A
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2018/0132849	12/2017	Miller et al.	N/A	N/A
2018/0132850	12/2017	Leimbach et al.	N/A	N/A
2018/0132926	12/2017	Asher et al.	N/A	N/A
2018/0132952	12/2017	Spivey et al.	N/A	N/A
2018/0133521	12/2017	Frushour et al.	N/A	N/A
2018/0140299	12/2017	Weaner et al.	N/A	N/A
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2018/0161034	12/2017	Scheib et al.	N/A	N/A
2018/0168572	12/2017	Burbank	N/A	N/A
2018/0168574	12/2017	Robinson et al.	N/A	N/A
2018/0168575	12/2017	Simms et al.	N/A	N/A
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2018/0168609	12/2017	Fanelli et al.	N/A	N/A
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2018/0168618	12/2017	Scott et al.	N/A	N/A
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2018/0236181	12/2017	Marlin et al.	N/A	N/A
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2018/0280026	12/2017	Zhang et al.	N/A	N/A
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2018/0310995	12/2017	Gliner et al.	N/A	N/A
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2019/0201045	12/2018	Yates et al.	N/A	N/A
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2020/0405292	12/2019	Shelton, IV et al.	N/A	N/A
2020/0405296	12/2019	Shelton, IV et al.	N/A	N/A
2020/0405302	12/2019	Shelton, IV et al.	N/A	N/A
2020/0405304	12/2019	Mozdzierz et al.	N/A	N/A
2020/0405306	12/2019	Shelton, IV et al.	N/A	N/A
2020/0405307	12/2019	Shelton, IV et al.	N/A	N/A
2020/0405308	12/2019	Shelton, IV	N/A	N/A
2020/0405375	12/2019	Shelton, IV et al.	N/A	N/A
2020/0405403	12/2019	Shelton, IV et al.	N/A	N/A
2020/0410177	12/2019	Shelton, IV	N/A	N/A

2021/0000466	12/2020	Leimbach et al.	N/A	N/A
2021/0000467	12/2020	Shelton, IV et al.	N/A	N/A
2021/0007742	12/2020	Rector et al.	N/A	N/A
2021/0015480	12/2020	Shelton, IV et al.	N/A	N/A
2021/0030416	12/2020	Shelton, IV et al.	N/A	N/A
2021/0045742	12/2020	Shelton, IV et al.	N/A	N/A
2021/0052271	12/2020	Harris et al.	N/A	N/A
2021/0059661	12/2020	Schmid et al.	N/A	N/A
2021/0059662	12/2020	Shelton, IV	N/A	N/A
2021/0059664	12/2020	Hensel et al.	N/A	N/A
2021/0059670	12/2020	Overmyer et al.	N/A	N/A
2021/0059672	12/2020	Giordano et al.	N/A	N/A
2021/0059673	12/2020	Shelton, IV et al.	N/A	N/A
2021/0068829	12/2020	Miller et al.	N/A	N/A
2021/0068835	12/2020	Shelton, IV et al.	N/A	N/A
2021/0077099	12/2020	Shelton, IV et al.	N/A	N/A
2021/0077109	12/2020	Harris et al.	N/A	N/A
2021/0084700	12/2020	Daniels	N/A	N/A
2021/0085313	12/2020	Morgan et al.	N/A	N/A
2021/0085315	12/2020	Aronhalt et al.	N/A	N/A
2021/0085316	12/2020	Harris et al.	N/A	N/A
2021/0085320	12/2020	Leimbach et al.	N/A	N/A
2021/0085321	12/2020	Shelton, IV et al.	N/A	N/A
2021/0085325	12/2020	Shelton, IV et al.	N/A	N/A
2021/0093321	12/2020	Auld et al.	N/A	N/A
2021/0093323	12/2020	Scirica et al.	N/A	N/A
2021/0100541	12/2020	Shelton, IV et al.	N/A	N/A
2021/0100982	12/2020	Laby et al.	N/A	N/A
2021/0107031	12/2020	Bales, Jr. et al.	N/A	N/A
2021/0121175	12/2020	Yates et al.	N/A	N/A
2021/0128153	12/2020	Sgroi	N/A	N/A
2021/0137522	12/2020	Shelton, IV et al.	N/A	N/A
2021/0145441	12/2020	Weir et al.	N/A	N/A
2021/0153866	12/2020	Knapp et al.	N/A	N/A
2021/0177401	12/2020	Abramek et al.	N/A	N/A
2021/0177411	12/2020	Williams	N/A	N/A
2021/0177528	12/2020	Cappelleri et al.	N/A	N/A
2021/0186492	12/2020	Shelton, IV et al.	N/A	N/A
2021/0186497	12/2020	Shelton, IV et al.	N/A	N/A
2021/0186499	12/2020	Shelton, IV et al.	N/A	N/A
2021/0186501	12/2020	Shelton, IV et al.	N/A	N/A
2021/0196265	12/2020	Shelton, IV et al.	N/A	N/A
2021/0196269	12/2020	Shelton, IV et al.	N/A	N/A
2021/0196270	12/2020	Shelton, IV et al.	N/A	N/A
2021/0196425	12/2020	Shelton, IV et al.	N/A	N/A
2021/0204941	12/2020	Dewaele et al.	N/A	N/A
2021/0204951	12/2020	Sgroi et al.	N/A	N/A
2021/0212671	12/2020	Ramadan et al.	N/A	N/A
2021/0212776	12/2020	Schmitt et al.	N/A	N/A
2021/0225140	12/2020	Adachi et al.	N/A	N/A

2021/0228209	12/2020	Shelton, IV et al.	N/A	N/A
2021/0236124	12/2020	Shelton, IV et al.	N/A	N/A
2021/0244407	12/2020	Shelton, IV et al.	N/A	N/A
2021/0244411	12/2020	Smith et al.	N/A	N/A
2021/0251720	12/2020	Jhaveri et al.	N/A	N/A
2021/0259681	12/2020	Shelton, IV et al.	N/A	N/A
2021/0259687	12/2020	Gonzalez et al.	N/A	N/A
2021/0259790	12/2020	Kaiser	N/A	N/A
2021/0259986	12/2020	Widenhouse et al.	N/A	N/A
2021/0259987	12/2020	Widenhouse et al.	N/A	N/A
2021/0267589	12/2020	Swayze et al.	N/A	N/A
2021/0267594	12/2020	Morgan et al.	N/A	N/A
2021/0267595	12/2020	Posada et al.	N/A	N/A
2021/0267596	12/2020	Fanelli et al.	N/A	N/A
2021/0275053	12/2020	Shelton, IV et al.	N/A	N/A
2021/0275172	12/2020	Harris et al.	N/A	N/A
2021/0275173	12/2020	Shelton, IV et al.	N/A	N/A
2021/0275175	12/2020	Vadali et al.	N/A	N/A
2021/0275176	12/2020	Beckman et al.	N/A	N/A
2021/0282767	12/2020	Shelton, IV et al.	N/A	N/A
2021/0282769	12/2020	Baxter, III et al.	N/A	N/A
2021/0282774	12/2020	Shelton, IV et al.	N/A	N/A
2021/0282776	12/2020	Overmyer et al.	N/A	N/A
2021/0290226	12/2020	Mandakolathur Vasudevan et al.	N/A	N/A
2021/0290231	12/2020	Baxter, III et al.	N/A	N/A
2021/0290232	12/2020	Harris et al.	N/A	N/A
2021/0290233	12/2020	Shelton, IV et al.	N/A	N/A
2021/0290236	12/2020	Moore et al.	N/A	N/A
2021/0290322	12/2020	Traina	N/A	N/A
2021/0298745	12/2020	Leimbach et al.	N/A	N/A
2021/0298746	12/2020	Leimbach et al.	N/A	N/A
2021/0307744	12/2020	Walcott et al.	N/A	N/A
2021/0307748	12/2020	Harris et al.	N/A	N/A
2021/0313975	12/2020	Shan et al.	N/A	N/A
2021/0315566	12/2020	Yates et al.	N/A	N/A
2021/0315570	12/2020	Shelton, IV	N/A	N/A
2021/0315571	12/2020	Swayze et al.	N/A	N/A
2021/0315573	12/2020	Shelton, IV et al.	N/A	N/A
2021/0315574	12/2020	Shelton, IV et al.	N/A	N/A
2021/0315576	12/2020	Shelton, IV et al.	N/A	N/A
2021/0315577	12/2020	Shelton, IV et al.	N/A	N/A
2021/0322009	12/2020	Huang et al.	N/A	N/A
2021/0330321	12/2020	Leimbach et al.	N/A	N/A
2021/0338233	12/2020	Shelton, IV et al.	N/A	N/A
2021/0338234	12/2020	Shelton, IV et al.	N/A	N/A
2021/0338260	12/2020	Le Rolland et al.	N/A	N/A
2021/0346082	12/2020	Adams et al.	N/A	N/A
2021/0353284	12/2020	Yang et al.	N/A	N/A
2021/0369271	12/2020	Schings et al.	N/A	N/A

2021/0378669	12/2020	Shelton, IV et al.	N/A	N/A
2021/0393261	12/2020	Harris et al.	N/A	N/A
2021/0393262	12/2020	Shelton, IV et al.	N/A	N/A
2021/0393366	12/2020	Shelton, IV et al.	N/A	N/A
2021/0401487	12/2020	Apostolopoulos et al.	N/A	N/A
2021/0401513	12/2020	Apostolopoulos et al.	N/A	N/A
2022/0000478	12/2021	Shelton, IV et al.	N/A	N/A
2022/0000479	12/2021	Shelton, IV et al.	N/A	N/A
2022/0015760	12/2021	Beardsley et al.	N/A	N/A
2022/0031315	12/2021	Bakos et al.	N/A	N/A
2022/0031319	12/2021	Witte et al.	N/A	N/A
2022/0031320	12/2021	Hall et al.	N/A	N/A
2022/0031322	12/2021	Parks	N/A	N/A
2022/0031323	12/2021	Witte	N/A	N/A
2022/0031324	12/2021	Hall et al.	N/A	N/A
2022/0031345	12/2021	Witte	N/A	N/A
2022/0031346	12/2021	Parks	N/A	N/A
2022/0031350	12/2021	Witte	N/A	N/A
2022/0031351	12/2021	Moubarak et al.	N/A	N/A
2022/0049593	12/2021	Groover et al.	N/A	N/A
2022/0054125	12/2021	Ji et al.	N/A	N/A
2022/0054130	12/2021	Overmyer et al.	N/A	N/A
2022/0061642	12/2021	Park et al.	N/A	N/A
2022/0061836	12/2021	Parihar et al.	N/A	N/A
2022/0061843	12/2021	Vendely et al.	N/A	N/A
2022/0061845	12/2021	Shelton, IV et al.	N/A	N/A
2022/0061862	12/2021	Shelton, IV et al.	N/A	N/A
2022/0071630	12/2021	Swayze et al.	N/A	N/A
2022/0071631	12/2021	Harris et al.	N/A	N/A
2022/0071632	12/2021	Patel et al.	N/A	N/A
2022/0071635	12/2021	Shelton, IV et al.	N/A	N/A
2022/0079580	12/2021	Vendely et al.	N/A	N/A
2022/0079586	12/2021	Shelton, IV et al.	N/A	N/A
2022/0079588	12/2021	Harris et al.	N/A	N/A
2022/0079589	12/2021	Harris et al.	N/A	N/A
2022/0079590	12/2021	Harris et al.	N/A	N/A
2022/0079595	12/2021	Huitema et al.	N/A	N/A
2022/0079596	12/2021	Huitema et al.	N/A	N/A
2022/0104695	12/2021	Russell	N/A	N/A
2022/0104814	12/2021	Shelton, IV et al.	N/A	N/A
2022/0104816	12/2021	Fernandes et al.	N/A	N/A
2022/0104820	12/2021	Shelton, IV et al.	N/A	N/A
2022/0110673	12/2021	Boronyak et al.	N/A	N/A
2022/0117602	12/2021	Wise et al.	N/A	N/A
2022/0125472	12/2021	Beckman et al.	N/A	N/A
2022/0133299	12/2021	Baxter, III	N/A	N/A
2022/0133300	12/2021	Leimbach et al.	N/A	N/A
2022/0133302	12/2021	Zerkle et al.	N/A	N/A
2022/0133303	12/2021	Huang	N/A	N/A
2022/0133304	12/2021	Leimbach et al.	N/A	N/A

2022/0133310	12/2021	Ross	N/A	N/A
2022/0133312	12/2021	Huang	N/A	N/A
2022/0133318	12/2021	Hudson et al.	N/A	N/A
2022/0142643	12/2021	Shelton, IV et al.	N/A	N/A
2022/0151611	12/2021	Shelton, IV et al.	N/A	N/A
2022/0151613	12/2021	Vendely et al.	N/A	N/A
2022/0151614	12/2021	Vendely et al.	N/A	N/A
2022/0151615	12/2021	Shelton, IV et al.	N/A	N/A
2022/0151616	12/2021	Shelton, IV et al.	N/A	N/A
2022/0160355	12/2021	Harris et al.	N/A	N/A
2022/0160358	12/2021	Wixey	N/A	N/A
2022/0167968	12/2021	Worthington et al.	N/A	N/A
2022/0167970	12/2021	Aronhalt et al.	N/A	N/A
2022/0167972	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167973	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167975	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167977	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167979	12/2021	Yates et al.	N/A	N/A
2022/0167981	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167982	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167983	12/2021	Shelton, IV et al.	N/A	N/A
2022/0167995	12/2021	Parfett et al.	N/A	N/A
2022/0168038	12/2021	Shelton, IV et al.	N/A	N/A
2022/0175370	12/2021	Shelton, IV et al.	N/A	N/A
2022/0175371	12/2021	Hess et al.	N/A	N/A
2022/0175372	12/2021	Shelton, IV et al.	N/A	N/A
2022/0175375	12/2021	Harris et al.	N/A	N/A
2022/0175378	12/2021	Leimbach et al.	N/A	N/A
2022/0175381	12/2021	Scheib et al.	N/A	N/A
2022/0183685	12/2021	Shelton, IV et al.	N/A	N/A
2022/0183687	12/2021	Wixey et al.	N/A	N/A
2022/0202487	12/2021	Shelton, IV et al.	N/A	N/A
2022/0211367	12/2021	Schmid et al.	N/A	N/A
2022/0218332	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218336	12/2021	Timm et al.	N/A	N/A
2022/0218338	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218344	12/2021	Leimbach et al.	N/A	N/A
2022/0218345	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218346	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218347	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218348	12/2021	Swensgard et al.	N/A	N/A
2022/0218349	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218350	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218351	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218376	12/2021	Shelton, IV et al.	N/A	N/A
2022/0218381	12/2021	Leimbach et al.	N/A	N/A
2022/0218382	12/2021	Leimbach et al.	N/A	N/A
2022/0225986	12/2021	Shelton, IV et al.	N/A	N/A
2022/0225993	12/2021	Huitema et al.	N/A	N/A
2022/0225994	12/2021	Setser et al.	N/A	N/A

2022/0226013	12/2021	Hall et al.	N/A	N/A
2022/0233186	12/2021	Timm et al.	N/A	N/A
2022/0233188	12/2021	Timm et al.	N/A	N/A
2022/0233194	12/2021	Baxter, III et al.	N/A	N/A
2022/0233195	12/2021	Shelton, IV et al.	N/A	N/A
2022/0233257	12/2021	Shelton, IV et al.	N/A	N/A
2022/0240928	12/2021	Timm et al.	N/A	N/A
2022/0240929	12/2021	Timm et al.	N/A	N/A
2022/0240930	12/2021	Yates et al.	N/A	N/A
2022/0240936	12/2021	Huitema et al.	N/A	N/A
2022/0240937	12/2021	Shelton, IV et al.	N/A	N/A
2022/0249095	12/2021	Shelton, IV et al.	N/A	N/A
2022/0265272	12/2021	Li et al.	N/A	N/A
2022/0273291	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273292	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273293	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273294	12/2021	Creamer et al.	N/A	N/A
2022/0273300	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273301	12/2021	Creamer et al.	N/A	N/A
2022/0273303	12/2021	Creamer et al.	N/A	N/A
2022/0273304	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273306	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273307	12/2021	Shelton, IV et al.	N/A	N/A
2022/0273308	12/2021	Shelton, IV et al.	N/A	N/A
2022/0278438	12/2021	Shelton, IV et al.	N/A	N/A
2022/0287711	12/2021	Ming et al.	N/A	N/A
2022/0296230	12/2021	Adams et al.	N/A	N/A
2022/0296231	12/2021	Adams et al.	N/A	N/A
2022/0296232	12/2021	Adams et al.	N/A	N/A
2022/0296233	12/2021	Morgan et al.	N/A	N/A
2022/0296234	12/2021	Shelton, IV et al.	N/A	N/A
2022/0296237	12/2021	Bakos et al.	N/A	N/A
2022/0304679	12/2021	Bakos et al.	N/A	N/A
2022/0304680	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304681	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304682	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304683	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304684	12/2021	Bakos et al.	N/A	N/A
2022/0304685	12/2021	Bakos et al.	N/A	N/A
2022/0304686	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304687	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304688	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304689	12/2021	Shelton, IV	N/A	N/A
2022/0304690	12/2021	Baxter, III et al.	N/A	N/A
2022/0304714	12/2021	Shelton, IV et al.	N/A	N/A
2022/0304715	12/2021	Shelton, IV	N/A	N/A
2022/0313253	12/2021	Shelton, IV et al.	N/A	N/A
2022/0313263	12/2021	Huitema et al.	N/A	N/A
2022/0313619	12/2021	Schmid et al.	N/A	N/A
2022/0323067	12/2021	Overmyer et al.	N/A	N/A

2022/0323070	12/2021	Ross et al.	N/A	N/A
2022/0330940	12/2021	Shelton, IV et al.	N/A	N/A
2022/0338870	12/2021	Swayze et al.	N/A	N/A
2022/0346774	12/2021	Hess et al.	N/A	N/A
2022/0346775	12/2021	Hess et al.	N/A	N/A
2022/0346776	12/2021	Aronhalt et al.	N/A	N/A
2022/0346781	12/2021	Shelton, IV et al.	N/A	N/A
2022/0346783	12/2021	Shelton, IV et al.	N/A	N/A
2022/0346785	12/2021	Aronhalt et al.	N/A	N/A
2022/0354492	12/2021	Baril	N/A	N/A
2022/0354493	12/2021	Shelton, IV et al.	N/A	N/A
2022/0354495	12/2021	Baxter, III et al.	N/A	N/A
2022/0361879	12/2021	Baxter, III et al.	N/A	N/A
2022/0370069	12/2021	Simms et al.	N/A	N/A
2022/0378418	12/2021	Huang et al.	N/A	N/A
2022/0378424	12/2021	Huang et al.	N/A	N/A
2022/0378425	12/2021	Huang et al.	N/A	N/A
2022/0378426	12/2021	Huang et al.	N/A	N/A
2022/0378427	12/2021	Huang et al.	N/A	N/A
2022/0378428	12/2021	Shelton, IV et al.	N/A	N/A
2022/0378435	12/2021	Dholakia et al.	N/A	N/A
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2022/0387031	12/2021	Yates et al.	N/A	N/A
2022/0387032	12/2021	Huitema et al.	N/A	N/A
2022/0387033	12/2021	Huitema et al.	N/A	N/A
2022/0387034	12/2021	Huitema et al.	N/A	N/A
2022/0387035	12/2021	Huitema et al.	N/A	N/A
2022/0387036	12/2021	Huitema et al.	N/A	N/A
2022/0387037	12/2021	Huitema et al.	N/A	N/A
2022/0387038	12/2021	Huitema et al.	N/A	N/A
2022/0387125	12/2021	Leimbach et al.	N/A	N/A
2023/0016171	12/2022	Yates et al.	N/A	N/A
2023/0018950	12/2022	Shelton, IV et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/580,493, entitled LOCAL DISPLAY OF TISSUE PARAMETER STABILIZATION, filed Sep. 24, 2019, which issued on May 23, 2023 as U.S. Pat. No. 11,653,918, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/459,531, entitled POWERED MEDICAL DEVICE INCLUDING MEASUREMENT OF CLOSURE STATE OF JAWS, filed Mar. 15, 2017, which issued on Apr. 26, 2022 as U.S. Pat. No. 11,311,294, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 14/479,108, entitled LOCAL DISPLAY OF TISSUE PARAMETER STABILIZATION, filed Sep. 5, 2014, now U.S. Patent Application Publication No. 2016/0066913, the entire disclosures of which are hereby incorporated by reference herein. (2) This application is also related to U.S. patent application Ser. No. 14/479,103 entitled CIRCUITRY AND SENSORS FOR POWERED MEDICAL DEVICE, now U.S. Pat. No. 10,111,679, 14/479,119 entitled ADJUNCT WITH INTEGRATED SENSORS TO QUANTIFY TISSUE COMPRESSION, now U.S. Pat. No. 9,724,094, Ser. No. 14/478,908 entitled MONITORING DEVICE DEGRADATION BASED ON COMPONENT EVALUATION, now U.S. Pat. No. 9,737,301, Ser. No. 14/478,895 entitled MULTIPLE SENSORS WITH ONE SENSOR AFFECTING A SECOND SENSOR'S OUTPUT OR INTERPRETATION, now U.S. Pat. No. 9,757,128, Ser. No. 14/479,110 entitled POLARITY OF HALL MAGNET TO DETECT MISLOADED CARTRIDGE, now U.S. patent Ser. No. 14/479,098 entitled SMART CARTRIDGE WAKE UP OPERATION AND DATA RETENTION, now U.S. Pat. No. 10,135,242, and Ser. No. 14/479,115 entitled MULTIPLE MOTOR CONTROL FOR POWERED MEDICAL DEVICE, now U.S. Pat. No. 9,788,836, each of which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The present embodiments of the invention relate to surgical instruments and, in various circumstances, to surgical stapling and cutting instruments and staple cartridges therefor that are designed to staple and cut tissue.

SUMMARY

(2) In one embodiment, a staple cartridge for use with a surgical stapler is provided. The staple cartridge comprises a cartridge body having a tissue-contacting surface; one or more LEDs

positioned at the edges of the tissue-contacting surface; and a plurality of staple drivers within the cartridge body each supporting a staple. In one embodiment the one or more LEDs emit ultraviolet light or infrared light. In one embodiment, the staples are coated in a fluorescing dye.

(3) In one embodiment, a surgical stapling system is provided. The surgical stapling system comprises an elongated shaft assembly configured to transmit actuation motions from an actuator; and an end effector for compressing and stapling tissue, the end effector operably coupled to the elongated shaft, the end effector comprising: an elongated channel; an anvil having a staple forming surface thereon, the anvil moveable relative to the elongated channel between an open position and a closed position; and a staple cartridge removably positioned within the elongated channel, the staple cartridge comprising: a cartridge body having a tissue-contacting surface in a confronting relationship with the anvil's staple forming surface when the anvil is in the closed position; one or more LEDs positioned to be visible when the anvil is in the closed position; and a plurality of staple drivers within the cartridge body each supporting a staple. In one embodiment, the one or more LEDs indicate that the compressed tissue is stable. In one embodiment, at least one LED is visible from either the left or the right side of the end effector, the at least one LED configured to flash at the rate of the tissue's stabilization and further configured to maintain a lit condition when the tissue is stable. In one embodiment, more than one LED is visible from either the left or the right side of the end effector. In one embodiment, the more than one LEDs light in sequence at the rate of the tissue's stabilization, such that when all the LEDs indicates that the tissue is stable. In one embodiment, the more than one LEDs light in sequence, the speed of the sequence indicating the rate of the tissue's stabilization, and wherein all the LEDs flash simultaneously when the tissue is stable. In one embodiment, the one or more LEDs indicate which portions of the end effector are in sufficient contact with the tissue.

(4) In one embodiment, the system further comprises a processor, the processor configured to compare the tissue's parameters against the acceptable parameters for all staple cartridges. In one embodiment, the processor is further configured to compare the tissue's parameters against the acceptable parameters for the staple cartridge currently present in the elongated channel. In one embodiment, the LEDs indicate that the staple cartridge is appropriate for the tissue. In one embodiment, the one or more LEDs indicate that the staple cartridge is not compatible with the stapling system. In one embodiment, the one or more LEDs indicate that the end effector is enclosing more tissue than is suitable for the staple cartridge. In one embodiment, a surgical stapling system is provided. The surgical stapling system comprises: an elongated shaft assembly configured to transmit actuation motions from an actuator; and an end effector for compressing and stapling tissue, the end effector operably coupled to the elongated shaft, the end effector comprising: an elongated channel; an anvil having a staple forming surface thereon, the anvil moveable relative to the elongated channel between an open position and a closed position; and a staple cartridge removably positioned within the elongated channel, the staple cartridge comprising: a cartridge body having an upper surface in a confronting relationship with the anvil's staple forming surface when the anvil is in a closed position; one or more LEDs positioned to provide illumination to the area between the anvil and the staple cartridge when the anvil is in a closed position; and a plurality of staple drivers within the cartridge body each supporting a staple. In one embodiment, the one or more LEDs emit ultraviolet light or infrared light. In one embodiment, the staples are coated in a fluorescing dye.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and advantages of the various embodiments of the invention, and the manner of attaining them, will become more apparent and the embodiment of the invention itself will be better

understood by reference to the following description of embodiments of the embodiment of the invention taken in conjunction with the accompanying drawings, wherein:

- (2) FIG. 1 is a perspective view of a surgical instrument that has an interchangeable shaft assembly operably coupled thereto;
- (3) FIG. 2 is an exploded assembly view of the interchangeable shaft assembly and surgical instrument of FIG. 1;
- (4) FIG. 3 is another exploded assembly view showing portions of the interchangeable shaft assembly and surgical instrument of FIGS. 1 and 2;
- (5) FIG. 4 is an exploded assembly view of a portion of the surgical instrument of FIGS. 1-3;
- (6) FIG. 5 is a cross-sectional side view of a portion of the surgical instrument of FIG. 4 with the firing trigger in a fully actuated position;
- (7) FIG. 6 is another cross-sectional view of a portion of the surgical instrument of FIG. 5 with the firing trigger in an unactuated position;
- (8) FIG. 7 is an exploded assembly view of one form of an interchangeable shaft assembly;
- (9) FIG. 8 is another exploded assembly view of portions of the interchangeable shaft assembly of FIG. 7;
- (10) FIG. 9 is another exploded assembly view of portions of the interchangeable shaft assembly of FIGS. 7 and 8;
- (11) FIG. 10 is a cross-sectional view of a portion of the interchangeable shaft assembly of FIGS. 7-9;
- (12) FIG. 11 is a perspective view of a portion of the shaft assembly of FIGS. 7-10 with the switch drum omitted for clarity;
- (13) FIG. 12 is another perspective view of the portion of the interchangeable shaft assembly of FIG. 11 with the switch drum mounted thereon;
- (14) FIG. 13 is a perspective view of a portion of the interchangeable shaft assembly of FIG. 11 operably coupled to a portion of the surgical instrument of FIG. 1 illustrated with the closure trigger thereof in an unactuated position;
- (15) FIG. 14 is a right side elevational view of the interchangeable shaft assembly and surgical instrument of FIG. 13;
- (16) FIG. 15 is a left side elevational view of the interchangeable shaft assembly and surgical instrument of FIGS. 13 and 14;
- (17) FIG. 16 is a perspective view of a portion of the interchangeable shaft assembly of FIG. 11 operably coupled to a portion of the surgical instrument of FIG. 1 illustrated with the closure trigger thereof in an actuated position and a firing trigger thereof in an unactuated position;
- (18) FIG. 17 is a right side elevational view of the interchangeable shaft assembly and surgical instrument of FIG. 16;
- (19) FIG. 18 is a left side elevational view of the interchangeable shaft assembly and surgical instrument of FIGS. 16 and 17;
- (20) FIG. 18A is a right side elevational view of the interchangeable shaft assembly of FIG. 11 operably coupled to a portion of the surgical instrument of FIG. 1 illustrated with the closure trigger thereof in an actuated position and the firing trigger thereof in an actuated position;
- (21) FIG. 19 is a schematic of a system for powering down an electrical connector of a surgical instrument handle when a shaft assembly is not coupled thereto;
- (22) FIG. 20 is an exploded view of one embodiment of an end effector of the surgical instrument of FIG. 1;
- (23) FIGS. 21A-21B is a circuit diagram of the surgical instrument of FIG. 1 spanning two drawings sheets;
- (24) FIG. 22 illustrates one instance of a power assembly comprising a usage cycle circuit configured to generate a usage cycle count of the battery back;
- (25) FIG. 23 illustrates one embodiment of a process for sequentially energizing a segmented

circuit;

(26) FIG. 24 illustrates one embodiment of a power segment comprising a plurality of daisy chained power converters;

(27) FIG. 25 illustrates one embodiment of a segmented circuit configured to maximize power available for critical and/or power intense functions;

(28) FIG. 26 illustrates one embodiment of a power system comprising a plurality of daisy chained power converters configured to be sequentially energized;

(29) FIG. 27 illustrates one embodiment of a segmented circuit comprising an isolated control section;

(30) FIG. 28 illustrates one embodiment of an end effector comprising a first sensor and a second sensor;

(31) FIG. 29 is a logic diagram illustrating one embodiment of a process for adjusting the measurement of the first sensor based on input from the second sensor of the end effector illustrated in FIG. 28;

(32) FIG. 30 is a logic diagram illustrating one embodiment of a process for determining a look-up table for a first sensor based on the input from a second sensor;

(33) FIG. 31 is a logic diagram illustrating one embodiment of a process for calibrating a first sensor in response to an input from a second sensor;

(34) FIG. 32A is a logic diagram illustrating one embodiment of a process for determining and displaying the thickness of a tissue section clamped between an anvil and a staple cartridge of an end effector;

(35) FIG. 32B is a logic diagram illustrating one embodiment of a process for determining and displaying the thickness of a tissue section clamped between the anvil and the staple cartridge of the end effector;

(36) FIG. 33 is a graph illustrating an adjusted Hall effect thickness measurement compared to an unmodified Hall effect thickness measurement;

(37) FIG. 34 illustrates one embodiment of an end effector comprising a first sensor and a second sensor;

(38) FIG. 35 illustrates one embodiment of an end effector comprising a first sensor and a plurality of second sensors;

(39) FIG. 36 is a logic diagram illustrating one embodiment of a process for adjusting a measurement of a first sensor in response to a plurality of secondary sensors;

(40) FIG. 37 illustrates one embodiment of a circuit configured to convert signals from a first sensor and a plurality of secondary sensors into digital signals receivable by a processor;

(41) FIG. 38 illustrates one embodiment of an end effector comprising a plurality of sensors;

(42) FIG. 39 is a logic diagram illustrating one embodiment of a process for determining one or more tissue properties based on a plurality of sensors;

(43) FIG. 40 illustrates one embodiment of an end effector comprising a plurality of sensors coupled to a second jaw member;

(44) FIG. 41 illustrates one embodiment of a staple cartridge comprising a plurality of sensors formed integrally therein;

(45) FIG. 42 is a logic diagram illustrating one embodiment of a process for determining one or more parameters of a tissue section clamped within an end effector;

(46) FIG. 43 illustrates one embodiment of an end effector comprising a plurality of redundant sensors;

(47) FIG. 44 is a logic diagram illustrating one embodiment of a process for selecting the most reliable output from a plurality of redundant sensors;

(48) FIG. 45 illustrates one embodiment of an end effector comprising a sensor comprising a specific sampling rate to limit or eliminate false signals;

(49) FIG. 46 is a logic diagram illustrating one embodiment of a process for generating a thickness

measurement for a tissue section located between an anvil and a staple cartridge of an end effector;

(50) FIG. **47** illustrates one embodiment of a circular stapler;

(51) FIGS. **48A-48D** illustrate a clamping process of the circular stapler illustrated in FIG. **47**, where FIG. **48A** illustrates the circular stapler in an initial position with the anvil and the body in a closed configuration, FIG. **48B** illustrates that the anvil is moved distally to disengage with the body and create a gap configured to receive a tissue section therein, once the circular stapler **3400** is positioned, FIG. **48C** illustrates the tissue section compressed to a predetermined compression between the anvil and the body, and FIG. **48D** illustrates the circular stapler in position corresponding to staple deployment;

(52) FIG. **49** illustrates one embodiment of a circular staple anvil and an electrical connector configured to interface therewith;

(53) FIG. **50** illustrates one embodiment of a surgical instrument comprising a sensor coupled to a drive shaft of the surgical instrument;

(54) FIG. **51** is a flow chart illustrating one embodiment of a process for determining uneven tissue loading in an end effector;

(55) FIG. **52** illustrates one embodiment of an end effector configured to determine one or more parameters of a tissue section during a clamping operation;

(56) FIGS. **53A** and **53B** illustrate an embodiment of an end effector configured to normalize a Hall effect voltage irrespective of a deck height of a staple cartridge;

(57) FIG. **54** is a logic diagram illustrating one embodiment of a process for determining when the compression of tissue within an end effector, such as, for example, the end effector illustrated in FIGS. **53A-53B**, has reached a steady state;

(58) FIG. **55** is a graph illustrating various Hall effect sensor readings;

(59) FIG. **56** is a logic diagram illustrating one embodiment of a process for determining when the compression of tissue within an end effector, such as, for example, the end effector illustrated in FIGS. **53A-53B**, has reached a steady state;

(60) FIG. **57** is a logic diagram illustrating one embodiment of a process for controlling an end effector to improve proper staple formation during deployment;

(61) FIG. **58** is a logic diagram illustrating one embodiment of a process for controlling an end effector to allow for fluid evacuation and provide improved staple formation;

(62) FIGS. **59A-59B** illustrate one embodiment of an end effector comprising a pressure sensor;

(63) FIG. **60** illustrates one embodiment of an end effector comprising a second sensor located between a staple cartridge and a second jaw member;

(64) FIG. **61** is a logic diagram illustrating one embodiment of a process for determining and displaying the thickness of a tissue section clamped in an end effector, according to FIGS. **59A-59B** or FIG. **60**;

(65) FIG. **62** illustrates one embodiment of an end effector comprising a plurality of second sensors located between a staple cartridge and an elongated channel;

(66) FIGS. **63A** and **63B** further illustrate the effect of a full versus partial bite of tissue;

(67) FIG. **64** illustrates one embodiment of an end effector comprising a coil and oscillator circuit for measuring the gap between the anvil and the staple cartridge;

(68) FIG. **65** illustrates and alternate view of the end effector. As illustrated, in some embodiments external wiring may supply power to the oscillator circuit;

(69) FIG. **66** illustrates examples of the operation of a coil to detect eddy currents in a target;

(70) FIG. **67** illustrates a graph of a measured quality factor, the measured inductance, and measure resistance of the radius of a coil as a function of the coil's standoff to a target;

(71) FIG. **68** illustrates one embodiment of an end effector comprising an emitter and sensor placed between the staple cartridge and the elongated channel;

(72) FIG. **69** illustrates an embodiment of an emitter and sensor in operation;

(73) FIG. **70** illustrates the surface of an embodiment of an emitter and sensor comprising a MEMS

transducer;

(74) FIG. **71** illustrates a graph of an example of the reflected signal that may be measured by the emitter and sensor of FIG. **69**;

(75) FIG. **72** illustrates an embodiment of an end effector that is configured to determine the location of a cutting member or knife;

(76) FIG. **73** illustrates an example of the code strip in operation with red LEDs and an infrared LEDs;

(77) FIG. **74** illustrates a partial perspective view of an end effector of a surgical instrument comprising a staple cartridge according to various embodiments described herein;

(78) FIG. **75** illustrates a elevational view of a portion of the end effector of FIG. **74** according to various embodiments described herein;

(79) FIG. **76** illustrates a logic diagram of a module of the surgical instrument of FIG. **74** according to various embodiments described herein;

(80) FIG. **77** illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. **74** according to various embodiments described herein;

(81) FIG. **78** illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. **74** according to various embodiments described herein;

(82) FIG. **79** illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. **74** according to various embodiments described herein;

(83) FIG. **80** illustrates a partial view of a cutting edge, optical sensors, and light sources of the surgical instrument of FIG. **74** according to various embodiments described herein;

(84) FIG. **81** illustrates a partial view of a cutting edge, an optical sensor, and a light source of the surgical instrument of FIG. **74** according to various embodiments described herein;

(85) FIG. **82** illustrates a partial view of a cutting edge between cleaning blades of the surgical instrument of FIG. **74** according to various embodiments described herein;

(86) FIG. **83** illustrates a partial view of a cutting edge between cleaning sponges of the surgical instrument of FIG. **74** according to various embodiments described herein;

(87) FIG. **84** illustrates a perspective view of a staple cartridge including a sharpness testing member according to various embodiments described herein;

(88) FIG. **85** illustrates a logic diagram of a module of a surgical instrument according to various embodiments described herein;

(89) FIG. **86** illustrates a logic diagram of a module of a surgical instrument according to various embodiments described herein;

(90) FIG. **87** illustrates a logic diagram outlining a method for evaluating sharpness of a cutting edge of a surgical instrument according to various embodiments described herein;

(91) FIG. **88** illustrates a chart of the forces applied against a cutting edge of a surgical instrument by the sharpness testing member of FIG. **84** at various sharpness levels according to various embodiments described herein;

(92) FIG. **89** illustrates a flow chart outlining a method for determining whether a cutting edge of a surgical instrument is sufficiently sharp to transect tissue captured by the surgical instrument according to various embodiments described herein; and

(93) FIG. **90** illustrates a table showing predefined tissue thicknesses and corresponding predefined threshold forces according to various embodiments described herein.

(94) FIG. **91** illustrates a perspective view of a surgical instrument including a handle, a shaft assembly, and an end effector;

(95) FIG. **92** illustrates a logic diagram of a common control module for use with a plurality of motors of the surgical instrument of FIG. **91**;

(96) FIG. **93** illustrates a partial elevational view of the handle of the surgical instrument of FIG. **91** with a removed outer casing;

(97) FIG. **94** illustrates a partial elevational view of the surgical instrument of FIG. **91** with a

removed outer casing.

(98) FIG. **95A** illustrates a side angle view of an end effector with the anvil in a closed position, illustrating one located on either side of the cartridge deck;

(99) FIG. **95B** illustrates a three-quarter angle view of the end effector with the anvil in an open position, and one LED located on either side of the cartridge deck;

(100) FIG. **96A** illustrates a side angle view of an end effector with the anvil in a closed position and a plurality of LEDs located on either side of the cartridge deck;

(101) FIG. **96B** illustrates a three-quarter angle view of the end effector with the anvil in an open position, and a plurality of LEDs located on either side of the cartridge deck;

(102) FIG. **97A** illustrates a side angle view of an end effector with the anvil in a closed position, and a plurality of LEDs from the proximal to the distal end of the staple cartridge, on either side of the cartridge deck; and

(103) FIG. **97B** illustrates a three-quarter angle view of the end effector with the anvil in an open position, illustrating a plurality of LEDs from the proximal to the distal end of the staple cartridge, and on either side of the cartridge deck.

(104) FIG. **98A** illustrates an embodiment wherein the tissue compensator is removably attached to the anvil portion of the end effector;

(105) FIG. **98B** illustrates a detail view of a portion of the tissue compensator shown in FIG. **98A**;

(106) FIG. **99** illustrates various example embodiments that use the layer of conductive elements and conductive elements in the staple cartridge to detect the distance between the anvil and the upper surface of the staple cartridge;

(107) FIGS. **100A** and **100B** illustrate an embodiment of the tissue compensator comprising a layer of conductive elements in operation;

(108) FIGS. **101A** and **101B** illustrate an embodiment of an end effector comprising a tissue compensator further comprising conductors embedded within;

(109) FIGS. **102A** and **102B** illustrate an embodiment of an end effector comprising a tissue compensator further comprising conductors embedded therein;

(110) FIG. **103** illustrates an embodiment of a staple cartridge and a tissue compensator wherein the staple cartridge provides power to the conductive elements that comprise the tissue compensator;

(111) FIGS. **104A** and **104B** illustrate an embodiment of a staple cartridge and a tissue compensator wherein the staple cartridge provides power to the conductive elements that comprise the tissue compensator;

(112) FIGS. **105A** and **105B** illustrate an embodiment of an end effector comprising position sensing elements and a tissue compensator;

(113) FIGS. **106A** and **106B** illustrate an embodiment of an end effector comprising position sensing elements and a tissue compensator;

(114) FIGS. **107A** and **107B** illustrate an embodiment of a staple cartridge and a tissue compensator that is operable to indicate the position of a cutting member or knife bar;

(115) FIG. **108** illustrates one embodiment of an end effector comprising a magnet and a Hall effect sensor wherein the detected magnetic field can be used to identify a staple cartridge;

(116) FIG. **109** illustrates one embodiment of an end effector comprising a magnet and a Hall effect sensor wherein the detected magnetic field can be used to identify a staple cartridge;

(117) FIG. **110** illustrates a graph of the voltage detected by a Hall effect sensor located in the distal tip of a staple cartridge, such as is illustrated in FIGS. **108** and **109**, in response to the distance or gap between a magnet located in the anvil and the Hall effect sensor in the staple cartridge, such as illustrated in FIGS. **108** and **109**;

(118) FIG. **111** illustrates one embodiment of the housing of the surgical instrument, comprising a display;

(119) FIG. **112** illustrates one embodiment of a staple retainer comprising a magnet;

(120) FIGS. **113A** and **113B** illustrate one embodiment of an end effector comprising a sensor for identifying staple cartridges of different types;

(121) FIG. **114** is a partial view of an end effector with sensor power conductors for transferring power and data signals between the connected components of the surgical instrument according to one embodiment.

(122) FIG. **115** is a partial view of the end effector shown in FIG. **114** showing sensors and/or electronic components located in an end effector.

(123) FIG. **116** is a block diagram of a surgical instrument electronic subsystem comprising a short circuit protection circuit for the sensors and/or electronic components according to one embodiment.

(124) FIG. **117** is a short circuit protection circuit comprising a supplementary power supply circuit **7014** coupled to a main power supply circuit, according to one embodiment.

(125) FIG. **118** is a block diagram of a surgical instrument electronic subsystem comprising a sample rate monitor to provide power reduction by limiting sample rates and/or duty cycle of the sensor components when the surgical instrument is in a non-sensing state, according to one embodiment.

(126) FIG. **119** is a block diagram of a surgical instrument electronic subsystem comprising an over current/voltage protection circuit for sensors and/or electronic components of a surgical instrument, according to one embodiment.

(127) FIG. **120** is an over current/voltage protection circuit for sensors and electronic components for a surgical instrument, according to one embodiment.

(128) FIG. **121** is a block diagram of a surgical instrument electronic subsystem with a reverse polarity protection circuit for sensors and/or electronic components according to one embodiment.

(129) FIG. **122** is a reverse polarity protection circuit for sensors and/or electronic components for a surgical instrument according to one embodiment.

(130) FIG. **123** is a block diagram of a surgical instrument electronic subsystem with power reduction utilizing a sleep mode monitor for sensors and/or electronic components according to one embodiment.

(131) FIG. **124** is a block diagram of a surgical instrument electronic subsystem comprising a temporary power loss circuit to provide protection against intermittent power loss for sensors and/or electronic components in modular surgical instruments.

(132) FIG. **125** illustrates one embodiment of a temporary power loss circuit implemented as a hardware circuit.

(133) FIG. **126A** illustrates a perspective view of one embodiment of an end effector comprising a magnet and a Hall effect sensor in communication with a processor;

(134) FIG. **126B** illustrates a sideways cross-sectional view of one embodiment of an end effector comprising a magnet and a Hall effect sensor in communication with processor;

(135) FIG. **127** illustrates one embodiment of the operable dimensions that relate to the operation of the Hall effect sensor;

(136) FIG. **128A** illustrates an external side view of an embodiment of a staple cartridge;

(137) FIG. **128B** illustrates various dimensions possible between the lower surface of the push-off lug and the top of the Hall effect sensor;

(138) FIG. **128C** illustrates an external side view of an embodiment of a staple cartridge;

(139) FIG. **128D** illustrates various dimensions possible between the lower surface of the push-off lug and the upper surface of the staple cartridge above the Hall effect sensor;

(140) FIG. **129A** further illustrates a front-end cross-sectional view **10054** of the anvil **10002** and the central axis point of the anvil;

(141) FIG. **129B** is a cross sectional view of a magnet shown in FIG. **129A**;

(142) FIGS. **130A-130E** illustrate one embodiment of an end effector that comprises a magnet where FIG. **130A** illustrates a front-end cross-sectional view of the end effector, FIG. **130B**

illustrates a front-end cutaway view of the anvil and the magnet in situ, FIG. 130C illustrates a perspective cutaway view of the anvil and the magnet, FIG. 130D illustrates a side cutaway view of the anvil and the magnet, and FIG. 130E illustrates a top cutaway view of the anvil and the magnet; (143) FIGS. 131A-131E illustrate another embodiment of an end effector that comprises a magnet where FIG. 131A illustrates a front-end cross-sectional view of the end effector, FIG. 131B illustrates a front-end cutaway view of the anvil and the magnet, in situ, FIG. 131C illustrates a perspective cutaway view of the anvil and the magnet, FIG. 131D illustrates a side cutaway view of the anvil and the magnet, and FIG. 131E illustrates a top cutaway view of the anvil and magnet; (144) FIG. 132 illustrates contact points between the anvil and either the staple cartridge and/or the elongated channel;

(145) FIGS. 133A and 133B illustrate one embodiment of an end effector that is operable to use conductive surfaces at the distal contact point to create an electrical connection;

(146) FIGS. 134A-134C illustrate one embodiment of an end effector that is operable to use conductive surfaces to form an electrical connection where FIG. 134A illustrates an end effector comprising an anvil, an elongated channel, and a staple cartridge, FIG. 134B illustrates the inside surface of the anvil further comprising first conductive surfaces located distally from the staple-forming indents, and FIG. 134C illustrates the staple cartridge comprising a cartridge body and first conductive surfaces located such that they can come into contact with a second conductive surface located on the staple cartridge;

(147) FIGS. 135A and 135B illustrate one embodiment of an end effector that is operable to use conductive surfaces to form an electrical connection where FIG. 135A illustrates an end effector comprising an anvil, an elongated channel, and a staple cartridge and FIGS. 109B 135B is a close-up view of the staple cartridge illustrating the first conductive surface located such that it can come into contact with second conductive surfaces;

(148) FIGS. 136A and 136B illustrate one embodiment of an end effector that is operable to use conductive surfaces to form an electrical connection where FIG. 136A illustrates an end effector comprising an anvil, an elongated channel, and a staple cartridge and FIG. 136B is a close-up view of the staple cartridge illustrating the anvil further comprising a magnet and an inside surface, which further comprises a number of staple-forming indents;

(149) FIGS. 137A-137C illustrate one embodiment of an end effector that is operable to use the proximal contact point to form an electrical connection where FIG. 137A illustrates the end effector, which comprises an anvil, an elongated channel, and a staple cartridge, FIG. 137B is a close-up view of a pin as it rests within an aperture defined in the elongated channel for that purpose, and FIG. 137C illustrates an alternate embodiment, with an alternate location for a second conductive surface on the surface of the aperture;

(150) FIG. 138 illustrates one embodiment of an end effector with a distal sensor plug;

(151) FIG. 139A illustrates the end effector shown in FIG. 138 with the anvil in an open position;

(152) FIG. 139B illustrates a cross-sectional view of the end effector shown in FIG. 139A with the anvil in an open position;

(153) FIG. 139C illustrates the end effector shown in FIG. 138 with the anvil in a closed position;

(154) FIG. 139D illustrates a cross sectional view of the end effector shown in FIG. 139C with the anvil in a closed position;

(155) FIG. 140 provides a close-up view of the cross section of the distal end of the end effector;

(156) FIG. 141 illustrates a close-up top view of the staple cartridge that comprises a distal sensor plug;

(157) FIG. 142A is a perspective view of the underside of a staple cartridge that comprises a distal sensor plug;

(158) FIG. 142B illustrates a cross sectional view of the distal end of the staple cartridge;

(159) FIGS. 143A-143C illustrate one embodiment of a staple cartridge that comprises a flex cable connected to a Hall effect sensor and processor where FIG. 143A is an exploded view of the staple

cartridge, FIG. 143B illustrates the assembly of the staple cartridge and the flex cable in greater detail, and FIG. 143C illustrates a cross sectional view of the staple cartridge to illustrate the placement of the Hall effect sensor, processor, and conductive coupling within the distal end of the staple cartridge, in accordance with the present embodiment;

(160) FIG. 144A-144F illustrate one embodiment of a staple cartridge that comprises a flex cable connected to a Hall effect sensor and a processor where FIG. 144A is an exploded view of the staple cartridge, FIG. 144B illustrates the assembly of the staple cartridge, FIG. 144C illustrates the underside of an assembled staple cartridge, and also illustrates the flex cable in greater detail, FIG. 144D illustrates a cross sectional view of the staple cartridge to illustrate the placement of the Hall effect sensor, processor, and conductive coupling, FIG. 144E illustrates the underside of the staple cartridge without the cartridge tray and including the wedge sled, in its most distal position, and FIG. 144F illustrates the staple cartridge without the cartridge tray in order to illustrate a possible placement for the cable traces;

(161) FIGS. 145A and 145B illustrates one embodiment of a staple cartridge that comprises a flex cable, a Hall effect sensor, and a processor where FIG. 145A is an exploded view of the staple cartridge and FIG. 145B illustrates the assembly of the staple cartridge and the flex cable in greater detail;

(162) FIG. 146A illustrates a perspective view of an end effector coupled to a shaft assembly;

(163) FIG. 146B illustrates a perspective view of an underside of the end effector and shaft assembly shown in FIG. 146A;

(164) FIG. 146C illustrates the end effector shown in FIGS. 146A and 146B with a flex cable and without the shaft assembly;

(165) FIGS. 146D and 146E illustrate an elongated channel portion of the end effector shown in FIGS. 146A and 146B without the anvil or the staple cartridge, to illustrate how the flex cable shown in FIG. 146C can be seated within the elongated channel;

(166) FIG. 146F illustrates the flex cable, shown in FIGS. 146C-120E 146C-146E, alone;

(167) FIG. 147 illustrates a close up view of the elongated channel shown in FIGS. 146D and 146E with a staple cartridge coupled thereto;

(168) FIGS. 148A-148D further illustrate one embodiment of a staple cartridge operative with the present embodiment of an end effector where FIG. 148A illustrates a close up view of the proximal end of the staple cartridge, FIG. 148B illustrates a close-up view of the distal end of the staple cartridge, with a space for a distal sensor plug, FIG. 148C further illustrates the distal sensor plug, and FIG. 148D illustrates the proximal-facing side of the distal sensor plug;

(169) FIGS. 149A and 149B illustrate one embodiment of a distal sensor plug where FIG. 149A illustrates a cutaway view of the distal sensor plug and FIG. 149B further illustrates the Hall effect sensor and the processor operatively coupled to the flex board such that they are capable of communicating;

(170) FIG. 150 illustrates an embodiment of an end effector with a flex cable operable to provide power to sensors and electronics in the distal tip of the anvil portion;

(171) FIGS. 151A-151C illustrate the operation of the articulation joint and flex cable of the end effector where FIG. 151A illustrates a top view of the end effector with the end effector pivoted -45 degrees with respect to the shaft assembly, FIG. 151B illustrates a top view of the end effector, and FIG. 151C illustrates a top view of the end effector with the end effector pivoted +45 degrees with respect to the shaft assembly;

(172) FIG. 152 illustrates cross-sectional view of the distal tip of an embodiment of an anvil with sensors and electronics; and

(173) FIG. 153 illustrates a cutaway view of the distal tip of the anvil.

DESCRIPTION

(174) Certain example embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed

herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those of ordinary skill in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting example embodiments. The features illustrated or described in connection with one example embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present embodiment of the invention.

(175) Reference throughout the specification to “various embodiments,” “some embodiments,” “one embodiment,” or “an embodiment”, or the like, means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in various embodiments,” “in some embodiments,” “in one embodiment”, or “in an embodiment”, or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment may be combined, in whole or in part, with the features structures, or characteristics of one or more other embodiments without limitation. Such modifications and variations are intended to be included within the scope of the present embodiment of the invention.

(176) The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” referring to the portion closest to the clinician and the term “distal” referring to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up,” and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

(177) Various example devices and methods are provided for performing laparoscopic and minimally invasive surgical procedures. However, the person of ordinary skill in the art will readily appreciate that the various methods and devices disclosed herein can be used in numerous surgical procedures and applications including, for example, in connection with open surgical procedures. As the present Detailed Description proceeds, those of ordinary skill in the art will further appreciate that the various instruments disclosed herein can be inserted into a body in any way, such as through a natural orifice, through an incision or puncture hole formed in tissue, etc. The working portions or end effector portions of the instruments can be inserted directly into a patient's body or can be inserted through an access device that has a working channel through which the end effector and elongated shaft of a surgical instrument can be advanced.

(178) FIGS. **1-6** depict a motor-driven surgical cutting and fastening instrument **10** that may or may not be reused. In the illustrated embodiment, the instrument **10** includes a housing **12** that comprises a handle **14** that is configured to be grasped, manipulated and actuated by the clinician. The housing **12** is configured for operable attachment to an interchangeable shaft assembly **200** that has a surgical end effector **300** operably coupled thereto that is configured to perform one or more surgical tasks or procedures. As the present Detailed Description proceeds, it will be understood that the various unique and novel arrangements of the various forms of interchangeable shaft assemblies disclosed herein may also be effectively employed in connection with robotically-controlled surgical systems. Thus, the term “housing” may also encompass a housing or similar portion of a robotic system that houses or otherwise operably supports at least one drive system that is configured to generate and apply at least one control motion which could be used to actuate the interchangeable shaft assemblies disclosed herein and their respective equivalents. The term “frame” may refer to a portion of a handheld surgical instrument. The term “frame” may also represent a portion of a robotically controlled surgical instrument and/or a portion of the robotic system that may be used to operably control a surgical instrument. For example, the interchangeable shaft assemblies disclosed herein may be employed with various robotic systems,

instruments, components and methods disclosed in U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Pat. No. 9,072,535. U.S. patent application Ser. No. 13/118,241, entitled SURGICAL STAPLING INSTRUMENTS WITH ROTATABLE STAPLE DEPLOYMENT ARRANGEMENTS, now U.S. Pat. No. 9,072,535, is incorporated by reference herein in its entirety.

(179) The housing **12** depicted in FIGS. **1-3** is shown in connection with an interchangeable shaft assembly **200** that includes an end effector **300** that comprises a surgical cutting and fastening device that is configured to operably support a surgical staple cartridge **304** therein. The housing **12** may be configured for use in connection with interchangeable shaft assemblies that include end effectors that are adapted to support different sizes and types of staple cartridges, have different shaft lengths, sizes, and types, etc. In addition, the housing **12** may also be effectively employed with a variety of other interchangeable shaft assemblies including those assemblies that are configured to apply other motions and forms of energy such as, for example, radio frequency (RF) energy, ultrasonic energy and/or motion to end effector arrangements adapted for use in connection with various surgical applications and procedures. Furthermore, the end effectors, shaft assemblies, handles, surgical instruments, and/or surgical instrument systems can utilize any suitable fastener, or fasteners, to fasten tissue. For instance, a fastener cartridge comprising a plurality of fasteners removably stored therein can be removably inserted into and/or attached to the end effector of a shaft assembly.

(180) FIG. **1** illustrates the surgical instrument **10** with an interchangeable shaft assembly **200** operably coupled thereto. FIGS. **2** and **3** illustrate attachment of the interchangeable shaft assembly **200** to the housing **12** or handle **14**. As shown in FIG. **4**, the handle **14** may comprise a pair of interconnectable handle housing segments **16** and **18** that may be interconnected by screws, snap features, adhesive, etc. In the illustrated arrangement, the handle housing segments **16, 18** cooperate to form a pistol grip portion **19** that can be gripped and manipulated by the clinician. As will be discussed in further detail below, the handle **14** operably supports a plurality of drive systems therein that are configured to generate and apply various control motions to corresponding portions of the interchangeable shaft assembly that is operably attached thereto.

(181) Referring now to FIG. **4**, the handle **14** may further include a frame **20** that operably supports a plurality of drive systems. For example, the frame **20** can operably support a “first” or closure drive system, generally designated as **30**, which may be employed to apply closing and opening motions to the interchangeable shaft assembly **200** that is operably attached or coupled thereto. In at least one form, the closure drive system **30** may include an actuator in the form of a closure trigger **32** that is pivotally supported by the frame **20**. More specifically, as illustrated in FIG. **4**, the closure trigger **32** is pivotally coupled to the housing **14** by a pin **33**. Such arrangement enables the closure trigger **32** to be manipulated by a clinician such that when the clinician grips the pistol grip portion **19** of the handle **14**, the closure trigger **32** may be easily pivoted from a starting or “unactuated” position to an “actuated” position and more particularly to a fully compressed or fully actuated position. The closure trigger **32** may be biased into the unactuated position by spring or other biasing arrangement (not shown). In various forms, the closure drive system **30** further includes a closure linkage assembly **34** that is pivotally coupled to the closure trigger **32**. As shown in FIG. **4**, the closure linkage assembly **34** may include a first closure link **36** and a second closure link **38** that are pivotally coupled to the closure trigger **32** by a pin **35**. The second closure link **38** may also be referred to herein as an “attachment member” and include a transverse attachment pin **37**.

(182) Still referring to FIG. **4**, it can be observed that the first closure link **36** may have a locking wall or end **39** thereon that is configured to cooperate with a closure release assembly **60** that is pivotally coupled to the frame **20**. In at least one form, the closure release assembly **60** may comprise a release button assembly **62** that has a distally protruding locking pawl **64** formed

thereon. The release button assembly **62** may be pivoted in a counterclockwise direction by a release spring (not shown). As the clinician depresses the closure trigger **32** from its unactuated position towards the pistol grip portion **19** of the handle **14**, the first closure link **36** pivots upward to a point wherein the locking pawl **64** drops into retaining engagement with the locking wall **39** on the first closure link **36** thereby preventing the closure trigger **32** from returning to the unactuated position. See FIG. **18**. Thus, the closure release assembly **60** serves to lock the closure trigger **32** in the fully actuated position. When the clinician desires to unlock the closure trigger **32** to permit it to be biased to the unactuated position, the clinician simply pivots the closure release button assembly **62** such that the locking pawl **64** is moved out of engagement with the locking wall **39** on the first closure link **36**. When the locking pawl **64** has been moved out of engagement with the first closure link **36**, the closure trigger **32** may pivot back to the unactuated position. Other closure trigger locking and release arrangements may also be employed.

(183) Further to the above, FIGS. **13-15** illustrate the closure trigger **32** in its unactuated position which is associated with an open, or unclamped, configuration of the shaft assembly **200** in which tissue can be positioned between the jaws of the shaft assembly **200**. FIGS. **16-18** illustrate the closure trigger **32** in its actuated position which is associated with a closed, or clamped, configuration of the shaft assembly **200** in which tissue is clamped between the jaws of the shaft assembly **200**. Upon comparing FIGS. **14** and **17**, the reader will appreciate that, when the closure trigger **32** is moved from its unactuated position (FIG. **14**) to its actuated position (FIG. **17**), the closure release button **62** is pivoted between a first position (FIG. **14**) and a second position (FIG. **17**). The rotation of the closure release button **62** can be referred to as being an upward rotation; however, at least a portion of the closure release button **62** is being rotated toward the circuit board **100**. Referring to FIG. **4**, the closure release button **62** can include an arm **61** extending therefrom and a magnetic element **63**, such as a permanent magnet, for example, mounted to the arm **61**. When the closure release button **62** is rotated from its first position to its second position, the magnetic element **63** can move toward the circuit board **100**. The circuit board **100** can include at least one sensor configured to detect the movement of the magnetic element **63**. In at least one embodiment, a Hall effect sensor **65**, for example, can be mounted to the bottom surface of the circuit board **100**. The Hall effect sensor **65** can be configured to detect changes in a magnetic field surrounding the Hall effect sensor **65** caused by the movement of the magnetic element **63**. The Hall effect sensor **65** can be in signal communication with a microcontroller **1500** (FIG. **19**), for example, which can determine whether the closure release button **62** is in its first position, which is associated with the unactuated position of the closure trigger **32** and the open configuration of the end effector, its second position, which is associated with the actuated position of the closure trigger **32** and the closed configuration of the end effector, and/or any position between the first position and the second position.

(184) In at least one form, the handle **14** and the frame **20** may operably support another drive system referred to herein as a firing drive system **80** that is configured to apply firing motions to corresponding portions of the interchangeable shaft assembly attached thereto. The firing drive system may **80** also be referred to herein as a “second drive system”. The firing drive system **80** may employ an electric motor **82**, located in the pistol grip portion **19** of the handle **14**. In various forms, the motor **82** may be a DC brushed driving motor having a maximum rotation of, approximately, 25,000 RPM, for example. In other arrangements, the motor may include a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor. The motor **82** may be powered by a power source **90** that in one form may comprise a removable power pack **92**. As shown in FIG. **4**, for example, the power pack **92** may comprise a proximal housing portion **94** that is configured for attachment to a distal housing portion **96**. The proximal housing portion **94** and the distal housing portion **96** are configured to operably support a plurality of batteries **98** therein. Batteries **98** may each comprise, for example, a Lithium Ion (“LI”) or other suitable battery. The distal housing portion **96** is configured for removable operable

attachment to a control circuit board assembly **100** which is also operably coupled to the motor **82**. A number of batteries **98** may be connected in series may be used as the power source for the surgical instrument **10**. In addition, the power source **90** may be replaceable and/or rechargeable. (185) As outlined above with respect to other various forms, the electric motor **82** can include a rotatable shaft (not shown) that operably interfaces with a gear reducer assembly **84** that is mounted in meshing engagement with a with a set, or rack, of drive teeth **122** on a longitudinally-movable drive member **120**. In use, a voltage polarity provided by the power source **90** can operate the electric motor **82** in a clockwise direction wherein the voltage polarity applied to the electric motor by the battery can be reversed in order to operate the electric motor **82** in a counter-clockwise direction. When the electric motor **82** is rotated in one direction, the drive member **120** will be axially driven in the distal direction “DD”. When the motor **82** is driven in the opposite rotary direction, the drive member **120** will be axially driven in a proximal direction “PD”. The handle **14** can include a switch which can be configured to reverse the polarity applied to the electric motor **82** by the power source **90**. As with the other forms described herein, the handle **14** can also include a sensor that is configured to detect the position of the drive member **120** and/or the direction in which the drive member **120** is being moved.

(186) Actuation of the motor **82** can be controlled by a firing trigger **130** that is pivotally supported on the handle **14**. The firing trigger **130** may be pivoted between an unactuated position and an actuated position. The firing trigger **130** may be biased into the unactuated position by a spring **132** or other biasing arrangement such that when the clinician releases the firing trigger **130**, it may be pivoted or otherwise returned to the unactuated position by the spring **132** or biasing arrangement. In at least one form, the firing trigger **130** can be positioned “outboard” of the closure trigger **32** as was discussed above. In at least one form, a firing trigger safety button **134** may be pivotally mounted to the closure trigger **32** by pin **35**. The safety button **134** may be positioned between the firing trigger **130** and the closure trigger **32** and have a pivot arm **136** protruding therefrom. See FIG. 4. When the closure trigger **32** is in the unactuated position, the safety button **134** is contained in the handle **14** where the clinician cannot readily access it and move it between a safety position preventing actuation of the firing trigger **130** and a firing position wherein the firing trigger **130** may be fired. As the clinician depresses the closure trigger **32**, the safety button **134** and the firing trigger **130** pivot down wherein they can then be manipulated by the clinician.

(187) As discussed above, the handle **14** can include a closure trigger **32** and a firing trigger **130**. Referring to FIGS. 14-18A, the firing trigger **130** can be pivotably mounted to the closure trigger **32**. The closure trigger **32** can include an arm **31** extending therefrom and the firing trigger **130** can be pivotably mounted to the arm **31** about a pivot pin **33**. When the closure trigger **32** is moved from its unactuated position (FIG. 14) to its actuated position (FIG. 17), the firing trigger **130** can descend downwardly, as outlined above. After the safety button **134** has been moved to its firing position, referring primarily to FIG. 18A, the firing trigger **130** can be depressed to operate the motor of the surgical instrument firing system. In various instances, the handle **14** can include a tracking system, such as system **800**, for example, configured to determine the position of the closure trigger **32** and/or the position of the firing trigger **130**. With primary reference to FIGS. 14, 17, and 18A, the tracking system **800** can include a magnetic element, such as permanent magnet **802**, for example, which is mounted to an arm **801** extending from the firing trigger **130**. The tracking system **800** can comprise one or more sensors, such as a first Hall effect sensor **803** and a second Hall effect sensor **804**, for example, which can be configured to track the position of the magnet **802**. Upon comparing FIGS. 14 and 17, the reader will appreciate that, when the closure trigger **32** is moved from its unactuated position to its actuated position, the magnet **802** can move between a first position adjacent the first Hall effect sensor **803** and a second position adjacent the second Hall effect sensor **804**. Upon comparing FIGS. 17 and 18A, the reader will further appreciate that, when the firing trigger **130** is moved from an unfired position (FIG. 17) to a fired position (FIG. 18A), the magnet **802** can move relative to the second Hall effect sensor **804**. The

sensors **803** and **804** can track the movement of the magnet **802** and can be in signal communication with a microcontroller on the circuit board **100**. With data from the first sensor **803** and/or the second sensor **804**, the microcontroller can determine the position of the magnet **802** along a predefined path and, based on that position, the microcontroller can determine whether the closure trigger **32** is in its unactuated position, its actuated position, or a position therebetween. Similarly, with data from the first sensor **803** and/or the second sensor **804**, the microcontroller can determine the position of the magnet **802** along a predefined path and, based on that position, the microcontroller can determine whether the firing trigger **130** is in its unfired position, its fully fired position, or a position therebetween.

(188) As indicated above, in at least one form, the longitudinally movable drive member **120** has a rack of teeth **122** formed thereon for meshing engagement with a corresponding drive gear **86** of the gear reducer assembly **84**. At least one form also includes a manually-actuatable “bailout” assembly **140** that is configured to enable the clinician to manually retract the longitudinally movable drive member **120** should the motor **82** become disabled. The bailout assembly **140** may include a lever or bailout handle assembly **142** that is configured to be manually pivoted into ratcheting engagement with teeth **124** also provided in the drive member **120**. Thus, the clinician can manually retract the drive member **120** by using the bailout handle assembly **142** to ratchet the drive member **120** in the proximal direction “PD”. U.S. Pat. No. 8,608,045 discloses bailout arrangements and other components, arrangements and systems that may also be employed with the various instruments disclosed herein. U.S. patent application Ser. No. 12/249,117, entitled POWERED SURGICAL CUTTING AND STAPLING APPARATUS WITH MANUALLY RETRACTABLE FIRING SYSTEM, now U.S. Pat. No. 8,608,045, is hereby incorporated by reference in its entirety.

(189) Turning now to FIGS. **1** and **7**, the interchangeable shaft assembly **200** includes a surgical end effector **300** that comprises an elongated channel **302** that is configured to operably support a staple cartridge **304** therein. The end effector **300** may further include an anvil **306** that is pivotally supported relative to the elongated channel **302**. The interchangeable shaft assembly **200** may further include an articulation joint **270** and an articulation lock **350** (FIG. **8**) which can be configured to releasably hold the end effector **300** in a desired position relative to a shaft axis SA-SA. Details regarding the construction and operation of the end effector **300**, the articulation joint **270** and the articulation lock **350** are set forth in U.S. patent application Ser. No. 13/803,086, filed Mar. 14, 2013, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541. The entire disclosure of U.S. patent application Ser. No. 13/803,086, filed Mar. 14, 2013, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541 is hereby incorporated by reference herein. As shown in FIGS. **7** and **8**, the interchangeable shaft assembly **200** can further include a proximal housing or nozzle **201** comprised of nozzle portions **202** and **203**. The interchangeable shaft assembly **200** can further include a closure tube **260** which can be utilized to close and/or open the anvil **306** of the end effector **300**. Primarily referring now to FIGS. **8** and **9**, the shaft assembly **200** can include a spine **210** which can be configured to fixably support a shaft frame portion **212** of the articulation lock **350**. See FIG. **8**. The spine **210** can be configured to, one, slidably support a firing member **220** therein and, two, slidably support the closure tube **260** which extends around the spine **210**. The spine **210** can also be configured to slidably support a proximal articulation driver **230**. The articulation driver **230** has a distal end **231** that is configured to operably engage the articulation lock **350**. The articulation lock **350** interfaces with an articulation frame **352** that is adapted to operably engage a drive pin (not shown) on the end effector frame (not shown). As indicated above, further details regarding the operation of the articulation lock **350** and the articulation frame may be found in U.S. patent application Ser. No. 13/803,086. In various circumstances, the spine **210** can comprise a proximal end **211** which is rotatably supported in a

chassis **240**. In one arrangement, for example, the proximal end **211** of the spine **210** has a thread **214** formed thereon for threaded attachment to a spine bearing **216** configured to be supported within the chassis **240**. See FIG. 7. Such an arrangement facilitates rotatable attachment of the spine **210** to the chassis **240** such that the spine **210** may be selectively rotated about a shaft axis SA-SA relative to the chassis **240**.

(190) Referring primarily to FIG. 7, the interchangeable shaft assembly **200** includes a closure shuttle **250** that is slidably supported within the chassis **240** such that it may be axially moved relative thereto. As shown in FIGS. 3 and 7, the closure shuttle **250** includes a pair of proximally-projecting hooks **252** that are configured for attachment to the attachment pin **37** that is attached to the second closure link **38** as will be discussed in further detail below. A proximal end **261** of the closure tube **260** is coupled to the closure shuttle **250** for relative rotation thereto. For example, a U shaped connector **263** is inserted into an annular slot **262** in the proximal end **261** of the closure tube **260** and is retained within vertical slots **253** in the closure shuttle **250**. See FIG. 7. Such an arrangement serves to attach the closure tube **260** to the closure shuttle **250** for axial travel therewith while enabling the closure tube **260** to rotate relative to the closure shuttle **250** about the shaft axis SA-SA. A closure spring **268** is journaled on the closure tube **260** and serves to bias the closure tube **260** in the proximal direction “PD” which can serve to pivot the closure trigger into the unactuated position when the shaft assembly is operably coupled to the handle **14**.

(191) In at least one form, the interchangeable shaft assembly **200** may further include an articulation joint **270**. Other interchangeable shaft assemblies, however, may not be capable of articulation. As shown in FIG. 7, for example, the articulation joint **270** includes a double pivot closure sleeve assembly **271**. According to various forms, the double pivot closure sleeve assembly **271** includes an end effector closure sleeve assembly **272** having upper and lower distally projecting tangs **273**, **274**. An end effector closure sleeve assembly **272** includes a horseshoe aperture **275** and a tab **276** for engaging an opening tab on the anvil **306** in the various manners described in U.S. patent application Ser. No. 13/803,086, filed Mar. 14, 2013, entitled ARTICULATABLE SURGICAL INSTRUMENT COMPRISING AN ARTICULATION LOCK, now U.S. Patent Application Publication No. 2014/0263541, which has been incorporated by reference herein. As described in further detail therein, the horseshoe aperture **275** and tab **276** engage a tab on the anvil when the anvil **306** is opened. An upper double pivot link **277** includes upwardly projecting distal and proximal pivot pins that engage respectively an upper distal pin hole in the upper proximally projecting tang **273** and an upper proximal pin hole in an upper distally projecting tang **264** on the closure tube **260**. A lower double pivot link **278** includes upwardly projecting distal and proximal pivot pins that engage respectively a lower distal pin hole in the lower proximally projecting tang **274** and a lower proximal pin hole in the lower distally projecting tang **265**. See also FIG. 8.

(192) In use, the closure tube **260** is translated distally (direction “DD”) to close the anvil **306**, for example, in response to the actuation of the closure trigger **32**. The anvil **306** is closed by distally translating the closure tube **260** and thus the shaft closure sleeve assembly **272**, causing it to strike a proximal surface on the anvil **306** in the manner described in the aforementioned reference U.S. patent application Ser. No. 13/803,086. As was also described in detail in that reference, the anvil **306** is opened by proximally translating the closure tube **260** and the shaft closure sleeve assembly **272**, causing tab **276** and the horseshoe aperture **275** to contact and push against the anvil tab to lift the anvil **306**. In the anvil-open position, the shaft closure tube **260** is moved to its proximal position.

(193) As indicated above, the surgical instrument **10** may further include an articulation lock **350** of the types and construction described in further detail in U.S. patent application Ser. No. 13/803,086 which can be configured and operated to selectively lock the end effector **300** in position. Such arrangement enables the end effector **300** to be rotated, or articulated, relative to the shaft closure tube **260** when the articulation lock **350** is in its unlocked state. In such an unlocked state, the end

effector **300** can be positioned and pushed against soft tissue and/or bone, for example, surrounding the surgical site within the patient in order to cause the end effector **300** to articulate relative to the closure tube **260**. The end effector **300** may also be articulated relative to the closure tube **260** by an articulation driver **230**.

(194) As was also indicated above, the interchangeable shaft assembly **200** further includes a firing member **220** that is supported for axial travel within the shaft spine **210**. The firing member **220** includes an intermediate firing shaft portion **222** that is configured for attachment to a distal cutting portion or knife bar **280**. The firing member **220** may also be referred to herein as a “second shaft” and/or a “second shaft assembly”. As shown in FIGS. **8** and **9**, the intermediate firing shaft portion **222** may include a longitudinal slot **223** in the distal end thereof which can be configured to receive a tab **284** on the proximal end **282** of the distal knife bar **280**. The longitudinal slot **223** and the proximal end **282** can be sized and configured to permit relative movement therebetween and can comprise a slip joint **286**. The slip joint **286** can permit the intermediate firing shaft portion **222** of the firing drive **220** to be moved to articulate the end effector **300** without moving, or at least substantially moving, the knife bar **280**. Once the end effector **300** has been suitably oriented, the intermediate firing shaft portion **222** can be advanced distally until a proximal sidewall of the longitudinal slot **223** comes into contact with the tab **284** in order to advance the knife bar **280** and fire the staple cartridge positioned within the channel **302**. As can be further seen in FIGS. **8** and **9**, the shaft spine **210** has an elongate opening or window **213** therein to facilitate assembly and insertion of the intermediate firing shaft portion **222** into the shaft frame **210**. Once the intermediate firing shaft portion **222** has been inserted therein, a top frame segment **215** may be engaged with the shaft frame **212** to enclose the intermediate firing shaft portion **222** and knife bar **280** therein. Further description of the operation of the firing member **220** may be found in U.S. patent application Ser. No. 13/803,086.

(195) Further to the above, the shaft assembly **200** can include a clutch assembly **400** which can be configured to selectively and releasably couple the articulation driver **230** to the firing member **220**. In one form, the clutch assembly **400** includes a lock collar, or sleeve **402**, positioned around the firing member **220** wherein the lock sleeve **402** can be rotated between an engaged position in which the lock sleeve **402** couples the articulation driver **360** to the firing member **220** and a disengaged position in which the articulation driver **360** is not operably coupled to the firing member **200**. When lock sleeve **402** is in its engaged position, distal movement of the firing member **220** can move the articulation driver **360** distally and, correspondingly, proximal movement of the firing member **220** can move the articulation driver **230** proximally. When lock sleeve **402** is in its disengaged position, movement of the firing member **220** is not transmitted to the articulation driver **230** and, as a result, the firing member **220** can move independently of the articulation driver **230**. In various circumstances, the articulation driver **230** can be held in position by the articulation lock **350** when the articulation driver **230** is not being moved in the proximal or distal directions by the firing member **220**.

(196) Referring primarily to FIG. **9**, the lock sleeve **402** can comprise a cylindrical, or an at least substantially cylindrical, body including a longitudinal aperture **403** defined therein configured to receive the firing member **220**. The lock sleeve **402** can comprise diametrically-opposed, inwardly-facing lock protrusions **404** and an outwardly-facing lock member **406**. The lock protrusions **404** can be configured to be selectively engaged with the firing member **220**. More particularly, when the lock sleeve **402** is in its engaged position, the lock protrusions **404** are positioned within a drive notch **224** defined in the firing member **220** such that a distal pushing force and/or a proximal pulling force can be transmitted from the firing member **220** to the lock sleeve **402**. When the lock sleeve **402** is in its engaged position, the second lock member **406** is received within a drive notch **232** defined in the articulation driver **230** such that the distal pushing force and/or the proximal pulling force applied to the lock sleeve **402** can be transmitted to the articulation driver **230**. In effect, the firing member **220**, the lock sleeve **402**, and the articulation driver **230** will move

together when the lock sleeve **402** is in its engaged position. On the other hand, when the lock sleeve **402** is in its disengaged position, the lock protrusions **404** may not be positioned within the drive notch **224** of the firing member **220** and, as a result, a distal pushing force and/or a proximal pulling force may not be transmitted from the firing member **220** to the lock sleeve **402**. Correspondingly, the distal pushing force and/or the proximal pulling force may not be transmitted to the articulation driver **230**. In such circumstances, the firing member **220** can be slid proximally and/or distally relative to the lock sleeve **402** and the proximal articulation driver **230**.

(197) As shown in FIGS. **8-12**, the shaft assembly **200** further includes a switch drum **500** that is rotatably received on the closure tube **260**. The switch drum **500** comprises a hollow shaft segment **502** that has a shaft boss **504** formed thereon for receive an outwardly protruding actuation pin **410** therein. In various circumstances, the actuation pin **410** extends through a slot **267** into a longitudinal slot **408** provided in the lock sleeve **402** to facilitate axial movement of the lock sleeve **402** when it is engaged with the articulation driver **230**. A rotary torsion spring **420** is configured to engage the boss **504** on the switch drum **500** and a portion of the nozzle housing **203** as shown in FIG. **10** to apply a biasing force to the switch drum **500**. The switch drum **500** can further comprise at least partially circumferential openings **506** defined therein which, referring to FIGS. **5** and **6**, can be configured to receive circumferential mounts **204**, **205** extending from the nozzle halves **202**, **203** and permit relative rotation, but not translation, between the switch drum **500** and the proximal nozzle **201**. As shown in those Figures, the mounts **204** and **205** also extend through openings **266** in the closure tube **260** to be seated in recesses **211** in the shaft spine **210**. However, rotation of the nozzle **201** to a point where the mounts **204**, **205** reach the end of their respective slots **506** in the switch drum **500** will result in rotation of the switch drum **500** about the shaft axis SA-SA. Rotation of the switch drum **500** will ultimately result in the rotation of the actuation pin **410** and the lock sleeve **402** between its engaged and disengaged positions. Thus, in essence, the nozzle **201** may be employed to operably engage and disengage the articulation drive system with the firing drive system in the various manners described in further detail in U.S. patent application Ser. No. 13/803,086.

(198) As also illustrated in FIGS. **8-12**, the shaft assembly **200** can comprise a slip ring assembly **600** which can be configured to conduct electrical power to and/or from the end effector **300** and/or communicate signals to and/or from the end effector **300**, for example. The slip ring assembly **600** can comprise a proximal connector flange **604** mounted to a chassis flange **242** extending from the chassis **240** and a distal connector flange **601** positioned within a slot defined in the shaft housings **202**, **203**. The proximal connector flange **604** can comprise a first face and the distal connector flange **601** can comprise a second face which is positioned adjacent to and movable relative to the first face. The distal connector flange **601** can rotate relative to the proximal connector flange **604** about the shaft axis SA-SA. The proximal connector flange **604** can comprise a plurality of concentric, or at least substantially concentric, conductors **602** defined in the first face thereof. A connector **607** can be mounted on the proximal side of the connector flange **601** and may have a plurality of contacts (not shown) wherein each contact corresponds to and is in electrical contact with one of the conductors **602**. Such an arrangement permits relative rotation between the proximal connector flange **604** and the distal connector flange **601** while maintaining electrical contact therebetween. The proximal connector flange **604** can include an electrical connector **606** which can place the conductors **602** in signal communication with a shaft circuit board **610** mounted to the shaft chassis **240**, for example. In at least one instance, a wiring harness comprising a plurality of conductors can extend between the electrical connector **606** and the shaft circuit board **610**. The electrical connector **606** may extend proximally through a connector opening **243** defined in the chassis mounting flange **242**. See FIG. **7**. U.S. patent application Ser. No. 13/800,067, entitled STAPLE CARTRIDGE TISSUE THICKNESS SENSOR SYSTEM, filed on Mar. 13, 2013, now U.S. Patent Application Publication No. 2014/0263552, is incorporated by reference in its entirety. U.S. patent application Ser. No. 13/800,025, entitled STAPLE

CARDIDGE THICKNESS SENSOR SYSTEM, filed on Mar. 13, 2013, now U.S. Patent Application Publication No. 2014/0263552, is incorporated by reference in its entirety. Further details regarding slip ring assembly **600** may be found in U.S. patent application Ser. No. 13/803,086.

(199) As discussed above, the shaft assembly **200** can include a proximal portion which is fixably mounted to the handle **14** and a distal portion which is rotatable about a longitudinal axis. The rotatable distal shaft portion can be rotated relative to the proximal portion about the slip ring assembly **600**, as discussed above. The distal connector flange **601** of the slip ring assembly **600** can be positioned within the rotatable distal shaft portion. Moreover, further to the above, the switch drum **500** can also be positioned within the rotatable distal shaft portion. When the rotatable distal shaft portion is rotated, the distal connector flange **601** and the switch drum **500** can be rotated synchronously with one another. In addition, the switch drum **500** can be rotated between a first position and a second position relative to the distal connector flange **601**. When the switch drum **500** is in its first position, the articulation drive system may be operably disengaged from the firing drive system and, thus, the operation of the firing drive system may not articulate the end effector **300** of the shaft assembly **200**. When the switch drum **500** is in its second position, the articulation drive system may be operably engaged with the firing drive system and, thus, the operation of the firing drive system may articulate the end effector **300** of the shaft assembly **200**. When the switch drum **500** is moved between its first position and its second position, the switch drum **500** is moved relative to distal connector flange **601**. In various instances, the shaft assembly **200** can comprise at least one sensor configured to detect the position of the switch drum **500**. Turning now to FIGS. **11** and **12**, the distal connector flange **601** can comprise a Hall effect sensor **605**, for example, and the switch drum **500** can comprise a magnetic element, such as permanent magnet **505**, for example. The Hall effect sensor **605** can be configured to detect the position of the permanent magnet **505**. When the switch drum **500** is rotated between its first position and its second position, the permanent magnet **505** can move relative to the Hall effect sensor **605**. In various instances, Hall effect sensor **605** can detect changes in a magnetic field created when the permanent magnet **505** is moved. The Hall effect sensor **605** can be in signal communication with the shaft circuit board **610** and/or the handle circuit board **100**, for example. Based on the signal from the Hall effect sensor **605**, a microcontroller on the shaft circuit board **610** and/or the handle circuit board **100** can determine whether the articulation drive system is engaged with or disengaged from the firing drive system.

(200) Referring again to FIGS. **3** and **7**, the chassis **240** includes at least one, and preferably two, tapered attachment portions **244** formed thereon that are adapted to be received within corresponding dovetail slots **702** formed within a distal attachment flange portion **700** of the frame **20**. Each dovetail slot **702** may be tapered or, stated another way, be somewhat V-shaped to seatingly receive the attachment portions **244** therein. As can be further seen in FIGS. **3** and **7**, a shaft attachment lug **226** is formed on the proximal end of the intermediate firing shaft **222**. As will be discussed in further detail below, when the interchangeable shaft assembly **200** is coupled to the handle **14**, the shaft attachment lug **226** is received in a firing shaft attachment cradle **126** formed in the distal end **125** of the longitudinal drive member **120**. See FIGS. **3** and **6**.

(201) Various shaft assembly embodiments employ a latch system **710** for removably coupling the shaft assembly **200** to the housing **12** and more specifically to the frame **20**. As shown in FIG. **7**, for example, in at least one form, the latch system **710** includes a lock member or lock yoke **712** that is movably coupled to the chassis **240**. In the illustrated embodiment, for example, the lock yoke **712** has a U-shape with two spaced downwardly extending legs **714**. The legs **714** each have a pivot lug **716** formed thereon that are adapted to be received in corresponding holes **245** formed in the chassis **240**. Such arrangement facilitates pivotal attachment of the lock yoke **712** to the chassis **240**. The lock yoke **712** may include two proximally protruding lock lugs **714** that are configured for releasable engagement with corresponding lock detents or grooves **704** in the distal attachment

flange **700** of the frame **20**. See FIG. 3. In various forms, the lock yoke **712** is biased in the proximal direction by spring or biasing member (not shown). Actuation of the lock yoke **712** may be accomplished by a latch button **722** that is slidably mounted on a latch actuator assembly **720** that is mounted to the chassis **240**. The latch button **722** may be biased in a proximal direction relative to the lock yoke **712**. As will be discussed in further detail below, the lock yoke **712** may be moved to an unlocked position by biasing the latch button the in distal direction which also causes the lock yoke **712** to pivot out of retaining engagement with the distal attachment flange **700** of the frame **20**. When the lock yoke **712** is in “retaining engagement” with the distal attachment flange **700** of the frame **20**, the lock lugs **716** are retainingly seated within the corresponding lock detents or grooves **704** in the distal attachment flange **700**.

(202) When employing an interchangeable shaft assembly that includes an end effector of the type described herein that is adapted to cut and fasten tissue, as well as other types of end effectors, it may be desirable to prevent inadvertent detachment of the interchangeable shaft assembly from the housing during actuation of the end effector. For example, in use the clinician may actuate the closure trigger **32** to grasp and manipulate the target tissue into a desired position. Once the target tissue is positioned within the end effector **300** in a desired orientation, the clinician may then fully actuate the closure trigger **32** to close the anvil **306** and clamp the target tissue in position for cutting and stapling. In that instance, the first drive system **30** has been fully actuated. After the target tissue has been clamped in the end effector **300**, it may be desirable to prevent the inadvertent detachment of the shaft assembly **200** from the housing **12**. One form of the latch system **710** is configured to prevent such inadvertent detachment.

(203) As can be most particularly seen in FIG. 7, the lock yoke **712** includes at least one and preferably two lock hooks **718** that are adapted to contact corresponding lock lug portions **256** that are formed on the closure shuttle **250**. Referring to FIGS. 13-15, when the closure shuttle **250** is in an unactuated position (i.e., the first drive system **30** is unactuated and the anvil **306** is open), the lock yoke **712** may be pivoted in a distal direction to unlock the interchangeable shaft assembly **200** from the housing **12**. When in that position, the lock hooks **718** do not contact the lock lug portions **256** on the closure shuttle **250**. However, when the closure shuttle **250** is moved to an actuated position (i.e., the first drive system **30** is actuated and the anvil **306** is in the closed position), the lock yoke **712** is prevented from being pivoted to an unlocked position. See FIGS. 16-18. Stated another way, if the clinician were to attempt to pivot the lock yoke **712** to an unlocked position or, for example, the lock yoke **712** was inadvertently bumped or contacted in a manner that might otherwise cause it to pivot distally, the lock hooks **718** on the lock yoke **712** will contact the lock lugs **256** on the closure shuttle **250** and prevent movement of the lock yoke **712** to an unlocked position.

(204) Attachment of the interchangeable shaft assembly **200** to the handle **14** will now be described with reference to FIG. 3. To commence the coupling process, the clinician may position the chassis **240** of the interchangeable shaft assembly **200** above or adjacent to the distal attachment flange **700** of the frame **20** such that the tapered attachment portions **244** formed on the chassis **240** are aligned with the dovetail slots **702** in the frame **20**. The clinician may then move the shaft assembly **200** along an installation axis IA that is perpendicular to the shaft axis SA-SA to seat the attachment portions **244** in “operable engagement” with the corresponding dovetail receiving slots **702**. In doing so, the shaft attachment lug **226** on the intermediate firing shaft **222** will also be seated in the cradle **126** in the longitudinally movable drive member **120** and the portions of pin **37** on the second closure link **38** will be seated in the corresponding hooks **252** in the closure yoke **250**. As used herein, the term “operable engagement” in the context of two components means that the two components are sufficiently engaged with each other so that upon application of an actuation motion thereto, the components may carry out their intended action, function and/or procedure.

(205) As discussed above, at least five systems of the interchangeable shaft assembly **200** can be operably coupled with at least five corresponding systems of the handle **14**. A first system can

comprise a frame system which couples and/or aligns the frame or spine of the shaft assembly **200** with the frame **20** of the handle **14**. Another system can comprise a closure drive system **30** which can operably connect the closure trigger **32** of the handle **14** and the closure tube **260** and the anvil **306** of the shaft assembly **200**. As outlined above, the closure tube attachment yoke **250** of the shaft assembly **200** can be engaged with the pin **37** on the second closure link **38**. Another system can comprise the firing drive system **80** which can operably connect the firing trigger **130** of the handle **14** with the intermediate firing shaft **222** of the shaft assembly **200**.

(206) As outlined above, the shaft attachment lug **226** can be operably connected with the cradle **126** of the longitudinal drive member **120**. Another system can comprise an electrical system which can signal to a controller in the handle **14**, such as microcontroller, for example, that a shaft assembly, such as shaft assembly **200**, for example, has been operably engaged with the handle **14** and/or, two, conduct power and/or communication signals between the shaft assembly **200** and the handle **14**. For instance, the shaft assembly **200** can include an electrical connector **1410** that is operably mounted to the shaft circuit board **610**. The electrical connector **1410** is configured for mating engagement with a corresponding electrical connector **1400** on the handle control board **100**. Further details regarding the circuitry and control systems may be found in U.S. patent application Ser. No. 13/803,086, the entire disclosure of which was previously incorporated by reference herein. The fifth system may consist of the latching system for releasably locking the shaft assembly **200** to the handle **14**.

(207) Referring again to FIGS. 2 and 3, the handle **14** can include an electrical connector **1400** comprising a plurality of electrical contacts. Turning now to FIG. 19, the electrical connector **1400** can comprise a first contact **1401a**, a second contact **1401b**, a third contact **1401c**, a fourth contact **1401d**, a fifth contact **1401e**, and a sixth contact **1401f**, for example. While the illustrated embodiment utilizes six contacts, other embodiments are envisioned which may utilize more than six contacts or less than six contacts.

(208) As illustrated in FIG. 19, the first contact **1401a** can be in electrical communication with a transistor **1408**, contacts **1401b-1401e** can be in electrical communication with a microcontroller **1500**, and the sixth contact **1401f** can be in electrical communication with a ground. In certain circumstances, one or more of the electrical contacts **1401b-1401e** may be in electrical communication with one or more output channels of the microcontroller **1500** and can be energized, or have a voltage potential applied thereto, when the handle **1042** is in a powered state. In some circumstances, one or more of the electrical contacts **1401b-1401e** may be in electrical communication with one or more input channels of the microcontroller **1500** and, when the handle **14** is in a powered state, the microcontroller **1500** can be configured to detect when a voltage potential is applied to such electrical contacts. When a shaft assembly, such as shaft assembly **200**, for example, is assembled to the handle **14**, the electrical contacts **1401a-1401f** may not communicate with each other. When a shaft assembly is not assembled to the handle **14**, however, the electrical contacts **1401a-1401f** of the electrical connector **1400** may be exposed and, in some circumstances, one or more of the contacts **1401a-1401f** may be accidentally placed in electrical communication with each other. Such circumstances can arise when one or more of the contacts **1401a-1401f** come into contact with an electrically conductive material, for example. When this occurs, the microcontroller **1500** can receive an erroneous input and/or the shaft assembly **200** can receive an erroneous output, for example. To address this issue, in various circumstances, the handle **14** may be unpowered when a shaft assembly, such as shaft assembly **200**, for example, is not attached to the handle **14**.

(209) In other circumstances, the handle **1042** can be powered when a shaft assembly, such as shaft assembly **200**, for example, is not attached thereto. In such circumstances, the microcontroller **1500** can be configured to ignore inputs, or voltage potentials, applied to the contacts in electrical communication with the microcontroller **1500**, i.e., contacts **1401b-1401e**, for example, until a shaft assembly is attached to the handle **14**. Even though the microcontroller **1500** may be supplied

with power to operate other functionalities of the handle **14** in such circumstances, the handle **14** may be in a powered-down state. In a way, the electrical connector **1400** may be in a powered-down state as voltage potentials applied to the electrical contacts **1401b-1401e** may not affect the operation of the handle **14**. The reader will appreciate that, even though contacts **1401b-1401e** may be in a powered-down state, the electrical contacts **1401a** and **1401f**, which are not in electrical communication with the microcontroller **1500**, may or may not be in a powered-down state. For instance, sixth contact **1401f** may remain in electrical communication with a ground regardless of whether the handle **14** is in a powered-up or a powered-down state.

(210) Furthermore, the transistor **1408**, and/or any other suitable arrangement of transistors, such as transistor **1410**, for example, and/or switches may be configured to control the supply of power from a power source **1404**, such as a battery **90** within the handle **14**, for example, to the first electrical contact **1401a** regardless of whether the handle **14** is in a powered-up or a powered-down state. In various circumstances, the shaft assembly **200**, for example, can be configured to change the state of the transistor **1408** when the shaft assembly **200** is engaged with the handle **14**. In certain circumstances, further to the below, a Hall effect sensor **1402** can be configured to switch the state of transistor **1410** which, as a result, can switch the state of transistor **1408** and ultimately supply power from power source **1404** to first contact **1401a**. In this way, both the power circuits and the signal circuits to the connector **1400** can be powered down when a shaft assembly is not installed to the handle **14** and powered up when a shaft assembly is installed to the handle **14**.

(211) In various circumstances, referring again to FIG. **19**, the handle **14** can include the Hall effect sensor **1402**, for example, which can be configured to detect a detectable element, such as a magnetic element **1407** (FIG. **3**), for example, on a shaft assembly, such as shaft assembly **200**, for example, when the shaft assembly is coupled to the handle **14**. The Hall effect sensor **1402** can be powered by a power source **1406**, such as a battery, for example, which can, in effect, amplify the detection signal of the Hall effect sensor **1402** and communicate with an input channel of the microcontroller **1500** via the circuit illustrated in FIG. **19**. Once the microcontroller **1500** has received an input indicating that a shaft assembly has been at least partially coupled to the handle **14**, and that, as a result, the electrical contacts **1401a-1401f** are no longer exposed, the microcontroller **1500** can enter into its normal, or powered-up, operating state. In such an operating state, the microcontroller **1500** will evaluate the signals transmitted to one or more of the contacts **1401b-1401e** from the shaft assembly and/or transmit signals to the shaft assembly through one or more of the contacts **1401b-1401e** in normal use thereof. In various circumstances, the shaft assembly **200** may have to be fully seated before the Hall effect sensor **1402** can detect the magnetic element **1407**. While a Hall effect sensor **1402** can be utilized to detect the presence of the shaft assembly **200**, any suitable system of sensors and/or switches can be utilized to detect whether a shaft assembly has been assembled to the handle **14**, for example. In this way, further to the above, both the power circuits and the signal circuits to the connector **1400** can be powered down when a shaft assembly is not installed to the handle **14** and powered up when a shaft assembly is installed to the handle **14**.

(212) In various embodiments, any number of magnetic sensing elements may be employed to detect whether a shaft assembly has been assembled to the handle **14**, for example. For example, the technologies used for magnetic field sensing include search coil, fluxgate, optically pumped, nuclear precession, SQUID, Hall-effect, anisotropic magnetoresistance, giant magnetoresistance, magnetic tunnel junctions, giant magnetoimpedance, magnetostrictive/piezoelectric composites, magnetodiode, magnetotransistor, fiber optic, magnetooptic, and microelectromechanical systems-based magnetic sensors, among others.

(213) Referring to FIG. **19**, the microcontroller **1500** may generally comprise a microprocessor ("processor") and one or more memory units operationally coupled to the processor. By executing instruction code stored in the memory, the processor may control various components of the surgical instrument, such as the motor, various drive systems, and/or a user display, for example.

The microcontroller **1500** may be implemented using integrated and/or discrete hardware elements, software elements, and/or a combination of both. Examples of integrated hardware elements may include processors, microprocessors, microcontrollers, integrated circuits, application specific integrated circuits (ASIC), programmable logic devices (PLD), digital signal processors (DSP), field programmable gate arrays (FPGA), logic gates, registers, semiconductor devices, chips, microchips, chip sets, microcontrollers, system-on-chip (SoC), and/or system-in-package (SIP). Examples of discrete hardware elements may include circuits and/or circuit elements such as logic gates, field effect transistors, bipolar transistors, resistors, capacitors, inductors, and/or relays. In certain instances, the microcontroller **1500** may include a hybrid circuit comprising discrete and integrated circuit elements or components on one or more substrates, for example.

(214) Referring to FIG. **19**, the microcontroller **1500** may be an LM 4F230H5QR, available from Texas Instruments, for example. In certain instances, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle serial random access memory (SRAM), internal read-only memory (ROM) loaded with StellarisWare® software, 2 KB electrically erasable programmable read-only memory (EEPROM), one or more pulse width modulation (PWM) modules, one or more quadrature encoder inputs (QEI) analog, one or more 12-bit Analog-to-Digital Converters (ADC) with 12 analog input channels, among other features that are readily available. Other microcontrollers may be readily substituted for use with the present disclosure. Accordingly, the present disclosure should not be limited in this context.

(215) As discussed above, the handle **14** and/or the shaft assembly **200** can include systems and configurations configured to prevent, or at least reduce the possibility of, the contacts of the handle electrical connector **1400** and/or the contacts of the shaft electrical connector **1410** from becoming shorted out when the shaft assembly **200** is not assembled, or completely assembled, to the handle **14**. Referring to FIG. **3**, the handle electrical connector **1400** can be at least partially recessed within a cavity **1409** defined in the handle frame **20**. The six contacts **1401a-1401f** of the electrical connector **1400** can be completely recessed within the cavity **1409**. Such arrangements can reduce the possibility of an object accidentally contacting one or more of the contacts **1401a-1401f**. Similarly, the shaft electrical connector **1410** can be positioned within a recess defined in the shaft chassis **240** which can reduce the possibility of an object accidentally contacting one or more of the contacts **1411a-1411f** of the shaft electrical connector **1410**. With regard to the particular embodiment depicted in FIG. **3**, the shaft contacts **1411a-1411f** can comprise male contacts. In at least one embodiment, each shaft contact **1411a-1411f** can comprise a flexible projection extending therefrom which can be configured to engage a corresponding handle contact **1401a-1401f**, for example. The handle contacts **1401a-1401f** can comprise female contacts. In at least one embodiment, each handle contact **1401a-1401f** can comprise a flat surface, for example, against which the male shaft contacts **1401a-1401f** can wipe, or slide, against and maintain an electrically conductive interface therebetween. In various instances, the direction in which the shaft assembly **200** is assembled to the handle **14** can be parallel to, or at least substantially parallel to, the handle contacts **1401a-1401f** such that the shaft contacts **1411a-1411f** slide against the handle contacts **1401a-1401f** when the shaft assembly **200** is assembled to the handle **14**. In various alternative embodiments, the handle contacts **1401a-1401f** can comprise male contacts and the shaft contacts **1411a-1411f** can comprise female contacts. In certain alternative embodiments, the handle contacts **1401a-1401f** and the shaft contacts **1411a-1411f** can comprise any suitable arrangement of contacts.

(216) In various instances, the handle **14** can comprise a connector guard configured to at least partially cover the handle electrical connector **1400** and/or a connector guard configured to at least partially cover the shaft electrical connector **1410**. A connector guard can prevent, or at least reduce the possibility of, an object accidentally touching the contacts of an electrical connector when the

shaft assembly is not assembled to, or only partially assembled to, the handle. A connector guard can be movable. For instance, the connector guard can be moved between a guarded position in which it at least partially guards a connector and an unguarded position in which it does not guard, or at least guards less of, the connector. In at least one embodiment, a connector guard can be displaced as the shaft assembly is being assembled to the handle. For instance, if the handle comprises a handle connector guard, the shaft assembly can contact and displace the handle connector guard as the shaft assembly is being assembled to the handle. Similarly, if the shaft assembly comprises a shaft connector guard, the handle can contact and displace the shaft connector guard as the shaft assembly is being assembled to the handle. In various instances, a connector guard can comprise a door, for example. In at least one instance, the door can comprise a beveled surface which, when contacted by the handle or shaft, can facilitate the displacement of the door in a certain direction. In various instances, the connector guard can be translated and/or rotated, for example. In certain instances, a connector guard can comprise at least one film which covers the contacts of an electrical connector. When the shaft assembly is assembled to the handle, the film can become ruptured. In at least one instance, the male contacts of a connector can penetrate the film before engaging the corresponding contacts positioned underneath the film.

(217) As described above, the surgical instrument can include a system which can selectively power-up, or activate, the contacts of an electrical connector, such as the electrical connector **1400**, for example. In various instances, the contacts can be transitioned between an unactivated condition and an activated condition. In certain instances, the contacts can be transitioned between a monitored condition, a deactivated condition, and an activated condition. For instance, the microcontroller **1500**, for example, can monitor the contacts **1401a-1401f** when a shaft assembly has not been assembled to the handle **14** to determine whether one or more of the contacts **1401a-1401f** may have been shorted. The microcontroller **1500** can be configured to apply a low voltage potential to each of the contacts **1401a-1401f** and assess whether only a minimal resistance is present at each of the contacts. Such an operating state can comprise the monitored condition. In the event that the resistance detected at a contact is high, or above a threshold resistance, the microcontroller **1500** can deactivate that contact, more than one contact, or, alternatively, all of the contacts. Such an operating state can comprise the deactivated condition. If a shaft assembly is assembled to the handle **14** and it is detected by the microcontroller **1500**, as discussed above, the microcontroller **1500** can increase the voltage potential to the contacts **1401a-1401f**. Such an operating state can comprise the activated condition.

(218) The various shaft assemblies disclosed herein may employ sensors and various other components that require electrical communication with the controller in the housing. These shaft assemblies generally are configured to be able to rotate relative to the housing necessitating a connection that facilitates such electrical communication between two or more components that may rotate relative to each other. When employing end effectors of the types disclosed herein, the connector arrangements must be relatively robust in nature while also being somewhat compact to fit into the shaft assembly connector portion.

(219) Referring to FIG. **20**, a non-limiting form of the end effector **300** is illustrated. As described above, the end effector **300** may include the anvil **306** and the staple cartridge **304**. In this non-limiting embodiment, the anvil **306** is coupled to an elongate channel **198**. For example, apertures **199** can be defined in the elongate channel **198** which can receive pins **152** extending from the anvil **306** and allow the anvil **306** to pivot from an open position to a closed position relative to the elongate channel **198** and staple cartridge **304**. In addition, FIG. **20** shows a firing bar **172**, configured to longitudinally translate into the end effector **300**. The firing bar **172** may be constructed from one solid section, or in various embodiments, may include a laminate material comprising, for example, a stack of steel plates. A distally projecting end of the firing bar **172** can be attached to an E-beam **178** that can, among other things, assist in spacing the anvil **306** from a staple cartridge **304** positioned in the elongate channel **198** when the anvil **306** is in a closed

position. The E-beam **178** can also include a sharpened cutting edge **182** which can be used to sever tissue as the E-beam **178** is advanced distally by the firing bar **172**. In operation, the E-beam **178** can also actuate, or fire, the staple cartridge **304**. The staple cartridge **304** can include a molded cartridge body **194** that holds a plurality of staples **191** resting upon staple drivers **192** within respective upwardly open staple cavities **195**. A wedge sled **190** is driven distally by the E-beam **178**, sliding upon a cartridge tray **196** that holds together the various components of the replaceable staple cartridge **304**. The wedge sled **190** upwardly cams the staple drivers **192** to force out the staples **191** into deforming contact with the anvil **306** while a cutting surface **182** of the E-beam **178** severs clamped tissue.

(220) Further to the above, the E-beam **178** can include upper pins **180** which engage the anvil **306** during firing. The E-beam **178** can further include middle pins **184** and a bottom foot **186** which can engage various portions of the cartridge body **194**, cartridge tray **196** and elongate channel **198**. When a staple cartridge **304** is positioned within the elongate channel **198**, a slot **193** defined in the cartridge body **194** can be aligned with a slot **197** defined in the cartridge tray **196** and a slot **189** defined in the elongate channel **198**. In use, the E-beam **178** can slide through the aligned slots **193**, **197**, and **189** wherein, as indicated in FIG. **20**, the bottom foot **186** of the E-beam **178** can engage a groove running along the bottom surface of channel **198** along the length of slot **189**, the middle pins **184** can engage the top surfaces of cartridge tray **196** along the length of longitudinal slot **197**, and the upper pins **180** can engage the anvil **306**. In such circumstances, the E-beam **178** can space, or limit the relative movement between, the anvil **306** and the staple cartridge **304** as the firing bar **172** is moved distally to fire the staples from the staple cartridge **304** and/or incise the tissue captured between the anvil **306** and the staple cartridge **304**. Thereafter, the firing bar **172** and the E-beam **178** can be retracted proximally allowing the anvil **306** to be opened to release the two stapled and severed tissue portions (not shown).

(221) Having described a surgical instrument **10** in general terms, the description now turns to a detailed description of various electrical/electronic component of the surgical instrument **10**. Turning now to FIGS. **21A-21B**, where one embodiment of a segmented circuit **2000** comprising a plurality of circuit segments **2002a-2002g** is illustrated. The segmented circuit **2000** comprising the plurality of circuit segments **2002a-2002g** is configured to control a powered surgical instrument, such as, for example, the surgical instrument **10** illustrated in FIGS. **1-18A**, without limitation. The plurality of circuit segments **2002a-2002g** is configured to control one or more operations of the powered surgical instrument **10**. A safety processor segment **2002a** (Segment 1) comprises a safety processor **2004**. A primary processor segment **2002b** (Segment 2) comprises a primary processor **2006**. The safety processor **2004** and/or the primary processor **2006** are configured to interact with one or more additional circuit segments **2002c-2002g** to control operation of the powered surgical instrument **10**. The primary processor **2006** comprises a plurality of inputs coupled to, for example, one or more circuit segments **2002c-2002g**, a battery **2008**, and/or a plurality of switches **2058a-2070**. The segmented circuit **2000** may be implemented by any suitable circuit, such as, for example, a printed circuit board assembly (PCBA) within the powered surgical instrument **10**. It should be understood that the term processor as used herein includes any microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer's central processing unit (CPU) on an integrated circuit or at most a few integrated circuits. The processor is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system.

(222) In one embodiment, the main processor **2006** may be any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In one embodiment, the safety processor **2004** may be a safety microcontroller platform comprising two microcontroller-based families such as TMS570 and RM4x known under the trade name Hercules

ARM Cortex R4, also by Texas Instruments. Nevertheless, other suitable substitutes for microcontrollers and safety processor may be employed, without limitation. In one embodiment, the safety processor **2004** may be configured specifically for IEC 61508 and ISO 26262 safety critical applications, among others, to provide advanced integrated safety features while delivering scalable performance, connectivity, and memory options.

(223) In certain instances, the main processor **2006** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available for the product datasheet. Other processors may be readily substituted and, accordingly, the present disclosure should not be limited in this context.

(224) In one embodiment, the segmented circuit **2000** comprises an acceleration segment **2002c** (Segment 3). The acceleration segment **2002c** comprises an acceleration sensor **2022**. The acceleration sensor **2022** may comprise, for example, an accelerometer. The acceleration sensor **2022** is configured to detect movement or acceleration of the powered surgical instrument **10**. In some embodiments, input from the acceleration sensor **2022** is used, for example, to transition to and from a sleep mode, identify an orientation of the powered surgical instrument, and/or identify when the surgical instrument has been dropped. In some embodiments, the acceleration segment **2002c** is coupled to the safety processor **2004** and/or the primary processor **2006**.

(225) In one embodiment, the segmented circuit **2000** comprises a display segment **2002d** (Segment 4). The display segment **2002d** comprises a display connector **2024** coupled to the primary processor **2006**. The display connector **2024** couples the primary processor **2006** to a display **2028** through one or more display driver integrated circuits **2026**. The display driver integrated circuits **2026** may be integrated with the display **2028** and/or may be located separately from the display **2028**. The display **2028** may comprise any suitable display, such as, for example, an organic light-emitting diode (OLED) display, a liquid-crystal display (LCD), and/or any other suitable display. In some embodiments, the display segment **2002d** is coupled to the safety processor **2004**.

(226) In some embodiments, the segmented circuit **2000** comprises a shaft segment **2002e** (Segment 5). The shaft segment **2002e** comprises one or more controls for a shaft **2004** coupled to the surgical instrument **10** and/or one or more controls for an end effector **2006** coupled to the shaft **2004**. The shaft segment **2002e** comprises a shaft connector **2030** configured to couple the primary processor **2006** to a shaft PCBA **2031**. The shaft PCBA **2031** comprises a first articulation switch **2036**, a second articulation switch **2032**, and a shaft PCBA EEPROM **2034**. In some embodiments, the shaft PCBA EEPROM **2034** comprises one or more parameters, routines, and/or programs specific to the shaft **2004** and/or the shaft PCBA **2031**. The shaft PCBA **2031** may be coupled to the shaft **2004** and/or integral with the surgical instrument **10**. In some embodiments, the shaft segment **2002e** comprises a second shaft EEPROM **2038**. The second shaft EEPROM **2038** comprises a plurality of algorithms, routines, parameters, and/or other data corresponding to one or more shafts **2004** and/or end effectors **2006** which may be interfaced with the powered surgical instrument **10**.

(227) In some embodiments, the segmented circuit **2000** comprises a position encoder segment **2002f** (Segment 6). The position encoder segment **2002f** comprises one or more magnetic rotary position encoders **2040a-2040b**. The one or more magnetic rotary position encoders **2040a-2040b** are configured to identify the rotational position of a motor **2048**, a shaft **2004**, and/or an end effector **2006** of the surgical instrument **10**. In some embodiments, the magnetic rotary position encoders **2040a-2040b** may be coupled to the safety processor **2004** and/or the primary processor

2006.

(228) In some embodiments, the segmented circuit **2000** comprises a motor segment **2002g** (Segment 7). The motor segment **2002g** comprises a motor **2048** configured to control one or more movements of the powered surgical instrument **10**. The motor **2048** is coupled to the primary processor **2006** by an H-Bridge driver **2042** and one or more H-bridge field-effect transistors (FETs) **2044**. The H-bridge FETs **2044** are coupled to the safety processor **2004**. A motor current sensor **2046** is coupled in series with the motor **2048** to measure the current draw of the motor **2048**. The motor current sensor **2046** is in signal communication with the primary processor **2006** and/or the safety processor **2004**. In some embodiments, the motor **2048** is coupled to a motor electromagnetic interference (EMI) filter **2050**.

(229) The segmented circuit **2000** comprises a power segment **2002h** (Segment 8). A battery **2008** is coupled to the safety processor **2004**, the primary processor **2006**, and one or more of the additional circuit segments **2002c-2002g**. The battery **2008** is coupled to the segmented circuit **2000** by a battery connector **2010** and a current sensor **2012**. The current sensor **2012** is configured to measure the total current draw of the segmented circuit **2000**. In some embodiments, one or more voltage converters **2014a**, **2014b**, **2016** are configured to provide predetermined voltage values to one or more circuit segments **2002a-2002g**. For example, in some embodiments, the segmented circuit **2000** may comprise 3.3V voltage converters **2014a-2014b** and/or 5V voltage converters **2016**. A boost converter **2018** is configured to provide a boost voltage up to a predetermined amount, such as, for example, up to 13V. The boost converter **2018** is configured to provide additional voltage and/or current during power intensive operations and prevent brownout or low-power conditions.

(230) In some embodiments, the safety segment **2002a** comprises a motor power interrupt **2020**. The motor power interrupt **2020** is coupled between the power segment **2002h** and the motor segment **2002g**. The safety segment **2002a** is configured to interrupt power to the motor segment **2002g** when an error or fault condition is detected by the safety processor **2004** and/or the primary processor **2006** as discussed in more detail herein. Although the circuit segments **2002a-2002g** are illustrated with all components of the circuit segments **2002a-2002h** located in physical proximity, one skilled in the art will recognize that a circuit segment **2002a-2002h** may comprise components physically and/or electrically separate from other components of the same circuit segment **2002a-2002g**. In some embodiments, one or more components may be shared between two or more circuit segments **2002a-2002g**.

(231) In some embodiments, a plurality of switches **2056-2070** are coupled to the safety processor **2004** and/or the primary processor **2006**. The plurality of switches **2056-2070** may be configured to control one or more operations of the surgical instrument **10**, control one or more operations of the segmented circuit **2000**, and/or indicate a status of the surgical instrument **10**. For example, a bail-out door switch **2056** is configured to indicate the status of a bail-out door. A plurality of articulation switches, such as, for example, a left side articulation left switch **2058a**, a left side articulation right switch **2060a**, a left side articulation center switch **2062a**, a right side articulation left switch **2058b**, a right side articulation right switch **2060b**, and a right side articulation center switch **2062b** are configured to control articulation of a shaft **2004** and/or an end effector **2006**. A left side reverse switch **2064a** and a right side reverse switch **2064b** are coupled to the primary processor **2006**. In some embodiments, the left side switches comprising the left side articulation left switch **2058a**, the left side articulation right switch **2060a**, the left side articulation center switch **2062a**, and the left side reverse switch **2064a** are coupled to the primary processor **2006** by a left flex connector **2072a**. The right side switches comprising the right side articulation left switch **2058b**, the right side articulation right switch **2060b**, the right side articulation center switch **2062b**, and the right side reverse switch **2064b** are coupled to the primary processor **2006** by a right flex connector **2072b**. In some embodiments, a firing switch **2066**, a clamp release switch **2068**, and a shaft engaged switch **2070** are coupled to the primary processor **2006**.

(232) The plurality of switches **2056-2070** may comprise, for example, a plurality of handle controls mounted to a handle of the surgical instrument **10**, a plurality of indicator switches, and/or any combination thereof. In various embodiments, the plurality of switches **2056-2070** allow a surgeon to manipulate the surgical instrument, provide feedback to the segmented circuit **2000** regarding the position and/or operation of the surgical instrument, and/or indicate unsafe operation of the surgical instrument **10**. In some embodiments, additional or fewer switches may be coupled to the segmented circuit **2000**, one or more of the switches **2056-2070** may be combined into a single switch, and/or expanded to multiple switches. For example, in one embodiment, one or more of the left side and/or right side articulation switches **2058a-2064b** may be combined into a single multi-position switch.

(233) In one embodiment, the safety processor **2004** is configured to implement a watchdog function, among other safety operations. The safety processor **2004** and the primary processor **2006** of the segmented circuit **2000** are in signal communication. A microprocessor alive heartbeat signal is provided at output **2096**. The acceleration segment **2002c** comprises an accelerometer **2022** configured to monitor movement of the surgical instrument **10**. In various embodiments, the accelerometer **2022** may be a single, double, or triple axis accelerometer. The accelerometer **2022** may be employed to measure proper acceleration that is not necessarily the coordinate acceleration (rate of change of velocity). Instead, the accelerometer sees the acceleration associated with the phenomenon of weight experienced by a test mass at rest in the frame of reference of the accelerometer **2022**. For example, the accelerometer **2022** at rest on the surface of the earth will measure an acceleration $g=9.8\text{ m/s}^2$ (gravity) straight upwards, due to its weight. Another type of acceleration that accelerometer **2022** can measure is g-force acceleration. In various other embodiments, the accelerometer **2022** may comprise a single, double, or triple axis accelerometer. Further, the acceleration segment **2002c** may comprise one or more inertial sensors to detect and measure acceleration, tilt, shock, vibration, rotation, and multiple degrees-of-freedom (DoF). A suitable inertial sensor may comprise an accelerometer (single, double, or triple axis), a magnetometer to measure a magnetic field in space such as the earth's magnetic field, and/or a gyroscope to measure angular velocity.

(234) In one embodiment, the safety processor **2004** is configured to implement a watchdog function with respect to one or more circuit segments **2002c-2002h**, such as, for example, the motor segment **2002g**. In this regard, the safety processor **2004** employs the watchdog function to detect and recover from malfunctions of the primary processor **2006**. During normal operation, the safety processor **2004** monitors for hardware faults or program errors of the primary processor **2004** and to initiate corrective action or actions. The corrective actions may include placing the primary processor **2006** in a safe state and restoring normal system operation. In one embodiment, the safety processor **2004** is coupled to at least a first sensor. The first sensor measures a first property of the surgical instrument **10**. In some embodiments, the safety processor **2004** is configured to compare the measured property of the surgical instrument **10** to a predetermined value. For example, in one embodiment, a motor sensor **2040a** is coupled to the safety processor **2004**. The motor sensor **2040a** provides motor speed and position information to the safety processor **2004**. The safety processor **2004** monitors the motor sensor **2040a** and compares the value to a maximum speed and/or position value and prevents operation of the motor **2048** above the predetermined values. In some embodiments, the predetermined values are calculated based on real-time speed and/or position of the motor **2048**, calculated from values supplied by a second motor sensor **2040b** in communication with the primary processor **2006**, and/or provided to the safety processor **2004** from, for example, a memory module coupled to the safety processor **2004**.

(235) In some embodiments, a second sensor is coupled to the primary processor **2006**. The second sensor is configured to measure the first physical property. The safety processor **2004** and the primary processor **2006** are configured to provide a signal indicative of the value of the first sensor and the second sensor respectively. When either the safety processor **2004** or the primary processor

2006 indicates a value outside of an acceptable range, the segmented circuit **2000** prevents operation of at least one of the circuit segments **2002c-2002h**, such as, for example, the motor segment **2002g**. For example, in the embodiment illustrated in FIGS. **21A-21B**, the safety processor **2004** is coupled to a first motor position sensor **2040a** and the primary processor **2006** is coupled to a second motor position sensor **2040b**. The motor position sensors **2040a**, **2040b** may comprise any suitable motor position sensor, such as, for example, a magnetic angle rotary input comprising a sine and cosine output. The motor position sensors **2040a**, **2040b** provide respective signals to the safety processor **2004** and the primary processor **2006** indicative of the position of the motor **2048**.

(236) The safety processor **2004** and the primary processor **2006** generate an activation signal when the values of the first motor sensor **2040a** and the second motor sensor **2040b** are within a predetermined range. When either the primary processor **2006** or the safety processor **2004** to detect a value outside of the predetermined range, the activation signal is terminated and operation of at least one circuit segment **2002c-2002h**, such as, for example, the motor segment **2002g**, is interrupted and/or prevented. For example, in some embodiments, the activation signal from the primary processor **2006** and the activation signal from the safety processor **2004** are coupled to an AND gate. The AND gate is coupled to a motor power switch **2020**. The AND gate maintains the motor power switch **2020** in a closed, or on, position when the activation signal from both the safety processor **2004** and the primary processor **2006** are high, indicating a value of the motor sensors **2040a**, **2040b** within the predetermined range. When either of the motor sensors **2040a**, **2040b** detect a value outside of the predetermined range, the activation signal from that motor sensor **2040a**, **2040b** is set low, and the output of the AND gate is set low, opening the motor power switch **2020**. In some embodiments, the value of the first sensor **2040a** and the second sensor **2040b** is compared, for example, by the safety processor **2004** and/or the primary processor **2006**. When the values of the first sensor and the second sensor are different, the safety processor **2004** and/or the primary processor **2006** may prevent operation of the motor segment **2002g**.

(237) In some embodiments, the safety processor **2004** receives a signal indicative of the value of the second sensor **2040b** and compares the second sensor value to the first sensor value. For example, in one embodiment, the safety processor **2004** is coupled directly to a first motor sensor **2040a**. A second motor sensor **2040b** is coupled to a primary processor **2006**, which provides the second motor sensor **2040b** value to the safety processor **2004**, and/or coupled directly to the safety processor **2004**. The safety processor **2004** compares the value of the first motor sensor **2040a** to the value of the second motor sensor **2040b**. When the safety processor **2004** detects a mismatch between the first motor sensor **2040a** and the second motor sensor **2040b**, the safety processor **2004** may interrupt operation of the motor segment **2002g**, for example, by cutting power to the motor segment **2002g**.

(238) In some embodiments, the safety processor **2004** and/or the primary processor **2006** is coupled to a first sensor **2040a** configured to measure a first property of a surgical instrument and a second sensor **2040b** configured to measure a second property of the surgical instrument. The first property and the second property comprise a predetermined relationship when the surgical instrument is operating normally. The safety processor **2004** monitors the first property and the second property. When a value of the first property and/or the second property inconsistent with the predetermined relationship is detected, a fault occurs. When a fault occurs, the safety processor **2004** takes at least one action, such as, for example, preventing operation of at least one of the circuit segments, executing a predetermined operation, and/or resetting the primary processor **2006**. For example, the safety processor **2004** may open the motor power switch **2020** to cut power to the motor circuit segment **2002g** when a fault is detected.

(239) In one embodiment, the safety processor **2004** is configured to execute an independent control algorithm. In operation, the safety processor **2004** monitors the segmented circuit **2000** and is configured to control and/or override signals from other circuit components, such as, for

example, the primary processor **2006**, independently. The safety processor **2004** may execute a preprogrammed algorithm and/or may be updated or programmed on the fly during operation based on one or more actions and/or positions of the surgical instrument **10**. For example, in one embodiment, the safety processor **2004** is reprogrammed with new parameters and/or safety algorithms each time a new shaft and/or end effector is coupled to the surgical instrument **10**. In some embodiments, one or more safety values stored by the safety processor **2004** are duplicated by the primary processor **2006**. Two-way error detection is performed to ensure values and/or parameters stored by either of the processors **2004**, **2006** are correct.

(240) In some embodiments, the safety processor **2004** and the primary processor **2006** implement a redundant safety check. The safety processor **2004** and the primary processor **2006** provide periodic signals indicating normal operation. For example, during operation, the safety processor **2004** may indicate to the primary processor **2006** that the safety processor **2004** is executing code and operating normally. The primary processor **2006** may, likewise, indicate to the safety processor **2004** that the primary processor **2006** is executing code and operating normally. In some embodiments, communication between the safety processor **2004** and the primary processor **2006** occurs at a predetermined interval. The predetermined interval may be constant or may be variable based on the circuit state and/or operation of the surgical instrument **10**.

(241) FIG. 22 illustrates one example of a power assembly **2100** comprising a usage cycle circuit **2102** configured to monitor a usage cycle count of the power assembly **2100**. The power assembly **2100** may be coupled to a surgical instrument **2110**. The usage cycle circuit **2102** comprises a processor **2104** and a use indicator **2106**. The use indicator **2106** is configured to provide a signal to the processor **2104** to indicate a use of the battery back **2100** and/or a surgical instrument **2110** coupled to the power assembly **2100**. A “use” may comprise any suitable action, condition, and/or parameter such as, for example, changing a modular component of a surgical instrument **2110**, deploying or firing a disposable component coupled to the surgical instrument **2110**, delivering electrosurgical energy from the surgical instrument **2110**, reconditioning the surgical instrument **2110** and/or the power assembly **2100**, exchanging the power assembly **2100**, recharging the power assembly **2100**, and/or exceeding a safety limitation of the surgical instrument **2110** and/or the battery back **2100**.

(242) In some instances, a usage cycle, or use, is defined by one or more power assembly **2100** parameters. For example, in one instance, a usage cycle comprises using more than 5% of the total energy available from the power assembly **2100** when the power assembly **2100** is at a full charge level. In another instance, a usage cycle comprises a continuous energy drain from the power assembly **2100** exceeding a predetermined time limit. For example, a usage cycle may correspond to five minutes of continuous and/or total energy draw from the power assembly **2100**. In some instances, the power assembly **2100** comprises a usage cycle circuit **2102** having a continuous power draw to maintain one or more components of the usage cycle circuit **2102**, such as, for example, the use indicator **2106** and/or a counter **2108**, in an active state.

(243) The processor **2104** maintains a usage cycle count. The usage cycle count indicates the number of uses detected by the use indicator **2106** for the power assembly **2100** and/or the surgical instrument **2110**. The processor **2104** may increment and/or decrement the usage cycle count based on input from the use indicator **2106**. The usage cycle count is used to control one or more operations of the power assembly **2100** and/or the surgical instrument **2110**. For example, in some instances, a power assembly **2100** is disabled when the usage cycle count exceeds a predetermined usage limit. Although the instances discussed herein are discussed with respect to incrementing the usage cycle count above a predetermined usage limit, those skilled in the art will recognize that the usage cycle count may start at a predetermined amount and may be decremented by the processor **2104**. In this instance, the processor **2104** initiates and/or prevents one or more operations of the power assembly **2100** when the usage cycle count falls below a predetermined usage limit.

(244) The usage cycle count is maintained by a counter **2108**. The counter **2108** comprises any

suitable circuit, such as, for example, a memory module, an analog counter, and/or any circuit configured to maintain a usage cycle count. In some instances, the counter **2108** is formed integrally with the processor **2104**. In other instances, the counter **2108** comprises a separate component, such as, for example, a solid state memory module. In some instances, the usage cycle count is provided to a remote system, such as, for example, a central database. The usage cycle count is transmitted by a communications module **2112** to the remote system. The communications module **2112** is configured to use any suitable communications medium, such as, for example, wired and/or wireless communication. In some instances, the communications module **2112** is configured to receive one or more instructions from the remote system, such as, for example, a control signal when the usage cycle count exceeds the predetermined usage limit.

(245) In some instances, the use indicator **2106** is configured to monitor the number of modular components used with a surgical instrument **2110** coupled to the power assembly **2100**. A modular component may comprise, for example, a modular shaft, a modular end effector, and/or any other modular component. In some instances, the use indicator **2106** monitors the use of one or more disposable components, such as, for example, insertion and/or deployment of a staple cartridge within an end effector coupled to the surgical instrument **2110**. The use indicator **2106** comprises one or more sensors for detecting the exchange of one or more modular and/or disposable components of the surgical instrument **2110**.

(246) In some instances, the use indicator **2106** is configured to monitor single patient surgical procedures performed while the power assembly **2100** is installed. For example, the use indicator **2106** may be configured to monitor firings of the surgical instrument **2110** while the power assembly **2100** is coupled to the surgical instrument **2110**. A firing may correspond to deployment of a staple cartridge, application of electrosurgical energy, and/or any other suitable surgical event. The use indicator **2106** may comprise one or more circuits for measuring the number of firings while the power assembly **2100** is installed. The use indicator **2106** provides a signal to the processor **2104** when a single patient procedure is performed and the processor **2104** increments the usage cycle count.

(247) In some instances, the use indicator **2106** comprises a circuit configured to monitor one or more parameters of the power source **2114**, such as, for example, a current draw from the power source **2114**. The one or more parameters of the power source **2114** correspond to one or more operations performable by the surgical instrument **2110**, such as, for example, a cutting and sealing operation. The use indicator **2106** provides the one or more parameters to the processor **2104**, which increments the usage cycle count when the one or more parameters indicate that a procedure has been performed.

(248) In some instances, the use indicator **2106** comprises a timing circuit configured to increment a usage cycle count after a predetermined time period. The predetermined time period corresponds to a single patient procedure time, which is the time required for an operator to perform a procedure, such as, for example, a cutting and sealing procedure. When the power assembly **2100** is coupled to the surgical instrument **2110**, the processor **2104** polls the use indicator **2106** to determine when the single patient procedure time has expired. When the predetermined time period has elapsed, the processor **2104** increments the usage cycle count. After incrementing the usage cycle count, the processor **2104** resets the timing circuit of the use indicator **2106**.

(249) In some instances, the use indicator **2106** comprises a time constant that approximates the single patient procedure time. In one embodiment, the usage cycle circuit **2102** comprises a resistor-capacitor (RC) timing circuit **2506**. The RC timing circuit comprises a time constant defined by a resistor-capacitor pair. The time constant is defined by the values of the resistor and the capacitor. In one embodiment, the usage cycle circuit **2552** comprises a rechargeable battery and a clock. When the power assembly **2100** is installed in a surgical instrument, the rechargeable battery is charged by the power source. The rechargeable battery comprises enough power to run the clock for at least the single patient procedure time. The clock may comprise a real time clock, a

processor configured to implement a time function, or any other suitable timing circuit.

(250) Referring back to FIG. 2, in some instances, the use indicator **2106** comprises a sensor configured to monitor one or more environmental conditions experienced by the power assembly **2100**. For example, the use indicator **2106** may comprise an accelerometer. The accelerometer is configured to monitor acceleration of the power assembly **2100**. The power assembly **2100** comprises a maximum acceleration tolerance. Acceleration above a predetermined threshold indicates, for example, that the power assembly **2100** has been dropped. When the use indicator **2106** detects acceleration above the maximum acceleration tolerance, the processor **2104** increments a usage cycle count. In some instances, the use indicator **2106** comprises a moisture sensor. The moisture sensor is configured to indicate when the power assembly **2100** has been exposed to moisture. The moisture sensor may comprise, for example, an immersion sensor configured to indicate when the power assembly **2100** has been fully immersed in a cleaning fluid, a moisture sensor configured to indicate when moisture is in contact with the power assembly **2100** during use, and/or any other suitable moisture sensor.

(251) In some instances, the use indicator **2106** comprises a chemical exposure sensor. The chemical exposure sensor is configured to indicate when the power assembly **2100** has come into contact with harmful and/or dangerous chemicals. For example, during a sterilization procedure, an inappropriate chemical may be used that leads to degradation of the power assembly **2100**. The processor **2104** increments the usage cycle count when the use indicator **2106** detects an inappropriate chemical.

(252) In some instances, the usage cycle circuit **2102** is configured to monitor the number of reconditioning cycles experienced by the power assembly **2100**. A reconditioning cycle may comprise, for example, a cleaning cycle, a sterilization cycle, a charging cycle, routine and/or preventative maintenance, and/or any other suitable reconditioning cycle. The use indicator **2106** is configured to detect a reconditioning cycle. For example, the use indicator **2106** may comprise a moisture sensor to detect a cleaning and/or sterilization cycle. In some instances, the usage cycle circuit **2102** monitors the number of reconditioning cycles experienced by the power assembly **2100** and disables the power assembly **2100** after the number of reconditioning cycles exceeds a predetermined threshold.

(253) The usage cycle circuit **2102** may be configured to monitor the number of power assembly **2100** exchanges. The usage cycle circuit **2102** increments the usage cycle count each time the power assembly **2100** is exchanged. When the maximum number of exchanges is exceeded the usage cycle circuit **2102** locks out the power assembly **2100** and/or the surgical instrument **2110**. In some instances, when the power assembly **2100** is coupled the surgical instrument **2110**, the usage cycle circuit **2102** identifies the serial number of the power assembly **2100** and locks the power assembly **2100** such that the power assembly **2100** is usable only with the surgical instrument **2110**. In some instances, the usage cycle circuit **2102** increments the usage cycle each time the power assembly **2100** is removed from and/or coupled to the surgical instrument **2110**.

(254) In some instances, the usage cycle count corresponds to sterilization of the power assembly **2100**. The use indicator **2106** comprises a sensor configured to detect one or more parameters of a sterilization cycle, such as, for example, a temperature parameter, a chemical parameter, a moisture parameter, and/or any other suitable parameter. The processor **2104** increments the usage cycle count when a sterilization parameter is detected. The usage cycle circuit **2102** disables the power assembly **2100** after a predetermined number of sterilizations. In some instances, the usage cycle circuit **2102** is reset during a sterilization cycle, a voltage sensor to detect a recharge cycle, and/or any suitable sensor. The processor **2104** increments the usage cycle count when a reconditioning cycle is detected. The usage cycle circuit **2102** is disabled when a sterilization cycle is detected. The usage cycle circuit **2102** is reactivated and/or reset when the power assembly **2100** is coupled to the surgical instrument **2110**. In some instances, the use indicator comprises a zero power indicator. The zero power indicator changes state during a sterilization cycle and is checked by the

processor **2104** when the power assembly **2100** is coupled to a surgical instrument **2110**. When the zero power indicator indicates that a sterilization cycle has occurred, the processor **2104** increments the usage cycle count.

(255) A counter **2108** maintains the usage cycle count. In some instances, the counter **2108** comprises a non-volatile memory module. The processor **2104** increments the usage cycle count stored in the non-volatile memory module each time a usage cycle is detected. The memory module may be accessed by the processor **2104** and/or a control circuit, such as, for example, the control circuit **200**. When the usage cycle count exceeds a predetermined threshold, the processor **2104** disables the power assembly **2100**. In some instances, the usage cycle count is maintained by a plurality of circuit components. For example, in one instance, the counter **2108** comprises a resistor (or fuse) pack. After each use of the power assembly **2100**, a resistor (or fuse) is burned to an open position, changing the resistance of the resistor pack. The power assembly **2100** and/or the surgical instrument **2110** reads the remaining resistance. When the last resistor of the resistor pack is burned out, the resistor pack has a predetermined resistance, such as, for example, an infinite resistance corresponding to an open circuit, which indicates that the power assembly **2100** has reached its usage limit. In some instances, the resistance of the resistor pack is used to derive the number of uses remaining.

(256) In some instances, the usage cycle circuit **2102** prevents further use of the power assembly **2100** and/or the surgical instrument **2110** when the usage cycle count exceeds a predetermined usage limit. In one instance, the usage cycle count associated with the power assembly **2100** is provided to an operator, for example, utilizing a screen formed integrally with the surgical instrument **2110**. The surgical instrument **2110** provides an indication to the operator that the usage cycle count has exceeded a predetermined limit for the power assembly **2100**, and prevents further operation of the surgical instrument **2110**.

(257) In some instances, the usage cycle circuit **2102** is configured to physically prevent operation when the predetermined usage limit is reached. For example, the power assembly **2100** may comprise a shield configured to deploy over contacts of the power assembly **2100** when the usage cycle count exceeds the predetermined usage limit. The shield prevents recharge and use of the power assembly **2100** by covering the electrical connections of the power assembly **2100**.

(258) In some instances, the usage cycle circuit **2102** is located at least partially within the surgical instrument **2110** and is configured to maintain a usage cycle count for the surgical instrument **2110**. FIG. 22 illustrates one or more components of the usage cycle circuit **2102** within the surgical instrument **2110** in phantom, illustrating the alternative positioning of the usage cycle circuit **2102**. When a predetermined usage limit of the surgical instrument **2110** is exceeded, the usage cycle circuit **2102** disables and/or prevents operation of the surgical instrument **2110**. The usage cycle count is incremented by the usage cycle circuit **2102** when the use indicator **2106** detects a specific event and/or requirement, such as, for example, firing of the surgical instrument **2110**, a predetermined time period corresponding to a single patient procedure time, based on one or more motor parameters of the surgical instrument **2110**, in response to a system diagnostic indicating that one or more predetermined thresholds are met, and/or any other suitable requirement. As discussed above, in some instances, the use indicator **2106** comprises a timing circuit corresponding to a single patient procedure time. In other instances, the use indicator **2106** comprises one or more sensors configured to detect a specific event and/or condition of the surgical instrument **2110**.

(259) In some instances, the usage cycle circuit **2102** is configured to prevent operation of the surgical instrument **2110** after the predetermined usage limit is reached. In some instances, the surgical instrument **2110** comprises a visible indicator to indicate when the predetermined usage limit has been reached and/or exceeded. For example, a flag, such as a red flag, may pop-up from the surgical instrument **2110**, such as from the handle, to provide a visual indication to the operator that the surgical instrument **2110** has exceeded the predetermined usage limit. As another example, the usage cycle circuit **2102** may be coupled to a display formed integrally with the surgical

instrument **2110**. The usage cycle circuit **2102** displays a message indicating that the predetermined usage limit has been exceeded. The surgical instrument **2110** may provide an audible indication to the operator that the predetermined usage limit has been exceeded. For example, in one instance, the surgical instrument **2110** emits an audible tone when the predetermined usage limit is exceeded and the power assembly **2100** is removed from the surgical instrument **2110**. The audible tone indicates the last use of the surgical instrument **2110** and indicates that the surgical instrument **2110** should be disposed or reconditioned.

(260) In some instances, the usage cycle circuit **2102** is configured to transmit the usage cycle count of the surgical instrument **2110** to a remote location, such as, for example, a central database. The usage cycle circuit **2102** comprises a communications module **2112** configured to transmit the usage cycle count to the remote location. The communications module **2112** may utilize any suitable communications system, such as, for example, wired or wireless communications system. The remote location may comprise a central database configured to maintain usage information. In some instances, when the power assembly **2100** is coupled to the surgical instrument **2110**, the power assembly **2100** records a serial number of the surgical instrument **2110**. The serial number is transmitted to the central database, for example, when the power assembly **2100** is coupled to a charger. In some instances, the central database maintains a count corresponding to each use of the surgical instrument **2110**. For example, a bar code associated with the surgical instrument **2110** may be scanned each time the surgical instrument **2110** is used. When the use count exceeds a predetermined usage limit, the central database provides a signal to the surgical instrument **2110** indicating that the surgical instrument **2110** should be discarded.

(261) The surgical instrument **2110** may be configured to lock and/or prevent operation of the surgical instrument **2110** when the usage cycle count exceeds a predetermined usage limit. In some instances, the surgical instrument **2110** comprises a disposable instrument and is discarded after the usage cycle count exceeds the predetermined usage limit. In other instances, the surgical instrument **2110** comprises a reusable surgical instrument which may be reconditioned after the usage cycle count exceeds the predetermined usage limit. The surgical instrument **2110** initiates a reversible lockout after the predetermined usage limit is met. A technician reconditions the surgical instrument **2110** and releases the lockout, for example, utilizing a specialized technician key configured to reset the usage cycle circuit **2102**.

(262) In some embodiments, the segmented circuit **2000** is configured for sequential start-up. An error check is performed by each circuit segment **2002a-2002g** prior to energizing the next sequential circuit segment **2002a-2002g**. FIG. 23 illustrates one embodiment of a process for sequentially energizing a segmented circuit **2270**, such as, for example, the segmented circuit **2000**. When a battery **2008** is coupled to the segmented circuit **2000**, the safety processor **2004** is energized **2272**. The safety processor **2004** performs a self-error check **2274**. When an error is detected **2276a**, the safety processor stops energizing the segmented circuit **2000** and generates an error code **2278a**. When no errors are detected **2276b**, the safety processor **2004** initiates **2278b** power-up of the primary processor **2006**. The primary processor **2006** performs a self-error check. When no errors are detected, the primary processor **2006** begins sequential power-up of each of the remaining circuit segments **2278b**. Each circuit segment is energized and error checked by the primary processor **2006**. When no errors are detected, the next circuit segment is energized **2278b**. When an error is detected, the safety processor **2004** and/or the primary process stops energizing the current segment and generates an error **2278a**. The sequential start-up continues until all of the circuit segments **2002a-2002g** have been energized. In some embodiments, the segmented circuit **2000** transitions from sleep mode following a similar sequential power-up process **11250**.

(263) FIG. 24 illustrates one embodiment of a power segment **2302** comprising a plurality of daisy chained power converters **2314**, **2316**, **2318**. The power segment **2302** comprises a battery **2308**. The battery **2308** is configured to provide a source voltage, such as, for example, 12V. A current sensor **2312** is coupled to the battery **2308** to monitor the current draw of a segmented circuit

and/or one or more circuit segments. The current sensor **2312** is coupled to an FET switch **2313**. The battery **2308** is coupled to one or more voltage converters **2309**, **2314**, **2316**. An always on converter **2309** provides a constant voltage to one or more circuit components, such as, for example, a motion sensor **2322**. The always on converter **2309** comprises, for example, a 3.3V converter. The always on converter **2309** may provide a constant voltage to additional circuit components, such as, for example, a safety processor (not shown). The battery **2308** is coupled to a boost converter **2318**. The boost converter **2318** is configured to provide a boosted voltage above the voltage provided by the battery **2308**. For example, in the illustrated embodiment, the battery **2308** provides a voltage of 12V. The boost converter **2318** is configured to boost the voltage to 13V. The boost converter **2318** is configured to maintain a minimum voltage during operation of a surgical instrument, for example, the surgical instrument **10** illustrated in FIGS. **69-71**. Operation of a motor can result in the power provided to the primary processor **2306** dropping below a minimum threshold and creating a brownout or reset condition in the primary processor **2306**. The boost converter **2318** ensures that sufficient power is available to the primary processor **2306** and/or other circuit components, such as the motor controller **2343**, during operation of the surgical instrument **10**. In some embodiments, the boost converter **2318** is coupled directly one or more circuit components, such as, for example, an OLED display **2388**.

(264) The boost converter **2318** is coupled to one or more step-down converters to provide voltages below the boosted voltage level. A first voltage converter **2316** is coupled to the boost converter **2318** and provides a first stepped-down voltage to one or more circuit components. In the illustrated embodiment, the first voltage converter **2316** provides a voltage of 5V. The first voltage converter **2316** is coupled to a rotary position encoder **2340**. A FET switch **2317** is coupled between the first voltage converter **2316** and the rotary position encoder **2340**. The FET switch **2317** is controlled by the processor **2306**. The processor **2306** opens the FET switch **2317** to deactivate the position encoder **2340**, for example, during power intensive operations. The first voltage converter **2316** is coupled to a second voltage converter **2314** configured to provide a second stepped-down voltage. The second stepped-down voltage comprises, for example, 3.3V. The second voltage converter **2314** is coupled to a processor **2306**. In some embodiments, the boost converter **2318**, the first voltage converter **2316**, and the second voltage converter **2314** are coupled in a daisy chain configuration. The daisy chain configuration allows the use of smaller, more efficient converters for generating voltage levels below the boosted voltage level. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(265) FIG. **25** illustrates one embodiment of a segmented circuit **2400** configured to maximize power available for critical and/or power intense functions. The segmented circuit **2400** comprises a battery **2408**. The battery **2408** is configured to provide a source voltage such as, for example, 12V. The source voltage is provided to a plurality of voltage converters **2409**, **2418**. An always-on voltage converter **2409** provides a constant voltage to one or more circuit components, for example, a motion sensor **2422** and a safety processor **2404**. The always-on voltage converter **2409** is directly coupled to the battery **2408**. The always-on converter **2409** provides a voltage of 3.3V, for example. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(266) The segmented circuit **2400** comprises a boost converter **2418**. The boost converter **2418** provides a boosted voltage above the source voltage provided by the battery **2408**, such as, for example, 13V. The boost converter **2418** provides a boosted voltage directly to one or more circuit components, such as, for example, an OLED display **2488** and a motor controller **2443**. By coupling the OLED display **2488** directly to the boost converter **2418**, the segmented circuit **2400** eliminates the need for a power converter dedicated to the OLED display **2488**. The boost converter **2418** provides a boosted voltage to the motor controller **2443** and the motor **2448** during one or more power intensive operations of the motor **2448**, such as, for example, a cutting

operation. The boost converter **2418** is coupled to a step-down converter **2416**. The step-down converter **2416** is configured to provide a voltage below the boosted voltage to one or more circuit components, such as, for example, 5V. The step-down converter **2416** is coupled to, for example, a FET switch **2451** and a position encoder **2440**. The FET switch **2451** is coupled to the primary processor **2406**. The primary processor **2406** opens the FET switch **2451** when transitioning the segmented circuit **2400** to sleep mode and/or during power intensive functions requiring additional voltage delivered to the motor **2448**. Opening the FET switch **2451** deactivates the position encoder **2440** and eliminates the power draw of the position encoder **2440**. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(267) The step-down converter **2416** is coupled to a linear converter **2414**. The linear converter **2414** is configured to provide a voltage of, for example, 3.3V. The linear converter **2414** is coupled to the primary processor **2406**. The linear converter **2414** provides an operating voltage to the primary processor **2406**. The linear converter **2414** may be coupled to one or more additional circuit components. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(268) The segmented circuit **2400** comprises a bailout switch **2456**. The bailout switch **2456** is coupled to a bailout door on the surgical instrument **10**. The bailout switch **2456** and the safety processor **2404** are coupled to an AND gate **2419**. The AND gate **2419** provides an input to a FET switch **2413**. When the bailout switch **2456** detects a bailout condition, the bailout switch **2456** provides a bailout shutdown signal to the AND gate **2419**. When the safety processor **2404** detects an unsafe condition, such as, for example, due to a sensor mismatch, the safety processor **2404** provides a shutdown signal to the AND gate **2419**. In some embodiments, both the bailout shutdown signal and the shutdown signal are high during normal operation and are low when a bailout condition or an unsafe condition is detected. When the output of the AND gate **2419** is low, the FET switch **2413** is opened and operation of the motor **2448** is prevented. In some embodiments, the safety processor **2404** utilizes the shutdown signal to transition the motor **2448** to an off state in sleep mode. A third input to the FET switch **2413** is provided by a current sensor **2412** coupled to the battery **2408**. The current sensor **2412** monitors the current drawn by the circuit **2400** and opens the FET switch **2413** to shut-off power to the motor **2448** when an electrical current above a predetermined threshold is detected. The FET switch **2413** and the motor controller **2443** are coupled to a bank of FET switches **2445** configured to control operation of the motor **2448**.

(269) A motor current sensor **2446** is coupled in series with the motor **2448** to provide a motor current sensor reading to a current monitor **2447**. The current monitor **2447** is coupled to the primary processor **2406**. The current monitor **2447** provides a signal indicative of the current draw of the motor **2448**. The primary processor **2406** may utilize the signal from the motor current **2447** to control operation of the motor, for example, to ensure the current draw of the motor **2448** is within an acceptable range, to compare the current draw of the motor **2448** to one or more other parameters of the circuit **2400** such as, for example, the position encoder **2440**, and/or to determine one or more parameters of a treatment site. In some embodiments, the current monitor **2447** may be coupled to the safety processor **2404**.

(270) In some embodiments, actuation of one or more handle controls, such as, for example, a firing trigger, causes the primary processor **2406** to decrease power to one or more components while the handle control is actuated. For example, in one embodiment, a firing trigger controls a firing stroke of a cutting member. The cutting member is driven by the motor **2448**. Actuation of the firing trigger results in forward operation of the motor **2448** and advancement of the cutting member. During firing, the primary processor **2406** closes the FET switch **2451** to remove power from the position encoder **2440**. The deactivation of one or more circuit components allows higher power to be delivered to the motor **2448**. When the firing trigger is released, full power is restored to the deactivated components, for example, by closing the FET switch **2451** and reactivating the

position encoder **2440**.

(271) In some embodiments, the safety processor **2404** controls operation of the segmented circuit **2400**. For example, the safety processor **2404** may initiate a sequential power-up of the segmented circuit **2400**, transition of the segmented circuit **2400** to and from sleep mode, and/or may override one or more control signals from the primary processor **2406**. For example, in the illustrated embodiment, the safety processor **2404** is coupled to the step-down converter **2416**. The safety processor **2404** controls operation of the segmented circuit **2400** by activating or deactivating the step-down converter **2416** to provide power to the remainder of the segmented circuit **2400**.

(272) FIG. **26** illustrates one embodiment of a power system **2500** comprising a plurality of daisy chained power converters **2514**, **2516**, **2518** configured to be sequentially energized. The plurality of daisy chained power converters **2514**, **2516**, **2518** may be sequentially activated by, for example, a safety processor during initial power-up and/or transition from sleep mode. The safety processor may be powered by an independent power converter (not shown). For example, in one embodiment, when a battery voltage $V_{\text{sub.BATT}}$ is coupled to the power system **2500** and/or an accelerometer detects movement in sleep mode, the safety processor initiates a sequential start-up of the daisy chained power converters **2514**, **2516**, **2518**. The safety processor activates the 13V boost section **2518**. The boost section **2518** is energized and performs a self-check. In some embodiments, the boost section **2518** comprises an integrated circuit **2520** configured to boost the source voltage and to perform a self check. A diode **D** prevents power-up of a 5V supply section **2516** until the boost section **2518** has completed a self-check and provided a signal to the diode **D** indicating that the boost section **2518** did not identify any errors. In some embodiments, this signal is provided by the safety processor. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(273) The 5V supply section **2516** is sequentially powered-up after the boost section **2518**. The supply section **2516** performs a self-check during power-up to identify any errors in the 5V supply section **2516**. The 5V supply section **2516** comprises an integrated circuit **2515** configured to provide a step-down voltage from the boost voltage and to perform an error check. When no errors are detected, the supply section **2516** completes sequential power-up and provides an activation signal to the 3.3V supply section **2514**. In some embodiments, the safety processor provides an activation signal to the 3.3V supply section **2514**. The 3.3V supply section comprises an integrated circuit **2513** configured to provide a step-down voltage from the 5V supply section **2516** and perform a self-error check during power-up. When no errors are detected during the self-check, the 3.3V supply section **2514** provides power to the primary processor. The primary processor is configured to sequentially energize each of the remaining circuit segments. By sequentially energizing the power system **2500** and/or the remainder of a segmented circuit, the power system **2500** reduces error risks, allows for stabilization of voltage levels before loads are applied, and prevents large current draws from all hardware being turned on simultaneously in an uncontrolled manner. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(274) In one embodiment, the power system **2500** comprises an over voltage identification and mitigation circuit. The over voltage identification and mitigation circuit is configured to detect a monopolar return current in the surgical instrument and interrupt power from the power segment when the monopolar return current is detected. The over voltage identification and mitigation circuit is configured to identify ground floatation of the power system. The over voltage identification and mitigation circuit comprises a metal oxide varistor. The over voltage identification and mitigation circuit comprises at least one transient voltage suppression diode.

(275) FIG. **27** illustrates one embodiment of a segmented circuit **2600** comprising an isolated control section **2602**. The isolated control section **2602** isolates control hardware of the segmented circuit **2600** from a power section (not shown) of the segmented circuit **2600**. The control section **2602** comprises, for example, a primary processor **2606**, a safety processor (not shown), and/or

additional control hardware, for example, a FET Switch **2617**. The power section comprises, for example, a motor, a motor driver, and/or a plurality of motor MOSFETS. The isolated control section **2602** comprises a charging circuit **2603** and a rechargeable battery **2608** coupled to a 5V power converter **2616**. The charging circuit **2603** and the rechargeable battery **2608** isolate the primary processor **2606** from the power section. In some embodiments, the rechargeable battery **2608** is coupled to a safety processor and any additional support hardware. Isolating the control section **2602** from the power section allows the control section **2602**, for example, the primary processor **2606**, to remain active even when main power is removed, provides a filter, through the rechargeable battery **2608**, to keep noise out of the control section **2602**, isolates the control section **2602** from heavy swings in the battery voltage to ensure proper operation even during heavy motor loads, and/or allows for real-time operating system (RTOS) to be used by the segmented circuit **2600**. In some embodiments, the rechargeable battery **2608** provides a stepped-down voltage to the primary processor, such as, for example, 3.3V. The embodiments, however, are not limited to the particular voltage range(s) described in the context of this specification.

(276) Use of Multiple Sensors with One Sensor Affecting a Second Sensor's Output or Interpretation

(277) FIG. **28** illustrates one embodiment of an end effector **3000** comprising a first sensor **3008a** and a second sensor **3008b**. The end effector **3000** is similar to the end effector **300** described above. The end effector **3000** comprises a first jaw member, or anvil, **3002** pivotally coupled to a second jaw member **3004**. The second jaw member **3004** is configured to receive a staple cartridge **3006** therein. The staple cartridge **3006** comprises a plurality of staples (not shown). The plurality of staples is deployable from the staple cartridge **3006** during a surgical operation. The end effector **3000** comprises a first sensor **3008a**. The first sensor **3008a** is configured to measure one or more parameters of the end effector **3000**. For example, in one embodiment, the first sensor **3008a** is configured to measure the gap **3010** between the anvil **3002** and the second jaw member **3004**. The first sensor **3008a** may comprise, for example, a Hall effect sensor configured to detect a magnetic field generated by a magnet **3012** embedded in the second jaw member **3004** and/or the staple cartridge **3006**. As another example, in one embodiment, the first sensor **3008a** is configured to measure one or more forces exerted on the anvil **3002** by the second jaw member **3004** and/or tissue clamped between the anvil **3002** and the second jaw member **3004**.

(278) The end effector **3000** comprises a second sensor **3008b**. The second sensor **3008b** is configured to measure one or more parameters of the end effector **3000**. For example, in various embodiments, the second sensor **3008b** may comprise a strain gauge configured to measure the magnitude of the strain in the anvil **3002** during a clamped condition. The strain gauge provides an electrical signal whose amplitude varies with the magnitude of the strain. In various embodiments, the first sensor **3008a** and/or the second sensor **3008b** may comprise, for example, a magnetic sensor such as, for example, a Hall effect sensor, a strain gauge, a pressure sensor, a force sensor, an inductive sensor such as, for example, an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor for measuring one or more parameters of the end effector **3000**. The first sensor **3008a** and the second sensor **3008b** may be arranged in a series configuration and/or a parallel configuration. In a series configuration, the second sensor **3008b** may be configured to directly affect the output of the first sensor **3008a**. In a parallel configuration, the second sensor **3008b** may be configured to indirectly affect the output of the first sensor **3008a**.

(279) In one embodiment, the one or more parameters measured by the first sensor **3008a** are related to the one or more parameters measured by the second sensor **3008b**. For example, in one embodiment, the first sensor **3008a** is configured to measure the gap **3010** between the anvil **3002** and the second jaw member **3004**. The gap **3010** is representative of the thickness and/or compressibility of a tissue section clamped between the anvil **3002** and the staple cartridge **3006**. The first sensor **3008a** may comprise, for example, a Hall effect sensor configured to detect a

magnetic field generated by a magnet **3012** coupled to the second jaw member **3004** and/or the staple cartridge **3006**. Measuring at a single location accurately describes the compressed tissue thickness for a calibrated full bite of tissue, but may provide inaccurate results when a partial bite of tissue is placed between the anvil **3002** and the second jaw member **3004**. A partial bite of tissue, either a proximal partial bite or a distal partial bite, changes the clamping geometry of the anvil **3002**.

(280) In some embodiments, the second sensor **3008b** is configured to detect one or more parameters indicative of a type of tissue bite, for example, a full bite, a partial proximal bite, and/or a partial distal bite. The measurement of the second sensor **3008b** may be used to adjust the measurement of the first sensor **3008a** to accurately represent a proximal or distal positioned partial bite's true compressed tissue thickness. For example, in one embodiment, the second sensor **3008b** comprises a strain gauge, such as, for example, a micro-strain gauge, configured to monitor the amplitude of the strain in the anvil during a clamped condition. The amplitude of the strain of the anvil **3002** is used to modify the output of the first sensor **3008a**, for example, a Hall effect sensor, to accurately represent a proximal or distal positioned partial bite's true compressed tissue thickness. The first sensor **3008a** and the second sensor **3008b** may be measured in real-time during a clamping operation. Real-time measurement allows time based information to be analyzed, for example, by the primary processor **2006**, and used to select one or more algorithms and/or look-up tables to recognize tissue characteristics and clamping positioning to dynamically adjust tissue thickness measurements.

(281) In some embodiments, the thickness measurement of the first sensor **3008a** may be provided to an output device of a surgical instrument **10** coupled to the end effector **3000**. For example, in one embodiment, the end effector **3000** is coupled to the surgical instrument **10** comprising a display **2028**. The measurement of the first sensor **3008a** is provided to a processor, for example, the primary processor **2006**. The primary processor **2006** adjusts the measurement of the first sensor **3008a** based on the measurement of the second sensor **3008b** to reflect the true tissue thickness of a tissue section clamped between the anvil **3002** and the staple cartridge **3006**. The primary processor **2006** outputs the adjusted tissue thickness measurement and an indication of full or partial bite to the display **2028**. An operator may determine whether or not to deploy the staples in the staple cartridge **3006** based on the displayed values.

(282) In some embodiments, the first sensor **3008a** and the second sensor **3008b** may be located in different environments, such as, for example, the first sensor **3008a** being located within a patient at a treatment site and the second sensor **3008b** being located externally to the patient. The second sensor **3008b** may be configured to calibrate and/or modify the output of the first sensor **3008a**. The first sensor **3008a** and/or the second sensor **3008b** may comprise, for example, an environmental sensor. Environmental sensors may comprise, for example, temperature sensors, humidity sensors, pressure sensors, and/or any other suitable environmental sensor.

(283) FIG. **29** is a logic diagram illustrating one embodiment of a process **3020** for adjusting the measurement of a first sensor **3008a** based on input from a second sensor **3008b**. A first signal is captured **3022a** by the first sensor **3008a**. The first signal **3022a** may be conditioned based on one or more predetermined parameters, such as, for example, a smoothing function, a look-up table, and/or any other suitable conditioning parameters. A second signal is captured **3022b** by the second sensor **3008b**. The second signal **3022b** may be conditioned based on one or more predetermined conditioning parameters. The first signal **3022a** and the second signal **3022b** are provided to a processor, such as, for example, the primary processor **2006**. The processor **2006** adjusts the measurement of the first sensor **3022a**, as represented by the first signal **3022a**, based on the second signal **3022b** from the second sensor. For example, in one embodiment, the first sensor **3022a** comprises a Hall effect sensor and the second sensor **3022b** comprises a strain gauge. The distance measurement of the first sensor **3022a** is adjusted by the amplitude of the strain measured by the second sensor **3022b** to determine the fullness of the bite of tissue in the end effector **3000**.

The adjusted measurement is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(284) FIG. **30** is a logic diagram illustrating one embodiment of a process **3030** for determining a look-up table for a first sensor **3008a** based on the input from a second sensor **3008b**. The first sensor **3008a** captures **3022a** a signal indicative of one or more parameters of the end effector **3000**. The first signal **3022a** may be conditioned based on one or more predetermined parameters, such as, for example, a smoothing function, a look-up table, and/or any other suitable conditioning parameters. A second signal is captured **3022b** by the second sensor **3008b**. The second signal **3022b** may be conditioned based on one or more predetermined conditioning parameters. The first signal **3022a** and the second signal **3022b** are provided to a processor, such as, for example, the primary processor **2006**. The processor **2006** selects a look-up table from one or more available look-up tables **3034a**, **3034b** based on the value of the second signal. The selected look-up table is used to convert the first signal into a thickness measurement of the tissue located between the anvil **3002** and the staple cartridge **3006**. The adjusted measurement is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(285) FIG. **31** is a logic diagram illustrating one embodiment of a process **3040** for calibrating a first sensor **3008a** in response to an input from a second sensor **3008b**. The first sensor **3008a** is configured to capture **3022a** a signal indicative of one or more parameters of the end effector **3000**. The first signal **3022a** may be conditioned based on one or more predetermined parameters, such as, for example, a smoothing function, a look-up table, and/or any other suitable conditioning parameters. A second signal is captured **3022b** by the second sensor **3008b**. The second signal **3022b** may be conditioned based on one or more predetermined conditioning parameters. The first signal **3022a** and the second signal **3022b** are provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** calibrates **3042** the first signal **3022a** in response to the second signal **3022b**. The first signal **3022a** is calibrated **3042** to reflect the fullness of the bite of tissue in the end effector **3000**. The calibrated signal is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(286) FIG. **32A** is a logic diagram illustrating one embodiment of a process **3050** for determining and displaying the thickness of a tissue section clamped between the anvil **3002** and the staple cartridge **3006** of the end effector **3000**. The process **3050** comprises obtaining a Hall effect voltage **3052**, for example, through a Hall effect sensor located at the distal tip of the anvil **3002**. The Hall effect voltage **3052** is provided to an analog to digital convertor **3054** and converted into a digital signal. The digital signal is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** calibrates **3056** the curve input of the Hall effect voltage **3052** signal. A strain gauge **3058**, such as, for example, a micro-strain gauge, is configured to measure one or more parameters of the end effector **3000**, such as, for example, the amplitude of the strain exerted on the anvil **3002** during a clamping operation. The measured strain is converted **3060** to a digital signal and provided to the processor, such as, for example, the primary processor **2006**. The primary processor **2006** uses one or more algorithms and/or lookup tables to adjust the Hall effect voltage **3052** in response to the strain measured by the strain gauge **3058** to reflect the true thickness and fullness of the bite of tissue clamped by the anvil **3002** and the staple cartridge **3006**. The adjusted thickness is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(287) In some embodiments, the surgical instrument can further comprise a load cell or sensor **3082**. The load sensor **3082** can be located, for instance, in the shaft assembly **200**, described above, or in the housing **12**, also described above. FIG. **32B** is a logic diagram illustrating one embodiment of a process **3070** for determining and displaying the thickness of a tissue section clamped between the anvil **3002** and the staple cartridge **3006** of the end effector **3000**. The process comprises obtaining a Hall effect voltage **3072**, for example, through a Hall effect sensor located at the distal tip of the anvil **3002**. The Hall effect voltage **3072** is provided to an analog to digital

converter **3074** and converted into a digital signal. The digital signal is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** applies calibrates **3076** the curve input of the Hall effect voltage **3072** signal. A strain gauge **3078**, such as, for example, a micro-strain gauge, is configured to measure one or more parameters of the end effector **3000**, such as, for example, the amplitude of the strain exerted on the anvil **3002** during a clamping operation. The measured strain is converted **3080** to a digital signal and provided to the processor, such as, for example, the primary processor **2006**. The load sensor **3082** measures the clamping force of the anvil **3002** against the staple cartridge **3006**. The measured clamping force is converted **3084** to a digital signal and provided to the processor, such as for example, the primary processor **2006**. The primary processor **2006** uses one or more algorithms and/or lookup tables to adjust the Hall effect voltage **3072** in response to the strain measured by the strain gauge **3078** and the clamping force measured by the load sensor **3082** to reflect the true thickness and fullness of the bite of tissue clamped by the anvil **3002** and the staple cartridge **3006**. The adjusted thickness is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(288) FIG. **33** is a graph **3090** illustrating an adjusted Hall effect thickness measurement **3094** compared to an unmodified Hall effect thickness measurement **3092**. As shown in FIG. **33**, the unmodified Hall effect thickness measurement **3092** indicates a thicker tissue measurement, as the single sensor is unable to compensate for partial distal/proximal bites that result in incorrect thickness measurements. The adjusted thickness measurement **3094** is generated by, for example, the process **3050** illustrated in FIG. **32A**. The Hall effect thickness measurement **3092** is calibrated based on input from one or more additional sensors, such as, for example, a strain gauge. The adjusted Hall effect thickness **3094** reflects the true thickness of the tissue located between an anvil **3002** and a staple cartridge **3006**.

(289) FIG. **34** illustrates one embodiment of an end effector **3100** comprising a first sensor **3108a** and a second sensor **3108b**. The end effector **3100** is similar to the end effector **3000** illustrated in FIG. **28**. The end effector **3100** comprises a first jaw member, or anvil, **3102** pivotally coupled to a second jaw member **3104**. The second jaw member **3104** is configured to receive a staple cartridge **3106** therein. The end effector **3100** comprises a first sensor **3108a** coupled to the anvil **3102**. The first sensor **3108a** is configured to measure one or more parameters of the end effector **3100**, such as, for example, the gap **3110** between the anvil **3102** and the staple cartridge **3106**. The gap **3110** may correspond to, for example, a thickness of tissue clamped between the anvil **3102** and the staple cartridge **3106**. The first sensor **3108a** may comprise any suitable sensor for measuring one or more parameters of the end effector. For example, in various embodiments, the first sensor **3108a** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

(290) In some embodiments, the end effector **3100** comprises a second sensor **3108b**. The second sensor **3108b** is coupled to second jaw member **3104** and/or the staple cartridge **3106**. The second sensor **3108b** is configured to detect one or more parameters of the end effector **3100**. For example, in some embodiments, the second sensor **3108b** is configured to detect one or more instrument conditions such as, for example, a color of the staple cartridge **3106** coupled to the second jaw member **3104**, a length of the staple cartridge **3106**, a clamping condition of the end effector **3100**, the number of uses/number of remaining uses of the end effector **3100** and/or the staple cartridge **3106**, and/or any other suitable instrument condition. The second sensor **3108b** may comprise any suitable sensor for detecting one or more instrument conditions, such as, for example, a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

(291) The end effector **3100** may be used in conjunction with any of the processes shown in FIGS.

29-33. For example, in one embodiment, input from the second sensor **3108b** may be used to calibrate the input of the first sensor **3108a**. The second sensor **3108b** may be configured to detect one or more parameters of the staple cartridge **3106**, such as, for example, the color and/or length of the staple cartridge **3106**. The detected parameters, such as the color and/or the length of the staple cartridge **3106**, may correspond to one or more properties of the cartridge, such as, for example, the height of the cartridge deck, the thickness of tissue useable/optimal for the staple cartridge, and/or the pattern of the staples in the staple cartridge **3106**. The known parameters of the staple cartridge **3106** may be used to adjust the thickness measurement provided by the first sensor **3108a**. For example, if the staple cartridge **3106** has a higher deck height, the thickness measurement provided by the first sensor **3108a** may be reduced to compensate for the added deck height. The adjusted thickness may be displayed to an operator, for example, through a display **2026** coupled to the surgical instrument **10**.

(292) FIG. **35** illustrates one embodiment of an end effector **3150** comprising a first sensor **3158** and a plurality of second sensors **3160a**, **3160b**. The end effector **3150** comprises a first jaw member, or anvil, **3152** and a second jaw member **3154**. The second jaw member **3154** is configured to receive a staple cartridge **3156**. The anvil **3152** is pivotally moveable with respect to the second jaw member **3154** to clamp tissue between the anvil **3152** and the staple cartridge **3156**. The anvil comprises a first sensor **3158**. The first sensor **3158** is configured to detect one or more parameters of the end effector **3150**, such as, for example, the gap **3110** between the anvil **3152** and the staple cartridge **3156**. The gap **3110** may correspond to, for example, a thickness of tissue clamped between the anvil **3152** and the staple cartridge **3156**. The first sensor **3158** may comprise any suitable sensor for measuring one or more parameters of the end effector. For example, in various embodiments, the first sensor **3158** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

(293) In some embodiments, the end effector **3150** comprises a plurality of secondary sensors **3160a**, **3160b**. The secondary sensors **3160a**, **3160b** are configured to detect one or more parameters of the end effector **3150**. For example, in some embodiments, the secondary sensors **3160a**, **3160b** are configured to measure an amplitude of strain exerted on the anvil **3152** during a clamping procedure. In various embodiments, the secondary sensors **3160a**, **3160b** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor. The secondary sensors **3160a**, **3160b** may be configured to measure one or more identical parameters at different locations of the anvil **3152**, different parameters at identical locations on the anvil **3152**, and/or different parameters at different locations on the anvil **3152**.

(294) FIG. **36** is a logic diagram illustrating one embodiment of a process **3170** for adjusting a measurement of a first sensor **3158** in response to a plurality of secondary sensors **3160a**, **3160b**. In one embodiment, a Hall effect voltage is obtained **3172**, for example, by a Hall effect sensor. The Hall effect voltage is converted **3174** by an analog to digital convertor. The converted Hall effect voltage signal is calibrated **3176**. The calibrated curve represents the thickness of a tissue section located between the anvil **3152** and the staple cartridge **3156**. A plurality of secondary measurements are obtained **3178a**, **3178b** by a plurality of secondary sensors, such as, for example, a plurality of strain gauges. The input of the strain gauges is converted **3180a**, **3180b** into one or more digital signals, for example, by a plurality of electronic μ Strain conversion circuits. The calibrated Hall effect voltage and the plurality of secondary measurements are provided to a processor, such as, for example, the primary processor **2006**. The primary processor utilizes the secondary measurements to adjust **3182** the Hall effect voltage, for example, by applying an algorithm and/or utilizing one or more look-up tables. The adjusted Hall effect voltage represents the true thickness and fullness of the bite of tissue clamped by the anvil **3152** and the staple cartridge **3156**. The adjusted thickness is displayed **3026** to an operator by, for example, a display

2026 embedded in the surgical instrument **10**.

(295) FIG. **37** illustrates one embodiment of a circuit **3190** configured to convert signals from the first sensor **3158** and the plurality of secondary sensors **3160a**, **3160b** into digital signals receivable by a processor, such as, for example, the primary processor **2006**. The circuit **3190** comprises an analog-to-digital convertor **3194**. In some embodiments, the analog-to-digital convertor **3194** comprises a 4-channel, 18-bit analog to digital convertor. Those skilled in the art will recognize that the analog-to-digital convertor **3194** may comprise any suitable number of channels and/or bits to convert one or more inputs from analog to digital signals. The circuit **3190** comprises one or more level shifting resistors **3196** configured to receive an input from the first sensor **3158**, such as, for example, a Hall effect sensor. The level shifting resistors **3196** adjust the input from the first sensor, shifting the value to a higher or lower voltage depending on the input. The level shifting resistors **3196** provide the level-shifted input from the first sensor **3158** to the analog-to-digital convertor.

(296) In some embodiments, a plurality of secondary sensors **3160a**, **3160b** are coupled to a plurality of bridges **3192a**, **3192b** within the circuit **3190**. The plurality of bridges **3192a**, **3192b** may provide filtering of the input from the plurality of secondary sensors **3160a**, **3160b**. After filtering the input signals, the plurality of bridges **3192a**, **3192b** provide the inputs from the plurality of secondary sensors **3160a**, **3160b** to the analog-to-digital convertor **3194**. In some embodiments, a switch **3198** coupled to one or more level shifting resistors may be coupled to the analog-to-digital convertor **3194**. The switch **3198** is configured to calibrate one or more of the input signals, such as, for example, an input from a Hall effect sensor. The switch **3198** may be engaged to provide one or more level shifting signals to adjust the input of one or more of the sensors, such as, for example, to calibrate the input of a Hall effect sensor. In some embodiments, the adjustment is not necessary, and the switch **3198** is left in the open position to decouple the level shifting resistors. The switch **3198** is coupled to the analog-to-digital convertor **3194**. The analog-to-digital convertor **3194** provides an output to one or more processors, such as, for example, the primary processor **2006**. The primary processor **2006** calculates one or more parameters of the end effector **3150** based on the input from the analog-to-digital convertor **3194**. For example, in one embodiment, the primary processor **2006** calculates a thickness of tissue located between the anvil **3152** and the staple cartridge **3156** based on input from one or more sensors **3158**, **3160a**, **3160b**.

(297) FIG. **38** illustrates one embodiment of an end effector **3200** comprising a plurality of sensors **3208a-3208d**. The end effector **3200** comprises an anvil **3202** pivotally coupled to a second jaw member **3204**. The second jaw member **3204** is configured to receive a staple cartridge **3206** therein. The anvil **3202** comprises a plurality of sensors **3208a-3208d** thereon. The plurality of sensors **3208a-3208d** is configured to detect one or more parameters of the end effector **3200**, such as, for example, the anvil **3202**. The plurality of sensors **3208a-3208d** may comprise one or more identical sensors and/or different sensors. The plurality of sensors **3208a-3208d** may comprise, for example, magnetic sensors, such as a Hall effect sensor, strain gauges, pressure sensors, inductive sensors, such as an eddy current sensor, resistive sensors, capacitive sensors, optical sensors, and/or any other suitable sensors or combination thereof. For example, in one embodiment, the plurality of sensors **3208a-3208d** may comprise a plurality of strain gauges.

(298) In one embodiment, the plurality of sensors **3208a-3208d** allows a robust tissue thickness sensing process to be implemented. By detecting various parameters along the length of the anvil **3202**, the plurality of sensors **3208a-3208d** allow a surgical instrument, such as, for example, the surgical instrument **10**, to calculate the tissue thickness in the jaws regardless of the bite, for example, a partial or full bite. In some embodiments, the plurality of sensors **3208a-3208d** comprises a plurality of strain gauges. The plurality of strain gauges is configured to measure the strain at various points on the anvil **3202**. The amplitude and/or the slope of the strain at each of the various points on the anvil **3202** can be used to determine the thickness of tissue in between the

anvil **3202** and the staple cartridge **3206**. The plurality of strain gauges may be configured to optimize maximum amplitude and/or slope differences based on clamping dynamics to determine thickness, tissue placement, and/or material properties of the tissue. Time based monitoring of the plurality of sensors **3208a-3208d** during clamping allows a processor, such as, for example, the primary processor **2006**, to utilize algorithms and look-up tables to recognize tissue characteristics and clamping positions and dynamically adjust the end effector **3200** and/or tissue clamped between the anvil **3202** and the staple cartridge **3206**.

(299) FIG. **39** is a logic diagram illustrating one embodiment of a process **3220** for determining one or more tissue properties based on a plurality of sensors **3208a-3208d**. In one embodiment, a plurality of sensors **3208a-3208d** generate **3222a-3222d** a plurality of signals indicative of one or more parameters of the end effector **3200**. The plurality of generated signals is converted **3224a-3224d** to digital signals and provided to a processor. For example, in one embodiment comprising a plurality of strain gauges, a plurality of electronic μ Strain (micro-strain) conversion circuits convert **3224a-3224d** the strain gauge signals to digital signals. The digital signals are provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** determines **3226** one or more tissue characteristics based on the plurality of signals. The processor **2006** may determine the one or more tissue characteristics by applying an algorithm and/or a look-up table. The one or more tissue characteristics are displayed **3026** to an operator, for example, by a display **2026** embedded in the surgical instrument **10**.

(300) FIG. **40** illustrates one embodiment of an end effector **3250** comprising a plurality of sensors **3260a-3260d** coupled to a second jaw member **3254**. The end effector **3250** comprises an anvil **3252** pivotally coupled to a second jaw member **3254**. The anvil **3252** is moveable relative to the second jaw member **3254** to clamp one or more materials, such as, for example, a tissue section **3264**, therebetween. The second jaw member **3254** is configured to receive a staple cartridge **3256**. A first sensor **3258** is coupled to the anvil **3252**. The first sensor is configured to detect one or more parameters of the end effector **3150**, such as, for example, the gap **3110** between the anvil **3252** and the staple cartridge **3256**. The gap **3110** may correspond to, for example, a thickness of tissue clamped between the anvil **3252** and the staple cartridge **3256**. The first sensor **3258** may comprise any suitable sensor for measuring one or more parameters of the end effector. For example, in various embodiments, the first sensor **3258** may comprise a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

(301) A plurality of secondary sensors **3260a-3260d** is coupled to the second jaw member **3254**. The plurality of secondary sensors **3260a-3260d** may be formed integrally with the second jaw member **3254** and/or the staple cartridge **3256**. For example, in one embodiment, the plurality of secondary sensors **3260a-3260d** is disposed on an outer row of the staple cartridge **3256** (see FIG. **41**). The plurality of secondary sensors **3260a-3260d** are configured to detect one or more parameters of the end effector **3250** and/or a tissue section **3264** clamped between the anvil **3252** and the staple cartridge **3256**. The plurality of secondary sensors **3260a-3260d** may comprise any suitable sensors for detecting one or more parameters of the end effector **3250** and/or the tissue section **3264**, such as, for example, magnetic sensors, such as a Hall effect sensor, strain gauges, pressure sensors, inductive sensors, such as an eddy current sensor, resistive sensors, capacitive sensors, optical sensors, and/or any other suitable sensors or combination thereof. The plurality of secondary sensors **3260a-3260d** may comprise identical sensors and/or different sensors.

(302) In some embodiments, the plurality of secondary sensors **3260a-3260d** comprises dual purpose sensors and tissue stabilizing elements. The plurality of secondary sensors **3260a-3260d** comprise electrodes and/or sensing geometries configured to create a stabilized tissue condition when the plurality of secondary sensors **3260a-3260d** are engaged with a tissue section **3264**, such as, for example, during a clamping operation. In some embodiments, one or more of the plurality of secondary sensors **3260a-3260d** may be replaced with non-sensing tissue stabilizing elements. The

secondary sensors **3260a-3260d** create a stabilized tissue condition by controlling tissue flow, staple formation, and/or other tissue conditions during a clamping, stapling, and/or other treatment process.

(303) FIG. **41** illustrates one embodiment of a staple cartridge **3270** comprising a plurality of sensors **3272a-3272h** formed integrally therein. The staple cartridge **3270** comprises a plurality of rows containing a plurality of holes for storing staples therein. One or more of the holes in the outer row **3278** are replaced with one of the plurality of sensors **3272a-3272h**. A cut-away section **3274** is shown to illustrate a sensor **3272f** coupled to a sensor wire **3276b**. The sensor wires **3276a**, **3276b** may comprise a plurality of wires for coupling the plurality of sensors **3272a-3272h** to one or more circuits of a surgical instrument, such as, for example, the surgical instrument **10**. In some embodiments, one or more of the plurality of sensors **3272a-3272h** comprise dual purpose sensor and tissue stabilizing elements having electrodes and/or sensing geometries configured to provide tissue stabilization. In some embodiments, the plurality of sensors **3272a-3272h** may be replaced with and/or co-populated with a plurality of tissue stabilizing elements. Tissue stabilization may be provided by, for example, controlling tissue flow and/or staple formation during a clamping and/or stapling process. The plurality of sensors **3272a-3272h** provide signals to one or more circuits of the surgical instrument **10** to enhance feedback of stapling performance and/or tissue thickness sensing.

(304) FIG. **42** is a logic diagram illustrating one embodiment of a process **3280** for determining one or more parameters of a tissue section **3264** clamped within an end effector, such as, for example, the end effector **3250** illustrated in FIG. **40**. In one embodiment, a first sensor **3258** is configured to detect one or more parameters of the end effector **3250** and/or a tissue section **3264** located between the anvil **3252** and the staple cartridge **3256**. A first signal is generated **3282** by the first sensors **3258**. The first signal is indicative of the one or more parameters detected by the first sensor **3258**. One or more secondary sensors **3260** are configured to detect one or more parameters of the end effector **3250** and/or the tissue section **3264**. The secondary sensors **3260** may be configured to detect the same parameters, additional parameters, or different parameters as the first sensor **3258**. Secondary signals **3284** are generated by the secondary sensors **3260**. The secondary signals **3284** are indicative of the one or more parameters detected by the secondary sensors **3260**. The first signal and the secondary signals are provided to a processor, such as, for example, a primary processor **2006**. The processor **2006** adjusts **3286** the first signal generated by the first sensor **3258** based on input generated by the secondary sensors **3260**. The adjusted signal may be indicative of, for example, the true thickness of a tissue section **3264** and the fullness of the bite. The adjusted signal is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(305) FIG. **43** illustrates one embodiment of an end effector **3300** comprising a plurality of redundant sensors **3308a**, **3308b**. The end effector **3300** comprises a first jaw member, or anvil, **3302** pivotally coupled to a second jaw member **3304**. The second jaw member **3304** is configured to receive a staple cartridge **3306** therein. The anvil **3302** is moveable with respect to the staple cartridge **3306** to grasp a material, such as, for example, a tissue section, between the anvil **3302** and the staple cartridge **3306**. A plurality of sensors **3308a**, **3308b** is coupled to the anvil. The plurality of sensors **3308a**, **3308b** are configured to detect one or more parameters of the end effector **3300** and/or a tissue section located between the anvil **3302** and the staple cartridge **3306**. In some embodiments, the plurality of sensors **3308a**, **3308b** are configured to detect a gap **3310** between the anvil **3302** and the staple cartridge **3306**. The gap **3310** may correspond to, for example, the thickness of tissue located between the anvil **3302** and the staple cartridge **3306**. The plurality of sensors **3308a**, **3308b** may detect the gap **3310** by, for example, detecting a magnetic field generated by a magnet **3312** coupled to the second jaw member **3304**.

(306) In some embodiments, the plurality of sensors **3308a**, **3308b** comprise redundant sensors. The redundant sensors are configured to detect the same properties of the end effector **3300** and/or

a tissue section located between the anvil **3302** and the staple cartridge **3306**. The redundant sensors may comprise, for example, Hall effect sensors configured to detect the gap **3310** between the anvil **3302** and the staple cartridge **3306**. The redundant sensors provide signals representative of one or more parameters allowing a processor, such as, for example, the primary processor **2006**, to evaluate the multiple inputs and determine the most reliable input. In some embodiments, the redundant sensors are used to reduce noise, false signals, and/or drift. Each of the redundant sensors may be measured in real-time during clamping, allowing time-based information to be analyzed and algorithms and/or look-up tables to recognize tissue characteristics and clamping positioning dynamically. The input of one or more of the redundant sensors may be adjusted and/or selected to identify the true tissue thickness and bite of a tissue section located between the anvil **3302** and the staple cartridge **3306**.

(307) FIG. **44** is a logic diagram illustrating one embodiment of a process **3320** for selecting the most reliable output from a plurality of redundant sensors, such as, for example, the plurality of sensors **3308a**, **3308b** illustrated in FIG. **43**. In one embodiment, a first signal is generated by a first sensor **3308a**. The first signal is converted **3322a** by an analog-to-digital convertor. One or more additional signals are generated by one or more redundant sensors **3308b**. The one or more additional signals are converted **3322b** by an analog-to-digital convertor. The converted signals are provided to a processor, such as, for example, the primary processor **2006**. The primary processor evaluates **3324** the redundant inputs to determine the most reliable output. The most reliable output may be selected based on one or more parameters, such as, for example, algorithms, look-up tables, input from additional sensors, and/or instrument conditions. After selecting the most reliable output, the processor may adjust the output based on one or more additional sensors to reflect, for example, the true thickness and bite of a tissue section located between the anvil **3302** and the staple cartridge **3306**. The adjusted most reliable output is displayed **3026** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(308) FIG. **45** illustrates one embodiment of an end effector **3350** comprising a sensor **3358** comprising a specific sampling rate to limit or eliminate false signals. The end effector **3350** comprises a first jaw member, or anvil, **3352** pivotably coupled to a second jaw member **3354**. The second jaw member **3354** is configured to receive a staple cartridge **3356** therein. The staple cartridge **3356** contains a plurality of staples that may be delivered to a tissue section located between the anvil **3352** and the staple cartridge **3356**. A sensor **3358** is coupled to the anvil **3352**. The sensor **3358** is configured to detect one or more parameters of the end effector **3350**, such as, for example, the gap **3364** between the anvil **3352** and the staple cartridge **3356**. The gap **3364** may correspond to the thickness of a material, such as, for example, a tissue section, and/or the fullness of a bite of material located between the anvil **3352** and the staple cartridge **3356**. The sensor **3358** may comprise any suitable sensor for detecting one or more parameters of the end effector **3350**, such as, for example, a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

(309) In one embodiment, the sensor **3358** comprises a magnetic sensor configured to detect a magnetic field generated by an electromagnetic source **3360** coupled to the second jaw member **3354** and/or the staple cartridge **3356**. The electromagnetic source **3360** generates a magnetic field detected by the sensor **3358**. The strength of the detected magnetic field may correspond to, for example, the thickness and/or fullness of a bite of tissue located between the anvil **3352** and the staple cartridge **3356**. In some embodiments, the electromagnetic source **3360** generates a signal at a known frequency, such as, for example, 1 MHz. In other embodiments, the signal generated by the electromagnetic source **3360** may be adjustable based on, for example, the type of staple cartridge **3356** installed in the second jaw member **3354**, one or more additional sensor, an algorithm, and/or one or more parameters.

(310) In one embodiment, a signal processor **3362** is coupled to the end effector **3350**, such as, for

example, the anvil **3352**. The signal processor **3362** is configured to process the signal generated by the sensor **3358** to eliminate false signals and to boost the input from the sensor **3358**. In some embodiments, the signal processor **3362** may be located separately from the end effector **3350**, such as, for example, in the handle **14** of a surgical instrument **10**. In some embodiments, the signal processor **3362** is formed integrally with and/or comprises an algorithm executed by a general processor, such as, for example, the primary processor **2006**. The signal processor **3362** is configured to process the signal from the sensor **3358** at a frequency substantially equal to the frequency of the signal generated by the electromagnetic source **3360**. For example, in one embodiment, the electromagnetic source **3360** generates a signal at a frequency of 1 MHz. The signal is detected by the sensor **3358**. The sensor **3358** generates a signal indicative of the detected magnetic field which is provided to the signal processor **3362**. The signal is processed by the signal processor **3362** at a frequency of 1 MHz to eliminate false signals. The processed signal is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** correlates the received signal to one or more parameters of the end effector **3350**, such as, for example, the gap **3364** between the anvil **3352** and the staple cartridge **3356**.

(311) FIG. **46** is a logic diagram illustrating one embodiment of a process **3370** for generating a thickness measurement for a tissue section located between an anvil and a staple cartridge of an end effector, such as, for example, the end effector **3350** illustrated in FIG. **45**. In one embodiment of the process **3370**, a signal is generated **3372** by a modulated electromagnetic source **3360**. The generated signal may comprise, for example, a 1 MHz signal. A magnetic sensor **3358** is configured to detect **3374** the signal generated by the electromagnetic source **3360**. The magnetic sensor **3358** generates a signal indicative of the detected magnetic field and provides the signal to a signal processor **3362**. The signal processor **3362** processes **3376** the signal to remove noise, false signals, and/or to boost the signal. The processed signal is provided to an analog-to-digital convertor for conversion **3378** to a digital signal. The digital signal may be calibrated **3380**, for example, by application of a calibration curve input algorithm and/or look-up table. The signal processing **3376**, conversion **3378**, and calibration **3380** may be performed by one or more circuits. The calibrated signal is displayed **3026** to a user by, for example, a display **2026** formed integrally with a surgical instrument **10**.

(312) Although the various embodiments so far described comprise an end effector having first and second jaw members pivotally coupled, the described embodiments are not so limited. For example, in one embodiment, the end effector may comprise a circular stapler end effector. FIG. **47** illustrates one embodiment of a circular stapler **3400** configured to implement one or more of the processes described in FIGS. **28-46**. The circular stapler **3400** comprises a body **3402**. The body **3402** may be coupled to a shaft, such as, for example, the shaft assembly **200** of the surgical instrument **10**. The body **3402** is configured to receive a staple cartridge and/or one or more staples therein (not shown). An anvil **3404** is moveably coupled to the body **3402**. The anvil **3404** may be coupled to the body **3402** by, for example, a shaft **3406**. The shaft **3406** is receivable within a cavity within the body (not shown). In some embodiments, a breakaway washer **3408** is coupled to the anvil **3404**. The breakaway washer **3408** may comprise a buttress or reinforcing material during stapling.

(313) In some embodiments, the circular stapler **3400** comprises a plurality of sensors **3410a**, **3410b**. The plurality of sensor **3410a**, **3410b** is configured to detect one or more parameters of the circular stapler **3400** and/or a tissue section located between the body **3402** and the anvil **3404**. The plurality of sensors **3410a**, **3410b** may be coupled to any suitable portion of the anvil **3404**, such as, for example, being positioned under the breakaway washer **3408**. The plurality of sensors **3410a**, **3410b** may be arranged in any suitable arrangement, such as, for example, being equally spaced about the perimeter of the anvil **3404**. The plurality of sensors **3410a**, **3410b** may comprise any suitable sensors for detecting one or more parameters of the end effector **3400** and/or a tissue section located between the body **3402** and the anvil **3404**. For example, the plurality of sensors

3410a, **3410b** may comprise magnetic sensors, such as a Hall effect sensor, strain gauges, pressure sensors, inductive sensors, such as an eddy current sensor, resistive sensors, capacitive sensors, optical sensors, any combination thereof, and/or any other suitable sensor.

(314) In one embodiment, the plurality of sensors **3410a**, **3410b** comprise a plurality of pressure sensors positioned under the breakaway washer **3408**. Each of the sensors **3410a**, **3410b** is configured to detect a pressure generated by the presence of compressed tissue between the body **3402** and the anvil **3404**. In some embodiments the plurality of sensors **3410a**, **3410b** are configured to detect the impedance of a tissue section located between the anvil **3404** and the body **3402**. The detected impedance may be indicative of the thickness and/or fullness of tissue located between the anvil **3404** and the body **3402**. The plurality of sensors **3410a**, **3410b** generate a plurality of signals indicative of the detected pressure. The plurality of generated signals is provided to a processor, such as, for example, the primary processor **2006**. The primary processor **2006** applies one or more algorithms and/or look-up tables based on the input from the plurality of sensors **3410a**, **3410b** to determine one or more parameters of the end effector **3400** and/or a tissue section located between the body **3402** and the anvil **3404**. For example, in one embodiment comprising a plurality of pressure sensors, the processor **2006** is configured to apply an algorithm to quantitatively compare the output of the plurality of sensors **3410a**, **3410b** with respect to each other and with respect to a predetermined threshold. In one embodiment, if the delta, or difference, between the outputs of the plurality of sensors **3410a**, **3410b** is greater than a predetermined threshold, feedback is provided to the operator indicating a potential uneven loading condition. In some embodiments, the end effector **3400** may be coupled to a shaft comprising one or more additional sensors, such as, for example, the drive shaft **3504** described in connection to FIG. 50 below.

(315) FIGS. 48A-48D illustrate a clamping process of the circular stapler **3400** illustrated in FIG. 47. FIG. 48A illustrates the circular stapler **3400** in an initial position with the anvil **3404** and the body **3402** in a closed configuration. The circular stapler **3400** is positioned at a treatment site in the closed configuration. Once the circular stapler **3400** is positioned, the anvil **3404** is moved distally to disengage with the body **3402** and create a gap configured to receive a tissue section **3412** therein, as illustrated in FIG. 48B. The tissue section **3412** is compressed to a predetermined compression **3414** between the anvil **3404** and the body **3402**, as shown in FIG. 48C. The tissue section **3412** is further compressed between the anvil **3404** and the body **3402**. The additional compression deploys one or more staples from the body **3402** into the tissue section **3412**. The staples are shaped by the anvil **3404**. FIG. 48D illustrates the circular stapler **3400** in position corresponding to staple deployment. Proper staple deployment is dependent on obtaining a proper bite of tissue between the body **3402** and the anvil **3404**. The plurality of sensors **3410a**, **3410b** disposed on the anvil **3404** allow a processor to determine that a proper bite of tissue is located between the anvil **3404** and the body **3402** prior to deployment of the staples.

(316) FIG. 49 illustrates one embodiment of a circular staple anvil **3452** and an electrical connector **3466** configured to interface therewith. The anvil **3452** comprises an anvil head **3454** coupled to an anvil shaft **3456**. A breakaway washer **3458** is coupled to the anvil head **3452**. A plurality of pressure sensors **3460a**, **3460b** are coupled to the anvil head **3452** between the anvil head **3452** and the breakaway washer **3458**. A flex circuit **3462** is formed on the shaft **3456**. The flex circuit **3462** is coupled to the plurality of pressure sensors **3460a**, **3460b**. One or more contacts **3464** are formed on the shaft **3456** to couple the flex circuit **3462** to one or more circuits, such as, for example, the control circuit **2000** of the surgical instrument **10**. The flex circuit **3462** may be coupled to the one or more circuits by an electrical connector **3466**. The electrical connector **3466** is coupled to the anvil **3454**. For example, in one embodiment, the shaft **3456** is hollow and configured to receive the electrical connector **3466** therein. The electrical connector **3466** comprises a plurality of contacts **3468** configured to interface with the contacts **3464** formed on the anvil shaft **3456**. The plurality of contacts **3468** on the electrical connector **3466** are coupled to a flex circuit **3470** which

is coupled the one or more circuits, such as, for example, a control circuit **2000**.

(317) FIG. **50** illustrates one embodiment of a surgical instrument **3500** comprising a sensor **3506** coupled to a drive shaft **3504** of the surgical instrument **3500**. The surgical instrument **3500** may be similar to the surgical instrument **10** described above. The surgical instrument **3500** comprises a handle **3502** and a drive shaft **3504** coupled to a distal end of the handle. The drive shaft **3504** is configured to receive an end effector (not shown) at the distal end. A sensor **3506** is fixedly mounted in the drive shaft **3504**. The sensor **3506** is configured to detect one or more parameters of the drive shaft **3504**. The sensor **3506** may comprise any suitable sensor, such as, for example, a magnetic sensor, such as a Hall effect sensor, a strain gauge, a pressure sensor, an inductive sensor, such as an eddy current sensor, a resistive sensor, a capacitive sensor, an optical sensor, and/or any other suitable sensor.

(318) In some embodiments, the sensor **3506** comprises a magnetic Hall effect sensor. A magnet **3508** is located within the drive shaft **3504**. The sensor **3506** is configured to detect a magnetic field generated by the magnet **3508**. The magnet **3508** is coupled to a spring-backed bracket **3510**. The spring-backed bracket **3510** is coupled to the end effector. The spring-backed bracket **3510** is moveable in response to an action of the end effector, for example, compression of an anvil towards a body and/or second jaw member. The spring-backed bracket **3510** moves the magnet **3508** in response to the movement of the end effector. The sensor **3506** detects the change in the magnetic field generated by the magnet **3508** and generates a signal indicative of the movement of the magnet **3508**. The movement of the magnet **3508** may correspond to, for example, the thickness of tissue clamped by the end effector. The thickness of the tissue may be displayed to an operator by, for example, a display **3512** embedded in the handle **3502** of the surgical instrument **3500**. In some embodiments, the Hall effect sensor **3508** may be combined with one or more additional sensors, such as, for example, the pressure sensors illustrated in FIG. **47**.

(319) FIG. **51** is a flow chart illustrating one embodiment of a process **3550** for determining uneven tissue loading in an end effector, for example, the end effector **3400** illustrated in FIG. **47** coupled to the surgical instrument **3500** illustrated in FIG. **50**. In one embodiment, the process **3550** comprises utilizing one or more first sensors **3552**, such as, for example, a plurality of pressure sensors, to detect **3554** the presence of tissue within an end effector. During a clamping operation of the end effector **3400**, the input from the pressure sensors, P , is analyzed to determine the value of P . If P is less **3556** than a predetermined threshold, the end effector **3400** continues **3558** the clamping operation. If P is greater than or equal to **3560** the predetermined threshold, clamping is stopped. The delta (difference) between the plurality of sensors **3552** is compared **3562**. If the delta is greater than a predetermined delta, the surgical instrument **3500** displays **3564** a warning to the user. The warning may comprise, for example, a message indicating that there is uneven clamping in the end effector. If the delta is less than or equal to the predetermined delta, the input of the one or more sensors **3552** is compared to an input from an additional sensor **3566**.

(320) In some embodiments, a second sensor **3566** is configured to detect one or more parameters of the surgical instrument **3500**. For example, in one some embodiments, a magnetic sensor, such as, for example, a Hall effect sensor, is located in a shaft **3504** of the surgical instrument **3500**. The second sensor **3566** generates a signal indicative of the one or more parameters of the surgical instrument **3500**. A preset calibration curve is applied **3568** to the input from the second sensor **3566**. The preset calibration curve may adjust **3568** a signal generated by the second sensor **3566**, such as, for example, a Hall voltage generated by a Hall effect sensor. For example, in one embodiment, the Hall effect voltage is adjusted such that the generated Hall effect voltage is set at a predetermined value when the gap between the anvil **3404** and the body **3402**, $X1$, is equal to zero. The adjusted sensor **3566** input is used to calculate **3570** a distance, $X3$, between the anvil **3404** and the body **3402** when the pressure threshold P is met. The clamping process is continued **3572** to deploy a plurality of staples into the tissue section clamped in the end effector **3400**. The input from the second sensor **3566** changes dynamically during the clamping procedure and is used

to calculate the distance, X_2 , between the anvil **3404** and the body **3402** in real-time. A real-time percent compression is calculated **3574** and displayed to an operator. In one embodiment, the percent compression is calculated as: $(((X_3 - X_2)/X_3) * 100)$.

(321) In some embodiments, one or more of the sensors illustrated in FIGS. **28-50** are used to indicate: whether the anvil is attached to the body of the surgical device; the compressed tissue gap; and/or whether the anvil is in a proper position for removing the device, or any combination of these indicators.

(322) In some embodiments, one or more of the sensors illustrated in FIGS. **28-50** are used to affect device performance. One or more control parameters of a surgical device **10** may be adjusted by at least one sensor output. For example, in some embodiments, the speed control of a firing operation may be adjusted by the output of one or more sensors, such as, for example, a Hall effect sensor. In some embodiments, one or more the sensors may adjust a closure and/or clamping operation based on load and/or tissue type. In some embodiments, multiple stage compression sensors allow the surgical instrument **10** to stop closure at a predetermined load and/or a predetermined displacement. The control circuit **2000** may apply one or more predetermined algorithms to apply varying compression to a tissue section to determine a tissue type, for example, based on a tissue response. The algorithms may be varied based on closure rate and/or predetermined tissue parameters. In some embodiments, one or more sensors are configured to detect a tissue property and one or more sensors are configured to detect a device property and/or configuration parameter. For example, in one embodiment, capacitive blocks may be formed integrally with a staple cartridge to measure skew.

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(324) FIG. **52** illustrates one embodiment of an end effector **3600** configured to determine one or more parameters of a tissue section during a clamping operation. The end effector **3600** comprises a first jaw member, or anvil, **3602** pivotally coupled to a second jaw member **3604**. The second jaw member **3604** is configured to receive a staple cartridge **3606** therein. The staple cartridge **3606** contains a plurality of staples (not shown) configured to be deployed into a tissue section during a clamping and stapling operation. The staple cartridge **3606** comprises a staple cartridge deck **3622** having a predetermined height. The staple cartridge **3606** further comprises a slot **3624** defined within the body of the staple cartridge, similar to slot **193** described above. A Hall effect sensor **3608** is configured to detect the distance **3616** between the Hall effect sensor **3608** and a magnet **3610** coupled to the second jaw member **3604**. The distance **3616** between the Hall effect sensor **3608** and the magnet **3610** is indicative of a thickness of tissue located between the anvil **3602** and the staple cartridge deck **3622**.

(325) The second jaw member **3604** is configured to receive a plurality of staple cartridge **3606** types. The types of staple cartridge **3606** may vary by, for example, containing different length staples, comprising a buttress material, and/or containing different types of staples. In some embodiments, the height **3618** of the staple cartridge deck **3622** may vary based on the type of staple cartridge **3606** coupled to the second jaw member **3604**. The varying cartridge height **3618** may result in an inaccurate thickness measurement by the Hall effect sensor **3608**. For example, in one embodiment, a first cartridge comprises a first cartridge deck height X and a second cartridge comprises a second cartridge deck height Y , where $Y > X$. A fixed Hall effect sensor **3608** and fixed magnet will produce an accurate thickness measurement only for one of the two cartridge deck heights. In some embodiments, an adjustable magnet is used to compensate for various deck heights.

(326) In some embodiments, the second jaw member **3604** and the staple cartridge **3606** comprise a magnet cavity **3614**. The magnet cavity **3614** is configured to receive the magnet **3610** therein. The magnet is coupled to a spring-arm **3612**. The spring-arm **3612** is configured to bias the magnet towards the upper surface of the magnet cavity **3614**. A depth **3620** of the magnet cavity **3614** varies depending on the deck height **3618** of the staple cartridge **3606**. For example, each staple

cartridge **3606** may define a cavity depth **3620** such that the upper surface of the cavity **3614** is a set distance from the plane of the deck **3622**. The magnet **3610** is biased against the upper surface of the cavity **3614**. The magnetic reference of the magnet **3610**, as viewed by the Hall effect sensor **3608**, is consistent relative to all cartridge decks but variable relative to the slot **3624**. For example, in some embodiments, the upper-biased magnet **3610** and the cavity **3614** provide a set distance **3616** from the Hall effect sensor **3608** to the magnet **3610**, regardless of the staple cartridge **3606** inserted into the second jaw member **3604**. The set distance **3616** allows the Hall effect sensor **3608** to generate an accurate thickness measurement irrespective of the staple cartridge **3606** type. In some embodiments, the depth **3620** of the cavity **3614** may be adjusted to calibrate the Hall effect sensor **3608** for one or more surgical procedures.

(327) FIGS. **53A** and **53B** illustrate an embodiment of an end effector **3650** configured to normalize a Hall effect voltage irrespective of a deck height of a staple cartridge **3656**. FIG. **53A** illustrates one embodiment of the end effector **3650** comprising a first cartridge **3656a** inserted therein. The end effector **3650** comprises a first jaw member, or anvil, **3652** pivotally coupled to a second jaw member **3654** to grasp tissue therebetween. The second jaw member **3654** is configured to receive a staple cartridge **3656a**. The staple cartridge **3656a** may comprise a variety of staple lengths, buttress materials, and/or deck heights. A magnetic sensor **3658**, such as, for example, a Hall effect sensor, is coupled to the anvil **3652**. The magnetic sensor **3658** is configured to detect a magnetic field generated by a magnet **3660**. The detected magnetic field strength is indicative of the distance **3664** between the magnetic sensor **3658** and the magnet **3660**, which may be indicative of, for example, a thickness of a tissue section grasped between the anvil **3652** and the staple cartridge **3656**. As noted above, various staple cartridges **3656a** may comprise varying deck heights which create differences in the calibrated compression gap **3664**.

(328) In some embodiments, a magnetic attenuator **3662** is coupled to the staple cartridge **3656a**. The magnetic attenuator **3662** is configured to attenuate the magnetic flux generated to by the magnet **3660**. The magnetic attenuator **3662** is calibrated to produce a magnetic flux based on the height of the staple cartridge **3656a**. By attenuating the magnet **3660** based on the staple cartridge **3656** type, the magnetic attenuator **3662** normalizes the magnetic sensor **3658** signal to the same calibration level for various deck heights. The magnetic attenuator **3662** may comprise any suitable magnet attenuator, such as, for example, a ferrous metallic cap. The magnetic attenuator **3662** is molded into the staple cartridge **3656a** such that the magnetic attenuator **3662** is positioned above the magnet **3660** when the staple cartridge **3656** is inserted into the second jaw member **3654**.

(329) In some embodiments, attenuation of the magnet **3660** is not required for the deck height of the staple cartridge. FIG. **53B** illustrates one embodiment of the end effector **3650** comprising a second staple cartridge **3656b** coupled to the second jaw member **3654**. The second staple cartridge **3656b** comprises a deck height matching the calibration of the magnet **3660** and the Hall effect sensor **3658**, and therefore does not require attenuation. As shown in FIG. **53B**, the second staple cartridge **3656b** comprises a cavity **3666** in place of the magnetic attenuator **3662** of the first staple cartridge **3656a**. In some embodiments, larger and/or smaller attenuation members are provided depending on the height of the cartridge deck. The design of the attenuation member **3662** shape may be optimized to create features in the response signal generated by the Hall effect sensor **3658** that allow for the distinction of one or more additional cartridge attributes.

(330) FIG. **54** is a logic diagram illustrating one embodiment of a process **3670** for determining when the compression of tissue within an end effector, such as, for example, the end effector **3650** illustrated in FIGS. **53A-53B**, has reached a steady state. In some embodiments, a clinician initiates **3672** a clamping procedure to clamp tissue within the end effector, for example, between an anvil **3652** and staple cartridge **3656**. The end effector engages **3674** with tissue during the clamping procedure. Once the tissue has been engaged **3674**, the end effector begins **3676** real time gap monitoring. The real time gap monitoring monitors the gap between, for example, the anvil **3652** and the staple cartridge **3656** of the end effector **3650**. The gap may be monitored by, for example,

a sensor **3658**, such as a Hall effect sensor, coupled to the end effector **3650**. The sensor **3658** may be coupled to a processor, such as, for example, the primary processor **2006**. The processor determines **3678** when tissue clamping requirements of the end effector **3650** and/or the staple cartridge **3656** have been met. Once the processor determines that the tissue has stabilized, the process indicates **3680** to the user that the tissue has stabilized. The indication may be provided by, for example, a display embedded within a surgical instrument **10**.

(331) In some embodiments, the gap measurement is provided by a Hall effect sensor. The Hall effect sensor may be located, for example, at the distal tip of an anvil **3652**. The Hall effect sensor is configured to measure the gap between the anvil **3652** and a staple cartridge **3656** deck at the distal tip. The measured gap may be used to calculate a jaw closure gap and/or to monitor a change in tissue compression of a tissue section clamped in the end effector **3650**. In one embodiment, the Hall effect sensor is coupled to a processor, such as, for example, the primary processor **2006**. The processor is configured to receive real time measurements from the Hall effect sensor and compare the received signal to a predetermined set of criteria. For example, in one embodiment, a logic equation at equally spaced intervals, such as one second, is used to indicate stabilization of a tissue section to the user when a gap reading remains unchanged for a predetermined interval, such as, for example, 3.0 seconds. Tissue stabilization may also be indicated after a predetermined time period, such as, for example, 15.0 seconds. As another example, tissue stabilization may be indicated when $y_n = y_{n+1} = y_{n+2}$, where y equals a gap measurement of the Hall effect sensor and n is a predetermined measurement interval. A surgical instrument **10** may display an indication to a user, such as, for example, a graphical and/or numerical representation, when stabilization has occurred.

(332) FIG. **55** is a graph **3690** illustrating various Hall effect sensor readings **3692a-3692d**. As shown in graph **3690**, a thickness, or compression, of a tissue section stabilizes after a predetermined time period. A processor, such as, for example, the primary processor **2006**, may be configured to indicate when the calculated thickness from a sensor, such as a Hall effect sensor, is relatively consistent or constant over a predetermined time period. The processor **2006** may indicate to a user, for example, through a number display, that the tissue has stabilized.

(333) FIG. **56** is a logic diagram illustrating one embodiment of a process **3700** for determining when the compression of tissue within an end effector, such as, for example, the end effector **3650** illustrated in FIGS. **53A-53B**, has reached a steady state. In some embodiments, a clinician initiates **3702** a clamping procedure to clamp tissue within the end effector, for example, between an anvil **3652** and staple cartridge **3656**. The end effector engages **3704** with tissue during the clamping procedure. Once the tissue has been engaged **3704**, the end effector begins **3706** real time gap monitoring. The real time gap monitoring technique monitors **3706** the gap between, for example, the anvil **3652** and the staple cartridge **3656** of the end effector **3650**. The gap may be monitored **3706** by, for example, a sensor **3658**, such as a Hall effect sensor, coupled to the end effector **3650**. The sensor **3658** may be coupled to a processor, such as, for example, the primary processor **2006**. The processor is configured to execute one or more algorithms determine when tissue section compressed by the end effector **3650** has stabilized.

(334) For example, in the embodiment illustrated in FIG. **56**, the process **3700** is configured to utilize a slope calculation to determine stabilization of tissue. The processor calculates **3708** the slope, S , of an input from a sensor, such as a Hall effect sensor. The slope may be calculated **3708** by, for example, the equation $S = ((V_1 - V_2) / (T_1 - T_2))$. The processor compares **3710** the calculated slope to a predetermined value, such as, for example, 0.005 volts/sec. If the value of the calculated slope is greater than the predetermined value, the processor resets **3712** a count, C , to zero. If the calculated slope is less than or equal to the predetermined value, the processor increments **3714** the value of the count C . The count, C , is compared **3716** to a predetermined threshold value, such as, for example, 3. If the value of the count C is greater than or equal to the predetermined threshold value, the processor indicates **3718** to the user that the tissue section has stabilized. If the value of the count C is less than the predetermined threshold value, the processor

continues monitoring the sensor **3658**. In various embodiments, the slope of the sensor input, the change in the slope, and/or any other suitable change in the input signal may be monitored.

(335) In some embodiments, an end effector, such as for example, the end effectors **3600**, **3650** illustrated in FIGS. **52**, **53A**, and **53B** may comprise a cutting member deployable therein. The cutting member may comprise, for example, an I-Beam configured to simultaneously cut a tissue section located between an anvil **3602** and a staple cartridge **3608** and to deploy staples from the staple cartridge **3608**. In some embodiments, the I-Beam may comprise only a cutting member and/or may only deploy one or more staples. Tissue flow during firing may affect the proper formation of staples. For example, during I-Beam deployment, fluid in the tissue may cause the thickness of tissue to temporarily increase, causing improper deployment of staples.

(336) FIG. **57** is a logic diagram illustrating one embodiment of a process **3730** for controlling an end effector to improve proper staple formation during deployment. The control process **3730** comprises generating **3732** a sensor measurement indicative of the thickness of a tissue section within the end effector **3650**, such as for example, a Hall effect voltage generated by a Hall effect sensor. The sensor measurement is converted **3734** to a digital signal by an analog-to-digital convertor. The digital signal is calibrated **3736**. The calibration **3736** may be performed by, for example, a processor and/or a dedicated calibration circuit. The digital signal is calibrated **3736** based on one or more calibration curve inputs. The calibrated digital signal is displayed **3738** to an operator by, for example, a display **2026** embedded in a surgical instrument **10**. The calibrated signal may be displayed **3738** as a thickness measurement of a tissue section grasped between the anvil **3652** and the staple cartridge **3656** and/or as a unit-less range.

(337) In some embodiments, the generated **3732** Hall effect voltage is used to control an I-beam. For example, in the illustrated embodiment, the Hall effect voltage is provided to a processor configured to control deployment of an I-Beam within an end effector, such as, for example, the primary processor **2006**. The processor receives the Hall effect voltage and calculates the voltage rate of change over a predetermined time period. The processor compares **3740** the calculated rate of change to a predetermined value, $x1$. If the calculated rate of change is greater than the predetermined value, $x1$, the processor slows **3742** the speed of the I-Beam. The speed may be reduced by, for example, decrementing a speed variable by a predetermined unit. If the calculated voltage rate of change is less than or equal to the predetermine value, $x1$, the processor maintains **3744** the current speed of the I-Beam.

(338) In some embodiments, the processor may temporarily reduce the speed of the I-Beam to compensate, for example, for thicker tissue, uneven loading, and/or any other tissue characteristic. For example, in one embodiment, the processor is configured to monitor **3740** the rate of voltage change of a Hall effect sensor. If the rate of change monitored **3740** by the processor exceeds a first predetermine value, $x1$, the processor slows down or stops deployment of the I-Beam until the rate of change is less than a second predetermined value, $x2$. When the rate of change is less than the second predetermined value, $x2$, the processor may return the I-beam to normal speed. In some embodiments, the sensor input may be generated by for example, a pressure sensor, a strain gauge, a Hall effect sensor, and/or any other suitable sensor. In some embodiments, the processor may implement one or more pause points during deployment of an I-Beam. For example, in some embodiments, the processor may implement three predetermined pause points, at which the processor pauses deployment of the I-Beam for a predetermined time period. The pause points are configured to provide optimized tissue flow control.

(339) FIG. **58** is a logic diagram illustrating one embodiment of a process **3750** for controlling an end effector to allow for fluid evacuation and provide improved staple formation. The process **3750** comprises generating **3752** a sensor measurement, such as, for example, a Hall effect voltage. The sensor measurement may be indicative of, for example, the thickness of a tissue section grasped between an anvil **3652** and a staple cartridge **3656** of an end effector **3650**. The generated signal is provided to an analog-to-digital convertor for conversion **3754** to a digital signal. The converted

signal is calibrated **3756** based on one or more inputs, such as, for example, a second sensor input and/or a predetermined calibration curve. The calibrated signal is representative of one or more parameters of the end effector **3650**, such as, for example, the thickness of a tissue section grasped therein. The calibrated thickness measurement may be displayed to a user as a thickness and/or as a unit-less range. The calibrated thickness may be displayed by, for example, a display **2026** embedded in a surgical instrument **10** coupled to the end effector **3650**.

(340) In some embodiments, the calibrated thickness measurement is used to control deployment of an I-Beam and/or other firing member within the end effector **3650**. The calibrated thickness measurement is provided to a processor. The processor compares **3760** the change in the calibrated thickness measurement to a predetermined threshold percentage, x . If the rate of change of the thickness measurement is greater than x , the processor slows **3762** the speed, or rate of deployment, of the I-Beam within the end effector. The processor may slow **3762** the speed of the I-Beam by, for example, decrementing a speed variable by a predetermined unit. If the rate of change of the thickness measurement is less than or equal to x , the processor maintains **3764** the speed of the I-Beam within the end effector **3650**. The real time feedback of tissue thickness and/or compression allows the surgical instrument **10** to affect the firing speed to allow for fluid evacuation and/or provide improved staple form.

(341) In some embodiments, the sensor reading generated **3752** by the sensor, for example, a Hall effect voltage, may be adjusted by one or more additional sensor inputs. For example, in one embodiment, a generated **3752** Hall effect voltage may be adjusted by an input from a micro-strain gauge sensor on the anvil **3652**. The micro-strain gauge may be configured to monitor the strain amplitude of the anvil **3652**. The generated **3752** Hall effect voltage may be adjusted by the monitored strain amplitude to indicate, for example, partial proximal or distal tissue bites. Time based monitoring of the micro-strain and Hall effect sensor output during clamping allows one or more algorithms and/or look-up tables to recognize tissue characteristics and clamping positioning and dynamically adjust tissue thickness measurements to control firing speed of, for example, an I-Beam. In some embodiments, the processor may implement one or more pause points during deployment of an I-Beam. For example, in some embodiments, the processor may implement three predetermined pause points, at which the processor pauses deployment of the I-Beam for a predetermined time period. The pause points are configured to provide optimized tissue flow control.

(342) FIGS. **59A-59B** illustrate one embodiment of an end effector **3800** comprising a pressure sensor. The end effector **3800** comprises a first jaw member, or anvil, **3802** pivotally coupled to a second jaw member **3804**. The second jaw member **3804** is configured to receive a staple cartridge **3806** therein. The staple cartridge **3806** comprises a plurality of staples. A first sensor **3808** is coupled to the anvil **3802** at a distal tip. The first sensor **3808** is configured to detect one or more parameters of the end effector, such as, for example, the distance, or gap **3814**, between the anvil **3802** and the staple cartridge **3806**. The first sensor **3808** may comprise any suitable sensor, such as, for example, a magnetic sensor. A magnet **3810** may be coupled to the second jaw member **3804** and/or the staple cartridge **3806** to provide a magnetic signal to the magnetic sensor.

(343) In some embodiments, the end effector **3800** comprises a second sensor **3812**. The second sensor **3812** is configured to detect one or more parameters of the end effector **3800** and/or a tissue section located therebetween. The second sensor **3812** may comprise any suitable sensor, such as, for example, one or more pressure sensors. The second sensor **3812** may be coupled to the anvil **3802**, the second jaw member **3804**, and/or the staple cartridge **3806**. A signal from the second sensor **3812** may be used to adjust the measurement of the first sensor **3808** to adjust the reading of the first sensor to accurately represent proximal and/or distal positioned partial bites true compressed tissue thickness. In some embodiments, the second sensor **3812** may be surrogate with respect to the first sensor **3808**.

(344) In some embodiments, the second sensor **3812** may comprise, for example, a single

continuous pressure sensing film and/or an array of pressure sensing films. The second sensor **3812** is coupled to the deck of the staple cartridge **3806** along the central axis covering, for example, a slot **3816** configured to receive a cutting and/or staple deployment member. The second sensor **3812** provides signals indicate of the amplitude of pressure applied by the tissue during a clamping procedure. During firing of the cutting and/or deployment member, the signal from the second sensor **3812** may be severed, for example, by cutting electrical connections between the second sensor **3812** and one or more circuits. In some embodiments, a severed circuit of the second sensor **3812** may be indicative of a spent staple cartridge **3806**. In other embodiments, the second sensor **3812** may be positioned such that deployment of a cutting and/or deployment member does not sever the connection to the second sensor **3812**.

(345) FIG. **60** illustrates one embodiment of an end effector **3850** comprising a second sensor **3862** located between a staple cartridge **3806** and a second jaw member **3804**. The end effector **3850** comprises a first jaw member, or anvil, **3852** pivotally coupled to a second jaw member **3854**. The second jaw member **3854** is configured to receive a staple cartridge **3856** therein. A first sensor **3858** is coupled to the anvil **3852** at a distal tip. The first sensor **3858** is configured to detect one or more parameters of the end effector **3850**, such as, for example, the distance, or gap **3864**, between the anvil **3852** and the staple cartridge **3856**. The first sensor **3858** may comprise any suitable sensor, such as, for example, a magnetic sensor. A magnet **3860** may be coupled to the second jaw member **3854** and/or the staple cartridge **3856** to provide a magnetic signal to the magnetic sensor. In some embodiments, the end effector **3850** comprises a second sensor **3862** similar in all respect to the second sensor **3812** of FIGS. **59A-59B**, except that it is located between the staple cartridge **3856** and the second jaw member **3864**.

(346) FIG. **61** is a logic diagram illustrating one embodiment of a process **3870** for determining and displaying the thickness of a tissue section clamped in an end effector **3800** or **3850**, according to FIGS. **59A-59B** or FIG. **60**. The process comprises obtaining a Hall effect voltage **3872**, for example, through a Hall effect sensor located at the distal tip of the anvil **3802**. The Hall effect voltage **3872** is provided to an analog to digital converter **3874** and converted into a digital signal. The digital signal is provided to a process, such as for example the primary processor **2006**. The primary processor **2006** calibrates **3874** the curve input of the Hall effect voltage **3872** signal. Pressure sensors, such as for example second sensor **3812**, is configured to measure **3880** one or more parameters of, for example, the end effector **3800**, such as for example the amount of pressure being exerted by the anvil **3802** on the tissue clamped in the end effector **3800**. In some embodiments the pressure sensors may comprise a single continuous pressure sensing film and/or array of pressure sensing films. The pressure sensors may thus be operable determine variations in the measure pressure at different locations between the proximal and distal ends of the end effector **3800**. The measured pressure is provided to the processor, such as for example the primary processor **2006**. The primary processor **2006** uses one or more algorithms and/or lookup tables to adjust **3882** the Hall effect voltage **3872** in response to the pressure measured by the pressure sensors **3880** to more accurately reflect the thickness of the tissue clamped between, for example, the anvil **3802** and the staple cartridge **3806**. The adjusted thickness is displayed **3878** to an operator by, for example, a display **2026** embedded in the surgical instrument **10**.

(347) FIG. **62** illustrates one embodiment of an end effector **3900** comprising a plurality of second sensors **3192a-3192b** located between a staple cartridge **3906** and an elongated channel **3916**. The end effector **3900** comprises a first jaw member or anvil **3902** pivotally coupled to a second jaw member or elongated channel **3904**. The elongated channel **3904** is configured to receive a staple cartridge **3906** therein. The anvil **3902** further comprises a first sensor **3908** located in the distal tip. The first sensor **3908** is configured to detect one or more parameters of the end effector **3900**, such as, for example, the distance, or gap, between the anvil **3902** and the staple cartridge **3906**. The first sensor **3908** may comprise any suitable sensor, such as, for example, a magnetic sensor. A magnet **3910** may be coupled to the elongated channel **3904** and/or the staple cartridge **3906** to provide a

magnetic signal to the first sensor **3908**. In some embodiments, the end effector **3900** comprises a plurality of second sensors **3912a-3912c** located between the staple cartridge **3906** and the elongated channel **3904**. The second sensors **3912a-3912c** may comprise any suitable sensors, such as for instance piezo-resistive pressure film strips. In some embodiments, the second sensors **3912a-3912c** may be uniformly distributed between the distal and proximal ends of the end effector **3900**.

(348) In some embodiments, signals from the second sensors **3912a-3912c** may be used to adjust the measurement of the first sensor **3908**. For instance, the signals from the second sensors **3912a-3912c** may be used to adjust the reading of the first sensor **3908** to accurately represent the gap between the anvil **3908** and the staple cartridge **3906**, which may vary between the distal and proximal ends of the end effector **3900**, depending on the location and/or density of tissue **3920** between the anvil **3902** and the staple cartridge **3906**. FIG. **11** illustrates an example of a partial bite of tissue **3920**. As illustrated for purposes of this example, the tissue is located only in the proximal area of the end effector **3900**, creating a high pressure **3918** area near the proximal area of the end effector **3900** and a corresponding low pressure **3916** area near the distal end of the end effector.

(349) FIGS. **63A** and **63B** further illustrate the effect of a full versus partial bite of tissue **3920**. FIG. **63A** illustrates the end effector **3900** with a full bite of tissue **3920**, where the tissue **3920** is of uniform density. With a full bite of tissue **3920** of uniform density, the measured first gap **3914a** at the distal tip of the end effector **3900** may be approximately the same as the measured second gap **3922a** in the middle or proximal end of the end effector **3900**. For example, the first gap **3914a** may measure 2.4 mm, and the second gap may measure 2.3 mm. FIG. **63B** illustrates an end effector **3900** with a partial bite of tissue **3920**, or alternatively a full bite of tissue **3920** of non-uniform density. In this case, the first gap **3914b** will measure less than the second gap **3922b** measured at the thickest or densest portion of the tissue **3920**. For example, the first gap may measure 1.0 mm, while the second gap may measure 1.9 mm. In the conditions illustrated in FIGS. **63A** and **63B**, signals from the second sensors **3912a-3912c**, such as for instance measured pressure at different points along the length of the end effector **3900**, may be employed by the instrument to determine tissue **3920** placement and/or material properties of the tissue **3920**. The instrument may further be operable to use measured pressure over time to recognize tissue characteristics and tissue position, and dynamically adjust tissue thickness measurements.

(350) FIG. **64** illustrates one embodiment of an end effector **3950** comprising a coil **3958** and oscillator circuit **3962** for measuring the gap between the anvil and the staple cartridge **3956**. The end effector **3950** comprises a first jaw member or anvil **3952** pivotally coupled to a second jaw member or elongated channel **3954**. The elongated channel **3954** is configured to receive a staple cartridge **3956** therein. In some embodiments the staple cartridge **3954** further comprises a coil **3958** and an oscillator circuit **3962** located at the distal end. The coil **3958** and oscillator circuit **3962** are operable as eddy current sensors and/or inductive sensors. The coil **3958** and oscillator circuit **3962** can detect eddy currents and/or induction as a target **3960**, such as for instance the distal tip of the anvil **3952**, approaches the coil **3958**. The eddy current and/or induction detected by the coil **3958** and oscillator circuit **3962** can be used to detect the distance or gap between the anvil **3952** and staple cartridge **3956**.

(351) FIG. **65** illustrates and alternate view of the end effector **3950**. As illustrated, in some embodiments external wiring **3964** may supply power to the oscillator circuit **3962**. The external wiring **3964** may be placed along the outside of the elongated channel **3954**.

(352) FIG. **66** illustrates examples of the operation of a coil **3958** to detect eddy currents **3972** in a target **3960**. Alternating current flowing through the coil **3958** at a chosen frequency generates a magnetic field **3970** around the coil **3958**. When the coil **3958** is at its position **3976a** a certain distance away from the target **3960**, the coil **3958** will not induce an eddy current **3972**. When the coil **3958** is at a position **3976b** close to an electrically conductive target **3960** and eddy current

3972 is produced in the target **3960**. When the coil **3958** is at a position **3976c** near a flaw in the target **3960**, the flaw may disrupt the eddy current circulation; in this case, the magnetic coupling with the coil **3958** is changed and a defect signal **3974** can be read by measuring the coil impedance variation.

(353) FIG. **67** illustrates a graph **3980** of a measured quality factor **3984**, the measured inductance **3986**, and measure resistance **3988** of the radius of a coil **3958** as a function of the coil's **3958** standoff **3978** to a target **3960**. The quality factor **3984** depends on the standoff **3978**, while both the inductance **3986** and resistance **3988** are functions of displacement. A higher quality factor **3984** results in a more purely reactive sensor. The specific value of the inductance **3986** is constrained only by the need for a manufacturable coil **3958** and a practical circuit design that burns a reasonable amount of energy at a reasonable frequency. Resistance **3988** is a parasitic effect.

(354) The graph **3980** illustrates how inductance **3986**, resistance **3988**, and the quality factor **3984** depend on the target standoff **3978**. As the standoff **3978** increases, the inductance **3986** increases by a factor of four, the resistance **3988** decreases slightly and as a consequence the quality factor **3984** increases. The change in all three parameters is highly nonlinear and each curve tends to decay roughly exponentially as standoff **3978** increases. The rapid loss of sensitivity with distance strictly limits the range of an eddy current sensor to approximately $\frac{1}{2}$ the coil diameter.

(355) FIG. **68** illustrates one embodiment of an end effector **4000** comprising an emitter and sensor **4008** placed between the staple cartridge **4006** and the elongated channel **4004**. The end effector **4000** comprises a first jaw member or anvil **4002** pivotally coupled to a second jaw member or elongated channel **4004**. The elongated channel **3904** is configured to receive a staple cartridge **4006** therein. In some embodiments, the end effector **4000** further comprises an emitter and sensor **4008** located between the staple cartridge **4006** and the elongated channel **4004**. The emitter and sensor **4008** can be any suitable device, such as for instance a MEMS ultrasonic transducer. In some embodiments, the emitter and sensor may be placed along the length of the end effector **4000**.

(356) FIG. **69** illustrates an embodiment of an emitter and sensor **4008** in operation. The emitter and sensor **4008** may be operable to emit a pulse **4014** and sense the reflected signal **4016** of the pulse **4014**. The emitter and sensor **4008** may further be operable to measure the time of flight **4018** between the issuance of the pulse **4014** and the reception of the reflected signal **4016**. The measured time of flight **4018** can be used to determine the thickness of tissue compressed in the end effector **4000** along the entire length of the end effector **4000**. In some embodiments, the emitter and sensor **4008** may be coupled to a processor, such as for instance the primary processor **2006**. The processor **2006** may be operable to use the time of flight **4018** to determine additional information about the tissue, such as for instance whether the tissue was diseased, bunched, or damaged. The surgical instrument can further be operable to convey this information to the operator of the instrument.

(357) FIG. **70** illustrates the surface of an embodiment of an emitter and sensor **4008** comprising a MEMS transducer.

(358) FIG. **71** illustrates a graph **4020** of an example of the reflected signal **4016** that may be measured by the emitter and sensor **4008** of FIG. **69**. FIG. **71** illustrates the amplitude **4022** of the reflected signal **4016** as a function of time **4024**. As illustrated, the amplitude of the transmitted pulse **4026** is greater than the amplitude of the reflected pulses **4028a-4028c**. The amplitude of the transmitted pulse **4026** may be of a known or expected value. The first reflected pulse **4028a** may be, for example, from the tissue enclosed by the end effector **4000**. The second reflected pulse **4028b** may be, for example, from the lower surface of the anvil **4002**. The third reflected pulse **4028c** may be, for example, from the upper surface of the anvil **4002**.

(359) FIG. **72** illustrates an embodiment of an end effector **4050** that is configured to determine the location of a cutting member or knife **4058**. The end effector **4050** comprises a first jaw member or anvil **4052** pivotally coupled to a second jaw member or elongated channel **4054**. The elongated

channel **4054** is configured to receive a staple cartridge **4056** therein. The staple cartridge **4056** further comprises a slot **4058** (not shown) and a cutting member or knife **4062** located therein. The knife **4062** is operably coupled to a knife bar **4064**. The knife bar **4064** is operable to move the knife **4062** from the proximal end of the slot **4058** to the distal end. The end effector **4050** may further comprise an optical sensor **4060** located near the proximal end of the slot **4058**. The optical sensor may be coupled to a processor, such as for instance the primary processor **2006**. The optical sensor **4060** may be operable to emit an optical signal towards the knife bar **4064**. The knife bar **4064** may further comprise a code strip **4066** along its length. The code strip **4066** may comprise cut-outs, notches, reflective pieces, or any other configuration that is optically readable. The code strip **4066** is placed such that the optical signal from the optical sensor **4060** will reflect off or through the code strip **4066**. As the knife **4062** and knife bar **4064** move **4068** along the slot **4058**, the optical sensor **4060** will detect the reflection of the emitted optical signal coupled to the code strip **4066**. The optical sensor **4060** may be operable to communicate the detected signal to the processor **2006**. The processor **2006** may be configured to use the detected signal to determine the position of the knife **4062**. The position of the knife **4062** may be sensed more precisely by designing the code strip **4066** such that the detected optical signal has a gradual rise and fall.

(360) FIG. **73** illustrates an example of the code strip **4066** in operation with red LEDs **4070** and infrared LEDs **4072**. For purposes of this example only, the code strip **4066** comprises cut-outs. As the code strip **4066** moves **4068**, the light emitted by the red LEDs **4070** will be interrupted as the cut-outs passed before it. The infrared LEDs **4072** will therefore detect the motion **4068** of the code strip **4066**, and therefore, by extension, the motion of the knife **4062**.

(361) Monitoring Device Degradation Based on Component Evaluation

(362) FIG. **74** depicts a partial view of the end effector **300** of the surgical instrument **10**. In the example form depicted in FIG. **74**, the end effector **300** comprises a staple cartridge **1100** which is similar in many respects to the staple cartridge **304** (FIG. **20**). Several parts of the end effector **300** are omitted to enable a clearer understanding of the present disclosure. In certain instances, the end effector **300** may include a first jaw such as, for example, the anvil **306** (FIG. **20**) and a second jaw such as, for example, the channel **198** (FIG. **20**). In certain instances, as described above, the channel **198** may accommodate a staple cartridge such as, for example, the staple cartridge **304** or the staple cartridge **1100**, for example. At least one of the channel **198** and the anvil **306** may be movable relative to the other one of the channel **198** and the anvil **306** to capture tissue between the staple cartridge **1100** and the anvil **306**. Various actuation assemblies are described herein to facilitation motion of the channel **198** and/or the anvil **306** between an open configuration (FIG. **1**) and a closed configuration (FIG. **75**), for example

(363) In certain instances, as described above, the E-beam **178** can be advanced distally to deploy the staples **191** into the captured tissue and/or advance the cutting edge **182** between a plurality of positions to engage and cut the captured tissue. As illustrated in FIG. **74**, the cutting edge **182** can be advanced distally along a path defined by the slot **193**, for example. In certain instances, the cutting edge **182** can be advanced from a proximal portion **1102** of the staple cartridge **1100** to a distal portion **1104** of the staple cartridge **1100** to cut the captured tissue, as illustrated in FIG. **74**. In certain instances, the cutting edge **182** can be retracted proximally from the distal portion **1104** to the proximal portion **1102** by retraction of the E-beam **178** proximally, for example.

(364) In certain instances, the cutting edge **182** can be employed to cut tissue captured by the end effector **300** in multiple procedures. The reader will appreciate that repetitive use of the cutting edge **182** may affect the sharpness of the cutting edge **182**. The reader will also appreciate that as the sharpness of the cutting edge **182** decreases, the force required to cut the captured tissue with the cutting edge **182** may increase. Referring to FIGS. **74-76**, in certain instances, the surgical instrument **10** may comprise a module **1106** (FIG. **76**) for monitoring the sharpness of the cutting edge **182** during, before, and/or after operation of the surgical instrument **10** in a surgical procedure, for example. In certain instances, the module **1106** can be employed to test the

sharpness of the cutting edge **182** prior to utilizing the cutting edge **182** to cut the captured tissue. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **182** after the cutting edge **182** has been used to cut the captured tissue. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **182** prior to and after the cutting edge **182** is used to cut the captured tissue. In certain instances, the module **1106** can be employed to test the sharpness of the cutting edge **1106** at the proximal portion **1102** and/or at the distal portion **1104**.

(365) Referring to FIGS. **74-76**, the module **1106** may include one or more sensors such as, for example, an optical sensor **1108**; the optical sensor **1108** of the module **1106** can be employed to test the reflective ability of the cutting edge **182**, for example. In certain instances, the ability of the cutting edge **182** to reflect light may correlate with the sharpness of the cutting edge **182**. In other words, a decrease in the sharpness of the cutting edge **182** may result in a decrease in the ability of the cutting edge **182** to reflect the light. Accordingly, in certain instances, the dullness of the cutting edge **182** can be evaluated by monitoring the intensity of the light reflected from the cutting edge **182**, for example. In certain instances, the optical sensor **1108** may define a light sensing region. The optical sensor **1108** can be oriented such that the optical sensing region is disposed in the path of the cutting edge **182**, for example. The optical sensor **1108** may be employed to sense the light reflected from the cutting edge **182** while the cutting edge **182** is in the optical sensing region, for example. A decrease in intensity of the reflected light beyond a threshold can indicate that the sharpness of the cutting edge **182** has decreased beyond an acceptable level.

(366) Referring again to FIGS. **74-76**, the module **1106** may include one or more light sources such as, for example, a light source **1110**. In certain instances, the module **1106** may include a microcontroller **1112** (“controller”) which may be operably coupled to the optical sensor **1108**, as illustrated in FIG. **76**. In certain instances, the controller **1112** may include a microprocessor **1114** (“processor”) and one or more computer readable mediums or memory units **1116** (“memory”). In certain instances, the memory **1116** may store various program instructions, which when executed may cause the processor **1114** to perform a plurality of functions and/or calculations described herein. In certain instances, the memory **1116** may be coupled to the processor **1114**, for example. A power source **1118** can be configured to supply power to the controller **1112**, the optical sensors **1108**, and/or the light sources **1110**, for example. In certain instances, the power source **1118** may comprise a battery (or “battery pack” or “power pack”), such as a Li ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to the handle **14** for supplying power to the surgical instrument **10**. A number of battery cells connected in series may be used as the power source **4428**. In certain instances, the power source **1118** may be replaceable and/or rechargeable, for example.

(367) The controller **1112** and/or other controllers of the present disclosure may be implemented using integrated and/or discrete hardware elements, software elements, and/or a combination of both. Examples of integrated hardware elements may include processors, microprocessors, microcontrollers, integrated circuits, ASICs, PLDs, DSPs, FPGAs, logic gates, registers, semiconductor devices, chips, microchips, chip sets, microcontrollers, SoC, and/or SIP. Examples of discrete hardware elements may include circuits and/or circuit elements such as logic gates, field effect transistors, bipolar transistors, resistors, capacitors, inductors, and/or relays. In certain instances, the controller **1112** may include a hybrid circuit comprising discrete and integrated circuit elements or components on one or more substrates, for example.

(368) In certain instances, the controller **1112** and/or other controllers of the present disclosure may be an LM 4F230H5QR, available from Texas Instruments, for example. In certain instances, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile memory, up to 40 MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or

more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available. Other microcontrollers may be readily substituted for use with the present disclosure. Accordingly, the present disclosure should not be limited in this context.

(369) In certain instances, the light source **1110** can be employed to emit light which can be directed at the cutting edge **182** in the optical sensing region, for example. The optical sensor **1108** may be employed to measure the intensity of the light reflected from the cutting edge **182** while in the optical sensing region in response to exposure to the light emitted by the light source **1110**. In certain instances, the processor **1114** may receive one or more values of the measured intensity of the reflected light and may store the one or more values of the measured intensity of the reflected light on the memory **1116**, for example. The stored values can be detected and/or recorded before, after, and/or during a plurality of surgical procedures performed by the surgical instrument **10**, for example.

(370) In certain instances, the processor **1114** may compare the measured intensity of the reflected light to a predefined threshold values that may be stored on the memory **1116**, for example. In certain instances, the controller **1112** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level if the measured light intensity exceeds the predefined threshold value by 1%, 5%, 10%, 25%, 50%, 100% and/or more than 100%, for example. In certain instances, the processor **1114** can be employed to detect a decreasing trend in the stored values of the measured intensity of the light reflected from the cutting edge **182** while in the optical sensing region.

(371) In certain instances, the surgical instrument **10** may include one or more feedback systems such as, for example, the feedback system **1120**. In certain instances, the processor **1114** can employ the feedback system **1120** to alert a user if the measured light intensity of the light reflected from cutting edge **182** while in the optical sensing region is beyond the stored threshold value, for example. In certain instances, the feedback system **1120** may comprise one or more visual feedback systems such as display screens, backlights, and/or LEDs, for example. In certain instances, the feedback system **1120** may comprise one or more audio feedback systems such as speakers and/or buzzers, for example. In certain instances, the feedback system **1120** may comprise one or more haptic feedback systems, for example. In certain instances, the feedback system **1120** may comprise combinations of visual, audio, and/or tactile feedback systems, for example.

(372) In certain instances, the surgical instrument **10** may comprise a firing lockout mechanism **1122** which can be employed to prevent advancement of the cutting edge **182**. Various suitable firing lockout mechanisms are described in greater detail in U.S. Patent Publication No. 2014/0001231, entitled FIRING SYSTEM LOCKOUT ARRANGEMENTS FOR SURGICAL INSTRUMENTS, and filed Jun. 28, 2012, which is hereby incorporated by reference herein in its entirety. In certain instances, as illustrated in FIG. 76, the processor **1114** can be operably coupled to the lockout mechanism **1122**; the processor **1114** may employ the lockout mechanism **1122** to prevent advancement of the cutting edge **182** in the event it is determined that the measured intensity of the light reflected from the cutting edge **182** is beyond the stored threshold, for example. In other words, the processor **1114** may activate the lockout mechanism **1122** if the cutting edge is not sufficiently sharp to cut the tissue captured by the end effector **300**.

(373) In certain instances, the optical sensor **1108** and the light source **1110** can be housed at a distal portion of the shaft assembly **200**. In certain instances, the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108**, as described above, prior to transitioning the cutting edge **182** into the end effector **300**. The firing bar **172** (FIG. 20) may advance the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** while the cutting edge **182** is in the shaft assembly **182** and prior to entering the end effector **300**, for example. In certain instances, the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** after retracting the cutting edge **182** proximally from the end effector **300**. The firing bar **172** (FIG. 20) may retract the cutting edge **182** through the optical sensing region defined by the optical sensor

1108 after retracting the cutting edge **182** from the end effector **300** into the shaft assembly **200**, for example.

(374) In certain instances, the optical sensor **1108** and the light source **1110** can be housed at a proximal portion of the end effector **300** which can be proximal to the staple cartridge **1100**, for example. The sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** after transitioning the cutting edge **182** into the end effector **300** but prior to engaging the staple cartridge **1100**, for example. In certain instances, the firing bar **172** (FIG. 20) may advance the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** while the cutting edge **182** is in the end effector **300** but prior to engaging the staple cartridge **1100**, for example.

(375) In various instances, the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** as the cutting edge **182** is advanced by the firing bar **172** through the slot **193**. As illustrated in FIG. 74, the optical sensor **1108** and the light source **1110** can be housed at the proximal portion **1102** of the staple cartridge **1100**, for example; and the sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** at the proximal portion **1102**, for example. The firing bar **172** (FIG. 20) may advance the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** at the proximal portion **1102** before the cutting edge **182** engages tissue captured between the staple cartridge **1100** and the anvil **306**, for example. In certain instances, as illustrated in FIG. 74, the optical sensor **1108** and the light source **1110** can be housed at the distal portion **1104** of the staple cartridge **1100**, for example. The sharpness of cutting edge **182** can be evaluated by the optical sensor **1108** at the distal portion **1104**. In certain instances, the firing bar **172** (FIG. 20) may advance the cutting edge **182** through the optical sensing region defined by the optical sensor **1108** at the distal portion **1104** after the cutting edge **182** has passed through the tissue captured between the staple cartridge **1100** and the anvil **306**, for example.

(376) Referring again to FIG. 74, the staple cartridge **1100** may comprise a plurality of optical sensors **1108** and a plurality of corresponding light sources **1110**, for example. In certain instances, a pair of the optical sensor **1108** and the light source **1110** can be housed at the proximal portion **1102** of the staple cartridge **1100**, for example; and a pair of the optical sensor **1108** and the light source **1110** can be housed at the distal portion **1104** of the staple cartridge **1100**, for example. In such instances, the sharpness of the cutting edge **182** can be evaluated a first time at the proximal portion **1102** prior to engaging the tissue, for example, and a second time at the distal portion **1104** after passing through the captured tissue, for example.

(377) The reader will appreciate that an optical sensor **1108** may evaluate the sharpness of the cutting edge **182** a plurality of times during a surgical procedure. For example, the sharpness of the cutting edge can be evaluated a first time during advancement of the cutting edge **182** through the slot **193** in a firing stroke, and a second time during retraction of the cutting edge **182** through the slot **193** in a return stroke, for example. In other words, the light reflected from the cutting edge **182** can be measured by the same optical sensor **1108** once as the cutting edge is advanced through the optical sensing region, and once as the cutting edge **182** is retracted through the optical sensing region, for example.

(378) The reader will appreciate that the processor **1114** may receive a plurality of readings of the intensity of the light reflected from the cutting edge **182** from one or more of the optical sensors **1108**. In certain instances, the processor **1114** may be configured to discard outliers and calculate an average reading from the plurality of readings, for example. In certain instances, the average reading can be compared to a threshold stored in the memory **1116**, for example. In certain instances, the processor **1114** may be configured to alert a user through the feedback system **1120** and/or activate the lockout mechanism **1122** if it is determined that the calculated average reading is beyond the threshold stored in the memory **1116**, for example.

(379) In certain instances, as illustrated in FIGS. 75, 77, and 78, a pair of the optical sensor **1108** and the light source **1110** can be positioned on opposite sides of the staple cartridge **1100**. In other

words, the optical sensor **1108** can be positioned on a first side **1124** of the slot **193**, for example, and the light source **1110** can be positioned on a second side **1126**, opposite the first side **1124**, of the slot **193**, for example. In certain instances, the pair of the optical sensor **1108** and the light source **1110** can be substantially disposed in a plane transecting the staple cartridge **1100**, as illustrated in FIG. 75. The pair of the optical sensor **1108** and the light source **1110** can be oriented to define an optical sensing region that is positioned, or at least substantially positioned, on the plane transecting the staple cartridge **1100**, for example. Alternatively, the pair of the optical sensor **1108** and the light source **1110** can be oriented to define an optical sensing region that is positioned proximal to the plane transecting the staple cartridge **1100**, for example, as illustrated in FIG. 78.

(380) In certain instances, a pair of the optical sensor **1108** and the light source **1110** can be positioned on a same side of the staple cartridge **1100**. In other words, as illustrated in FIG. 79, the pair of the optical sensor **1108** and the light source **1110** can be positioned on a first side of the cutting edge **182**, e.g. the side **1128**, as the cutting edge **182** is advanced through the slot **193**. In such instances, the light source **1110** can be oriented to direct light at the side **1128** of the cutting edge **182**; and the intensity of the light reflected from the side **1128**, as measured by the optical sensor **1108**, may represent the sharpness of the side **1128**.

(381) In certain instances, as illustrated in FIG. 80, a second pair of the optical sensor **1108** and the light source **1110** can be positioned on a second side of the cutting edge **182**, e.g. the side **1130**, for example. The second pair can be employed to evaluate the sharpness of the side **1130**. For example, the light source **1110** of the second pair can be oriented to direct light at the side **1130** of the cutting edge **182**; and the intensity of the light reflected from the side **1130**, as measured by the optical sensor **1108** of the second pair, may represent the sharpness of the side **1130**. In certain instances, the processor can be configured to assess the sharpness of the cutting edge **182** based upon the measured intensities of the light reflected from the sides **1128** and **1130** of the cutting edge **182**, for example.

(382) In certain instances, as illustrated in FIG. 75, a pair of the optical sensor **1108** and the light source **1110** can be housed at the distal portion **1104** of the staple cartridge **1100**. As illustrated in FIG. 81, the light source **1108** can be positioned, or at least substantially positioned, on an axis LL which extends longitudinally along the path of the cutting edge **182** through the slot **193**, for example. In addition, the light source **1110** can be positioned distal to the cutting edge **182** and oriented to direct light at the cutting edge **182** as the cutting edge is advanced toward the light source **1110**, for example. Furthermore, the optical sensor **1108** can be positioned, or at least substantially positioned, along an axis AA that intersects the axis LL, as illustrated in FIG. 81. In certain instances, the axis AA may be perpendicular to the axis LL, for example. In any event, the optical sensor **1108** can be oriented to define an optical sensing region at the intersection of the axis LL and the axis AA, for example.

(383) The reader will appreciate that the position, orientation and/or number of optical sensors and corresponding light sources described herein in connection with the surgical instrument **10** are example embodiments intended for illustration purposes. Various other arrangements of optical sensors and light sources can be employed by the present disclosure to evaluate the sharpness of the cutting edge **182**.

(384) The reader will appreciate that advancement of the cutting edge **182** through the tissue captured by the end effector **300** may cause the cutting edge to collect tissue debris and/or bodily fluids during each firing of the surgical instrument **10**. Such debris may interfere with the ability of the module **1106** to accurately evaluate the sharpness of the cutting edge **182**. In certain instances, the surgical instrument **10** can be equipped with one or more cleaning mechanisms which can be employed to clean the cutting edge **182** prior to evaluating the sharpness of the cutting edge **182**, for example. In certain instances, as illustrated in FIG. 82, a cleaning mechanism **1131** may comprise one or more cleaning members **1132**, for example. In certain instances, the cleaning members **1132** can be disposed on opposite sides of the slot **193** to receive the cutting edge **182**

therebetween (See FIG. 82) as the cutting edge **182** is advanced through the slot **193**, for example. In certain instances, as illustrated in FIG. 82, the cleaning members **1132** may comprise wiper blades, for example. In certain instances, as illustrated in FIG. 830, the cleaning members **1132** may comprise sponges, for example. The reader will appreciate that various other cleaning members can be employed to clean the cutting edge **182**, for example.

(385) Referring to FIG. 74, in certain instances, the staple cartridge **1100** may include a first pair of the optical sensor **1108** and the light source **1110**, which can be housed in the proximal portion **1102** of the staple cartridge **1100**, for example. Furthermore, as illustrated in FIG. 74, the staple cartridge **1100** may include a first pair of the cleaning members **1132**, which can be housed in the proximal portion **1102** on opposite sides of the slot **193**. The first pair of the cleaning members **1132** can be positioned distal to the first pair of the optical sensor **1108** and the light source **1110**, for example. As illustrated in FIG. 74, the staple cartridge **1100** may include a second pair of the optical sensor **1108** and the light source **1110**, which can be housed in the distal portion **1104** of the staple cartridge **1100**, for example. As illustrated in FIG. 74, the staple cartridge **1100** may include a second pair of the cleaning members **1132**, which can be housed in the distal portion **1104** on opposite sides of the slot **193**. The second pair of the cleaning members **1132** can be positioned proximal to the second pair of the optical sensor **1108** and the light source **1110**.

(386) Further to the above, as illustrated in FIG. 74, the cutting edge **182** may be advanced distally in a firing stroke to cut tissue captured by the end effector **300**. As the cutting edge is advanced, a first evaluation of the sharpness of the cutting edge **182** can be performed by the first pair of the optical sensor **1108** and the light source **1110** prior to tissue engagement by the cutting edge **182**, for example. A second evaluation of the sharpness of the cutting edge **182** can be performed by the second pair of the optical sensor **1108** and the light source **1110** after the cutting edge **182** has transected the captured tissue, for example. The cutting edge **182** may be advanced through the second pair of the cleaning members **1132** prior to the second evaluation of the sharpness of the cutting edge **182** to remove any debris collected by the cutting edge **182** during the transection of the captured tissue.

(387) Further to the above, as illustrated in FIG. 74, the cutting edge **182** may be retracted proximally in a return stroke. As the cutting edge is retracted, a third evaluation of the sharpness of the cutting edge **182** can be performed by the first pair of the optical sensor **1108** and the light source **1110** during the return stroke. The cutting edge **182** may be retracted through the first pair of the cleaning members **1132** prior to the third evaluation of the sharpness of the cutting edge **182** to remove any debris collected by the cutting edge **182** during the transection of the captured tissue, for example.

(388) In certain instances, one or more of the lights sources **1110** may comprise one or more optical fiber cables. In certain instances, one or more flex circuits **1134** can be employed to transmit energy from the power source **1118** to the optical sensors **1108** and/or the light sources **1110**. In certain instances, the flex circuits **1134** may be configured to transmit one or more of the readings of the optical sensors **1108** to the controller **1112**, for example.

(389) Referring now to FIG. 84, a staple cartridge **4300** is depicted; the staple cartridge **4300** is similar in many respects to the staple cartridge **304** (FIG. 20). For example, the staple cartridge **4300** can be employed with the end effector **300**. In certain instances, as illustrated in FIG. 84, the staple cartridge **4300** may comprise a sharpness testing member **4302** which can be employed to test the sharpness of the cutting edge **182**. In certain instances, the sharpness testing member **4302** can be attached to and/or integrated with the cartridge body **194** of the staple cartridge **4300**, for example. In certain instances, the sharpness testing member **4302** can be disposed in the proximal portion **1102** of the staple cartridge **4300**, for example. In certain instances, as illustrated in FIG. 84, the sharpness testing member **4302** can be disposed onto a cartridge deck **4304** of the staple cartridge **4300**, for example.

(390) In certain instances, as illustrated in FIG. 84, the sharpness testing member **4302** can extend

across the slot **193** of the staple cartridge **4300** to bridge, or at least partially bridge, the gap defined by the slot **193**, for example. In certain instances, the sharpness testing member **4302** may interrupt, or at least partially interrupt, the path of the cutting edge **182**. The cutting edge **182** may engage, cut, and/or pass through the sharpness testing member **4302** as the cutting edge **182** is advanced during a firing stroke, for example. In certain instances, the cutting edge **182** may be configured to engage, cut, and/or pass through the sharpness testing member **4302** prior to engaging tissue captured by the end effector **300** in a firing stroke, for example. In certain instances, the cutting edge **182** may be configured to engage the sharpness testing member **4302** at a proximal end **4306** of the sharpness testing member **4302**, and exit and/or disengage the sharpness testing member **4302** at a distal end **4308** of the sharpness testing member **4302**, for example. In certain instances, the cutting edge **182** can travel and/or cut through the sharpness testing member **4302** a distance (D) between the proximal end **4306** and the distal end **4308**, for example, as the cutting edge **182** is advanced during a firing stroke.

(391) Referring primarily to FIGS. **84** and **85**, the surgical instrument **10** may comprise a sharpness testing module **4310** for testing the sharpness of the cutting edge **182**, for example. In certain instances, the module **4310** can evaluate the sharpness of the cutting edge **182** by testing the ability of the cutting edge **182** to be advanced through the sharpness testing member **4302**. For example, the module **4310** can be configured to observe the time period the cutting edge **182** takes to fully transect and/or completely pass through at least a predetermined portion of the sharpness testing member **4302**. If the observed time period exceeds a predetermined threshold, the module **4310** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level, for example.

(392) In certain instances, the module **4310** may include a microcontroller **4312** (“controller”) which may include a microprocessor **4314** (“processor”) and one or more computer readable mediums or memory units **4316** (“memory”). In certain instances, the memory **4316** may store various program instructions, which when executed may cause the processor **4314** to perform a plurality of functions and/or calculations described herein. In certain instances, the memory **4316** may be coupled to the processor **4314**, for example. A power source **4318** can be configured to supply power to the controller **4312**, for example. In certain instances, the power source **4318** may comprise a battery (or “battery pack” or “power pack”), such as a Li ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to the handle **14**. A number of battery cells connected in series may be used as the power source **4318**. In certain instances, the power source **4318** may be replaceable and/or rechargeable, for example.

(393) In certain instances, the processor **4313** can be operably coupled to the feedback system **1120** and/or the lockout mechanism **1122**, for example.

(394) Referring to FIGS. **84** and **85**, the module **4310** may comprise one or more position sensors. Example position sensors and positioning systems suitable for use with the present disclosure are described in U.S. patent application Ser. No. 13/803,210, entitled SENSOR ARRANGEMENTS FOR ABSOLUTE POSITIONING SYSTEM FOR SURGICAL INSTRUMENTS, and filed Mar. 14, 2013, now U.S. Pat. No. 9,808,244, the disclosure of which is hereby incorporated by reference herein in its entirety. In certain instances, the module **4310** may include a first position sensor **4320** and a second position sensor **4322**. In certain instances, the first position sensor **4320** can be employed to detect a first position of the cutting edge **182** at the proximal end **4306** of the sharpness testing member **4302**, for example; and the second position sensor **4322** can be employed to detect a second position of the cutting edge **182** at the distal end **4308** of the sharpness cutting member **4302**, for example.

(395) In certain instances, the position sensors **4320** and **4322** can be employed to provide first and second position signals, respectively, to the microcontroller **4312**. It will be appreciated that the position signals may be analog signals or digital values based on the interface between the microcontroller **4312** and the position sensors **4320** and **4322**. In one embodiment, the interface

between the microcontroller **4312** and the position sensors **4320** and **4322** can be a standard serial peripheral interface (SPI), and the position signals can be digital values representing the first and second positions of the cutting edge **182**, as described above.

(396) Further to the above, the processor **4314** may determine the time period between receiving the first position signal and receiving the second position signal. The determined time period may correspond to the time it takes the cutting edge **182** to advance through the sharpness testing member **4302** from the first position at the proximal end **4306** of the sharpness testing member **4302**, for example, to the second position at the distal end **4308** of the sharpness testing member **4302**, for example. In at least one example, the controller **4312** may include a time element which can be activated by the processor **4314** upon receipt of the first position signal, and deactivated upon receipt of the second position signal. The time period between the activation and deactivation of the time element may correspond to the time it takes the cutting edge **182** to advance from the first position to the second position, for example. The time element may comprise a real time clock, a processor configured to implement a time function, or any other suitable timing circuit.

(397) In various instances, the controller **4312** can compare the time period it takes the cutting edge **182** to advance from the first position to the second position to a predefined threshold value to assess whether the sharpness of the cutting edge **182** has dropped below an acceptable level, for example. In certain instances, the controller **4312** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level if the measured time period exceeds the predefined threshold value by 1%, 5%, 10%, 25%, 50%, 100% and/or more than 100%, for example.

(398) Referring to FIG. **86**, in various instances, an electric motor **4330** can drive the firing bar **172** (FIG. **20**) to advance the cutting edge **182** during a firing stroke and/or to retract the cutting edge **182** during a return stroke, for example. A motor driver **4332** can control the electric motor **4330**; and a microcontroller such as, for example, the microcontroller **4312** can be in signal communication with the motor driver **4332**. As the electric motor **4330** advances the cutting edge **182**, the microcontroller **4312** can determine the current drawn by the electric motor **4330**, for example. In such instances, the force required to advance the cutting edge **182** can correspond to the current drawn by the electric motor **4330**, for example. Referring still to FIG. **86**, the microcontroller **4312** of the surgical instrument **10** can determine if the current drawn by the electric motor **4330** increases during advancement of the cutting edge **182** and, if so, can calculate the percentage increase of the current.

(399) In certain instances, the current drawn by the electric motor **4330** may increase significantly while the cutting edge **182** is in contact with the sharpness testing member **4302** due to the resistance of the sharpness testing member **4302** to the cutting edge **182**. For example, the current drawn by the electric motor **4330** may increase significantly as the cutting edge **182** engages, passes and/or cuts through the sharpness testing member **4302**. The reader will appreciate that the resistance of the sharpness testing member **4302** to the cutting edge **182** depends, in part, on the sharpness of the cutting edge **182**; and as the sharpness of the cutting edge **182** decreases from repetitive use, the resistance of the sharpness testing member **4302** to the cutting edge **182** will increase. Accordingly, the value of the percentage increase of the current drawn by the motor **4330** while the cutting edge is in contact with the sharpness testing member **4302** can increase as the sharpness of the cutting edge **182** decreases from repetitive use, for example.

(400) In certain instances, the determined value of the percentage increase of the current drawn by the motor **4330** can be the maximum detected percentage increase of the current drawn by the motor **4330**. In various instances, the microcontroller **4312** can compare the determined value of the percentage increase of the current drawn by the motor **4330** to a predefined threshold value of the percentage increase of the current drawn by the motor **4330**. If the determined value exceeds the predefined threshold value, the microcontroller **4312** may conclude that the sharpness of the cutting edge **182** has dropped below an acceptable level, for example.

(401) In certain instances, as illustrated in FIG. 86, the processor 4314 can be in communication with the feedback system 1120 and/or the lockout mechanism 1122, for example. In certain instances, the processor 4314 can employ the feedback system 1120 to alert a user if the determined value of the percentage increase of the current drawn by the motor 4330 exceeds the predefined threshold value, for example. In certain instances, the processor 4314 may employ the lockout mechanism 1122 to prevent advancement of the cutting edge 182 if the determined value of the percentage increase of the current drawn by the motor 4330 exceeds the predefined threshold value, for example.

(402) In various instances, the microcontroller 43312 can utilize an algorithm to determine the change in current drawn by the electric motor 4330. For example, a current sensor can detect the current drawn by the electric motor 4330 during the firing stroke. The current sensor can continually detect the current drawn by the electric motor and/or can intermittently detect the current draw by the electric motor. In various instances, the algorithm can compare the most recent current reading to the immediately proceeding current reading, for example. Additionally or alternatively, the algorithm can compare a sample reading within a time period X to a previous current reading. For example, the algorithm can compare the sample reading to a previous sample reading within a previous time period X, such as the immediately proceeding time period X, for example. In other instances, the algorithm can calculate the trending average of current drawn by the motor. The algorithm can calculate the average current draw during a time period X that includes the most recent current reading, for example, and can compare that average current draw to the average current draw during an immediately proceeding time period time X, for example.

(403) Referring to FIG. 87, a method is depicted for evaluating the sharpness of the cutting edge 182 of the surgical instrument 10; and various responses are outlined in the event the sharpness of the cutting edge 182 drops to and/or below an alert threshold and/or a high severity threshold, for example. In various instances, a microcontroller such as, for example, the microcontroller 4312 can be configured to implement the method depicted in FIG. 87. In certain instances, the surgical instrument 10 may include a load cell 4334 (FIG. 86); as illustrated in FIG. 86, the microcontroller 4312 may be in communication with the load cell 4334. In certain instances, the load cell 4334 may include a force sensor such as, for example, a strain gauge, which can be operably coupled to the firing bar 172, for example. In certain instances, the microcontroller 4312 may employ the load cell 4334 to monitor the force (F_x) applied to the cutting edge 182 as the cutting edge 182 is advanced during a firing stroke.

(404) In certain instances, as illustrated in FIG. 88, the load cell 4334 can be configured to monitor the force (F_x) applied to the cutting edge 182 while the cutting edge 182 is engaged and/or in contact with the sharpness testing member 4302, for example. The reader will appreciate that the force (F_x) applied by the sharpness testing member 4302 to the cutting edge 182 while the cutting edge 182 is engaged and/or in contact with the sharpness testing member 4302 may depend, at least in part, on the sharpness of the cutting edge 182. In certain instances, a decrease in the sharpness of the cutting edge 182 can result in an increase in the force (F_x) required for the cutting edge 182 to cut or pass through the sharpness testing member 4302. For example, as illustrated in FIG. 88, graphs 4336, 4338, and 4340 represent the force (F_x) applied to the cutting edge 182 while the cutting edge 182 travels a predefined distance (D) through three identical, or at least substantially identical, sharpness testing members 4302. The graph 4336 corresponds to a first sharpness of the cutting edge 182; the graph 4338 corresponds to a second sharpness of the cutting edge 182; and the graph 4340 corresponds to a third sharpness of the cutting edge 182. The first sharpness is greater than the second sharpness, and the second sharpness is greater than the third sharpness.

(405) In certain instances, the microcontroller 4312 may compare a maximum value of the monitored force (F_x) applied to the cutting edge 182 to one or more predefined threshold values. In certain instances, as illustrated in FIG. 88, the predefined threshold values may include an alert threshold (F_1) and/or a high severity threshold (F_2). In certain instances, as illustrated in the graph

4336 of FIG. **88**, the monitored force (Fx) can be less than the alert threshold (F1), for example. In such instances, as illustrated in FIG. **87**, the sharpness of the cutting edge **182** is at a good level and the microcontroller **4312** may take no action to alert a user as to the status of the cutting edge **182** or may inform the user that the sharpness of the cutting edge **182** is within an acceptable range.

(406) In certain instances, as illustrated in the graph **4338** of FIG. **88**, the monitored force (Fx) can be more than the alert threshold (F1) but less than the high severity threshold (F2), for example. In such instances, as illustrated in FIG. **87**, the sharpness of the cutting edge **182** can be dulling but still within an acceptable level. The microcontroller **4312** may take no action to alert a user as to the status of the cutting edge **182**. Alternatively, the microcontroller **4312** may inform the user that the sharpness of the cutting edge **182** is within an acceptable range. Alternatively or additionally, the microcontroller **4312** may determine or estimate the number of cutting cycles remaining in the lifecycle of the cutting edge **182** and may alert the user accordingly.

(407) In certain instances, the memory **4316** may include a database or a table that correlates the number of cutting cycles remaining in the lifecycle of the cutting edge **182** to predetermined values of the monitored force (Fx). The processor **4314** may access the memory **4316** to determine the number of cutting cycles remaining in the lifecycle of the cutting edge **182** which correspond to a particular measured value of the monitored force (Fx) and may alert the user to the number of cutting cycles remaining in the lifecycle of the cutting edge **182**, for example.

(408) In certain instances, as illustrated in the graph **4340** of FIG. **88**, the monitored force (Fx) can be more than the high severity threshold (F2), for example. In such instances, as illustrated in FIG. **87**, the sharpness of the cutting edge **182** can be below an acceptable level. In response, the microcontroller **4312** may employ the feedback system **1120** to warn the user that the cutting edge **182** is too dull for safe use, for example. In certain instances, the microcontroller **4312** may employ the lockout mechanism **1122** to prevent advancement of the cutting edge **182** upon detection that the monitored force (Fx) exceeds the high severity threshold (F2), for example. In certain instances, the microcontroller **4312** may employ the feedback system **1122** to provide instructions to the user for overriding the lockout mechanism **1122**, for example.

(409) Referring to FIG. **89**, a method is depicted for determining whether a cutting edge such as, for example, the cutting edge **182** is sufficiently sharp to be employed in transecting a tissue of a particular tissue thickness that is captured by the end effector **300**, for example. In certain instances, the microcontroller **4312** can be implemented to perform the method depicted in FIG. **16**, for example. As described above, repetitive use of the cutting edge **182** may dull or reduce the sharpness of the cutting edge **182** which may increase the force required for the cutting edge **182** to transect the captured tissue. In other words, the sharpness level of the cutting edge **182** can be defined by the force required for the cutting edge **182** to transect the captured tissue, for example. The reader will appreciate that the force required for the cutting edge **182** to transect a captured tissue may also depend on the thickness of the captured tissue. In certain instances, the greater the thickness of the captured tissue, the greater the force required for the cutting edge **182** to transect the captured tissue at the same sharpness level, for example.

(410) In certain instances, the cutting edge **182** may be sufficiently sharp for transecting a captured tissue comprising a first thickness but may not be sufficiently sharp for transecting a captured tissue comprising a second thickness greater than the first thickness, for example. In certain instances, a sharpness level of the cutting edge **182**, as defined by the force required for the cutting edge **182** to transect a captured tissue, may be adequate for transecting the captured tissue if the captured tissue comprises a tissue thickness that is in a particular range of tissue thicknesses, for example. In certain instances, as illustrated in FIG. **90**, the memory **4316** can store one or more predefined ranges of tissue thicknesses of tissue captured by the end effector **300**; and predefined threshold forces associated with the predefined ranges of tissue thicknesses. In certain instances, each predefined threshold force may represent a minimum sharpness level of the cutting edge **182** that is suitable for transecting a captured tissue comprising a tissue thickness (Tx) encompassed by

the range of tissue thicknesses that is associated with the predefined threshold force. In certain instances, if the force (Fx) required for the cutting edge **182** to transect the captured tissue, comprising the tissue thickness (Tx), exceeds the predefined threshold force associated with the predefined range of tissue thicknesses that encompasses the tissue thickness (Tx), the cutting edge **182** may not be sufficiently sharp to transect the captured tissue, for example.

(411) In certain instances, the predefined threshold forces and their corresponding predefined ranges of tissue thicknesses can be stored in a database and/or a table on the memory **4316** such as, for example, a table **4342**, as illustrated in FIG. **90**. In certain instances, the processor **4314** can be configured to receive a measured value of the force (Fx) required for the cutting edge **182** to transect a captured tissue and a measured value of the tissue thickness (Tx) of the captured tissue. The processor **4314** may access the table **4342** to determine the predefined range of tissue thicknesses that encompasses the measured tissue thickness (Tx). In addition, the processor **4314** may compare the measured force (Fx) to the predefined threshold force associated with the predefined range of tissue thicknesses that encompasses the tissue thickness (Tx). In certain instances, if the measured force (Fx) exceeds the predefined threshold force, the processor **4314** may conclude that the cutting edge **182** may not be sufficiently sharp to transect the captured tissue, for example.

(412) Further to the above, the processor **4314** may employ one or more tissue thickness sensing modules such as, for example, a tissue thickness sensing module **4336** to determine the thickness of the captured tissue. Various suitable tissue thickness sensing modules are described in the present disclosure. In addition, various tissue thickness sensing devices and methods, which are suitable for use with the present disclosure, are disclosed in U.S. Publication No. US 2011/0155781, entitled SURGICAL CUTTING INSTRUMENT THAT ANALYZES TISSUE THICKNESS, and filed Dec. 24, 2009, now U.S. Pat. No. 8,851,354, the entire disclosure of which is hereby incorporated by reference herein.

(413) In certain instances, the processor **4314** may employ the load cell **4334** to measure the force (Fx) required for the cutting edge **182** to transect a captured tissue comprising a tissue thickness (Tx). The reader will appreciate that that the force applied to the cutting edge **182** by the captured tissue, while the cutting edge **182** is engaged and/or in contact with the captured tissue, may increase as the cutting edge **182** is advanced against the captured tissue up to the force (Fx) at which the cutting edge **182** may transect the captured tissue. In certain instances, the processor **4314** may employ the load cell **4334** to continually monitor the force applied by the captured tissue against the cutting edge **182** as the cutting edge **182** is advanced against the captured tissue. The processor **4314** may continually compare the monitored force to the predefined threshold force associated with the predefined tissue thickness range encompassing the tissue thickness (Tx) of the captured tissue. In certain instances, if the monitored force exceeds the predefined threshold force, the processor **4314** may conclude that the cutting edge is not sufficiently sharp to safely transect the captured tissue, for example.

(414) The method described in FIG. **89** outline various example actions that can be taken by the processor **4313** in the event it is determined that the cutting edge **182** is not be sufficiently sharp to safely transect the captured tissue, for example. In certain instances, the microcontroller **4312** may warn the user that the cutting edge **182** is too dull for safe use, for example, through the feedback system **1120**, for example. In certain instances, the microcontroller **4312** may employ the lockout mechanism **1122** to prevent advancement of the cutting edge **182** upon concluding that the cutting edge **182** is not sufficiently sharp to safely transect the captured tissue, for example. In certain instances, the microcontroller **4312** may employ the feedback system **1122** to provide instructions to the user for overriding the lockout mechanism **1122**, for example.

(415) Multiple Motor Control for Powered Medical Device

(416) FIGS. **91-93** illustrate various embodiments of an apparatus, system, and method for employing a common control module with a plurality of motors in connection with a surgical

instrument such as, for example, a surgical instrument **4400**. The surgical instrument **4400** is similar in many respects to other surgical instruments described by the present disclosure such as, for example, the surgical instrument **10** of FIG. **1** which is described in greater detail above. For example, as illustrated in FIG. **91**, the surgical instrument **4400** includes the housing **12**, the handle **14**, the closure trigger **32**, the shaft assembly **200**, and the surgical end effector **300**. Accordingly, for conciseness and clarity of disclosure, a detailed description of certain features of the surgical instrument **4400**, which are common with the surgical instrument **10**, will not be repeated here.

(417) Referring primarily to FIG. **92**, the surgical instrument **4400** may include a plurality of motors which can be activated to perform various functions in connection with the operation of the surgical instrument **4400**. In certain instances, a first motor can be activated to perform a first function; a second motor can be activated to perform a second function; and a third motor can be activated to perform a third function. In certain instances, the plurality of motors of the surgical instrument **4400** can be individually activated to cause articulation, closure, and/or firing motions in the end effector **300**. The articulation, closure, and/or firing motions can be transmitted to the end effector **300** through the shaft assembly **200**, for example.

(418) In certain instances, as illustrated in FIG. **92**, the surgical instrument **4400** may include a firing motor **4402**. The firing motor **4402** may be operably coupled to a firing drive assembly **4404** which can be configured to transmit firing motions generated by the motor **4402** to the end effector **300**. In certain instances, the firing motions generated by the motor **4402** may cause the staples **191** to be deployed from the staple cartridge **304** into tissue captured by the end effector **300** and/or the cutting edge **182** to be advanced to cut the captured tissue, for example.

(419) In certain instances, as illustrated in FIG. **92**, the surgical instrument **4400** may include an articulation motor **4406**, for example. The motor **4406** may be operably coupled to an articulation drive assembly **4408** which can be configured to transmit articulation motions generated by the motor **4406** to the end effector **300**. In certain instances, the articulation motions may cause the end effector **300** to articulate relative to the shaft assembly **200**, for example. In certain instances, the surgical instrument **4400** may include a closure motor, for example. The closure motor may be operably coupled to a closure drive assembly which can be configured to transmit closure motions to the end effector **300**. In certain instances, the closure motions may cause the end effector **300** to transition from an open configuration to an approximated configuration to capture tissue, for example. The reader will appreciate that the motors described herein and their corresponding drive assemblies are intended as examples of the types of motors and/or driving assemblies that can be employed in connection with the present disclosure. The surgical instrument **4400** may include various other motors which can be utilized to perform various other functions in connection with the operation of the surgical instrument **4400**.

(420) As described above, the surgical instrument **4400** may include a plurality of motors which may be configured to perform various independent functions. In certain instances, the plurality of motors of the surgical instrument **4400** can be individually or separately activated to perform one or more functions while the other motors remain inactive. For example, the articulation motor **4406** can be activated to cause the end effector **300** to be articulated while the firing motor **4402** remains inactive. Alternatively, the firing motor **4402** can be activated to fire the plurality of staples **191** and/or advance the cutting edge **182** while the articulation motor **4406** remains inactive.

(421) In certain instances, the surgical instrument **4400** may include a common control module **4410** which can be employed with a plurality of motors of the surgical instrument **4400**. In certain instances, the common control module **4410** may accommodate one of the plurality of motors at a time. For example, the common control module **4410** can be separably couplable to the plurality of motors of the surgical instrument **4400** individually. In certain instances, a plurality of the motors of the surgical instrument **4400** may share one or more common control modules such as the module **4410**. In certain instances, a plurality of motors of the surgical instrument **4400** can be individually and selectively engaged the common control module **4410**. In certain instances, the

module **4410** can be selectively switched from interfacing with one of a plurality of motors of the surgical instrument **4400** to interfacing with another one of the plurality of motors of the surgical instrument **4400**.

(422) In at least one example, the module **4410** can be selectively switched between operable engagement with the articulation motor **4406** and operable engagement with the firing motor **4402**. In at least one example, as illustrated in FIG. **92**, a switch **4414** can be moved or transitioned between a plurality of positions and/or states such as a first position **4416** and a second position **4418**, for example. In the first position **4416**, the switch **4414** may electrically couple the module **4410** to the articulation motor **4406**; and in the second position **4418**, the switch **4414** may electrically couple the module **4410** to the firing motor **4402**, for example. In certain instances, the module **4410** can be electrically coupled to the articulation motor **4406**, while the switch **4414** is in the first position **4416**, to control the operation of the motor **4406** to articulate the end effector **300** to a desired position. In certain instances, the module **4410** can be electrically coupled to the firing motor **4402**, while the switch **4414** is in the second position **4418**, to control the operation of the motor **4402** to fire the plurality of staples **191** and/or advance the cutting edge **182**, for example. In certain instances, the switch **4414** may be a mechanical switch, an electromechanical switch, a solid state switch, or any suitable switching mechanism.

(423) Referring now to FIG. **93**, an outer casing of the handle **14** of the surgical instrument **4400** is removed and several features and elements of the surgical instrument **4400** are also removed for clarity of disclosure. In certain instances, as illustrated in FIG. **93**, the surgical instrument **4400** may include an interface **4412** which can be selectively transitioned between a plurality of positions and/or states. In a first position and/or state, the interface **4412** may couple the module **4410** to a first motor such as, for example, the articulation motor **4406**; and in a second position and/or state, the interface **4412** may couple the module **4410** to a second motor such as, for example, the firing motor **4402**. Additional positions and/or states of the interface **4412** are contemplated by the present disclosure.

(424) In certain instances, the interface **4412** is movable between a first position and a second position, wherein the module **4410** is coupled to a first motor in the first position and a second motor in the second position. In certain instances, the module **4410** is decoupled from first motor as the interface **4412** is moved from the first position; and the module **4410** is decoupled from second motor as the interface **4412** is moved from the second position. In certain instances, a switch or a trigger can be configured to transition the interface **4412** between the plurality of positions and/or states. In certain instances, a trigger can be movable to simultaneously effectuate the end effector and transition the control module **4410** from operable engagement with one of the motors of the surgical instrument **4400** to operable engagement with another one of the motors of the surgical instrument **4400**.

(425) In at least one example, as illustrated in FIG. **93**, the closure trigger **32** can be operably coupled to the interface **4412** and can be configured to transition the interface **4412** between a plurality of positions and/or states. As illustrated in FIG. **93**, the closure trigger **32** can be movable, for example during a closure stroke, to transition the interface **4412** from a first position and/or state to a second position and/or state while transitioning the end effector **300** to an approximated configuration to capture tissue by the end effector, for example.

(426) In certain instances, in the first position and/or state, the module **4410** can be electrically coupled to a first motor such as, for example, the articulation motor **4406**, and in the second position and/or state, the module **4410** can be electrically coupled to a second motor such as, for example, the firing motor **4402**. In the first position and/or state, the module **4410** may be engaged with the articulation motor **4406** to allow the user to articulate the end effector **300** to a desired position; and the module **4410** may remain engaged with the articulation motor **4406** until the trigger **32** is actuated. As the user actuates the closure trigger **32** to capture tissue by the end effector **300** at the desired position, the interface **4412** can be transitioned or shifted to transition

the module **4410** from operable engagement with the articulation motor **4406**, for example, to operable engagement with the firing motor **4402**, for example. Once operable engagement with the firing motor **4402** is established, the module **4410** may take control of the firing motor **4402**; and the module **4410** may activate the motor **4402**, in response to user input, to fire the plurality of staples **191** and/or advance the cutting edge **182**, for example.

(427) In certain instances, as illustrated in FIG. **93**, the module **4410** may include a plurality of electrical and/or mechanical contacts **4411** adapted for coupling engagement with the interface **4412**. The plurality of motors of the surgical instrument **4400**, which share the module **4410**, may each comprise one or more corresponding electrical and/or mechanical contacts **4413** adapted for coupling engagement with the interface **4412**, for example.

(428) In various instances, the motors of the surgical instrument **4400** can be electrical motors. In certain instances, one or more of the motors of the surgical instrument **4400** can be a DC brushed driving motor having a maximum rotation of, approximately, 25,000 RPM, for example. In other arrangements, the motors of the surgical instrument **4400** may include one or more motors selected from a group of motors comprising a brushless motor, a cordless motor, a synchronous motor, a stepper motor, or any other suitable electric motor.

(429) In various instances, as illustrated in FIG. **92**, the common control module **4410** may comprise a motor driver **4426** which may comprise one or more H-Bridge field-effect transistors (FETs). The motor driver **4426** may modulate the power transmitted from a power source **4428** to a motor coupled to the module **4410** based on input from a microcontroller **4420** (“controller”), for example. In certain instances, the controller **4420** can be employed to determine the current drawn by the motor, for example, while the motor is coupled to the module **4410**, as described above.

(430) In certain instances, the controller **4420** may include a microprocessor **4422** (“processor”) and one or more computer readable mediums or memory units **4424** (“memory”). In certain instances, the memory **4424** may store various program instructions, which when executed may cause the processor **4422** to perform a plurality of functions and/or calculations described herein. In certain instances, one or more of the memory units **4424** may be coupled to the processor **4422**, for example.

(431) In certain instances, the power source **4428** can be employed to supply power to the controller **4420**, for example. In certain instances, the power source **4428** may comprise a battery (or “battery pack” or “power pack”), such as a Li ion battery, for example. In certain instances, the battery pack may be configured to be releasably mounted to the handle **14** for supplying power to the surgical instrument **4400**. A number of battery cells connected in series may be used as the power source **4428**. In certain instances, the power source **4428** may be replaceable and/or rechargeable, for example.

(432) In various instances, the processor **4422** may control the motor driver **4426** to control the position, direction of rotation, and/or velocity of a motor that is coupled to the module **4410**. In certain instances, the processor **4422** can signal the motor driver **4426** to stop and/or disable a motor that is coupled to the module **4410**. It should be understood that the term processor as used herein includes any suitable microprocessor, microcontroller, or other basic computing device that incorporates the functions of a computer's central processing unit (CPU) on an integrated circuit or at most a few integrated circuits. The processor is a multipurpose, programmable device that accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output. It is an example of sequential digital logic, as it has internal memory. Processors operate on numbers and symbols represented in the binary numeral system.

(433) In one instance, the processor **4422** may be any single core or multicore processor such as those known under the trade name ARM Cortex by Texas Instruments. In certain instances, the microcontroller **4420** may be an LM 4F230H5QR, available from Texas Instruments, for example. In at least one example, the Texas Instruments LM4F230H5QR is an ARM Cortex-M4F Processor Core comprising on-chip memory of 256 KB single-cycle flash memory, or other non-volatile

memory, up to MHz, a prefetch buffer to improve performance above 40 MHz, a 32 KB single-cycle SRAM, internal ROM loaded with StellarisWare® software, 2 KB EEPROM, one or more PWM modules, one or more QEI analog, one or more 12-bit ADC with 12 analog input channels, among other features that are readily available for the product datasheet. Other microcontrollers may be readily substituted for use with the module **4410**. Accordingly, the present disclosure should not be limited in this context.

(434) In certain instances, the memory **4424** may include program instructions for controlling each of the motors of the surgical instrument **4400** that are couplable to the module **4410**. For example, the memory **4424** may include program instructions for controlling the articulation motor **4406**. Such program instructions may cause the processor **4422** to control the articulation motor **4406** to articulate the end effector **300** in accordance with user input while the articulation motor **4406** is coupled to the module **4410**. In another example, the memory **4424** may include program instructions for controlling the firing motor **4402**. Such program instructions may cause the processor **4422** to control the firing motor **4402** to fire the plurality of staples **191** and/or advance the cutting edge **182** in accordance with user input while the firing motor **4402** is coupled to the module **4410**.

(435) In certain instances, one or more mechanisms and/or sensors such as, for example, sensors **4430** can be employed to alert the processor **4422** to the program instructions that should be used in a particular setting. For example, the sensors **4430** may alert the processor **4422** to use the program instructions associated with articulation of the end effector **300** while the module **4410** is coupled to the articulation motor **4406**; and the sensors **4430** may alert the processor **4422** to use the program instructions associated with firing the surgical instrument **4400** while the module **4410** is coupled to the firing motor **4402**. In certain instances, the sensors **4430** may comprise position sensors which can be employed to sense the position of the switch **4414**, for example. Accordingly, the processor **4422** may use the program instructions associated with articulation of the end effector **300** upon detecting, through the sensors **4430** for example, that the switch **4414** is in the first position **4416**; and the processor **4422** may use the program instructions associated with firing the surgical instrument **4400** upon detecting, through the sensors **4430** for example, that the switch **4414** is in the second position **4418**.

(436) Referring now to FIG. **94**, an outer casing of the surgical instrument **4400** is removed and several features and elements of the surgical instrument **4400** are also removed for clarity of disclosure. As illustrated in FIG. **94**, the surgical instrument **4400** may include a plurality of sensors which can be employed to perform various functions in connection with the operation of the surgical instrument **4400**. For example, as illustrated in FIG. **94**, the surgical instrument **4400** may include sensors A, B, and/or C. In certain instances, the sensor A can be employed to perform a first function, for example; the sensor B can be employed to perform a second function, for example; and the sensor C can be employed to perform a third function, for example. In certain instances, the sensor A can be employed to sense a thickness of the tissue captured by the end effector **300** during a first segment of a closure stroke; the sensor B can be employed to sense the tissue thickness during a second segment of the closure stroke following the first segment; and the sensor C can be employed to sense the tissue thickness during a third segment of the closure stroke following the second segment, for example. In certain instances, the sensors A, B, and C can be disposed along the end effector **300**, for example.

(437) In certain instances, the sensors A, B, and C can be arranged, as illustrated in FIG. **94**, such that the sensor A is disposed proximal to the sensor B, and the sensor C is disposed proximal to the sensor B, for example. In certain instances, as illustrated in FIG. **94**, the sensor A can sense the tissue thickness of the tissue captured by the end effector **300** at a first position; the sensor B can sense the tissue thickness of the tissue captured by the end effector **300** at a second position distal to the first position; and the sensor C can sense the tissue thickness of the tissue captured by the end effector **300** at a third position distal to the second position, for example. The reader will

appreciate that the sensors described herein are intended as examples of the types of sensors which can be employed in connection with the present disclosure. Other suitable sensors and sensing arrangements can be employed by the present disclosure.

(438) In certain instances, the surgical instrument **4400** may include a common control module **4450** which can be similar in many respects to the module **4410**. For example, the module **4450**, like the module **4410**, may comprise the controller **4420**, the processor **4422**, and/or the memory **4424**. In certain instances, the power source **4428** can supply power to the module **4450**, for example. In certain instances, the surgical instrument **4400** may include a plurality of sensors such as the sensors A, B, and C, for example, which can be activated to perform various functions in connection with the operation of the surgical instrument **4400**. In certain instances, one of the sensors A, B, and C, for example, can be individually or separately activated to perform one or more functions while the other sensors remain inactive. In certain instances, a plurality of sensors of the surgical instrument **4400** such as, for example, the sensors A, B, and C may share the module **4450**. In certain instances, only one of the sensors A, B, and C can be coupled to the module **4450** at a time. In certain instances, the plurality of sensors of the surgical instrument **4400** can be individually and separately couplable to the module **4450**, for example. In at least one example, the module **4450** can be selectively switched between operable engagement with sensor A, Sensor B, and/or Sensor C.

(439) In certain instances, as illustrated in FIG. **94**, the module **4450** can be disposed in the handle **14**, for example, and the sensors that share the module **4450** can be disposed in the end effector **300**, for example. The reader will appreciate that the module **4450** and/or the sensors that share the module **4450** are not limited to the above identified positions. In certain instances, the module **4450** and the sensors that share the module **4450** can be disposed in the end effector **300**, for example. Other arrangements for the positions of the module **4450** and/or the sensors that share the module **4450** are contemplated by the present disclosure.

(440) In certain instances, as illustrated in FIG. **94**, an interface **4452** can be employed to manage the coupling and/or decoupling of the sensors of the surgical instrument **4400** to the module **4450**. In certain instances, the interface **4452** can be selectively transitioned between a plurality of positions and/or states. In a first position and/or state, the interface **4452** may couple the module **4450** to the sensor A, for example; in a second position and/or state, the interface **4452** may couple the module **4450** to the sensor B, for example; and in a third position and/or state, the interface **4452** may couple the module **4450** to the sensor C, for example. Additional positions and/or states of the interface **4452** are contemplated by the present disclosure.

(441) In certain instances, the interface **4452** is movable between a first position, a second position, and/or a third position, for example, wherein the module **4450** is coupled to a first sensor in the first position, a second sensor in the second position, and a third sensor in the third position. In certain instances, the module **4450** is decoupled from first sensor as the interface **4452** is moved from the first position; the module **4450** is decoupled from second sensor as the interface **4452** is moved from the second position; and the module **4450** is decoupled from third sensor as the interface **4452** is moved from the third position. In certain instances, a switch or a trigger can be configured to transition the interface **4452** between the plurality of positions and/or states. In certain instances, a trigger can be movable to simultaneously effectuate the end effector and transition the control module **4450** from operable engagement with one of the sensors that share the module **4450** to operable engagement with another one of the sensors that share the module **4450**, for example.

(442) In at least one example, as illustrated in FIG. **94**, the closure trigger **32** can be operably coupled to the interface **4450** and can be configured to transition the interface **4450** between a plurality of positions and/or states. As illustrated in FIG. **94**, the closure trigger **32** can be moveable between a plurality of positions, for example during a closure stroke, to transition the interface **4450** between a first position and/or state wherein the module **4450** is electrically coupled to the

sensor A, for example, a second position and/or state wherein the module **4450** is electrically coupled to the sensor B, for example, and/or a third position and/or state wherein the module **4450** is electrically coupled to the sensor C, for example.

(443) In certain instances, a user may actuate the closure trigger **32** to capture tissue by the end effector **300**. Actuation of the closure trigger may cause the interface **4452** to be transitioned or shifted to transition the module **4450** from operable engagement with the sensor A, for example, to operable engagement with the sensor B, for example, and/or from operable engagement with sensor B, for example, to operable engagement with sensor C, for example.

(444) In certain instances, the module **4450** may be coupled to the sensor A while the trigger **32** is in a first actuated position. As the trigger **32** is actuated past the first actuated position and toward a second actuated position, the module **4450** may be decoupled from the sensor A. Alternatively, the module **4450** may be coupled to the sensor A while the trigger **32** is in an unactuated position. As the trigger **32** is actuated past the unactuated position and toward a second actuated position, the module **4450** may be decoupled from the sensor A. In certain instances, the module **4450** may be coupled to the sensor B while the trigger **32** is in the second actuated position. As the trigger **32** is actuated past the second actuated position and toward a third actuated position, the module **4450** may be decoupled from the sensor B. In certain instances, the module **4450** may be coupled to the sensor C while the trigger **32** is in the third actuated position.

(445) In certain instances, as illustrated in FIG. **94**, the module **4450** may include a plurality of electrical and/or mechanical contacts **4451** adapted for coupling engagement with the interface **4452**. The plurality of sensors of the surgical instrument **4400**, which share the module **4450**, may each comprise one or more corresponding electrical and/or mechanical contacts **4453** adapted for coupling engagement with the interface **4452**, for example.

(446) In certain instances, the processor **4422** may receive input from the plurality of sensors that share the module **4450** while the sensors are coupled to the module **4452**. For example, the processor **4422** may receive input from the sensor A while the sensor A is coupled to the module **4450**; the processor **4422** may receive input from the sensor B while the sensor B is coupled to the module **4450**; and the processor **4422** may receive input from the sensor C while the sensor C is coupled to the module **4450**. In certain instances, the input can be a measurement value such as, for example, a measurement value of a tissue thickness of tissue captured by the end effector **300**. In certain instances, the processor **4422** may store the input from one or more of the sensors A, B, and C on the memory **4426**. In certain instances, the processor **4422** may perform various calculations based on the input provided by the sensors A, B, and C, for example.

(447) Local Display of Tissue Parameter Stabilization

(448) FIGS. **95A** and **1B** illustrate one embodiment of an end effector **5300** comprising a staple cartridge **5306** that further comprises two light-emitting diodes (LEDs) **5310**. The end effector **5300** is similar to the end effector **300** described above. The end effector comprises a first jaw member or anvil **5302**, pivotally coupled to a second jaw member or elongated channel **5304**. The elongated channel **5304** is configured to receive the staple cartridge **5306** therein. The staple cartridge **5306** comprises a plurality of staples (not shown). The plurality of staples are deployable from the staple cartridge **5306** during a surgical operation. The staple cartridge **5306** further comprises two LEDs **5310** mounted on the upper surface, or cartridge deck **5308** of the staple cartridge **5306**. The LEDs **5310** are mounted such that they will be visible when the anvil **5304** is in a closed position. Furthermore, the LEDs **5310** can be sufficiently bright to be visible through any tissue that may be obscuring a direct view of the LEDs **5310**. Additionally, one LED **5310** can be mounted on either side of the staple cartridge **5306** such that at least one LED **5310** is visible from either side of the end effector **5300**. The LED **5310** can be mounted near the proximal end of the staple cartridge **530**, as illustrated, or may be mounted at the distal end of the staple cartridge **5306**.

(449) The LEDs **5310** may be in communication with a processor or microcontroller, such as for

instance microcontroller **1500** of FIG. **19**. The microcontroller **1500** can be configured to detect a property of tissue compressed by the anvil **5304** against the cartridge deck **5308**. Tissue that is enclosed by the end effector **5300** may change height as fluid within the tissue is exuded from the tissue's layers. Stapling the tissue before it has sufficiently stabilized may affect the effectiveness of the staples. Tissue stabilization is typically communicated as a rate of change, where the rate of change indicates how rapidly the tissue enclosed by the end effector is changing height.

(450) The LEDs **5310** mounted to the staple cartridge **5306**, in the view of the operator of the instrument, can be used to indicate rate at which the enclosed tissue is stabilizing and/or whether the tissue has reached a stable state. The LEDs **5310** can, for example, be configured to flash at a rate that directly correlates to the rate of stabilization of the tissue, that is, can flash quickly initially, flash slower as the tissue stabilizes, and remain steady when the tissue is stable. Alternatively, the LEDs **5310** can flash slowly initially, flash more quickly as the tissue stabilizes, and turn off when the tissue is stable.

(451) The LEDs **5310** mounted on the staple cartridge **5306** can be used additionally or optionally to indicate other information. Examples of other information include, but are not limited to: whether the end effector **5300** is enclosing a sufficient amount of tissue, whether the staple cartridge **5306** is appropriate for the enclosed tissue, whether there is more tissue enclosed than is appropriate for the staple cartridge **5306**, whether the staple cartridge **5306** is not compatible with the surgical instrument, or any other indicator that would be useful to the operator of the instrument. The LEDs **5310** can indicate information by either flashing at a particular rate, turning on or off at a particular instance, lighting in different colors for different information. The LEDs **5310** can alternatively or additionally be used to illuminate the area of operation. In some embodiments the LEDs **5310** can be selected to emit ultraviolet or infrared light to illuminate information not visible under normal light, where that information is printed on the staple cartridge **5300** or on a tissue compensator (not illustrated). Alternatively or additionally, the staples can be coated with a fluorescing dye and the wavelength of the LEDs **5310** chosen so that the LEDs **5310** cause the fluorescing dye to glow. By illuminating the staples with the LEDs **5310** allows the operator of the instrument to see the staples after they have been driven.

(452) Returning to FIGS. **95A** and **95B**, FIG. **95A** illustrates a side angle view of the end effector **5300** with the anvil **5304** in a closed position. The illustrated embodiment comprises, by way of example, one LED **5310** located on either side of the cartridge deck **5308**. FIG. **95B** illustrates a three-quarter angle view of the end effector **5300** with the anvil **5304** in an open position, and one LED **5310** located on either side of the cartridge deck **5308**.

(453) FIGS. **96A** and **96B** illustrate one embodiment of the end effector **5300** comprising a staple cartridge **5356** that further comprises a plurality of LEDs **5360**. The staple cartridge **5356** comprises a plurality of LEDs **5360** mounted on the cartridge deck **5358** of the staple cartridge **5356**. The LEDs **5360** are mounted such that they will be visible when the anvil **5304** is in a closed position. Furthermore, the LEDs **5360** can be sufficiently bright to be visible through any tissue that may be obscuring a direct view of the LEDs **5360**. Additionally, the same number of LEDs **5360** can be mounted on either side of the staple cartridge **5356** such that the same number of LEDs **5360** is visible from either side of the end effector **5300**. The LEDs **5360** can be mounted near the proximal end of the staple cartridge **5356**, as illustrated, or may be mounted at the distal end of the staple cartridge **5356**.

(454) The LEDs **5360** may be in communication with a processor or microcontroller, such as for instance microcontroller **1500** of FIG. **15**. The microcontroller **1500** can be configured to detect a property of tissue compressed by the anvil **5304** against the cartridge deck **5358**, such as the rate of stabilization of the tissue, as described above. The LEDs **5360** can be used to indicate the rate at which the enclosed tissue is stabilizing and/or whether the tissue has reached a stable state. The LEDs **5360** can be configured, for instance, to light in sequence starting at the proximal end of the staple cartridge **5356** with each subsequent LED **5360** lighting at the rate at which the enclosed

tissue is stabilizing; when the tissue is stable, all the LEDs 5360 can be lit. Alternatively, the LEDs 5360 can light in sequence beginning at the distal end of the staple cartridge 5356. Yet another alternative is for the LEDs 5360 to light in a sequential, repeating sequence, with the sequence starting at either the proximal or distal end of the LEDs 5360. The rate at which the LEDs 5360 light and/or the speed of the repeat can indicate the rate at which the enclosed tissue is stabilizing. It is understood that these are only examples of how the LEDs 5360 can indicate information about the tissue, and that other combinations of the sequence in which the LEDs 5360 light, the rate at which they light, and or their on or off state are possible. It is also understood that the LEDs 5360 can be used to communicate some other information to the operator of the surgical instrument, or to light the work area, as described above.

(455) Returning to FIGS. 96A and 96B, FIG. 96A illustrates a side angle view of the end effector 5300 with the anvil 5304 in a closed position. The illustrated embodiment comprises, by way of example, a plurality of LEDs 5360 located on either side of the cartridge deck 5358. FIG. 96B illustrates a three-quarter angle view of the end effector 5300 with the anvil 5304 in an open position, illustrating a plurality of LEDs 5360 located on either side of the cartridge deck 5358.

(456) FIGS. 97A and 97B illustrate one embodiment of the end effector 5300 comprising a staple cartridge 5406 that further comprises a plurality of LEDs 5410. The staple cartridge 5406 comprises a plurality of LEDs 5410 mounted on the cartridge deck 5408 of the staple cartridge 5406, with the LEDs 5410 placed continuously from the proximal to the distal end of the staple cartridge 5406. The LEDs 5410 are mounted such that they will be visible when the anvil 5302 is in a closed position. The same number of LEDs 5410 can be mounted on either side of the staple cartridge 5406 such that the same number of LEDs 5410 is visible from either side of the end effector 5300.

(457) The LEDs 5410 can be in communication with a processor or microcontroller, such as for instance microcontroller 1500 of FIG. 15. The microcontroller 1500 can be configured to detect a property of tissue compressed by the anvil 5304 against the cartridge deck 5408, such as the rate of stabilization of the tissue, as described above. The LEDs 5410 can be configured to be turned on or off in sequences or groups as desired to indicate the rate of stabilization of the tissue and/or that the tissue is stable. The LEDs 5410 can further be configured communicate some other information to the operator of the surgical instrument, or to light the work area, as described above. Additionally or alternatively, the LEDs 5410 can be configured to indicate which areas of the end effector 5300 contain stable tissue, and or what areas of the end effector 5300 are enclosing tissue, and/or if those areas are enclosing sufficient tissue. The LEDs 5410 can further be configured to indicate if any portion of the enclosed tissue is unsuitable for the staple cartridge 5406.

(458) Returning to FIGS. 97A and 97B, FIG. 97A illustrates a side angle view of the end effector 5300 with the anvil 5304 in a closed position. The illustrated embodiment comprises, by way of example, a plurality of LEDs 5410 from the proximal to the distal end of the staple cartridge 5406, on either side of the cartridge deck 5408. FIG. 97B illustrates a three-quarter angle view of the end effector 5300 with the anvil 5304 in an open position, illustrating a plurality of LEDs 5410 from the proximal to the distal end of the staple cartridge 5406, and on either side of the cartridge deck 5408.

(459) Adjunct with Integrated Sensors to Quantify Tissue Compression

(460) FIG. 98A illustrates an embodiment of an end effector 5500 comprising a tissue compensator 5510 that further comprises a layer of conductive elements 5512. The end effector 5500 is similar to the end effector 300 described above. The end effector 5500 comprises a first jaw member, or anvil, 5502 pivotally coupled to a second jaw member 5504 (not shown). The second jaw member 5504 is configured to receive a staple cartridge 5506 therein (not shown). The staple cartridge 5506 comprises a plurality of staples (not shown). The plurality of staples 191 is deployable from the staple cartridge 3006 during a surgical operation. In some embodiments, the end effector 5500 further comprises a tissue compensator 5510 removably positioned on the anvil 5502 or on the

staple cartridge **5506**. FIG. **98B** illustrates a detail view of a portion of the tissue compensator **5510** shown in FIG. **98A**.

(461) As described above, the plurality of staples **191** can be deployed between an unfired position and a fired position, such that staple legs **5530** move through and penetrate tissue **5518** compressed between the anvil **5502** and the staple cartridge **5506**, and contact the anvil's **5502** staple-forming surface. In embodiments that include a tissue compensator **5510**, the staple legs **5530** also penetrate and puncture the tissue compensator **5510**. As the staple legs **5530** are deformed against the anvil's staple-forming surface, each staple **191** can capture a portion of the tissue **5518** and the tissue compensator **5510** and apply a compressive force to the tissue **5518**. The tissue compensator **5510** thus remains in place with the staples **191** after the surgical instrument **10** is withdrawn from the patient's body. Because they are to be retained by the patient's body, the tissue compensators **5510** are composed of biodurable and/or biodegradable materials. The tissue compensators **5510** are described in further detail in U.S. Pat. No. 8,657,176, entitled TISSUE THICKNESS COMPENSATOR FOR SURGICAL STAPLER, which is incorporated herein by reference in its entirety.

(462) Returning to FIG. **98A**, in some embodiments, the tissue compensator **5510** comprises a layer of conductive elements **5512**. The conductive elements **5512** can comprise any combination of conductive materials in any number of configurations, such as for instance coils of wire, a mesh or grid of wires, conductive strips, conductive plates, electrical circuits, microprocessors, or any combination thereof. The layer containing conductive elements **5512** can be located on the anvil-facing surface **5514** of the tissue compensator **5510**. Alternatively or additionally, the layer of conductive elements **5512** can be located on the staple cartridge-facing surface **5516** of the tissue compensator **5510**. Alternatively or additionally, the layer of conductive elements **5512** can be embedded within the tissue compensator **5510**. Alternatively, the layer of conductive elements **5512** can comprise all of the tissue compensator **5510**, such as when a conductive material is uniformly or non-uniformly distributed in the material comprising the tissue compensator **5510**.

(463) FIG. **98A** illustrates an embodiment wherein the tissue compensator **5510** is removably attached to the anvil **5502** portion of the end effector **5500**. The tissue compensator **5510** would be so attached before the end effector **5500** would be inserted into a patient's body. Additionally or alternatively, a tissue compensator **5610** can be attached to a staple cartridge **5506** (not illustrated) after or before the staple cartridge **5506** is applied to the end effector **6600** and before the device is inserted into a patient's body.

(464) FIG. **99** illustrates various example embodiments that use the layer of conductive elements **5512** and conductive elements **5524**, **5526**, and **5528** in the staple cartridge **5506** to detect the distance between the anvil **5502** and the upper surface of the staple cartridge **5506**. The distance between the anvil **5502** and the staple cartridge **5506** indicates the amount and/or density of tissue **5518** compressed therebetween. This distance can additionally or alternatively indicate which areas of the end effector **5500** contain tissue. The tissue **5518** thickness, density, and/or location can be communicated to the operator of the surgical instrument **10**.

(465) In the illustrated example embodiments, the layer of conductive elements **5512** is located on the anvil-facing surface **5514** of the tissue compensator **5510**, and comprises one or more coils of wire **5522** in communication with a microprocessor **5520**. The microprocessor **5500** can be located in the end effector **5500** or any component thereof, or can be located in the housing **12** of the instrument, or can comprise any microprocessor or microcontroller previously described. In the illustrated example embodiments, the staple cartridge **5506** also includes conductive elements, which can be any one of: one or more coils of wire **5524**, one or more conductive plates **5526**, a mesh of wires **5528**, or any other convenient configuration, or any combination thereof. The staple cartridge's **5506** conductive elements can be in communication with the same microprocessor **5520** or some other microprocessor in the instrument.

(466) When the anvil **5502** is in a closed position and thus is compressing tissue **5518** against

staple cartridge **5506**, the layer of conductive elements **5512** of the tissue compensator **5510** can capacitively couple with the conductors in staple cartridge **5506**. The strength of the capacitive field between the layer of conductive elements **5512** and the conductive elements of the staple cartridge **5506** can be used to determine the amount of tissue **5518** being compressed. Alternatively, the staple cartridge **5506** can comprise eddy current sensors in communication with a microprocessor **5520**, wherein the eddy current sensors are operable to sense the distance between the anvil **5502** and the upper surface of the staple cartridge **5506** using eddy currents.

(467) It is understood that other configurations of conductive elements are possible, and that the embodiments of FIG. **99** are by way of example only, and not limitation. For example, in some embodiments the layer of conductive elements **5512** can be located on the staple cartridge-facing surface **5516** of the tissue compensator **5510**. Also, in some embodiments the conductive elements **5524**, **5526**, and/or **5528** can be located on or within the anvil **5502**. Thus in some embodiments, the layer of conductive elements **5512** can capacitively couple with conductive elements in the anvil **5502** and thereby sense properties of tissue **5518** enclosed within the end effector.

(468) It can also be recognized that tissue compensator **5512** can comprise a layer of conductive elements **5512** on both the anvil-facing surface **5514** and the cartridge-facing surface **5516**. A system to detect the amount, density, and/or location of tissue **5518** compressed by the anvil **5502** against the staple cartridge **5506** can comprise conductors or sensors either in the anvil **5502**, the staple cartridge **5506**, or both. Embodiments that include conductors or sensors in both the anvil **5502** and the staple cartridge **5506** can optionally achieve enhanced results by allowing differential analysis of the signals that can be achieved by this configuration.

(469) FIGS. **100A** and **100B** illustrate an embodiment of the tissue compensator **5510** comprising a layer of conductive elements **5512** in operation. FIG. **100A** illustrates one of the plurality of staples **191** after it has been deployed. As illustrated, the staple **191** has penetrated both the tissue **5518** and the tissue compensator **5510**. The layer of conductive elements **5512** may comprise, for example, mesh wires. Upon penetrating the layer of conductive elements **5512**, the staple legs **5530** may puncture the mesh of wires, thus altering the conductivity of the layer of conductive elements **5512**. This change in the conductivity can be used to indicate the locations of each of the plurality of staples **191**. The location of the staples **191** can be compared against the expected location of the staples, and this comparison can be used to determine if any staples did not fire or if any staples are not where they are expected to be.

(470) FIG. **100A** also illustrates staple legs **5530** that failed to completely deform. FIG. **100B** illustrates staple legs **5530** that have properly and completely deformed. As illustrated in FIG. **100B**, the layer of conductive elements **5512** can be punctured by the staple legs **5530** a second time, such as when the staple legs **5530** deform against the staple-forming surface of the anvil **5502** and turn back towards the tissue **5518**. The secondary breaks in the layer of conductive elements **5512** can be used to indicate complete staple **191** formation, as illustrated in FIG. **100B**, or incomplete staple **191** formation, as in FIG. **100A**.

(471) FIGS. **101A** and **101B** illustrate an embodiment of an end effector **5600** comprising a tissue compensator **5610** further comprising conductors **5620** embedded within. The end effector **5600** comprises a first jaw member, or anvil, **5602** pivotally coupled to a second jaw member **5604**. The second jaw member **5604** is configured to receive a staple cartridge **5606** therein. In some embodiments, the end effector **5600** further comprises a tissue compensator **5610** removably positioned on the anvil **5602** or the staple cartridge **5606**.

(472) Turning first to FIG. **4B**, FIG. **4B** illustrates a cutaway view of the tissue compensator **5610** removably positioned on the staple cartridge **5606**. The cutaway view illustrates an array of conductors **5620** embedded within the material that comprises the tissue compensator **5610**. The array of conductors **5620** can be arranged in an opposing configuration, and the opposing elements can be separated by insulating material. The array of conductors **5620** are each coupled to one or more conductive wires **5622**. The conductive wires **5622** allow the array of conductors **5620** to

communicate with a microprocessor, such as for instance microprocessor **1500**. The array of conductors **5620** may span the width of the tissue compensator **5610** such that they will be in the path of a cutting member or knife bar **280**. As the knife bar **280** advances, it will sever, destroy, or otherwise disable the conductors **5620**, and thereby indicate its position within the end effector **5600**. The array of conductors **5610** can comprise conductive elements, electric circuits, microprocessors, or any combination thereof.

(473) Turning now to FIG. **101A**, FIG. **101A** illustrates a close-up cutaway view of the end effector **5600** with the anvil **5602** in a closed position. In a closed position, the anvil **5602** can compress tissue **5618** and the tissue compensator **5610** against the staple cartridge **5606**. In some cases, only a part of the end effector **5600** may be enclosing the tissue **5618**. In areas of the end effector **5600** that are enclosing tissue **5618**, the tissue compensator **5610** may be compressed **5624** a greater amount than areas that do not enclose tissue **5618**, where the tissue compensator **5618** may remain uncompressed **5626** or be less compressed. In areas of greater compression **5624**, the array of conductors **5620** will also be compressed, while in uncompressed **5626** areas, the array of conductors **5620** will be further apart. Hence, the conductivity, resistance, capacitance, and/or some other electrical property between the array of conductors **5620** can indicate which areas of the end effector **5600** contain tissue.

(474) FIGS. **102A** and **102B** illustrate an embodiment of an end effector **5650** comprising a tissue compensator **5660** further comprising conductors **5662** embedded therein. The end effector **5650** comprises a first jaw member, or anvil, **5652** pivotally coupled to a second jaw member **5654**. The second jaw member **5654** is configured to receive a staple cartridge **5656** therein. In some embodiments, the end effector **5650** further comprises a tissue compensator **5660** removably positioned on the anvil **5652** or the staple cartridge **5656**.

(475) FIG. **102A** illustrates a cutaway view of the tissue compensator **5660** removably positioned on the staple cartridge **5656**. The cutaway view illustrates conductors **5670** embedded within the material that comprises the tissue compensator **5660**. Each of the conductors **5672** is coupled to a conductive wire **5672**. The conductive wires **5672** allow the array of conductors **5672** to communicate with a microprocessor, such as for instance microprocessor **1500**. The conductors **5672** may comprise conductive elements, electric circuits, microprocessors, or any combination thereof.

(476) FIG. **102A** illustrates a close-up side view of the end effector **5650** with the anvil **5652** in a closed position. In a closed position, the anvil **5652** can compress tissue **5658** and the tissue compensator **5660** against the staple cartridge **5656**. The conductors **5672** embedded within the tissue compensator **5660** can be operable to apply pulses of electrical current **5674**, at predetermined frequencies, to the tissue **5658**. The same or additional conductors **5672** can detect the response of the tissue **5658** and transmit this response to a microprocessor or microcontroller located in the instrument. The response of the tissue **5658** to the electrical pulses **5674** can be used to determine a property of the tissue **5658**. For example, the galvanic response of the tissue **5658** indicates the tissue's **5658** moisture content. As another example, measurement of the electrical impedance through the tissue **5658** could be used to determine the conductivity of the tissue **5648**, which is an indicator of the tissue type. Other properties that can be determined include by way of example and not limitation: oxygen content, salinity, density, and/or the presence of certain chemicals. By combining data from several sensors, other properties could be determined, such as blood flow, blood type, the presence of antibodies, etc.

(477) FIG. **103** illustrates an embodiment of a staple cartridge **5706** and a tissue compensator **5710** wherein the staple cartridge **5706** provides power to the conductive elements **5720** that comprise the tissue compensator **5710**. As illustrated, the staple cartridge **5706** comprises electrical contacts **5724** in the form of patches, spokes, bumps, or some other raised configuration. The tissue compensator **5710** comprises mesh or solid contact points **5722** that can electrically couple to the contacts **5724** on the staple cartridge **5706**.

(478) FIGS. **104A** and **104B** illustrate an embodiment of a staple cartridge **5756** and a tissue compensator **5760** wherein the staple cartridge provides power to the conductive elements **5770** that comprise the tissue compensator **5710**. As illustrated in FIG. **104A**, the tissue compensator **5760** comprises an extension or tab **5772** configured to come into contact with the staple cartridge **5756**. The tab **5772** may contact and adhere to an electrical contact (not shown) on the staple cartridge **5756**. The tab **5772** further comprises a break point **5774** located in a wire comprising the conductive elements **5770** of the tissue compensator **5760**. When the tissue compensator **5760** is compressed, such as when an anvil is in a closed position towards the staple cartridge **5756**, the break point **5774** will break, thus allowing the tissue compensator **5756** to become free from the staple cartridge **5756**. FIG. **104B** illustrates another embodiment employing a break point **5774** positioned in the tab **5772**.

(479) FIGS. **105A** and **F8B** illustrate an embodiment of an end effector **5800** comprising position sensing elements **5824** and a tissue compensator **5810**. The end effector **5800** comprises a first jaw member, or anvil, **5802** pivotally coupled to a second jaw member **5804** (not shown). The second jaw member **5804** is configured to receive a staple cartridge **5806** (not shown) therein. In some embodiments, the end effector **5800** further comprises a tissue compensator **5810** removably positioned on the anvil **5802** or the staple cartridge **5806**.

(480) FIG. **105A** illustrates the anvil **5804** portion of the end effector **5800**. In some embodiments the anvil **5804** comprises position sensing elements **5824**. The position sensing elements **5824** can comprise, for example, electrical contacts, magnets, RF sensors, etc. The position sensing elements **5824** can be located in key locations, such as for instance the corner points where the tissue compensator **5810** will be attached, or along the exterior edges of the anvil's **5802** tissue-facing surface. In some embodiments, the tissue compensator **5810** can comprise position indicating elements **5820**. The position indicating elements **5820** can be located in corresponding locations to the position sensing elements **5824** on the anvil **5802**, or in proximal locations, or in overlapping locations. The tissue compensator **5810** optionally further comprises a layer of conductive elements **5812**. The layer of conductive elements **5812** and/or the position indicating elements **5820** can be electrically coupled to conductive wires **5822**. The conductive wires **5822** can provide communication with a microprocessor, such as for instance microprocessor **1500**.

(481) FIG. **F8B** illustrates an embodiment the position sensing elements **5824** and position indicating elements **5820** in operation. When the tissue compensator **5810** is positioned, the anvil **5802** can sense **5826** that the tissue compensator **5810** is properly position. When the tissue compensator **5810** is misaligned or missing entirely, the anvil **5802** (or some other component) can sense **5826** that the tissue compensator **5810** is misaligned. If the misalignment is above a threshold magnitude, a warning can be signaled to the operator of the instrument, and/or a function of the instrument can be disabled to prevent the staples from being fired.

(482) In FIGS. **105A** and **105B** the position sensing elements **5824** are illustrated as a part of the anvil **5804** by way of example only. It is understood that the position sensing elements **5824** can be located instead or additionally on the staple cartridge **5806**. It is also understood that the location of the position sensing elements **5824** and the position indicating elements **5820** can be reversed, such that the tissue compensator **5810** is operable to indicate whether it is properly aligned.

(483) FIGS. **106A** and **F9B** illustrate an embodiment of an end effector **5850** comprising position sensing elements **5874** and a tissue compensator **5860**. The end effector **5850** comprises a first jaw member, or anvil, **5852** pivotally coupled to a second jaw member **5854** (not shown). The second jaw member **5854** is configured to receive a staple cartridge **5856** (not show) therein. In some embodiments, the end effector **5850** further comprises a tissue compensator **5860** removably positioned on the anvil **5852** or the staple cartridge **5856**.

(484) FIG. **106A** illustrates the anvil **5852** portion of the end effector **5850**. In some embodiments, the anvil **5854** comprises an array of conductive elements **5474**. The array of conductive elements **5474** can comprise, for example, electrical contacts, magnets, RF sensors, etc. The array of

conductive elements **5474** are arrayed along the length of the tissue-facing surface of the anvil **5852**. In some embodiments, the tissue compensator **5860** can comprise a layer of conductive elements **5862**, wherein the conductive elements comprise a grid or mesh of wires. The layer of conductive elements **5862** may be coupled to conductive wires **5876**. The conductive wires **5862** can provide communication with a microprocessor, such as for instance microprocessor **1500**. (485) FIG. **106A** illustrates an embodiment wherein of the conductive elements **5474** of the anvil **5852** and the layer of conductive elements **5862** are operable to indicate whether the tissue compensator **5860** is misaligned or missing. As illustrated, the array of conductive elements **5874** is operable to electrically couple with the layer of conductive elements **5862**. When the tissue compensator **5860** is misaligned or missing, the electrical coupling will be incomplete. If the misalignment is above a threshold magnitude, a warning can be signaled to the operator of the instrument, and/or a function of the instrument can be disabled to prevent the staples from being fired.

(486) It is understood that the array of conductive elements **5874** may additionally or alternatively be located on the staple cartridge **5856**. It is also understood that the any of the anvil **5852**, staple cartridge **5856**, and/or tissue compensator **5860** may be operable to indicate misalignment of the tissue compensator **5860**.

(487) FIGS. **107A** and **107B** illustrate an embodiment of a staple cartridge **5906** and a tissue compensator **5910** that is operable to indicate the position of a cutting member or knife bar **280**. FIG. **107A** is a top-down view of the staple cartridge **5906** that has a tissue compensator **5920** placed on its upper surface **5916**. The staple cartridge **5906** further comprises a cartridge channel **5918** operable to accept a cutting member or knife bar **280**. FIG. **107A** illustrates only the layer of conductive elements **5922** of the tissue compensator **5910**, for clarity. As illustrated, the layer of conductive elements **5922** comprises a lengthwise segment **5930** that is located off-center. The lengthwise segment **5930** is coupled to conductive wires **5926**. The conductive wires **5926** allow the layer of conductive elements **5922** to communicate with a microprocessor, such as for instance microprocessor **1500**. The layer of conductive elements **5922** further comprises horizontal elements **5932** coupled to the lengthwise segment **5930** and spanning the width of the tissue compensator **5910**, and thus crossing the path of the knife bar **280**. As the knife bar **280** advances, it will sever the horizontal elements **5932** and thereby alter an electrical property of the layer of conductive elements **5922**. For example, the advancing of the knife bar **280** may alter the resistance, capacitance, conductivity, or some other electrical property of the layer of conductive elements **5922**. As each horizontal element **5932** is severed by the knife bar **280**, the change in the electrical properties of the layer of conductive elements **5922** will indicate the position of the knife bar **280**.

(488) FIG. **107B** illustrates an alternate configuration for the layer of conductive elements **5922**. As illustrated, the layer of conductive elements **5922** comprises a lengthwise segment **5934** on either side of the cartridge channel **5918**. The layer of conductive elements **5922** further comprises horizontal elements **5936** coupled to both of the lengthwise segments **5934**, thus spanning the path of the knife bar **280**. As the knife bar **280**, the resistance, for example between the knife bar and the horizontal elements **5396** can be measured and used to determine the location of the knife bar **280**. Other configurations of the layer of conductive elements **5922** can be used to accomplish the same result, such as for instance any of the arrangements illustrated in FIGS. **98A** through **102B**. For example, the layer of conductive elements **5922** can comprise a wire mesh or grid, such that as the knife bar **280** advances it can sever the wire mesh and thereby change the conductivity in the wire mesh. This change in conductivity can be used to indicate the position of the knife bar **280**.

(489) Other uses for the layer of conductive elements **5922** can be imagined. For example, a specific resistance can be created in the layer of conductive elements **592**, or a binary ladder of resistors or conductors can be implemented, such that simple data can be stored in the tissue compensator **5910**. This data can be extracted from the tissue compensator **5910** by conductive elements in the anvil and/or staple cartridge when either electrically couple with the layer of

conductive elements **5922**. The data can represent, for example, a serial number, a “use by” date, etc.

(490) Polarity of Hall Magnet to Detect Misloaded Cartridge

(491) FIG. **108** illustrates one embodiment of an end effector **6000** comprising a magnet **6008** and a Hall effect sensor **6010** wherein the detected magnetic field **6016** can be used to identify a staple cartridge **6006**. The end effector **6000** is similar to the end effector **300** described above. The end effector **6000** comprises a first jaw member or anvil **6002**, pivotally coupled to a second jaw member or elongated channel **6004**. The elongated channel **6004** is configured to operably support a staple cartridge **6006** therein. The staple cartridge **6006** is similar to the staple cartridge **304** described above. The anvil **6002** further comprises a magnet **6008**. The staple cartridge **6006** further comprises a Hall effect sensor **6010** and a processor **6012**. The Hall effect sensor **6010** is operable to communicate with the processor **6012** through a conductive coupling **6014**. The Hall effect sensor **6010** is positioned within the staple cartridge **6006** to operatively couple with the magnet **6008** when the anvil **6002** is in a closed position. The Hall effect sensor **6010** can be operable to detect the magnetic field **6016** produced by the magnet **6008**. The polarity of the magnetic field **6016** can be one of north or south depending on the orientation of the magnet **6008** within the anvil **6002**. In the illustrated embodiment of FIG. **108**, the magnet **6008** is oriented such that its south pole is directed towards the staple cartridge **6006**. The Hall effect sensor **6010** can be operable to detect the magnetic field **6016** produced by a south pole. If the Hall effect sensor **6010** detects a magnetic south pole, then the staple cartridge **6006** can be identified as of a first type.

(492) FIG. **109** illustrates one embodiment of an end effector **6050** comprising a magnet **6058** and a Hall effect sensor **6060** wherein the detected magnetic field **6066** can be used to identify a staple cartridge **6056**. The end effector **6050** comprises a first jaw member or anvil **6052**, pivotally coupled to a second jaw member or elongated channel **6054**. The elongated channel **6054** is configured to operably support a staple cartridge **6056** therein. The anvil **6052** further comprises a magnet **6058**. The staple cartridge **6056** further comprises a Hall effect sensor **6060** in communication with a processor **6062** over a conductive coupling **6064**. The Hall effect sensor **6060** is positioned such that it will operatively couple with the magnet **6058** when the anvil **6052** is in a closed position. The Hall effect sensor **6060** can be operable to detect the magnetic field **6066** produced by the magnet **6058**. In the illustrated embodiment, the magnet **6058** is oriented such that its north magnetic pole is directed towards the staple cartridge **6056**. The Hall effect sensor **6060** can be operable to detect the magnetic field **6066** produced by a north pole. If the Hall effect sensor **6060** detects a north magnetic pole, then the staple cartridge **6056** can be identified as a second type.

(493) It can be recognized that the second type staple cartridge **6056** of FIG. **109** can be substituted for the first type staple cartridge **6006** of FIG. **108**, and vice versa. In FIG. **108**, the second type staple cartridge **6056** would be operable to detect a magnetic north pole, but will detect a magnetic south pole instead. In this case, end effector **6000** will identify the staple cartridge **6056** as being of the second type. If the end effector **6000** did not expect a staple cartridge **6056** of the second type, the operator of the instrument can be alerted, and/or a function of the instrument can be disabled. The type of the staple cartridge **6056** can additionally or alternatively be used to identify some parameter of the staple cartridge **6056**, such as for instance the length of the cartridge and/or the height and length of the staples.

(494) Similarly, as shown in FIG. **109**, the first type staple cartridge **6006** can be substituted for the second staple cartridge **6056**. The first type staple cartridge **6006** would be operable to detect a south magnetic pole, but will instead detect a north magnetic pole. In this case, the end effector **6050** will identify the staple cartridge **6006** as being of the first type.

(495) FIG. **110** illustrates a graph **6020** of the voltage **6022** detected by a Hall effect sensor located in the distal tip of a staple cartridge, such as is illustrated in FIGS. **108** and **109**, in response to the distance or gap **6024** between a magnet located in the anvil and the Hall effect sensor in the staple cartridge, such as illustrated in FIGS. **108** and **109**. As illustrated FIG. **110**, when the magnet in the

anvil is oriented such that its north pole is towards the staple cartridge, the voltage will tend towards a first value as the magnet comes in proximity to the Hall effect sensor; when the magnet is oriented with its south pole towards the staple cartridge, the voltage will tend towards a second, different value. The measured voltage can be used by the instrument to identify the staple cartridge.

(496) FIG. **111** illustrates one embodiment of the housing **6100** of the surgical instrument, comprising a display **6102**. The housing **6100** is similar to the housing **12** described above. The display **6102** can be operable to convey information to the operator of the instrument, such as for instance, that the staple cartridge coupled to the end effector is inappropriate for the present application. Additionally or alternatively, the display **6102** can display the parameters of the staple cartridge, such as the length of the cartridge and/or the height and length of the staples.

(497) FIG. **112** illustrates one embodiment of a staple retainer **6160** comprising a magnet **6162**. The staple retainer **6160** can be operably coupled to a staple cartridge **6156** and functions to prevent staples from exiting of the staple cartridge **6156**. The staple retainer **6160** can be left in place when the staple cartridge **6156** is applied to an end effector. In some embodiments, the staple retainer **6160** comprises a magnet **6162** located in the distal area of the staple retainer **6160**. The anvil of the end effector can comprise a Hall effect sensor operable to couple with the magnet **6162** in the staple retainer **6160**. The Hall effect sensor can be operable to detect the properties of the magnet **6162**, such as for instance the magnetic field strength and magnetic polarity. The magnetic field strength can be varied by, for example, placing the magnet **6162** in different locations and/or depths on or in the staple retainer **6160**, or by selecting magnets **6162** of different compositions. The different properties of the magnet **6162** can be used to identify staple cartridges of different types.

(498) FIGS. **113A** and **113B** illustrate one embodiment of an end effector **6200** comprising a sensor **6208** for identifying staple cartridges **6206** of different types. The end effector **6200** comprises a first jaw member or anvil **6202**, pivotally coupled to a second jaw member or elongated channel **6204**. The elongated channel **6204** is configured to operably support a staple cartridge **6206** therein. The end effector **6200** further comprises a sensor **6208** located in the proximal area. The sensor **6208** can be any of an optical sensor, a magnetic sensor, an electrical sensor, or any other suitable sensor.

(499) The sensor **6208** can be operable to detect a property of the staple cartridge **6206** and thereby identify the staple cartridge **6206** type. FIG. **113B** illustrates an example where the sensor **6208** is an optical emitter and detector **6210**. The body of the staple cartridge **6206** can be different colors, such that the color identifies the staple cartridge **6206** type. An optical emitter and detector **6210** can be operable to interrogate the color of the staple cartridge **6206** body. In the illustrated example, the optical emitter and detector **6210** can detect white **6212** by receiving reflected light in the red, green, and blue spectrums in equal intensity. The optical emitter and detector **6210** can detect red **6214** by receiving very little reflected light in the green and blue spectrums while receiving light in the red spectrum in greater intensity.

(500) Alternately or additionally, the optical emitter and detector **6210**, or another suitable sensor **6208**, can interrogate and identify some other symbol or marking on the staple cartridge **6206**. The symbol or marking can be any one of a barcode, a shape or character, a color-coded emblem, or any other suitable marking. The information read by the sensor **6208** can be communicated to a microcontroller in the surgical device **10**, such as for instance microcontroller **1500**. The microcontroller **1500** can be configured to communicate information about the staple cartridge **6206** to the operator of the instrument. For instance, the identified staple cartridge **6206** may not be appropriate for a given application; in such case, the operator of the instrument can be informed, and/or a function of the instrument s inappropriate. In such instance, microcontroller **1500** can optionally be configured to disable a function of surgical instrument can be disabled. Alternatively or additionally, microcontroller **1500** can be configured to inform the operator of the surgical instrument **10** of the parameters of the identified staple cartridge **6206** type, such as for instance the length of the staple cartridge **6206**, or information about the staples, such as the height and length.

(501) Smart Cartridge Wake Up Operation and Data Retention

(502) In one embodiment the surgical instrument described herein comprises short circuit protection techniques for sensors and/or electronic components. To enable such sensors and other electronic technology both power and data signals are transferred between modular components of the surgical instrument. During assembly of modular sensor components electrical conductors that when connected are used to transfer power and data signals between the connected components are typically exposed.

(503) FIG. **114** is a partial view of an end effector **7000** with electrical conductors **7002**, **7004** for transferring power and data signals between the connected components of the surgical instrument according to one embodiment. There is potential for these electrical conductors **7002**, **7004** to become shorted and thus damage critical system electronic components. FIG. **115** is a partial view of the end effector **7000** shown in FIG. **114** showing sensors and/or electronic components **7005** located in the end effector. With reference now to both FIGS. **114** and **115**, in various embodiments the surgical instruments disclosed throughout the present disclosure provide real time feedback about the compressibility and thickness of tissue using electronic sensors. Modular architectures will enable the configuration of custom modular shafts to employ job specific technologies. To enable sensors and other electronic circuit components in surgical instruments it is necessary to transfer both power and data signals between a secondary circuit comprising the modular sensor and/or electronic circuit components **7005**. During the assembly of the modular sensors and/or electronic components **7005** the electrical conductors **7002**, **7004** are exposed such that when connected, they are used to transfer power and data signals between the connected sensors and/or electronic components **7005**. Because there is a potential for these electrical conductors **7002**, **7004** to become short circuited during the assembly process and thus damage other system electronic circuits, various embodiments of the surgical instruments described herein comprise short circuit protection techniques for the sensors and/or electronic components **7005**

(504) In one embodiment, the present disclosure provides a short circuit protection circuit **7012** for the sensors and/or electronic components **7005** of the secondary circuits of the surgical instrument. FIG. **116** is a block diagram of a surgical instrument electronic subsystem **7006** comprising a short circuit protection circuit **7012** for the sensors and/or electronic components **7005** according to one embodiment. A main power supply circuit **7010** is connected to a primary circuit comprising a microprocessor and other electronic components **7008** (processor **7008** hereinafter) through main power supply terminals **7018**, **7020**. The main power supply circuit **7010** also is connected to a short circuit protection circuit **7012**. The short circuit protection circuit **7012** is coupled to a supplementary power supply circuit **7014**, which supplies power to the sensors and/or electronic components **7005** via the electrical conductors **7002**, **7004**.

(505) To reduce damage to the processor **7008** connected to the main power supply terminals **7018**, **7020**, during a short circuit between the electrical conductors **7002**, **7004** of the power supply terminals feeding the sensors and/or electronic components **7005**, a self isolating/restoring short circuit protection circuit **7012** is provided. In one embodiment, the short circuit protection circuit **7012** may be implemented by coupling a supplementary power supply circuit **7014** to the main power supply circuit **7010**. In circumstances when the supplementary power supply circuit **7014** power conductors **7002**, **7004** are shorted, the supplementary power supply circuit **7014** isolates itself from the main power supply circuit **7010** to prevent damage to the processor **7008** of the surgical instrument. Thus, there is virtually no effect to the processor **7008** and other electronic circuit components coupled to the main power supply terminals **7018**, **7020** when a short circuit occurs in the electrical conductors **7002**, **7004** of the supplementary power supply circuit **7014**. Accordingly, in the event that a short circuit occurs between the electrical conductors **7002**, **7004** of the supplementary power supply circuit **7014**, the main power supply circuit **7010** is unaffected and remains active to supply power to the protected processor **7008** such that the processor **7008** can monitor the short circuit condition. When the short circuit between the electrical conductors **7002**,

7004 of the supplementary power supply circuit **7014** is remedied, the supplementary power supply circuit **7014** rejoins the main power supply circuit **7010** and is available once again to supply power to the sensor components **7005**. The short circuit protection circuit **7012** also may be monitored to indicate one or more short circuit conditions to the end user of the surgical instrument. The short circuit protection circuit **7012** also may be monitored to lockout the firing of the surgical instrument when a short circuit event is indicated. Many supplementary protection circuits may be networked together to isolate, detect, or protect other circuit functions.

(506) Accordingly, in one aspect, the present disclosure provides a short circuit protection circuit **7012** for electrical conductors **7002**, **7004** in the end effector **7000** (FIGS. **114** and **115**) or other elements of the surgical instrument. In one embodiment, the short circuit protection circuit **7012** employs a supplementary self-isolating/restoring power supply circuit **7014** coupled to the main power supply circuit **7010**. The short circuit protection circuit **7012** may be monitored to indicate one or more short circuit conditions to the end user of the surgical instrument. In the event of a short circuit, the short circuit protection circuit **7012** may be employed to lock-out the surgical instrument from being fired or other device operations. Many other supplementary protection circuits may be networked together to isolate, detect, or protect other circuit functions.

(507) FIG. **117** is a short circuit protection circuit **7012** comprising a supplementary power supply circuit **7014** coupled to a main power supply circuit **7010**, according to one embodiment. The main power supply circuit **7010** comprises a transformer **7023** (X1) coupled to a full wave rectifier **7025** implemented with diodes **91-94**. The full wave rectifier **7025** is coupled to the voltage regulator **7027**. The output (OUT) of the voltage regulator **7027** is coupled to both the output terminals **7018**, **7020** of the main power supply circuit **7010** (OP1) and the supplementary power supply circuit **7014**. An input capacitor C1 filters the input voltage in the voltage regulator **7027** and one or more capacitors C2 filter the output of the voltage regulator **7027**.

(508) In the embodiment illustrated in FIG. **117**, the supplementary power supply circuit **7014** comprises a pair of transistors T1, T2 configured to control the power supply output OP2 between the electrical conductors **7002**, **7004**. During normal operation when the electrical conductors **7002**, **7004** are not shorted, the output OP2 supplies power to the sensor components **7005**. Once the transistors T1 and T2 are turned ON (activated) and begin conducting current, the current from the output of the voltage regulator **7027** is shunted by the first transistor T1 such that no current flows through R1 and $i_{sub.R1}=0$. The output voltage of the regulator +V is applied at the node such the $V_{sub.n}+V$, which is then the output voltage OP2 of the supplementary power supply circuit **7014** and the first transistor T1 drives the current to the sensor components **7005** through the output terminal **7002**, where output terminal **7004** is the current return path. A portion of the output current $i_{sub.R5}$ is diverted through R5 to drive the output indicator LED2. The current through the LED2 is $i_{sub.R5}$. As long as the node voltage $V_{sub.n}$ is above the threshold necessary to turn ON (activate) the second transistor T2, the supplementary power supply circuit **7014** operates as a power supply circuit to feed the sensors and/or electronic components **7005**.

(509) When the electrical conductors **7002**, **7004** of the secondary circuit are shorted, the node voltage $V_{sub.n}$ drops to ground or zero and the second transistor T2 turns OFF and stops conducting, which turns OFF the first transistor T1. When the first transistor T1 is cut-OFF, the output voltage +V of the voltage regulator **7027** causes current $i_{sub.R1}$ to flow through the short circuit indicator LED1 and through to ground via the short circuit between the electrical conductors **7002**, **7004**. Thus, no current flows through R5 and $i_{sub.R5}=0A$ and $+V_{sub.OP2}=0V$. The supplementary power supply circuit **7014** isolates itself from the main power supply circuit **7010** until the short circuit is removed. During the short circuit only the short circuit indicator LED1 is energized while the output indicator LED2 is not. When the short circuit between the electrical conductors **7002**, **7004** is removed, the node voltage $V_{sub.n}$ rises until T2 turns ON and subsequently turning T1 ON. When T1 and T2 are turned ON (are biased in a conducting state such as saturation), until the node voltage $V_{sub.n}$ reaches $+V_{sub.OP2}$ and the supplementary power

supply circuit **7014** resumes its power supply function for the sensor components **7005**. Once the supplementary power supply circuit **7014** restores its power supply function, the short circuit indicator LED1 turns OFF and the output indicator LED2 turns ON. The cycle is repeated in the event of another short circuit between the supplementary power supply circuit **7014** electrical conductors **7002**, **7004**.

(510) In one embodiment, a sample rate monitor is provided to enable power reduction by limiting sample rates and/or duty cycle of the sensor components when the surgical instrument is in a non-sensing state. FIG. **118** is a block diagram of a surgical instrument electronic subsystem **7022** comprising a sample rate monitor **7024** to provide power reduction by limiting sample rates and/or duty cycle of the sensors and/or electronic components **7005** of the secondary circuit when the surgical instrument is in a non-sensing state, according to one embodiment. As shown in FIG. **118**, the surgical instrument electronic subsystem **7022** comprises a processor **7008** coupled to a main power supply circuit **7010**. The main power supply circuit **7010** is coupled to a sample rate monitor circuit **7024**. A supplementary power supply circuit **7014** is coupled to the sample rate **7024** as powers the sensors and/or electronic components **7005** via the electrical conductors **7002**, **7004**. The primary circuit comprising the processor **7008** is coupled to a device state monitor **7026**. In various embodiments, the surgical instrument electronic subsystem **7022** provides real time feedback about the compressibility and thickness of tissue using the sensors and/or electronic components **7005** as previously described herein. The modular architecture of the surgical instrument enables the configuration of custom modular shafts to employ function job specific technologies. To enable such additional functionality, electronic connection points and components are employed to transfer both power and signal between modular components of the surgical instrument. An increase in the number of sensors and/or electronic components **7005** increases the power consumption of the surgical instrument system **7022** and creates the need for various techniques for reducing power consumption of the surgical instrument system **7022**.

(511) In one embodiment, to reduce power consumption, a surgical instrument configured with sensors and/or electronic components **7005** (secondary circuit) comprises a sample rate monitor **7024**, which can be implemented as a hardware circuit or software technique to reduce the sample rate and/or duty cycle for the sensors and/or electronic components **7005**. The sample rate monitor **7024** operates in conjunction with the device state monitor **7026**. The device state monitor **7026** senses the state of various electrical/mechanical subsystems of the surgical instrument. In the embodiment illustrated in FIG. **118**, the device state monitor **7026** whether the state of the end effector is in an unclamped (State 1), a clamping (State 2), or a clamped (State 3) state of operation.

(512) The sample rate monitor **7024** sets the sample rate and/or duty cycle for the sensor components **7005** based on the state of the end effector determined by the device state monitor **7026**. In one aspect, the sample rate monitor **7024** may set the duty cycle to about 10% when the end effector is in State 1, to about 50% when the end effector is in State 2, or about 20% when the end effector is in State 3. In various other embodiments, the duty cycle and/or sample rate set by the sample rate monitor **7024** may take on ranges of values. For example, in another aspect, the sample rate monitor **7024** may set the duty cycle to a value between about 5% to about 15% when the end effector is in State 1, to a value of about 45% to about 55% when the end effector is in State 2, or to a value of about 15% to about 25% when the end effector is in State 3. In various other embodiments, the duty cycle and/or sample rate set by the sample rate monitor **7024** may take on additional ranges of values. For example, in another aspect, the sample rate monitor **7024** may set the duty cycle to a value between about 1% to about 20% when the end effector is in State 1, to a value of about 20% to about 80% when the end effector is in State 2, or to a value of about 1% to about 50% when the end effector is in State 3. In various other embodiments, the duty cycle and/or sample rate set by the sample rate monitor **7024** may take on additional ranges of values.

(513) In one aspect, the sample rate monitor **7024** may be implemented by creating a supplementary circuit/software coupled to a main circuit/software. When the supplementary

circuit/software determines that the surgical instrument system **7022** is in a non-sensing condition, the sample rate monitor **7024** enters the sensors and/or electronic components **7005** into a reduced sampling or duty cycle mode reducing the power load on the main circuit. The main power supply circuit **7010** will still be active to supply power, so that the protected processor **7008** of the primary circuit can monitor the condition. When the surgical instrument system **7022** enters a condition requiring more rigorous sensing activity the sample rate monitor **7024** increases the supplementary circuit sample rate or duty cycle. The circuit could utilize a mixture of integrated circuits, solid state components, microprocessors, and firmware. The reduced sample rate or duty cycle mode circuit also may be monitored to indicate the condition to the end user of the surgical instrument system **7022**. The circuit/software might also be monitored to lockout the firing or function of the device in the event the device is in the power saving mode.

(514) In one embodiment, the sample rate monitor **7024** hardware circuit or software technique reduce the sample rate and/or duty cycle for the sensors and/or electronic components **7005** to reduce power consumption of the surgical instrument. The reduced sample rate and/or duty cycle may be monitored to indicate one or more conditions to the end user of the surgical instrument. In the event of a reduced sample rate and/or duty cycle condition in the surgical instrument the protection circuit/software may be configured to lock-out the surgical instrument from being fired or otherwise operated.

(515) In one embodiment, the present disclosure provides an over current and/or a voltage protection circuit for sensors and/or electronic components of a surgical instrument. FIG. **119** is a block diagram of a surgical instrument electronic subsystem **7028** comprising an over current and/or over voltage protection circuit **7030** for sensors and/or electronic components **7005** of the secondary circuit of a surgical instrument, according to one embodiment. In various embodiments, the surgical instrument electronic subsystem **7028** provides real time feedback about the compressibility and thickness of tissue using the sensors and/or electronic components **7005** of the secondary circuit as previously described herein. The modular architecture of the surgical instrument enables the configuration of custom modular shafts to employ function job specific technologies. To enable the sensors and/or electronic components **7005**, additional electronic connection points and components to transfer both power and signal between modular components are added. There is potential for these additional conductors for the sensors and/or electronic components **7005** from the modular pieces to be shorted and or damaged causing large draws of current that could damage fragile processor **7008** circuits or and other electronic components of the primary circuit. In one embodiment, the over current/voltage protection circuit **7030** protects the conductors for the sensors and/or electronic components **7005** on a surgical instrument using a supplementary self-isolating/restoring circuit **7014** coupled to the main power supply circuit **7010**. The operation of one embodiment of the supplementary self-isolating/restoring circuit **7014** is described in connection with FIG. **117** and will not be repeated here for conciseness and clarity of disclosure.

(516) In one embodiment, to reduce electronic damage during large current draws in a sensing surgical instrument, the electronic subsystem **7028** of the surgical instrument comprises an over current/voltage protection circuit **7030** for the conductors for the sensors and/or electronic components **7005**. The over current/voltage protection circuit **7030** may be implemented by creating a supplementary circuit coupled to a main power supply circuit **7010** circuit. In the case that the supplementary circuit electrical conductors **7002**, **7004** experience higher levels of current than expected, the over current/voltage protection circuit **7030** isolates the current from the main power supply circuit **7010** circuit to prevent damage. The main power supply circuit **7010** circuit will still be active to supply power, so that the protected main processor **7008** can monitor the condition. When a large current draw in the supplementary power supply circuit **7014** is remedied, the supplementary power supply circuit **7014** rejoins the main power supply circuit **7010** and is available to supply power to the sensors and/or electronic components **7005** (e.g., the secondary

circuit). The over current/voltage protection circuit **7030** may utilize a mixture of integrated circuits, solid state components, micro-processors, firmware, circuit breaker, fuses, or PTC (positive temperature coefficient) type technologies.

(517) In various embodiments, the over current/voltage protection circuit **7030** also may be monitored to indicate the over current/voltage condition to the end user of the device. The over current/voltage protection circuit **7030** also may be monitored to lockout the firing of the surgical instrument when the over current/voltage condition event is indicated. The over current/voltage protection circuit **7030** also may be monitored to indicate one or more over current/voltage conditions to the end user of the device. In the event of over current/voltage condition in the device the over current/voltage protection circuit **7030** may lock-out the surgical instrument from being fired or lock-out other operations of the surgical instrument.

(518) FIG. **120** is an over current/voltage protection circuit **7030** for sensors and electronic components **7005** (FIG. **119**) of the secondary circuit of a surgical instrument, according to one embodiment. The over current/voltage protection circuit **7030** provides a current path during a hard short circuit (SHORT) at the output of the over current/voltage protection circuit **7030**, and also provides a path for follow-through current through a bypass capacitor C.sub.BYPASS driven by stray inductance L.sub.STRAY.

(519) In one embodiment, the over current/voltage protection circuit **7030** comprises a current limited switch **7032** with autoreset. The current limited switch **7032** comprises a current sense resistor R.sub.CS coupled to an amplifier A. When the amplifier A senses a surge current above a predetermined threshold, the amplifier activates a circuit breaker CB to open the current path to interrupt the surge current. In one embodiment, the current limited switch **7032** with autoreset may be implemented with a MAX1558 integrated circuit by Maxim. The current limited switch **7032** with autoreset. Autoreset latches the switch **7032** off if it is shorted for more than 20 ms, saving system power. The shorted output (SHORT) is then tested to determine when the short is removed to automatically restart the channel Low quiescent supply current (45 μ A) and standby current (3 μ A) conserve battery power in the surgical instrument. The current limited switch **7032** with autoreset safety features ensure that the surgical instrument is protected. Built-in thermal-overload protection limits power dissipation and junction temperature. Accurate, programmable current-limiting circuits, protects the input supply against both overload and short-circuit conditions. Fault blanking of 20 ms duration enables the circuit to ignore transient faults, such as those caused when hot swapping a capacitive load, preventing false alarms to the host system. In one embodiment, the current limited switch **7032** with autoreset also features a reverse-current protection circuitry to block current flow from the output to the input when the switch **7032** is off.

(520) In one embodiment, the present disclosure provides a reverse polarity protection for sensors and/or electronic components in a surgical instrument. FIG. **121** is a block diagram of a surgical instrument electronic subsystem **7040** with a reverse polarity protection circuit **7042** for sensors and/or electronic components **7005** of the secondary circuit according to one embodiment. Reverse polarity protection is provided for exposed leads (electrical conductors **7002**, **7004**) of a surgical instrument using a supplementary self-isolating/restoring circuit referred to herein as a supplementary power supply circuit **7014** coupled to the main power supply circuit **7010**. The reverse polarity protection circuit **7042** may be monitored to indicate one or more reverse polarity conditions to the end user of the device. In the event of reverse polarity applied to the device the protection circuit **7042** might lock-out the device from being fired or other device critical operations.

(521) In various embodiments, the surgical instruments described herein provide real time feedback about the compressibility and thickness of tissue using sensors and/or electronic components **7005**. The modular architecture of the surgical instrument enables the configuration of custom modular shafts to employ job specific technologies. To enable sensors and/or electronic components **7005**, both power and data signals are transferred between the modular components. During the assembly

of modular components there are typically exposed electrical conductors that when connected are used to transfer power and data signals between the connected components. There is potential for these conductors to become powered with reverse polarity.

(522) Accordingly, in one embodiment, the surgical instrument electronic subsystem **7040** is configured to reduce electronic damage during the application of a reverse polarity connection **7044** in a sensing surgical instrument. The surgical instrument electronic subsystem **7040** employs a polarity protection circuit **7042** inline with the exposed electrical conductors **7002**, **7004**. In one embodiment, the polarity protection circuit **7042** may be implemented by creating a supplementary power supply circuit **7014** coupled to a main power supply circuit **7010**. In the case that the supplementary power supply circuit **7014** electrical conductors **7002**, **7004** become powered with reverse polarity it isolates the power from the main power supply circuit **7010** to prevent damage. The main power supply circuit **7010** will still be active to supply power, so that the protected processor **7008** of the main circuit can monitor the condition. When the reverse polarity in the supplementary power supply circuit **7014** is remedied, the supplementary power supply circuit **7014** rejoins the main power supply circuit **7010** and is available to supply power to the secondary circuit. The reverse polarity protection circuit **7042** also may be monitored to indicate that the reverse polarity condition to the end user of the device. The reverse polarity protection circuit **7042** also may be monitored to lockout the firing of the device if a reverse polarity event is indicated.

(523) FIG. **122** is a reverse polarity protection circuit **7042** for sensors and/or electronic components **7005** of the secondary circuit of a surgical instrument according to one embodiment. During normal operation, the relay switch **S1** comprises output contacts in the normally closed (NC) position and the battery voltage B.sub.1 of the main power supply circuit **7010** (FIG. **121**) is applied to V.sub.OUT coupled to the secondary circuit. The diode D.sub.1 blocks current from flowing through the coil **7046** (inductor) of the relay switch S.sub.1. When the polarity of the battery B.sub.1 is reversed, diode D.sub.1 conducts and current flows through the coil **7046** of the relay switch S.sub.1 energizing the relay switch **S1** to place the output contacts in the normally open (NO) position and thus disconnecting the reverse voltage from V.sub.OUT coupled to the secondary circuit. Once the switch S.sub.1 is in the NO position, current from the positive terminal of the battery B.sub.1 flows through LED D.sub.3 and resistor R.sub.1 to prevent the battery B.sub.1 from shorting out. Diode D.sub.2 is a clamping diode to protect from spikes generated by the coil **7046** during switching.

(524) In one embodiment, the surgical instruments described herein provide a power reduction technique utilizing a sleep mode for sensors on a modular device. FIG. **123** is a block diagram of a surgical instrument electronic subsystem **7050** with power reduction utilizing a sleep mode monitor **7052** for sensors and/or electronic components **7005** according to one embodiment. In one embodiment, the sleep mode monitor **7052** for the sensors and/or electronic components **7005** of the secondary circuit may be implemented as a circuit and/or as a software routine to reduce the power consumption of a surgical instrument. The sleep mode monitor **7052** protection circuit may be monitored to indicate one or more sleep mode conditions to the end user of the device. In the event of a sleep mode condition in the device, the sleep mode monitor **7052** protection circuit/software may be configured to lock-out the device from being fired or operated by the user.

(525) In various embodiments, the surgical instruments described herein provide real time feedback about the compressibility and thickness of tissue using electronic sensors **7005**. The modular architecture enables the surgical instrument to be configured with custom modular shafts to employ job specific technologies. To enable sensors and/or electronic components **7005**, additional electronic connection points and components may be employed to transfer both power and data signal between the modular components. As the number of sensors and/or electronic components **7005** increases, the power consumption of the surgical instrument increases, thus creating a need for techniques to reduce the power consumption of the surgical instrument.

(526) In one embodiment, the electronic subsystem **7050** comprises a sleep mode monitor **7052**

circuit and/or software for the sensors **7005** to reduce power consumption of the sensing surgical instrument. The sleep mode monitor **7052** may be implemented by creating a supplementary power supply circuit **7014** coupled to a main power supply circuit **7010**. A device state monitor **7054** monitors whether the surgical instrument is in a 1=Unclamped State, 2=Clamping State, or a 3=Clamped State. When the sleep mode monitor **7052** software determines that the surgical instrument is in a non-sensing (1=Unclamped State) condition the sleep mode monitor **7052** enters the sensors and/or electronic components **7005** of the secondary circuit into a sleep mode to reduce the power load on the main power supply circuit **7010**. The main power supply circuit **7010** will still be active to supply power, so that the protected processor **7008** of the primary circuit can monitor the condition. When the surgical instrument enters a condition requiring sensor activity the supplementary power supply circuit **7014** is awakened and rejoins the main power supply circuit **7010**. The sleep mode monitor **7051** circuit can utilize a mixture of integrated circuits, solid state components, micro-processors, and/or firmware. The sleep mode monitor **7051** circuit also may be monitored to indicate the condition to the end user of the device. The sleep mode monitor **7051** circuit may also be monitored to lockout the firing or function of the device in the event the device is in a sleep mode.

(527) In one embodiment the present disclosure provides protection against intermittent power loss for sensors and/or electronic components in modular surgical instruments. FIG. **124** is a block diagram of a surgical instrument electronic subsystem **7060** comprising a temporary power loss circuit **7062** to provide protection against intermittent power loss for sensors and/or electronic components **7005** of the secondary circuit in modular surgical instruments.

(528) In various embodiments, the surgical instruments described herein provide real time feedback about the compressibility and thickness of tissue using sensors and/or electronic components **7005**. The modular architecture enables the surgical instrument to be configured with custom modular shafts to employ job specific technologies. To enable sensors and/or electronic components **7005** additional electronic connection points and components may be employed to transfer both power and signal between the modular components. As the number of electrical connection points increase, the potential for sensors and/or electronic components **7005** to experience short term intermittent power loss increases.

(529) In accordance with one embodiment, the temporary power loss circuit **7062** is configured to reduce device operation error from short term intermittent power loss in a sensing surgical instrument. The temporary power loss circuit **7062** has the capacity to deliver continuous power for short periods of time in the event the power from the main power supply circuit **7010** is interrupted. The temporary power loss circuit **7062** may comprises capacitive elements, batteries, and/or other electronic elements capable of leveling, detecting, or storing power.

(530) As shown in FIG. **124**, the temporary power loss circuit **7062** may be implemented by creating a supplementary circuit/software coupled to a main circuit/software. In the case that the supplementary circuit/software experiences a sudden power loss from the main power source, the sensors and/or electronic components **7005** powered by the supplementary power supply circuit **7014** would be unaffected for short period times. During the power loss the supplementary power supply circuit **7014** may be powered by capacitive elements, batteries, and/or other electronic elements that are capable of leveling or storing power. The temporary power loss circuit **7062** implemented either in hardware or software also may be monitored to lockout the firing or function of the surgical instrument in the event the device is in the power saving mode. In the event of an intermittent power loss condition in the surgical instrument the temporary power loss circuit **7062** implemented either in hardware or software may lock-out the surgical instrument from being fired or operated.

(531) FIG. **125** illustrates one embodiment of a temporary power loss circuit **7062** implemented as a hardware circuit. The temporary power loss circuit **7062** hardware circuit is configured to reduce surgical instrument operation error from short term intermittent power loss. The temporary power

loss circuit **7062** has the capacity to deliver continuous power for short periods of time in the event the power from the main power supply circuit **7010** (FIG. **124**) is interrupted. The temporary power loss circuit **7062** employs capacitive elements, batteries, and/or other electronic elements that are capable of leveling, detecting, or storing power. The temporary power loss circuit **7062** may be monitored to indicate one or more conditions to the end user of the surgical instrument. In the event of an intermittent power loss condition in the surgical instrument, the temporary power loss circuit **7062** protection circuit/software might lock-out the device from being fired or operated.

(532) In the illustrated embodiment, the temporary power loss circuit **7062** comprises an analog switch integrated circuit **U1**. In one embodiment, the analog switch integrated circuit **U1** is a single-pole/single-throw (SPST), low-voltage, single-supply, CMOS analog switch such as the MAX4501 provided by Maxim. In one embodiment, the analog switch integrated circuit **U1** is normally open (NO). In other embodiments, the analog switch integrated circuit **U1** may be normally closed (NC). The input **IN** activates the NO analog switch **7064** to connect the output of a step-up DC-DC converter **U3** to the input of a linear regulator **U2** via a standby “RESERVE CAPACITOR.” The output of the linear regulator **U2** is coupled to the input of the DC-DC converter **U3**. The linear regulator **U2** maximizes battery life by combining ultra-low supply currents and low dropout voltages. In one embodiment, the linear regulator **U2** is a MAX882 integrated circuit provided by Maxim.

(533) The batteries are also coupled to the input of the step-up DC-DC converter **U3**. The step-up DC-DC converter **U3** may be a compact, high-efficiency, step-up DC-DC converter with a built-in synchronous rectifier to improve efficiency and reduce size and cost by eliminating the need for an external Schottky diode. In one embodiment, the step-up DC-DC converter **U3** is a MAX1674 integrated circuit by Maxim.

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(535) FIGS. **126A** and **126B** illustrate one embodiment of an end effector **10000** comprising a magnet **10008** and a Hall effect sensor **10010** in communication with a processor **10012**. The end effector **10000** is similar to the end effector **300** described above. The end effector comprises a first jaw member, or anvil **10002**, pivotally coupled to a second jaw member, or elongated channel **10004**. The elongated channel **10004** is configured to operably support a staple cartridge **10006** therein. The staple cartridge **10006** is similar to the staple cartridge **304** described above. The anvil **10008** comprises a magnet **10008**. The staple cartridge comprises a Hall effect sensor **10010** and a processor **10012**. The Hall effect sensor **10010** is operable to communicate with the processor **10012** through a conductive coupling **10014**. The Hall effect sensor **10010** is positioned within the staple cartridge **10006** to operatively couple with the magnet **10008** when the anvil **10002** is in a closed position. The Hall effect sensor **10010** can be configured to detect changes in the magnetic field surrounding the Hall effect sensor **10010** caused by the movement of or location of magnet **10008**.

(536) FIG. **127** illustrates one embodiment of the operable dimensions that relate to the operation of the Hall effect sensor **10010**. A first dimension **10020** is between the bottom of the center of the magnet **10008** and the top of the staple cartridge **10006**. The first dimension **10020** can vary with the size and shape of the staple cartridge **10006**, such as for instance between 0.0466 inches, 0.0325 inches, 0.0154 inches, or 0.0154 inches, or any reasonable value. A second dimension **10022** is between the bottom of the center of the magnet **10008** and the top of the Hall effect sensor **10010**. The second dimension can also vary with the size and shape of the staple cartridge **10006**, such as for instance 0.0666 inches, 0.0525 inches, 0.0354 inches, 0.0347 inches, or any reasonable value. A third dimension **10024** is between the top of the processor **10012** and the lead-in surface **10028** of the staple cartridge **10006**. The third dimension can also vary with the size and the shape of the staple cartridge, such as for instance 0.0444 inches, 0.0440 inches, 0.0398 inches, 0.0356 inches, or any reasonable value. An angle **10026** is the angle between the anvil **10002** and the top of the staple cartridge **10006**. The angle **10026** also can vary with the size and shape of the staple cartridge

10006, such as for instance 0.91 degrees, 0.68 degrees, 0.62 degrees, 0.15 degrees, or any reasonable value.

(537) FIGS. **128A** through **128D** further illustrate dimensions that can vary with the size and shape of a staple cartridge **10006** and effect the operation of the Hall effect sensor **10010**. FIG. **128A** illustrates an external side view of an embodiment of a staple cartridge **10006**. The staple cartridge **10006** comprises a push-off lug **10036**. When the staple cartridge **10006** is operatively coupled with the end effector **10000** as illustrated in FIG. **126A**, the push-off lug **10036** rests on the side of the elongated channel **10004**.

(538) FIG. **128B** illustrates various dimensions possible between the lower surface **10038** of the push-off lug **10036** and the top of the Hall effect sensor **10010** (not pictured). A first dimension **10030a** is possible with black, blue, green or gold staple cartridges **10006**, where the color of the body of the staple cartridge **10006** may be used to identify various aspects of the staple cartridge **10006**. The first dimension **10030a** can be, for instance, 0.005 inches below the lower surface **10038** of the push-off lug **10036**. A second dimension **10030b** is possible with gray staple cartridges **10006**, and can be 0.060 inches above the lower surface **10038** of the push-off lug **10036**. A third dimension **10030c** is possible with white staple cartridges **10006**, and can be 0.030 inches above the lower-surface **10038** of the push-off lug **10036**.

(539) FIG. **128C** illustrates an external side view of an embodiment of a staple cartridge **10006**. The staple cartridge **10006** comprises a push-off lug **10036** with a lower surface **10038**. The staple cartridge **10006** further comprises an upper surface **10046** immediately above the Hall effect sensor **10010** (not pictured). FIG. **128D** illustrates various dimensions possible between the lower surface **10038** of the push-off lug **10036** and the upper surface **10046** of the staple cartridge **10006** above the Hall effect sensor **10010**. A first dimension **10040** is possible for black, blue, green or gold staple cartridges **10006**, and can be, for instance, 0.015 inches above the lower surface **10038** of the push-off lug **10036**. A second dimension **10042** is possible for gray staple cartridges **10006**, and can be, for instance, 0.080 inches. A third dimension **10044** is possible for white staple cartridges **10006**, and can be, for instance, 0.050.

(540) It is understood that the references to the color of the body of a staple cartridge **10006** is for convenience and by way of example only. It is understood that other staple cartridge **10006** body colors are possible. It is also understood that the dimensions given for FIGS. **128A** through **128D** are also example and non-limiting.

(541) FIG. **129A** illustrates various embodiments of magnets **10058a-10058d** of various sizes, according to how each magnet **10058a-10058d** may fit in the distal end of an anvil, such as anvil **10002** illustrated in FIGS. **126A-126B**. A magnet **10058a-10058d** can be positioned in the distal tip of the anvil **10002** at a given distance **10050** from the anvil's pin or pivot point **10052**. It is understood that this distance **10050** may vary with the construction of the end effector and staple cartridge and/or the desired position of the magnet. FIG. **129B** further illustrates a front-end cross-sectional view **10054** of the anvil **10002** and the central axis point of the anvil **10002**. FIG. **129A** also illustrates an example **10056** of how various embodiments of magnets **10058a-10058d** may fit within the same anvil **10002**.

(542) FIGS. **130A-130E** illustrate one embodiment of an end effector **10100** that comprises, by way of example, a magnet **10058a** as illustrated in FIGS. **129A-129B**. FIG. **130A** illustrates a front-end cross-sectional view of the end effector **10100**. The end effector **10100** is similar to the end effector **300** described above. The end effector **10100** comprises a first jaw member or anvil **10102**, a second jaw member or elongated channel **10104**, and a staple cartridge **10106** operatively coupled to the elongated channel **10104**. The anvil **10102** further comprises the magnet **10058a**. The staple cartridge **10106** further comprises a Hall effect sensor **10110**. The anvil **10102** is here illustrated in a closed position. FIG. **130B** illustrates a front-end cutaway view of the anvil **10102** and the magnet **10058a**, in situ. FIG. **130C** illustrates a perspective cutaway view of the anvil **10102** and the magnet **10058a**, in an optional location. FIG. **130D** illustrates a side cutaway view

of the anvil **10102** and the magnet **10058a**, in an optional location. FIG. **130E** illustrates a top cutaway view of the anvil **10102** and the magnet **10058a**, in an optional location.

(543) FIGS. **131A-131E** illustrate one embodiment of an end effector **10150** that comprises, by way of example, a magnet **10058d** as illustrated in FIGS. **129A-129B**. FIG. **131A** illustrates a front-end cross-sectional view of the end effector **10150**. The end effector **10150** comprises an anvil **10152**, an elongated channel **10154**, and a staple cartridge **10156**. The anvil **10152** further comprises magnet **10058d**. The staple cartridge **10156** further comprises a Hall effect sensor **10160**. FIG. **131B** illustrates a front-end cutaway view of the anvil **10150** and the magnet **10058d**, in situ. FIG. **131C** illustrates a perspective cutaway view of the anvil **10152** and the magnet **10058d** in an optional location. FIG. **131D** illustrates a side cutaway view of the anvil **10152** and the magnet **10058d** in an optional location. FIG. **131E** illustrates a top cutaway view of the anvil **10152** and magnet **10058d** in an optional location.

(544) FIG. **132** illustrates an end effector **300** as described above, and illustrates contact points between the anvil **306** and either the staple cartridge **304** and/or the elongated channel **302**. Contact points between the anvil **306** and the staple cartridge **304** and/or the elongated channel **302** can be used to determine the position of the anvil **306** and/or provide a point for an electrical contact between the anvil **306** and the staple cartridge **304**, and/or the anvil **306** and the elongated channel **302**. Distal contact point **10170** can provide a contact point between the anvil **306** and the staple cartridge **304**. Proximal contact point **10172** can provide a contact point between the anvil **306** and the elongated channel **302**.

(545) FIGS. **133A** and **133B** illustrate one embodiment of an end effector **10200** that is operable to use conductive surfaces at the distal contact point to create an electrical connection. The end effector **10200** is similar to the end effector **300** described above. The end effector comprises an anvil **10202**, an elongated channel **10204**, and a staple cartridge **10206**. The anvil **10202** further comprises a magnet **10208** and an inside surface **10210**, which further comprises a number of staple-forming indents **10212**. In some embodiments, the inside surface **10210** of the anvil **10202** further comprises a first conductive surface **10214** surrounding the staple-forming indents **10212**. The first conductive surface **10214** can come into contact with second conductive surfaces **10222** on the staple cartridge **10206**, as illustrated in FIG. **107B**. FIG. **107B** illustrates a close-up view of the cartridge body **10216** of the staple cartridge **10206**. The cartridge body **10216** comprises a number of staple cavities **10218** designed to hold staples (not pictured). In some embodiments the staple cavities **10218** further comprise staple cavity extensions **10220** that protrude above the surface of the cartridge body **10216**. The staple cavity extensions **10220** can be coated with the second conductive surfaces **10222**. Because the staple cavity extensions **10222** protrude above the surface of the cartridge body **10216**, the second conductive surfaces **10222** will come into contact with the first conductive surfaces **10214** when the anvil **10202** is in a closed position. In this manner the anvil **10202** can form an electrical contact with the staple cartridge **10206**.

(546) FIGS. **134A-134C** illustrate one embodiment of an end effector **10250** that is operable to use conductive surfaces to form an electrical connection. FIG. **134A** illustrates the end effector **10250** comprises an anvil **10252**, an elongated channel **10254**, and a staple cartridge **10256**. The anvil further comprises a magnet **10258** and an inside surface **10260**, which further comprises staple-forming indents **10262**. In some embodiments the inside surface **10260** of the anvil **10250** can further comprise first conductive surfaces **10264**, located, by way of example, distally from the staple-forming indents **10262**, as illustrated in FIG. **134B**. The first conductive surfaces **10264** are located such that they can come into contact with a second conductive surface **10272** located on the staple cartridge **10256**, as illustrated in FIG. **134C**. FIG. **134C** illustrates the staple cartridge **10256**, which comprises a cartridge body **10266**. The cartridge body **10266** further comprises an upper surface **10270**, which in some embodiments can be coated with the second conductive surface **10272**. The first conductive surfaces **10264** are located on the inside surface **10260** of the anvil **10252** such that they come into contact with the second conductive surface **10272** when the anvil

10252 is in a closed position. In this manner the anvil **10250** can form an electrical contact with the staple cartridge **10256**.

(547) FIGS. **135A** and **135B** illustrate one embodiment of an end effector **10300** that is operable to use conductive surfaces to form an electrical connection. The end effector **10300** comprises an anvil **10302**, an elongated channel **10304**, and a staple cartridge **10306**. The anvil **10302** further comprises a magnet **10308** and an inside surface **10310**, which further comprises a number of staple-forming indents **10312**. In some embodiments the inside surface **10310** further comprises a first conductive surface **10314** surrounding some of the staple-forming indents **10312**. The first conductive surface is located such that it can come into contact with second conductive surfaces **10322** as illustrated in FIG. **109B**. FIG. **109B** illustrates a close-up view of the staple cartridge **10306**. The staple cartridge **10306** comprises a cartridge body **10316** which further comprises an upper surface **10320**. In some embodiments, the leading edge of the upper surface **10320** can be coated with second conductive surfaces **10322**. The first conductive surface **10312** is positioned such that it will come into contact with the second conductive surfaces **10322** when the anvil **10302** is in a closed position. In this manner the anvil **10302** can form an electrical connection with the staple cartridge **10306**.

(548) FIGS. **136A** and **136B** illustrate one embodiment of an end effector **10350** that is operable to use conductive surfaces to form an electrical connection. FIG. **136A** illustrates an end effector **10350** comprising an anvil **10352**, an elongated channel **10354**, and a staple cartridge **10356**. The anvil **10352** further comprises a magnet **10358** and an inside surface **10360**, which further comprises a number of staple-forming indents **10362**. In some embodiments the inside surface **10360** further comprises a first conductive surface **10364** surrounding some of the staple-forming indents **10362**. The first conductive surface is located such that it can come into contact with second conductive surfaces **10372** as illustrated in FIG. **136B**. FIG. **136B** illustrates a close-up view of the staple cartridge **10356**. The staple cartridge **10356** comprises a cartridge body **10366** which further comprises an upper surface **10370**. In some embodiments, the leading edge of the upper surface **10370** can be coated with second conductive surfaces **10372**. The first conductive surface **10362** is positioned such that it will come into contact with the second conductive surfaces **10372** when the anvil **10352** is in a closed position. In this manner the anvil **10352** can form an electrical connection with the staple cartridge **10356**.

(549) FIGS. **137A-137C** illustrate one embodiment of an end effector **10400** that is operable to use the proximal contact point **10408** to form an electrical connection. FIG. **137A** illustrate the end effector **10400**, which comprises an anvil **10402**, an elongated channel **10404**, and a staple cartridge **10406**. The anvil **10402** further comprises pins **10410** that extend from the anvil **10402** and allow the anvil to pivot between an open and a closed position relative to the elongated channel **10404** and the staple cartridge **10406**. FIG. **137B** is a close-up view of a pin **10410** as it rests within an aperture **10418** defined in the elongated channel **10404** for that purpose. In some embodiments, pin **10410** further comprises a first conductive surface **10412** located on the exterior of the pin **10410**. In some embodiments the aperture **10418** further comprises a second conductive surface **10414** on its outside surface. As the anvil **10402** moves between a closed and an open position, the first conductive surface **10412** on the pin **10410** rotates and comes into contact with the second conductive surface **10414** on the surface of the aperture **10418**, thus forming an electrical contact. FIG. **137C** illustrates an alternate embodiment, with an alternate location for a second conductive surface **10416** on the surface of the aperture **10418**.

(550) FIG. **138** illustrates one embodiment of an end effector **10450** with a distal sensor plug **10466**. End effector **10450** comprises a first jaw member or anvil **10452**, a second jaw member or elongated channel **10454**, and a staple cartridge **10466**. The staple cartridge **10466** further comprises the distal sensor plug **10466**, located at the distal end of the staple cartridge **10466**.

(551) FIG. **139A** illustrates the end effector **10450** with the anvil **10452** in an open position. FIG. **139B** illustrates a cross-sectional view of the end effector **10450** with the anvil **10452** in an open

position. As illustrated, the anvil **10452** may further comprise a magnet **10458**, and the staple cartridge **10456** may further comprise the distal sensor plug **10466** and a wedge sled, **10468**, which is similar to the wedge sled **190** described above. FIG. **139C** illustrates the end effector **10450** with the anvil **10452** in a closed position. FIG. **139D** illustrates a cross sectional view of the end effector **10450** with the anvil **10452** in a closed position. As illustrated, the anvil **10452** may further comprise a magnet **10458**, and the staple cartridge **10456** may further comprise the distal sensor plug **10466** and a wedge sled **10468**. As illustrated, when the anvil **10452** is in a closed position relative to the staple cartridge **10456**, the magnet **10458** is in proximity to the distal sensor plug **10466**.

(552) FIG. **140** provides a close-up view of the cross section of the distal end of the end effector **10450**. As illustrated, the distal sensor plug **10466** may further comprise a Hall effect sensor **10460** in communication with a processor **10462**. The Hall effect sensor **10460** can be operatively connected to a flex board **10464**. The processor **10462** can also be operatively connect to the flex board **10464**, such that the flex board **10464** provides a communication path between the Hall effect sensor **10460** and the processor **10462**. The anvil **10452** is illustrated in a closed position, and as illustrated, when the anvil **10452** is in a closed position the magnet **10458** is in proximity to the Hall effect sensor **10460**.

(553) FIG. **141** illustrates a close-up top view of the staple cartridge **10456** that comprises a distal sensor plug **10466**. Staple cartridge **10456** further comprises a cartridge body **10470**. The cartridge body **10470** further comprises electrical traces **10472**. Electrical traces **10472** provide power to the distal sensor plug **10466**, and are connected to a power source at the proximal end of the staple cartridge **10456** as described in further detail below. Electrical traces **10472** can be placed in the cartridge body **10470** by various methods, such as for instance laser etching.

(554) FIGS. **142A** and **142B** illustrate one embodiment of a staple cartridge **10506** with a distal sensor plug **10516**. FIG. **142A** is a perspective view of the underside of the staple cartridge **10506**. The staple cartridge **10506** comprises a cartridge body **10520** and a cartridge tray **10522**. The staple cartridge **10506** further comprises a distal sensor cover **10524** that encloses the lower area of the distal end of the staple cartridge **10506**. The cartridge tray **10522** further comprises an electrical contact **10526**. FIG. **142B** illustrates a cross sectional view of the distal end of the staple cartridge **10506**. As illustrated, the staple cartridge **10506** can further comprise a distal sensor plug **10516** located within the cartridge body **10520**. The distal sensor plug **10516** further comprises a Hall effect sensor **10510** and a processor **10512**, both operatively connected to a flex board **10514**. The distal sensor plug **10516** can be connected to the electrical contact **10526**, and can thus use conductivity in the cartridge tray **10522** as a source of power. FIG. **142B** further illustrates the distal sensor cover **10524**, which encloses the distal sensor plug **10516** within the cartridge body **10520**.

(555) FIGS. **143A-143C** illustrate one embodiment of a staple cartridge **10606** that comprises a flex cable **10630** connected to a Hall effect sensor **10610** and processor **10612**. The staple cartridge **10606** is similar to the staple cartridge **10606** is similar to the staple cartridge **306** described above. FIG. **143A** is an exploded view of the staple cartridge **10606**. The staple cartridge comprises **10606** a cartridge body **10620**, a wedge sled **10618**, a cartridge tray **10622**, and a flex cable **10630**. The flex cable **10630** further comprises electrical contacts **10632** at the proximal end of the staple cartridge **10606**, placed to make an electrical connection when the staple cartridge **10606** is operatively coupled with an end effector, such as end effector **10800** described below. The electrical contacts **10632** are integrated with cable traces **10634**, which extend along some of the length of the staple cartridge **10606**. The cable traces **10634** connect **10636** near the distal end of the staple cartridge **10606** and this connection **10636** joins with a conductive coupling **10614**. A Hall effect sensor **10610** and a processor **10612** are operatively coupled to the conductive coupling **10614** such that the Hall effect sensor **10610** and the processor **10612** are able to communicate.

(556) FIG. **143B** illustrates the assembly of the staple cartridge **10606** and the flex cable **10630** in

greater detail. As illustrated, the cartridge tray **10622** encloses the underside of the cartridge body **10620**, thereby enclosing the wedge sled **10618**. The flex cable **10630** can be located on the exterior of the cartridge tray **10622**, with the conductive coupling **10614** positioned within the distal end of the cartridge body **10620** and the electrical contacts **10632** located on the outside near the proximal end. The flex cable **10630** can be placed on the exterior of the cartridge tray **10622** by any appropriate means, such as for instance bonding or laser etching.

(557) FIG. **143C** illustrates a cross sectional view of the staple cartridge **10606** to illustrate the placement of the Hall effect sensor **10610**, processor **10612**, and conductive coupling **10614** within the distal end of the staple cartridge, in accordance with the present embodiment.

(558) FIG. **144A-144F** illustrate one embodiment of a staple cartridge **10656** that comprises a flex cable **10680** connected to a Hall effect sensor **10660** and a processor **10662**. FIG. **144A** is an exploded view of the staple cartridge **10656**. The staple cartridge comprises a cartridge body **10670**, a wedge sled **10668**, a cartridge tray **10672**, and a flex cable **10680**. The flex cable **10680** further comprises cable traces **10684** that extend along some of the length of the staple cartridge **10656**. Each of the cable traces **10684** have an angle **10686** near their distal end, and connect therefrom to a conductive coupling **10664**. A Hall effect sensor **10660** and a processor **10662** are operatively coupled to the conductive coupling **10664** such that the Hall effect sensor **10660** and the processor **10662** are able to communicate.

(559) FIG. **144B** illustrates the assembly of the staple cartridge **10656**. The cartridge tray **10672** encloses the underside of the cartridge body **10670**, thereby enclosing the wedge sled **10668**. The flex cable **10680** is located between the cartridge body **10670** and the cartridge tray **10672**. As such, in the illustration only the angle **10686** and the conductive coupling **10664** are visible.

(560) FIG. **144C** illustrates the underside of an assembled staple cartridge **10656**, and also illustrates the flex cable **10680** in greater detail. In an assembled staple cartridge **10656**, the conductive coupling **10664** is located in the distal end of the staple cartridge **10656**. Because the flex cable **10680** can be located between the cartridge body **10670** and the cartridge tray **10672**, only the angle **10686** ends of the cable traces **10684** would be visible from the underside of the staple cartridge **10656**, as well as the conductive coupling **10664**.

(561) FIG. **144D** illustrates a cross sectional view of the staple cartridge **10656** to illustrate the placement of the Hall effect sensor **10660**, processor **10662**, and conductive coupling **10664**. Also illustrated is an angle **10686** of a cable trace **10684**, to illustrate where the angle **10686** could be placed. The cable traces **10684** are not pictured.

(562) FIG. **144E** illustrates the underside of the staple cartridge **10656** without the cartridge tray **10672** and including the wedge sled **10668**, in its most distal position. The staple cartridge **10656** is illustrated without the cartridge tray **10672** in order to illustrate a possible placement for the cable traces **10684**, which are otherwise obscured by the cartridge tray **10672**. As illustrated, the cable traces **10684** can be placed inside the cartridge body **10670**. The angle **10686** optionally allows the cable traces **10684** to occupy a narrower space in the distal end of the cartridge body **10670**.

(563) FIG. **144F** also illustrates the staple cartridge **10656** without the cartridge tray **10672** in order to illustrate a possible placement for the cable traces **10684**. As illustrated the cable traces **10684** can be placed along the length of the exterior of cartridge body **10670**. Furthermore, the cable traces **10684** can form an angle **10686** to enter the interior of the distal end of the cartridge body **10670**.

(564) FIGS. **145A** and **145B** illustrates one embodiment of a staple cartridge **10706** that comprises a flex cable **10730**, a Hall effect sensor **10710**, and a processor **10712**. FIG. **145A** is an exploded view of the staple cartridge **10706**. The staple cartridge **10706** comprises a cartridge body **10720**, a wedge sled **10718**, a cartridge tray **10722**, and a flex cable **10730**. The flex cable **10730** further comprises electrical contacts **10732** placed to make an electrical connection when the staple cartridge **10706** is operatively coupled with an end effector. The electrical contacts **10732** are integrated with cable traces **10734**. The cable traces connect **10736** near the distal end of the staple

cartridge **10706**, and this connection **10736** joins with a conductive coupling **10714**. A Hall effect sensor **10710** and a processor **10712** are operatively connected to the conductive coupling **10714** such that they are able to communicate.

(565) FIG. **145B** illustrates the assembly of the staple cartridge **10706** and the flex cable **10730** in greater detail. As illustrated, the cartridge tray **10722** encloses the underside of the cartridge body **10720**, thereby enclosing the wedge sled **10718**. The flex cable **10730** can be located on the exterior of the cartridge tray **10722** with the conductive coupling **10714** positioned within the distal end of the cartridge body **10720**. The flex cable **10730** can be placed on the exterior of the cartridge tray **10722** by any appropriate means, such as for instance bonding or laser etching.

(566) FIGS. **146A-146F** illustrate one embodiment of an end effector **10800** with a flex cable **10840** operable to provide power to a staple cartridge **10806** that comprises a distal sensor plug **10816**. The end effector **10800** is similar to the end effector **300** described above. The end effector **10800** comprises a first jaw member or anvil **10802**, a second jaw member or elongated channel **10804**, and a staple cartridge **10806** operatively coupled to the elongated channel **10804**. The end effector **10800** is operatively coupled to a shaft assembly **10900**. The shaft assembly **10900** is similar to shaft assembly **200** described above. The shaft assembly **10900** further comprises a closure tube **10902** that encloses the exterior of the shaft assembly **10900**. In some embodiments the shaft assembly **10900** further comprises an articulation joint **10904**, which includes a double pivot closure sleeve assembly **10906**. The double pivot closure sleeve assembly **10906** includes an end effector closure sleeve assembly **10908** that is operable to couple with the end effector **10800**.

(567) FIG. **146A** illustrates a perspective view of the end effector **10800** coupled to the shaft assembly **10900**. In various embodiments, the shaft assembly **10900** further comprises a flex cable **10830** that is configured to not interfere with the function of the articulation joint **10904**, as described in further detail below. FIG. **146B** illustrates a perspective view of the underside of the end effector **10800** and shaft assembly **10900**. In some embodiments, the closure tube **10902** of the shaft assembly **10900** further comprises a first aperture **10908**, through which the flex cable **10908** can extend. The closure sleeve assembly **10908** further comprises a second aperture **10910**, through which the flex cable **10908** can also pass.

(568) FIG. **146C** illustrates the end effector **10800** with the flex cable **10830** and without the shaft assembly **10900**. As illustrated, in some embodiments the flex cable **10830** can include a single coil **10832** operable to wrap around the articulation joint **10904**, and thereby be operable to flex with the motion of the articulation joint **10904**.

(569) FIGS. **146D** and **146E** illustrate the elongated channel **10804** portion of the end effector **10800** without the anvil **10802** or the staple cartridge **10806**, to illustrate how the flex cable **10830** can be seated within the elongated channel **10804**. In some embodiments, the elongated channel **10804** further comprises a third aperture **10824** for receiving the flex cable **10830**. Within the body of the elongated channel **10804** the flex cable splits **10834** to form extensions **10836** on either side of the elongated channel **10804**. FIG. **146E** further illustrates that connectors **10838** can be operatively coupled to the flex cable extensions **10836**.

(570) FIG. **146F** illustrates the flex cable **10830** alone. As illustrated, the flex cable **10830** comprises a single coil **10832** operative to wrap around the articulation joint **10904**, and a split **10834** that attaches to extensions **10836**. The extensions can be coupled to connectors **10838** that have on their distal facing surfaces prongs **10840** for coupling to the staple cartridge **10806**, as described below.

(571) FIG. **147** illustrates a close up view of the elongated channel **10804** with a staple cartridge **10806** coupled thereto. The staple cartridge **10806** comprises a cartridge body **10822** and a cartridge tray **10820**. In some embodiments the staple cartridge **10806** further comprises electrical traces **10828** that are coupled to proximal contacts **10856** at the proximal end of the staple cartridge **10806**. The proximal contacts **10856** can be positioned to form a conductive connection with the prongs **10840** of the connectors **10838** that are coupled to the flex cable extensions **10836**. Thus,

when the staple cartridge **10806** is operatively coupled with the elongated channel **10804**, the flex cable **10830**, through the connectors **10838** and the connector prongs **10840**, can provide power to the staple cartridge **10806**.

(572) FIGS. **148A-148D** further illustrate one embodiment of a staple cartridge **10806** operative with the present embodiment of an end effector **10800**. FIG. **148A** illustrates a close up view of the proximal end of the staple cartridge **10806**. As discussed above, the staple cartridge **10806** comprises electrical traces **10828** that, at the proximal end of the staple cartridge **10806**, form proximal contacts **10856** that are operable to couple with the flex cable **10830** as described above. FIG. **148B** illustrates a close-up view of the distal end of the staple cartridge **10806**, with a space for a distal sensor plug **10816**, described below. As illustrated, the electrical traces **10828** can extend along the length of the staple cartridge body **10822** and, at the distal end, form distal contacts **10856**. FIG. **148C** further illustrates the distal sensor plug **10816**, which in some embodiments is shaped to be received by the space formed for it in the distal end of the staple cartridge **10806**. FIG. **148D** illustrates the proximal-facing side of the distal sensor plug **10816**. As illustrated, the distal sensor plug **10816** has sensor plug contacts **10854**, positioned to couple with the distal contacts **10858** of the staple cartridge **10806**. Thus, in some embodiments the electrical traces **10828** can be operative to provide power to the distal sensor plug **10816**.

(573) FIGS. **149A** and **149B** illustrate one embodiment of a distal sensor plug **10816**. FIG. **149A** illustrates a cutaway view of the distal sensor plug **10816**. As illustrated, the distal sensor plug **10816** comprises a Hall effect sensor **10810** and a processor **10812**. The distal sensor plug **10816** further comprises a flex board **10814**. As further illustrated in FIG. **149B**, the Hall effect sensor **10810** and the processor **10812** are operatively coupled to the flex board **10814** such that they are capable of communicating.

(574) FIG. **150** illustrates an embodiment of an end effector **10960** with a flex cable **10980** operable to provide power to sensors and electronics **10972** in the distal tip of the anvil **19052** portion. The end effector **10950** comprises a first jaw member or anvil **10962**, a second jaw member or elongated channel **10964**, and a staple cartridge **10956** operatively coupled to the elongated channel **10952**. The end effector **10960** is operatively coupled to a shaft assembly **10960**. The shaft assembly **10960** further comprises a closure tube **10962** that encloses the shaft assembly **10960**. In some embodiments the shaft assembly **10960** further comprises an articulation joint **10964**, which includes a double pivot closure sleeve assembly **10966**.

(575) In various embodiments, the end effector **10950** further comprises a flex cable **19080** that is configured to not interfere with the function of the articulation joint **10964**. In some embodiments, the closure tube **10962** comprises a first aperture **10968** through which the flex cable **10980** can extend. In some embodiments, flex cable **10980** further comprises a loop or coil **10982** that wraps around the articulation joint **10964** such that the flex cable **10980** does not interfere with the operation of the articulation joint **10964**, as further described below. In some embodiments, the flex cable **10980** extends along the length of the anvil **10951** to a second aperture **10970** in the distal tip of the anvil **10951**.

(576) FIGS. **151A-151C** illustrate the operation of the articulation joint **10964** and flex cable **19080** of the end effector **10950**. FIG. **151A** illustrates a top view of the end effector **10952** with the end effector **109650** pivoted -45 degrees with respect to the shaft assembly **10960**. As illustrated, the coil **10982** of the flex cable **10980** flexes with the articulation joint **10964** such that the flex cable **10980** does not interfere with the operation of the articulation joint **10964**. FIG. **151B** illustrates a top view of the end effector **10950**. As illustrated, the coil **10982** wraps around the articulation joint **10964** once. FIG. **151C** illustrates a top view of the end effector **10950** with the end effector **10950** pivoted +45 degrees with respect to the shaft assembly **10960**. As illustrated, the coil **10982** of the flex cable **10980** flexes with the articulation joint **10964** such that the flex cable **10980** does not interfere with the operation of the articulation joint **10964**.

(577) FIG. **152** illustrates cross-sectional view of the distal tip of an embodiment of an anvil **10952**

with sensors and electronics **10972**. The anvil **10952** comprises a flex cable **10980**, as described with respect to FIGS. **150** and **151A-151C**. As illustrated in FIG. **152**, the anvil **10952** further comprises a second aperture **10970** through which the flex cable **10980** can pass such that the flex cable **10980** can enter a housing **10974** in the within the anvil **10952**. Within the housing **10974** the flex cable **10980** can operably couple to sensors and electronics **10972** located within the housing **10974** and thereby provide power to the sensors and electronics **10972**.

(578) FIG. **153** illustrates a cutaway view of the distal tip of the anvil **10952**. FIG. **153** illustrates an embodiment of the housing **10974** that can contain sensors and electronics **10972** as illustrated by FIG. **152**.

(579) In accordance with various embodiments, the surgical instruments described herein may comprise one or more processors (e.g., microprocessor, microcontroller) coupled to various sensors. In addition, to the processor(s), a storage (having operating logic) and communication interface, are coupled to each other.

(580) As described earlier, the sensors may be configured to detect and collect data associated with the surgical device. The processor processes the sensor data received from the sensor(s).

(581) The processor may be configured to execute the operating logic. The processor may be any one of a number of single or multi-core processors known in the art. The storage may comprise volatile and non-volatile storage media configured to store persistent and temporal (working) copy of the operating logic.

(582) In various embodiments, the operating logic may be configured to perform the initial processing, and transmit the data to the computer hosting the application to determine and generate instructions. For these embodiments, the operating logic may be further configured to receive information from and provide feedback to a hosting computer. In alternate embodiments, the operating logic may be configured to assume a larger role in receiving information and determining the feedback. In either case, whether determined on its own or responsive to instructions from a hosting computer, the operating logic may be further configured to control and provide feedback to the user.

(583) In various embodiments, the operating logic may be implemented in instructions supported by the instruction set architecture (ISA) of the processor, or in higher level languages and compiled into the supported ISA. The operating logic may comprise one or more logic units or modules. The operating logic may be implemented in an object oriented manner. The operating logic may be configured to be executed in a multi-tasking and/or multi-thread manner. In other embodiments, the operating logic may be implemented in hardware such as a gate array.

(584) In various embodiments, the communication interface may be configured to facilitate communication between a peripheral device and the computing system. The communication may include transmission of the collected biometric data associated with position, posture, and/or movement data of the user's body part(s) to a hosting computer, and transmission of data associated with the tactile feedback from the host computer to the peripheral device. In various embodiments, the communication interface may be a wired or a wireless communication interface. An example of a wired communication interface may include, but is not limited to, a Universal Serial Bus (USB) interface. An example of a wireless communication interface may include, but is not limited to, a Bluetooth interface.

(585) For various embodiments, the processor may be packaged together with the operating logic. In various embodiments, the processor may be packaged together with the operating logic to form a SiP. In various embodiments, the processor may be integrated on the same die with the operating logic. In various embodiments, the processor may be packaged together with the operating logic to form a System on Chip (SoC).

(586) Various embodiments may be described herein in the general context of computer executable instructions, such as software, program modules, and/or engines being executed by a processor. Generally, software, program modules, and/or engines include any software element arranged to

perform particular operations or implement particular abstract data types. Software, program modules, and/or engines can include routines, programs, objects, components, data structures and the like that perform particular tasks or implement particular abstract data types. An implementation of the software, program modules, and/or engines components and techniques may be stored on and/or transmitted across some form of computer-readable media. In this regard, computer-readable media can be any available medium or media useable to store information and accessible by a computing device. Some embodiments also may be practiced in distributed computing environments where operations are performed by one or more remote processing devices that are linked through a communications network. In a distributed computing environment, software, program modules, and/or engines may be located in both local and remote computer storage media including memory storage devices. A memory such as a random access memory (RAM) or other dynamic storage device may be employed for storing information and instructions to be executed by the processor. The memory also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor.

(587) Although some embodiments may be illustrated and described as comprising functional components, software, engines, and/or modules performing various operations, it can be appreciated that such components or modules may be implemented by one or more hardware components, software components, and/or combination thereof. The functional components, software, engines, and/or modules may be implemented, for example, by logic (e.g., instructions, data, and/or code) to be executed by a logic device (e.g., processor). Such logic may be stored internally or externally to a logic device on one or more types of computer-readable storage media. In other embodiments, the functional components such as software, engines, and/or modules may be implemented by hardware elements that may include processors, microprocessors, circuits, circuit elements (e.g., transistors, resistors, capacitors, inductors, and so forth), integrated circuits, ASICs, PLDs, DSPs, FPGAs, logic gates, registers, semiconductor device, chips, microchips, chip sets, and so forth.

(588) Examples of software, engines, and/or modules may include software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, functions, methods, procedures, software interfaces, application program interfaces (API), instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof. Determining whether an embodiment is implemented using hardware elements and/or software elements may vary in accordance with any number of factors, such as desired computational rate, power levels, heat tolerances, processing cycle budget, input data rates, output data rates, memory resources, data bus speeds and other design or performance constraints.

(589) One or more of the modules described herein may comprise one or more embedded applications implemented as firmware, software, hardware, or any combination thereof. One or more of the modules described herein may comprise various executable modules such as software, programs, data, drivers, application APIs, and so forth. The firmware may be stored in a memory of the controller and/or the controller which may comprise a nonvolatile memory (NVM), such as in bit-masked ROM or flash memory. In various implementations, storing the firmware in ROM may preserve flash memory. The NVM may comprise other types of memory including, for example, programmable ROM (PROM), erasable programmable ROM (EPROM), EEPROM, or battery backed RAM such as dynamic RAM (DRAM), Double-Data-Rate DRAM (DDRAM), and/or synchronous DRAM (SDRAM).

(590) In some cases, various embodiments may be implemented as an article of manufacture. The article of manufacture may include a computer readable storage medium arranged to store logic, instructions and/or data for performing various operations of one or more embodiments. In various

embodiments, for example, the article of manufacture may comprise a magnetic disk, optical disk, flash memory or firmware containing computer program instructions suitable for execution by a general purpose processor or application specific processor. The embodiments, however, are not limited in this context.

(591) The functions of the various functional elements, logical blocks, modules, and circuits elements described in connection with the embodiments disclosed herein may be implemented in the general context of computer executable instructions, such as software, control modules, logic, and/or logic modules executed by the processing unit. Generally, software, control modules, logic, and/or logic modules comprise any software element arranged to perform particular operations. Software, control modules, logic, and/or logic modules can comprise routines, programs, objects, components, data structures and the like that perform particular tasks or implement particular abstract data types. An implementation of the software, control modules, logic, and/or logic modules and techniques may be stored on and/or transmitted across some form of computer-readable media. In this regard, computer-readable media can be any available medium or media useable to store information and accessible by a computing device. Some embodiments also may be practiced in distributed computing environments where operations are performed by one or more remote processing devices that are linked through a communications network. In a distributed computing environment, software, control modules, logic, and/or logic modules may be located in both local and remote computer storage media including memory storage devices.

(592) Additionally, it is to be appreciated that the embodiments described herein illustrate example implementations, and that the functional elements, logical blocks, modules, and circuits elements may be implemented in various other ways which are consistent with the described embodiments. Furthermore, the operations performed by such functional elements, logical blocks, modules, and circuits elements may be combined and/or separated for a given implementation and may be performed by a greater number or fewer number of components or modules. As will be apparent to those of skill in the art upon reading the present disclosure, each of the individual embodiments described and illustrated herein has discrete components and features which may be readily separated from or combined with the features of any of the other several aspects without departing from the scope of the present disclosure. Any recited method can be carried out in the order of events recited or in any other order which is logically possible.

(593) It is worthy to note that any reference to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is comprised in at least one embodiment. The appearances of the phrase “in one embodiment” or “in one aspect” in the specification are not necessarily all referring to the same embodiment.

(594) Unless specifically stated otherwise, it may be appreciated that terms such as “processing,” “computing,” “calculating,” “determining,” or the like, refer to the action and/or processes of a computer or computing system, or similar electronic computing device, such as a general purpose processor, a DSP, ASIC, FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein that manipulates and/or transforms data represented as physical quantities (e.g., electronic) within registers and/or memories into other data similarly represented as physical quantities within the memories, registers or other such information storage, transmission or display devices.

(595) It is worthy to note that some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. These terms are not intended as synonyms for each other. For example, some embodiments may be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, also may mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. With respect to software elements, for example, the term “coupled” may refer to interfaces, message interfaces,

API, exchanging messages, and so forth.

(596) It should be appreciated that any patent, publication, or other disclosure material, in whole or in part, that is said to be incorporated by reference herein is incorporated herein only to the extent that the incorporated material does not conflict with existing definitions, statements, or other disclosure material set forth in this disclosure. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

(597) The disclosed embodiments have application in conventional endoscopic and open surgical instrumentation as well as application in robotic-assisted surgery.

(598) Embodiments of the devices disclosed herein can be designed to be disposed of after a single use, or they can be designed to be used multiple times. Embodiments may, in either or both cases, be reconditioned for reuse after at least one use. Reconditioning may include any combination of the steps of disassembly of the device, followed by cleaning or replacement of particular pieces, and subsequent reassembly. In particular, embodiments of the device may be disassembled, and any number of the particular pieces or parts of the device may be selectively replaced or removed in any combination. Upon cleaning and/or replacement of particular parts, embodiments of the device may be reassembled for subsequent use either at a reconditioning facility, or by a surgical team immediately prior to a surgical procedure. Those skilled in the art will appreciate that reconditioning of a device may utilize a variety of techniques for disassembly, cleaning/replacement, and reassembly. Use of such techniques, and the resulting reconditioned device, are all within the scope of the present application.

(599) By way of example only, embodiments described herein may be processed before surgery. First, a new or used instrument may be obtained and when necessary cleaned. The instrument may then be sterilized. In one sterilization technique, the instrument is placed in a closed and sealed container, such as a plastic or TYVEK bag. The container and instrument may then be placed in a field of radiation that can penetrate the container, such as gamma radiation, x-rays, or high-energy electrons. The radiation may kill bacteria on the instrument and in the container. The sterilized instrument may then be stored in the sterile container. The sealed container may keep the instrument sterile until it is opened in a medical facility. A device may also be sterilized using any other technique known in the art, including but not limited to beta or gamma radiation, ethylene oxide, or steam.

(600) One skilled in the art will recognize that the herein described components (e.g., operations), devices, objects, and the discussion accompanying them are used as examples for the sake of conceptual clarity and that various configuration modifications are contemplated. Consequently, as used herein, the specific exemplars set forth and the accompanying discussion are intended to be representative of their more general classes. In general, use of any specific exemplar is intended to be representative of its class, and the non-inclusion of specific components (e.g., operations), devices, and objects should not be taken limiting.

(601) With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations are not expressly set forth herein for sake of clarity.

(602) The herein described subject matter sometimes illustrates different components contained within, or connected with, different other components. It is to be understood that such depicted architectures are merely examples and that in fact many other architectures may be implemented which achieve the same functionality. In a conceptual sense, any arrangement of components to achieve the same functionality is effectively “associated” such that the desired functionality is

achieved. Hence, any two components herein combined to achieve a particular functionality can be seen as “associated with” each other such that the desired functionality is achieved, irrespective of architectures or intermedial components. Likewise, any two components so associated can also be viewed as being “operably connected,” or “operably coupled,” to each other to achieve the desired functionality, and any two components capable of being so associated can also be viewed as being “operably couplable,” to each other to achieve the desired functionality. Specific examples of operably couplable include but are not limited to physically mateable and/or physically interacting components, and/or wirelessly interactable, and/or wirelessly interacting components, and/or logically interacting, and/or logically interactable components.

(603) Some aspects may be described using the expression “coupled” and “connected” along with their derivatives. It should be understood that these terms are not intended as synonyms for each other. For example, some aspects may be described using the term “connected” to indicate that two or more elements are in direct physical or electrical contact with each other. In another example, some aspects may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, also may mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

(604) In some instances, one or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

(605) While particular aspects of the present subject matter described herein have been shown and described, it will be apparent to those skilled in the art that, based upon the teachings herein, changes and modifications may be made without departing from the subject matter described herein and its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications as are within the true scope of the subject matter described herein. It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that when a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

(606) In addition, even when a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together,

and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

(607) With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flows are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simultaneous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

(608) In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more embodiments has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more embodiments were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

Claims

1. A surgical stapling assembly, comprising: a shaft; an articulation joint; an electrical connector; and an end effector attached to a distal end of the shaft, wherein the end effector is articulatable relative to the shaft, and wherein the shaft comprises: a first jaw; a second jaw movable relative to the first jaw; and a replaceable staple cartridge assembly, comprising: a cartridge body comprising a proximal end; a plurality of staples; and electrical contacts positioned at the proximal end of the cartridge body and electrically coupleable to the electrical connector upon installation of the replaceable staple cartridge assembly in the end effector and electrically decouplable from the electrical connector upon removal of the replaceable staple cartridge assembly from the end effector, wherein the electrical contacts are positioned on a proximal face of the cartridge body adjacent a longitudinal slot of the cartridge body, and wherein the electrical connector faces distally toward the proximal face of the cartridge body.
2. The surgical stapling assembly of claim 1, wherein the electrical connector comprises a flex cable attached to a sidewall of the first jaw.
3. The surgical stapling assembly of claim 2, wherein the flex cable comprises a coiled section extending across the articulation joint.
4. The surgical stapling assembly of claim 1, wherein the electrical connector is positioned distal to a pivot about which the second jaw is movable relative to the first jaw.
5. The surgical stapling assembly of claim 1, wherein the replaceable staple cartridge assembly further comprises a cartridge nose and a distal sensor plug positioned within the cartridge nose, wherein the cartridge nose comprises a distal-facing surface comprising distal electrical contacts electrically coupled with the electrical contacts positioned at the proximal end of the cartridge

body, and wherein the distal sensor plug is in electrical communication with the distal electrical contacts.

6. The surgical stapling assembly of claim 5, wherein the distal sensor plug comprises a sensor and a processor.

7. A surgical stapling assembly, comprising: a shaft; and an end effector attached to the shaft, wherein the end effector comprises: a first jaw; a second jaw movable relative to the first jaw; a channel; an electrical connector; and an anvil comprising a proximal end, wherein the proximal end comprises downwardly-extending flanges, wherein each downwardly-extending flange comprises a distal end, and wherein the electrical connector is proximal to the distal ends of the flanges; a replaceable staple cartridge assembly configured to be installed into the channel, wherein the replaceable staple cartridge assembly comprises: a cartridge body, comprising: a deck surface; a proximal cartridge body end; and a plurality of staple cavities defined in the deck surface; and electrical contacts positioned at the proximal cartridge body end of the cartridge body and electrically coupleable to the electrical connector upon installation of the replaceable staple cartridge assembly into the channel and electrically decoupleable from the electrical connector upon removal of the replaceable staple cartridge assembly from the channel, wherein the electrical connector comprises a flex cable attached to a sidewall of the channel, and wherein the flex cable comprises a coiled section.

8. The surgical stapling assembly of claim 7, wherein the electrical contacts are positioned on a proximal face of the cartridge body adjacent a longitudinal slot of the cartridge body, and wherein the electrical connector faces distally toward the proximal face of the cartridge body.

9. The surgical stapling assembly of claim 7, wherein the electrical connector is positioned distal to a pivot about which the second jaw is movable relative to the first jaw.

10. The surgical stapling assembly of claim 7, wherein the replaceable staple cartridge assembly further comprises a cartridge nose and a distal sensor plug positioned within the cartridge nose, wherein the cartridge nose comprises a distal-facing surface comprising distal electrical contacts electrically coupled with the electrical contacts positioned at the proximal cartridge body end of the cartridge body, and wherein the distal sensor plug is in electrical communication with the distal electrical contacts.

11. The surgical stapling assembly of claim 10, wherein the distal sensor plug comprises a sensor and a processor.

12. A replaceable staple cartridge assembly for use with a surgical stapling end effector having a cartridge channel and an electrical connector, wherein the replaceable staple cartridge assembly is configured to be installed into the cartridge channel, the replaceable staple cartridge assembly comprising: a cartridge body comprising: a proximal end; a plurality of staples; and electrical contacts positioned at the proximal end of the cartridge body and electrically coupleable to the electrical connector of the end effector upon installation of the replaceable staple cartridge assembly into the cartridge channel of the end effector and electrically decoupleable from the electrical connector upon removal of the replaceable staple cartridge assembly from the cartridge channel, wherein the replaceable staple cartridge assembly further comprises a cartridge nose and a distal sensor plug positioned within the cartridge nose, wherein the cartridge nose comprises a distal-facing surface comprising distal electrical contacts electrically coupled with the electrical contacts positioned at the proximal end of the cartridge body, and wherein the distal sensor plug is in electrical communication with the distal electrical contacts.

13. The replaceable staple cartridge assembly of claim 12, wherein the electrical contacts are positioned on a proximal face of the cartridge body adjacent a longitudinal slot of the cartridge body, and wherein the electrical connector faces distally toward the proximal face of the cartridge body.

14. The replaceable staple cartridge assembly of claim 12, wherein the distal sensor plug comprises a sensor and a processor.

